

Hong Kong Institute of Information Technology (HKIIT)

HD in Software engineering

Requirement Specification Report

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All names, contact details, addresses, credentials, and business records shown in this report are fictional demonstration data.

Abstract

Smile & Sunshine Toy Co., Ltd., established in 2008 and headquartered in Hong Kong, is a global leader in the Original Design Manufacturing (ODM) toy industry. The company is renowned for delivering creative designs and high-quality manufacturing solutions to some of the world's most recognized toy brands. With state-of-the-art manufacturing facilities in mainland China, Vietnam, and Thailand, Smile & Sunshine Toy optimizes production efficiency, reduces costs, and ensures timely shipments to customers worldwide.

Currently, Smile & Sunshine Toy is transitioning from ODM to Original Brand Manufacturing (OBM) to directly engage with consumers, enhance its market competitiveness, and elevate brand value. This strategic shift includes the streamlining of its computer systems, which are built using C# and MySQL, and integrated with VX to optimize operations.

The company's production process encompasses both in-house designs and customer-requested products, with a collaborative effort across various departments, from design to delivery. In addition, Smile & Sunshine Toy has developed a robust distribution and after-sales service structure to guarantee customer satisfaction.

Through these initiatives, the company aims for sustainable growth and intends to retain its strong position within the toy industry. Smile & Sunshine Toy's commitment to innovation, quality, and market insights, coupled with its ongoing expansion, positions it to capitalize on emerging trends in the global toy market. With a projected compound annual growth rate (CAGR) of 1.1% from 2025 to 2030, the global toy market is expected to reach a size of \$94 billion by 2030. By focusing on consumer needs and leveraging its vast experience and resources, Smile & Sunshine Toy is well-equipped to strengthen its reputation as a trusted and innovative leader in the industry.

Table of Content

Abstract	1
Table of Content	2
1 Introduction.....	1
1.1 Company Background	1
1.2 Project Background.....	1
1.3 Assumption	2
2 Problem Findings.....	4
2.1 User Login	4
2.1.1 Problem Description	4
2.1.2 Role Management	4
2.1.3 Product Storage Status	4
2.1.4 Product Logistics Query	5
2.2 E - commerce - related Problems.....	5
2.2.1 Shopping Cart	5
2.2.2 Payment System.....	6
2.2.3 Customer Customization Order	6
2.2.4 Customer Relationship Management (CRM)	6
2.2.5 Quality Control	7
2.2.6 Communication and Collaboration	7
3 Functional Requirements	9
3.1 User Login System.....	9
3.1.1 Secure Authentication.....	9
3.1.2 Role - Based Access Control (RBAC).....	9
3.1.3 Single Sign - On (SSO).....	9
3.2 Real - Time Inventory Updates.....	10
3.3 Inventory Visibility	10
3.4 Batch and Expiry Tracking	11
3.5 Product Logistics Query	11
3.5.1 Shipment Tracking.....	11

3.5.2	Logistics Integration	11
3.5.3	Customer Notifications	12
3.6	Shopping Cart Management	12
3.6.1	User - Friendly Interface	12
3.6.2	Cart Persistence.....	13
3.7	Promotions and Discounts	13
3.8	Customer Customization Order Handling	14
3.8.1	Custom Order Management	14
3.9	Design Approval	14
3.10	Payment Management.....	15
3.10.1	Secure Payment Processing	15
3.10.2	Invoice Generation.....	15
3.11	Inventory Management Enhancement	15
3.11.1	Automated Reordering.....	15
3.11.2	Supplier Management	16
3.11.3	Stock Audits.....	16
3.12	Quality Control Monitoring	17
3.12.1	Quality Metrics Tracking	17
3.12.2	Non - Conformance Management.....	17
3.12.3	Supplier Quality Assessment	17
3.13	Communication and Collaboration Platform	18
3.13.1	Centralized Communication	18
3.13.2	Instant Messaging	18
3.14	Order Tracking.....	19
3.14.1	Real - Time Updates	19
3.15	Customer Relationship Management (CRM)	19
3.15.1	Customer Segmentation	19
4	Non-Functional Requirements	20
4.1	Security	20
4.2	Performance	20
4.3	Usability	20
4.4	Reliability.....	20
4.5	Scalability	21
4.6	Interoperability.....	21
4.7	Maintainability	21
4.8	Compliance	22
4.9	Cost - Effectiveness	22
4.10	Flexibility.....	22
5	Software and hardware	23
5.1.1	Computer.....	23
5.1.2	Server	24
5.1.3	Core Switch.....	26
5.1.4	AccessSwitch	27
5.1.5	windows 11	28
5.1.6	Visual Studio 2019.....	29
5.1.7	360Qihoo.....	30
5.1.8	Ubuntu Server	31
5.1.9	UPS	32
5.1.10	Order Overview	33
6	Network topology diagram	34

7	Benefits of a Centralized Computerized Management System	36
7.1	Data Security.....	36
7.2	Inventory Management	36
7.3	Logistics Visibility.....	36
7.4	Customer Experience	36
7.5	E - commerce Trust.....	37
7.6	Communication and Collaboration	37
7.7	Decision - making Insights	37
7.8	Customer Relationships	37
8	Constraints and Limitations	38
8.1	Initial Cost and Investment:	38
8.1.1	High Implementation Cost:.....	38
8.1.2	Training Costs:.....	38
8.2	Complexity:.....	38
8.2.1	System Complexity:.....	38
8.2.2	Integration Challenges:	38
8.3	Data Security and Privacy:.....	38
8.3.1	Security Risks:	38
8.3.2	Privacy Concerns:	38
8.4	Scalability:	38
8.4.1	Scalability Issues:.....	38
8.4.2	Performance Bottlenecks:	38
8.5	Dependence on Technology:.....	39
8.5.1	System Downtime:.....	39
8.5.2	Reliance on IT Support:	39
8.6	User Resistance:.....	39
8.6.1	Resistance to Change:	39
8.6.2	Learning Curve:	39
8.7	Customization and Flexibility:.....	39
8.7.1	Limited Customization:	39
8.7.2	Flexibility Constraints:.....	39
8.8	Maintenance and Upgrades:.....	39
8.8.1	Ongoing Maintenance:.....	39
8.8.2	Upgrade Challenges:	39
8.9	Compatibility Issues.....	40
8.9.1	Software Compatibility:.....	40
8.9.2	Hardware Compatibility:	40
8.10	Regulatory and Compliance Changes	40
8.10.1	Industry - Specific Regulations:.....	40
8.10.2	International Compliance Variations:	40
9	Actor Description.....	41
10	Use case diagram and description.....	44
10.1	Customer service and support system.....	44
10.2	Order and payment management system	49
10.3	Order processing system.	53
10.4	Order system	57
10.5	Procurement system.	61
10.6	Production management system	65
10.7	Research & development system	69
11	Class Diagram.....	73

11.1	Customer service and support system.....	74
11.2	Order and payment management system	75
11.3	Order processing system	76
11.4	Order system	77
11.5	Procurement system	78
11.6	Production Management System	79
11.7	Research & Development system	80
12	Sequence Diagram	81
12.1	System Level Sequence Diagram for Customer service and support	81
12.1.1	System Level Sequence Diagram for Customer request service	81
12.1.2	System Level Sequence Diagram for Customer service Agent Update case details	82
12.1.3	System Level Sequence Diagram for Customer receive feedback	83
12.1.4	System Level Sequence Diagram for Technician enter repair details	84
12.2	System Level Sequence Diagram for Order and payment management	85
12.2.1	System Level Sequence Diagram for System Administrator enter product details	85
12.2.2	System Level Sequence Diagram for Customer submit order.....	86
12.2.3	System Level Sequence Diagram for Customer submit payment	87
12.2.4	System Level Sequence Diagram for Sales Manager submit invoice	88
12.3	System Level Sequence Diagram for Order processing	89
12.3.1	System Level Sequence Diagram for Customer submit order.....	89
12.3.2	System Level Sequence Diagram for Customer submit payment	90
12.3.3	System Level Sequence Diagram for Shipping Clerk	91
12.3.4	System Level Sequence Diagram for Delivery Personnel	92
12.4	System Level Sequence Diagram for Order system	93
12.4.1	System Level Sequence Diagram for System Administrator update product details	93
12.4.2	System Level Sequence Diagram for Customer submit order.....	94
12.4.3	System Level Sequence Diagram for Sales Manager confirm Order	95
12.4.4	System Level Sequence Diagram for Customer submit payment	96
12.5	System Level Sequence Diagram for Procurement system	97
12.5.1	System Level Sequence Diagram for Procurement Officer submit purchase order	97
12.5.2	System Level Sequence Diagram for Material Manager update material information	98
12.5.3	System Level Sequence Diagram for Inventory Manager update inventory record	99
12.5.4	System Level Sequence Diagram for Procurement Officer manage supplier information	100
12.6	System Level Sequence Diagram for Production Management	101
12.6.1	System Level Sequence Diagram for Production Manager submit production plan	101
12.6.2	System Level Sequence Diagram for Production Engineer define process details	102
12.6.3	System Level Sequence Diagram for Inventory Manager update inventory record	103
12.6.4	System Level Sequence Diagram for Quality Inspector submit inspection report	104
12.7	System Level Sequence Diagram for Research & Development	105

12.7.1	System Level Sequence Diagram for Project Manager submit project timeline	105
12.7.2	System Level Sequence Diagram for Market Researcher submit marketing report	106
12.7.3	System Level Sequence Diagram for Prototype Designer submit draft prototype	107
12.7.4	System Level Sequence Diagram for Project Manager production approval	108
13	3-Tier sequence diagrams	109
13.1	3-Tier Sequence Diagram for Customer service and support	109
13.1.1	3-Tier Sequence Diagram for Customer request service	109
13.1.2	3-Tier Sequence Diagram for Customer service Agent Update case details	110
13.1.3	3-Tier Sequence Diagram for Customer receive feedback	111
13.1.4	3-Tier Sequence Diagram for Technician enter repair details	112
13.2	3-Tier Sequence Diagram for Order and payment management	113
13.2.1	3-Tier Sequence Diagram for System Administrator enter product details	113
13.2.2	3-Tier Sequence Diagram for Customer submit order	114
13.2.3	3-Tier Sequence Diagram for Customer submit payment	115
13.2.4	3-Tier Sequence Diagram for Sales Manager submit invoice	116
13.3	3-Tier Sequence Diagram for Order processing	117
13.3.1	3-Tier Sequence Diagram for Customer submit order	117
13.3.2	3-Tier Sequence Diagram for Customer submit payment	118
13.3.3	3-Tier Sequence Diagram for Shipping Clerk	119
13.3.4	3-Tier Sequence Diagram for Delivery Personnel	120
13.4	3-Tier Sequence Diagram for Order system	121
13.4.1	3-Tier Sequence Diagram for System Administrator update product details	121
13.4.2	3-Tier Sequence Diagram for Customer submit order	122
13.4.3	3-Tier Sequence Diagram for Sales Manager confirm Order	123
13.4.4	3-Tier Sequence Diagram for Customer submit payment	124
13.5	3-Tier Sequence Diagram for Procurement system	125
13.5.1	3-Tier Sequence Diagram for Procurement Officer submit purchase order	125
13.5.2	3-Tier Sequence Diagram for Material Manager update material information	126
13.5.3	3-Tier Sequence Diagram for Inventory Manager update inventory record	127
13.5.4	3-Tier Sequence Diagram for Procurement Officer managerial information	128
13.6	3-Tier Sequence Diagram for Production Management	129
13.6.1	3-Tier Sequence Diagram for Production Manager submit production plan	129
13.6.2	3-Tier Sequence Diagram for Production Engineer define process details	130
13.6.3	3-Tier Sequence Diagram for Inventory Manager update inventory record	131
13.6.4	3-Tier Sequence Diagram for Quality Inspector submit inspection report	132
13.7	3-Tier Sequence Diagram for Research & Development	133
13.7.1	3-Tier Sequence Diagram for Project Manager submit project timeline	133

13.7.2	3-Tier Sequence Diagram for Market Researcher submit marketing report	134
13.7.3	3-Tier Sequence Diagram for Prototype Designer submit draft prototype	135
13.7.4	3-Tier Sequence Diagram for Project Manager production approval.....	136
14	Entity Relation Diagram (ERD)	137
14.1	department.....	138
14.2	user	139
14.3	role	140
14.4	permission.....	141
14.5	user_role.....	141
14.6	role_permission.....	142
14.7	supplier.....	143
14.8	material	144
14.9	procurement_order	145
14.10	inventory_record	146
14.11	design_concept.....	147
14.12	prototype	148
14.13	prototype_comment	149
14.14	product	150
14.15	material_of_product.....	152
14.16	production_plan	153
14.17	production_process	154
14.18	prerequisite_of_process	155
14.19	customer	156
14.20	sales_order	157
14.21	order_item	159
14.22	payment.....	160
14.23	shipping_method.....	161
14.24	shipping_record.....	162
14.25	customer_service_request.....	163
14.26	technical_support	164
14.27	warranty_claim	165
14.28	refund_return_request.....	166
15	User Interface and Report Design.....	167
15.1	Login Page	167
15.2	Dashboard Page	168
15.3	Order Management Page.....	169
15.4	Edit Order Page.....	170
15.5	Save Order PDF Page	171
15.6	Product Manage Page	172
15.7	Add Product Page	173
15.8	Edit Product Page.....	174
15.9	Delete Product Page.....	174
15.10	Edit Product Page.....	176
15.11	Confirm Product Image Migration Page.....	177
15.12	Customer Manage Page	178
15.13	Customer Add Customer Page.....	179
15.14	Customer Edit Customer Page.....	180
15.15	Customer Confirm Delete Page	181
15.16	Customer Manage Search Page	182

15.17	Department Manage Page	183
15.18	Department Add Department Page	184
15.19	Department Edit Department Page	185
15.20	Department Confirm Delete Page	186
15.21	Department Refresh Page	187
15.22	Role Manage Page	188
15.23	Role Add Role Page.....	189
15.24	Role Modify Role Page.....	190
15.25	Role Confirm Delete Page	191
15.26	Role Of Department Navigation Menu Page	192
15.27	User Manage Page	193
15.28	Add User Page	194
15.29	Edit User Page.....	195
15.30	Confirm deletion Page	196
15.31	Confirm Password Reset Page	197
15.32	Permission Manage Page	198
16	TeseCase	199
17	User Guide	254
17.1	How to Log In to Your Account	254
17.2	Admin System Actions: Logging Out, Editing Orders, and Exporting PDFs ..	255
17.3	How to Add a New Product to the System	256
17.4	How to Edit Product Information in the System.....	257
17.5	Deleting a Product: Steps and Confirmation	258
17.6	How to Refresh the Product List and Update Information	259
17.7	How to Add a New Customer: Step-by-Step Guide	260
17.8	Customer Management Operations: Search, Add, Edit, and Delete	261
17.9	How to Add a New Department: Step-by-Step Guide.....	262
17.10	How to Edit Department Details: Step-by-Step Guide	263
17.11	Department Management: Refreshing, Logging Out, and Deleting Departments 264	
17.12	How to Refresh Department Data and Confirm Updates	265
17.13	Creating a New Role: Step-by-Step Guide	266
17.14	How to Update Role Information and Permissions: Step-by-Step Guide	267
17.15	Role Management: Deleting a Role and Logging Out.....	268
17.16	How to View and Manage Roles by Department	269
17.17	How to View and Manage Users by Department	270
17.18	How to Add a New User Account: Step-by-Step Guide.....	271
17.19	How to Edit an Existing User Account: Step-by-Step Guide	272
17.20	How to Delete a User Account: Steps and Confirmation	274
17.21	How to Reset a User's Password: Steps and Confirmation	275
17.22	Accessing Permission Management in the System Menu	276
17.23	How to Add a New Permission in the System.....	277
18	Projec Schedule	278
19	Conclusion	278
20	Project Log.....	280
21	Appendices	280

1 Introduction

1.1 Company Background

Smile & Sunshine Toy Co., Ltd., a globally recognized leader in the Original Design Manufacturing (ODM) toy sector, has been steadfastly dedicated to delivering innovative designs and premier manufacturing solutions to esteemed toy brands worldwide since its inception in 2008. The company's influence extends beyond mere design and production, as it plays a pivotal role in shaping trends and meeting the ever-evolving demands of the global toy market. With a strong presence in regions like mainland China, Vietnam, and Thailand, Smile & Sunshine Toy has established a robust network of state-of-the-art manufacturing facilities. These facilities not only enhance production efficiency and reduce costs but also ensure timely shipments to customers across the globe, solidifying the company's reputation as a reliable and efficient partner in the industry.

Despite its success in the ODM sector, Smile & Sunshine Toy recognizes the need to adapt to the changing dynamics of the market. To achieve long-term growth and sustainability, the company has embarked on a strategic transformation from ODM to Original Brand Manufacturing (OBM). This shift is not merely a change in business model but a profound evolution that aims to position the company as a direct and active participant in the consumer marketplace. By establishing its own brand identity, Smile & Sunshine Toy intends to build a stronger emotional connection with its customers, gain a deeper understanding of their preferences, and ultimately drive market competitiveness.

1.2 Project Background

The transition from ODM to OBM is a complex and multifaceted process that requires a comprehensive overhaul of the company's existing systems and operations. To support this strategic transformation and enhance its overall operational efficiency and competitiveness, Smile & Sunshine Toy has initiated a series of system development projects. These initiatives cover a wide array of critical business areas, each designed to address specific challenges and opportunities that arise during the transition.

The user login system, for instance, is being enhanced to provide a seamless and secure access point for employees, customers, and partners. This system will not only streamline the authentication process but also ensure the protection of sensitive data and user privacy. In terms of product storage status, the company is developing a sophisticated inventory management system that provides real-time visibility into stock levels, locations, and movement. This will enable more accurate demand forecasting, reduce stockouts, and optimize warehouse operations. For product logistics query, a user-friendly interface is being created to allow customers and internal stakeholders to track shipments in real time, enhancing transparency and customer satisfaction.

The shopping cart and customer customization order features are being integrated into the company's e-commerce platform to offer a more personalized and intuitive shopping experience. Customers will be able to customize products according to their preferences,

which is a key differentiator in the OBM model. The payment system is being upgraded to support multiple payment methods and ensure secure transactions, while maintaining compliance with international financial regulations. Inventory management systems are being refined to better synchronize with production schedules and sales forecasts, minimizing excess inventory and reducing carrying costs.

Quality control processes are being strengthened through the implementation of advanced tracking and testing systems. This will ensure that all products meet the highest standards of safety and durability, which is crucial for maintaining consumer trust and brand reputation. Communication and collaboration tools are being introduced to facilitate seamless interaction between different departments and teams, both internally and externally. This will enhance the company's ability to respond quickly to market changes and customer feedback.

Order tracking systems are being enhanced to provide detailed updates on the status of orders, from initial placement to final delivery. This will improve customer satisfaction by keeping them informed throughout the process and ensuring that any issues are resolved promptly. Data analytics capabilities are being expanded to extract valuable insights from vast amounts of operational and customer data. These insights will inform strategic decision-making, drive innovation, and optimize business processes. Lastly, the Customer Relationship Management (CRM) system is being overhauled to provide a centralized platform for managing customer interactions, preferences, and feedback. This will enable the company to deliver more personalized and targeted marketing efforts, fostering stronger customer loyalty.

1.3 Assumption

It is assumed that the successful implementation of these system development initiatives will be a catalyst for Smile & Sunshine Toy's strategic transformation into the OBM market. By enhancing data security, the company will protect its intellectual property and customer information, which is essential for maintaining trust and credibility in the digital age. The optimization of inventory management will lead to more efficient use of resources, reduced waste, and improved profitability. Enhanced logistics visibility will enable the company to respond more effectively to supply chain disruptions and ensure timely deliveries. By offering a superior customer shopping experience, Smile & Sunshine Toy aims to differentiate itself from competitors and attract a loyal customer base.

The ability to better meet customer needs through personalized customization options will foster deeper customer relationships and increase customer satisfaction. Trust in the e-commerce platform will be strengthened through secure transactions and transparent communications, which is vital for success in the digital marketplace. Improved inventory management efficiency will allow the company to react more quickly to market trends and consumer demands, while maintaining high-quality standards will ensure customer loyalty and positive brand perception.

Inter-departmental communication and collaboration will be bolstered by the new systems, leading to more cohesive and effective teamwork, which is essential for the smooth

execution of complex projects and the rapid resolution of issues. Increased order fulfillment transparency will reduce customer frustration and inquiries, streamlining customer service operations. Valuable insights generated by data analytics will empower the company to make informed decisions, identify new opportunities, and mitigate risks. Lastly, the enhanced CRM system will enable more effective customer relationship management, leading to higher customer retention rates and a stronger brand presence in the market.

In summary, the implementation of these system development initiatives is expected to be a transformative force for Smile & Sunshine Toy Co., Ltd., enabling it to achieve its long-term growth objectives and establish a strong foothold in the OBM market. By leveraging cutting-edge technology and innovative business practices, the company is poised to redefine its role in the global toy industry and continue its legacy of excellence.

2 Problem Findings

2.1 User Login

2.1.1 Problem Description

The existing user login system has a significant security flaw as it fails to provide a secure mechanism that can effectively differentiate between departments, employees, and customers. This deficiency means that unauthorized users may gain access to sensitive information and functionalities beyond their designated roles. For example, a customer might inadvertently access employees - only dashboards, which could lead to potential data breaches and operational disruptions.

Solutions

To address the user login issue, a more secure authentication mechanism should be implemented. This could involve the use of multi-factor authentication, such as requiring a password and a verification code sent to the user's registered email or phone number. Additionally, different user groups should be assigned unique login credentials and access levels. For instance, employees could have a separate login portal from customers, and their access rights should be strictly limited to the areas relevant to their job functions. Furthermore, the system should use SSL/TLS protocols to encrypt data transmission and prevent sensitive information from being intercepted during the login process.

2.1.2 Role Management

Problem Description

Without clear role definitions, ordinary employees may be able to modify critical system configurations, which may cause potential system failures and compromise the integrity of the system.

Solution

A comprehensive role-based access control (RBAC) system should be established to address role management issues. The system should clearly define the permissions of each user group. For example, administrators should have the highest level of access, including the ability to manage user accounts and system settings. Managers can access certain reports and employee performance data, while ordinary employees should only have access to the tools and information required for their daily tasks. And user permissions should be audited regularly to ensure that the system remains secure and users cannot exceed their authorized access levels. The system should also implement strict authentication and authorization processes to prevent unauthorized access and employee permissions modification.

2.1.3 Product Storage Status

Problem Description

If the system fails to update inventory levels after a large sale, a company may mistakenly believe that it has sufficient inventory, only to find out later that it needs to

restock urgently. This can lead to either out-of-stock situations, where customers cannot buy the products they want, or over-stock situations, where the company has too much inventory, tying up capital and increasing warehousing costs.

Solutions

To address the product storage status issue, an advanced inventory management system needs to be implemented to track inventory levels in real time. The system uses technologies such as IoT sensors and RFID tags to continuously monitor inventory levels and update the system in real time. The system should have features such as automatic alerts when inventory is low, notifying the purchasing department to reorder the product before it sells out. Inventory audits should be conducted regularly to ensure the accuracy of inventory records and identify any discrepancies. These measures can prevent out-of-stocks and overstocking, maintain optimal inventory levels, and effectively meet customer needs.

2.1.4 Product Logistics Query

Problem Description

If customers cannot track the status of their order, they may contact customer service multiple times to get updates. This not only increases the workload of the customer service team, but also leads to increased operating costs.

Solution

In order to solve the problem of product logistics inquiry, a logistics management system should be made. The system should provide real-time tracking of shipments, allowing users to easily check the status of their orders. The system can also send automatic updates to customers, reducing their need to contact customer service for updates. If possible, use technologies such as GPS and telematics for better route planning and tracking. This will make the logistics process more efficient and reduce operating costs.

2.2 E - commerce - related Problems

2.2.1 Shopping Cart

Problem description

If customers cannot easily modify the quantity of items in their shopping cart, they may simply abandon the purchase. This results in lost sales and customers will question the company's ability to pay attention to small details

Solution

To solve the problems of the shopping cart system, the following measures should be taken. Design the shopping cart for the average customer to make it more intuitive and user-friendly. It must be simple and straightforward, and add features such as drag-and-drop functionality for easy modification of items. Then simplify the checkout process, reduce the number of steps, and let customers pay easily. In addition, customer shopping information needs to be saved for future purchases.

2.2.2 Payment System

Problem Description

Payment methods should be expanded. If some users can only support one payment method, they will face the situation of being unable to pay, which will not only lose the sale of an item, but also face the permanent loss of a customer

Solution

Expand payment method support: Integrate multiple payment methods into the payment system, such as credit cards, debit cards, digital wallets, and online banking. This will provide customers with more convenient payment options and improve the overall user experience. This will allow customers to choose their favorite payment method during the checkout process, thereby reducing the possibility of abandoning the shopping cart due to payment method restrictions.

2.2.3 Customer Customization Order

Problem description

When customers request customized toys with specific functions, relying solely on language instructions may lead to inefficiency and difficulty in understanding. At the same time, if there are no restrictions, the toys that need to be customized may be too complex, and the misunderstood customized toys may not meet customer needs, which may lead to customer loss

Solution

Create an online platform where customers can submit customized toy requests in detail. The system should allow customers to specify functions, materials, colors, and any other relevant details. Include as many options as possible to reduce the difficulty of toy customization

2.2.4 Customer Relationship Management (CRM)

Problem Description

When a customer has a complaint about a product, the customer service team may not have a clear record of the interaction. Lack of a proper CRM system can lead to poor customer experience. Without a centralized system to manage customer information and interactions, it is difficult for companies to provide consistent and high-quality service.

Solution

Customer Information Management: The system should be able to collect and store detailed customer contact details, purchase history, and communication records. This will provide a 360-degree view of each customer, allowing companies to better understand their needs and preferences. The system should track all customer interactions, including phone calls, emails, and online chats. This will ensure that the customer service team has a clear record of each interaction and can provide

consistent and personalized service. There needs to be a dedicated module to manage customer queries and complaints. The module should automatically assign queries and complaints to the relevant department or person and track their resolution progress. The system should provide various reports and analytics on customer interactions, feedback, and after-sales service support. This will help companies monitor the performance of their customer service team and identify areas for improvement.

2.2.5 Quality Control

Problem Description

If the quality of a certain product does not meet the standards during the production process, the system should alert the quality control department to take corrective measures. If unqualified products flow to customers, it will be a huge problem.

Solution

Establish a quality control system that can monitor and report the quality of materials and products at the production stage. Import the color, appearance and weight of the inspected products into the system in advance, and then manually screen the data of the toys that are about to be completed. If the data does not match the initial data, a warning will be issued.

2.2.6 Communication and Collaboration

- **Problem Description**

Communication and collaboration between departments are not effective due to the lack of a centralized communication platform. This situation can lead to delays in project execution and a failure to respond quickly to market changes. For example, if different departments are unable to easily share documents and track project progress, the overall efficiency of the organization may be significantly reduced. Without a centralized platform, information silos may form, making it difficult for teams to access the latest updates and collaborate effectively. This can result in duplicated efforts, missed deadlines, and a lack of alignment on project goals.

- **Solutions**

To improve communication and collaboration between departments, the following solutions can be implemented:

- **Establish a Centralized Communication Platform:** Implement a unified communication platform that allows all departments to easily share information, documents, and project updates. This platform can include features such as instant messaging, video conferencing, and shared document repositories.
- **Integrate Communication Tools:** Integrate various communication tools into the centralized platform to ensure seamless communication across different

departments. This can include email, chat applications, and project management software.

- **Define Clear Communication Protocols:** Establish clear protocols for communication and collaboration, including guidelines for sharing information, responding to queries, and updating project status. This will help ensure that all team members are on the same page and that communication is consistent and efficient.
- **Provide Training and Support:** Offer training sessions and ongoing support to help employees effectively use the centralized communication platform. This will ensure that all team members are comfortable with the tools and can leverage them to their full potential.
- **Monitor and Optimize:** Regularly monitor the effectiveness of the communication platform and adjust as needed to optimize its functionality. This can involve gathering feedback from users, tracking key performance indicators, and implementing updates to improve the user experience.
- **Data Analytics:** The company does not fully utilize data analytics to gain insights into production efficiency, sales trends, and customer preferences. This results in missed opportunities for informed strategic decisions. For example, if the company is unable to analyze sales data to identify the best-selling products, it may miss out on opportunities to optimize its product offerings and marketing strategies.

3 Functional Requirements

3.1 User Login System

3.1.1 Secure Authentication

- **Multi - Factor Authentication (MFA):** The system should support MFA, requiring users to provide two or more verification factors to authenticate their identity. For example, in addition to a password, users may need to enter a verification code sent to their registered email or phone number. This adds an extra layer of security, making it more difficult for unauthorized users to gain access.
- **Role - Based Authentication:** The system should differentiate between departments, employees, and customers during the login process. For instance, an admin user might be required to enter a username, password, and a security question answer, while a customer may only need to enter a username and password. This ensures that users only access information and functionalities relevant to their roles.

3.1.2 Role - Based Access Control (RBAC)

- **Role Creation and Management:** Administrators should be able to create and manage user roles within the system. For example, they can define a "Manager" role with access to certain reports and a "Sales Staff" role with access to customer information and order processing. Each role should have specific permissions assigned to it, ensuring that users have the appropriate level of access.
- **Permission Assignment:** The system should allow administrators to assign permissions to each role. For example, the "Manager" role may have permission to view and edit employee records, while the "Sales Staff" role may only have permission to view customer contact information. This granular control over permission helps to prevent unauthorized access and data breaches.
- **Role Hierarchy:** The system should support a role hierarchy, where higher - level roles inherit the permissions of lower - level roles. For example, an "Admin" role may have all the permissions of the "Manager" and "Sales Staff" roles, in addition to system - wide administrative privileges. This simplifies permission management and ensures that users have the necessary access to perform their duties.

3.1.3 Single Sign - On (SSO)

- **Centralized Authentication:** The system should provide a centralized authentication platform that allows users to log in once and access multiple services without re-entering their credentials. For example, a user who logs into the main system should be able to access the inventory management module, the order tracking system, and the customer relationship management (CRM) system without having to log in again. This improves user convenience and reduces the risk of password fatigue.
- **Single Logout:** The system should also support single logout, allowing users to log out of all services with a single action. This ensures that when a user ends their session, they are logged out of all integrated systems, enhancing security and preventing unauthorized access.

3.2 Real - Time Inventory Updates

- **Automatic Inventory Adjustment:** The system should automatically update inventory levels when products are added or removed from the storage locations. For example, when a product is sold, the system should decrease the inventory count for that product in real - time. Similarly, when new stock is received, the system should increase the inventory count. This ensures that the inventory records are always accurate and up to date.
- **Low Stock Alerts:** The system should monitor inventory levels and send alerts when stock levels fall below a predefined threshold. For example, if a certain product's inventory level drops below 10 units, the system should send an alert to the procurement department, prompting them to act and replenish the stock. This helps to prevent stockouts and ensures that the company can meet customer demand.
- **Discrepancy Detection:** The system should also detect discrepancies between the physical inventory and the recorded inventory levels. For example, if a product is recorded as having 50 units in stock, but a physical count shows only 45 units, the system should flag this discrepancy and notify the relevant department for investigation. This helps to maintain the accuracy of inventory records and prevent potential losses.

3.3 Inventory Visibility

- **Comprehensive Inventory View:** The system should provide a comprehensive view of inventory across all storage locations, including warehouses and distribution centers. For example, a user should be able to see the total inventory level of a product, as well as the inventory levels at each specific location. This allows users to quickly identify the availability of products and make informed decisions about stock allocation and replenishment.
- **Location - Based Inventory Management:** The system should allow users to manage inventory at different locations separately. For example, a user can set different safety stock levels for a product at different warehouses based on the demand and lead times at each location. This enables more precise inventory management and reduces the risk of overstocking or understocking.
- **Inventory Transfer Tracking:** The system should track inventory transfers between different storage locations. For example, when a product is moved from one warehouse to another, the system should record the transfer and update the inventory levels at both locations accordingly. This ensures that the inventory records are accurate and provide visibility into the movement of products within the company.

3.4 Batch and Expiry Tracking

- **Batch Number Management:** For products with batch numbers, the system should track and manage these details. For example, when a product is received into inventory, the system should record the batch number and associate it with the product. This allows users to track the production date, expiration date, and other relevant information for each batch of products.
- **Expire Date Tracking:** The system should monitor the expiry dates of products and send alerts when products are approaching their expiration dates. For example, if a product's expiry date is within 30 days, the system should notify the sales department to prioritize selling this product to avoid waste. This helps to ensure product quality and reduces losses due to expired products.
- **First - In - First - Out (FIFO) Management:** The system should enforce the FIFO principle for products with expiry dates. For example, when a product is sold, the system should automatically select the oldest batch of products to fulfill the order. This ensures that the oldest stock is used first, reducing waste and maintaining product quality.

3.5 Product Logistics Query

3.5.1 Shipment Tracking

- **Real - Time Location Updates:** The system should provide real - time updates on the location and status of shipments. For example, a user should be able to see the current location of a shipment on a map, as well as its estimated time of arrival. This allows users to track the progress of their orders and plan accordingly.
- **Delay and Issue Notification:** The system should notify users of any delays or issues with their shipments. For example, if a shipment is delayed due to bad weather, the system should send a notification to the customer, providing an estimated time of arrival and any alternative arrangements. This helps to manage customer expectations and maintain customer satisfaction.
- **Shipment History:** The system should maintain a history of all shipments, allowing users to review past shipping records. For example, a user can view the shipping history of a particular product to identify any patterns or issues with the logistics process. This helps to improve logistics management and enhance customer service.

3.5.2 Logistics Integration

- **Automatic Data Retrieval:** The system should integrate with logistics providers to automatically retrieve shipment status information. For example, the system can connect with the logistics provider's API to fetch real - time data on shipment locations and statuses. This reduces the need for manual data entry and ensures that users must - date information.
- **Multiple Carrier Support:** The system should support multiple logistics providers, allowing users to choose the most suitable carrier for their shipments. For example, a user can select a carrier based on factors such as cost, delivery time, and service

quality. This enhances customer convenience and provides more options for logistics management.

- **Logistics Provider Management:** The system should allow users to manage their logistics providers, including adding new providers, updating provider information, and setting default carriers for specific products or regions. This helps to streamline logistics operations and improve efficiency.

3.5.3 Customer Notifications

- **Automatic Shipping Notifications:** The system should send automatic notifications to customers about the status of shipments. When an order is shipped, the system can send an email or SMS notification to the customer with the tracking number and estimated delivery date. This keeps customers informed so that they don't need to ask customer service for shipping information, and customer service can focus on product issues.
- **Customizable Notification Templates:** The system should allow users to customize the content and format of shipping notifications. For example, users can add company branding, contact information, and special instructions to the notification. This enhances customer experience and ensures that customers receive relevant and useful information.

3.6 Shopping Cart Management

3.6.1 User - Friendly Interface

- **Easy Item Management:** The system should provide a user - friendly interface for customers to manage their shopping carts. For example, customers should be able to easily add items to their cart by clicking an "Add to Cart" button on the product page. They should also be able to remove items from their cart by clicking a "Remove" button next to each item. Additionally, customers should be able to modify the quantity of each item in their cart by entering a new number in the quantity field.
- **Smooth Checkout Process:** The system should allow customers to proceed to check out smoothly from their shopping cart. For example, after adding items to their cart, customers can click a "Checkout" button to be taken to the checkout page. The checkout process should be simple and intuitive, guiding customers through the necessary steps to complete their purchase.
- **Cart Summary:** The system should provide a summary of the items in the shopping cart, including the product name, quantity, price, and total cost. This allows customers to review their cart and make any necessary changes before proceeding to checkout. The cart summary should also display the total cost of the order, including any applicable taxes and shipping fees.

3.6.2 Cart Persistence

- **Session - Based Cart Storage:** The system should save the contents of the shopping cart across sessions, allowing customers to return to their cart at a later time without losing their selected items. For example, if a customer adds items to their cart but does not complete the purchase, they can come back to the website later and find their items still in the cart. This improves the shopping experience and reduces cart abandonment.
- **Cart Recovery:** The system should provide a cart recovery feature, allowing customers to recover their abandoned carts. For example, if a customer leaves items in their cart without completing the purchase, the system can send an email notification reminding them of their cart and providing a link to recover it. This helps to convert abandoned carts into completed sales and improve customer retention.
- **Cart Expiry:** The system should set an expiry time for the shopping cart, after which the items in the cart will be removed. For example, if a customer leaves items in their cart for more than 24 hours, the system can automatically remove the items to free up inventory. This ensures that the inventory is accurate and prevents customers from holding onto items for an extended period.

3.7 Promotions and Discounts

- **Automatic Discount Application:** The system should support the application of promotions and discounts to the shopping cart. For example, if a customer adds multiple items to their cart that qualify for a bulk purchase discount, the system should automatically apply the discount to the total cost of the order. This encourages customers to buy more and improves sales.
- **Promotion Rules Configuration:** The system should allow administrators to configure promotion rules, such as the minimum purchase amount, the discount percentage, and the applicable products. For example, an administrator can set up a promotion that offers a 10% discount on all toys purchased in the month of December. This provides flexibility in creating and managing promotions to attract customers and increase sales.
- **Discount Display:** The system should clearly display the discounts applied to the shopping cart, including the discount amount and the total cost after the discount. This allows customers to see the savings they are receiving and encourages them to complete the purchase. The discount display should be prominent and easy to understand, enhancing the customer experience.

3.8 Customer Customization Order Handling

3.8.1 Custom Order Management

- **Customer Communication:** The system should facilitate communication between customers and R&D. For example, when a customer submits a custom order request, the system can automatically send a notification to R&D with all the necessary details of the request. R&D can then communicate with the customer through the system to clarify any requirements and provide order progress updates.
- **Order Tracking:** The system should track the progress of custom orders from initial request to final delivery. For example, the system can display the current status of an order, such as "In Design," "In Production," or "Shipped." This allows customers to track the progress of their custom orders.
- **Confirming Orders:** The system should manage all aspects of custom orders, including order details, customer requirements, and production schedules. For example, the system can generate a detailed order summary that includes all specifications for the custom product, the agreed price, and the expected delivery date. This ensures that all parties involved in the order have a clear understanding of the requirements and expectations.

3.9 Design Approval

- **Design Submission:** The system should allow the R&D department to submit design proposals for custom orders. For example, after discussing the customer's requirements, the R&D department can upload design files, such as 3D models or sketches, to the system for the customer's review.
- **Customer Approval:** The system should enable customers to review and approve the design proposals. For example, the customer can view the design files in the system and provide feedback or approve the design. The system can then record the customer's approval and notify the R&D department to proceed with production.
- **Revision Management:** The system should manage the revision process for designs that require changes. For example, if the customer requests modifications to the design, the R&D department can make the changes and resubmit the design for approval. The system can track the revision history and ensure that all parties are aware of the latest version of the design.

3.10 Payment Management

3.10.1 Secure Payment Processing

- **Multiple Payment Methods:** The system should support multiple payment methods, such as credit cards, debit cards, and online payment platforms. For example, customers can choose to pay with a Visa or Mastercard credit card, or use an online payment platform such as PayPal. This provides customers with a wide range of payment options to avoid losing orders due to customers being unable to pay.
- **Payment Security:** The system should ensure the security of customer financial information during payment processing. For example, the payment system can use SSL encryption to protect customer payment information during transmission. In addition, the system can securely store payment information and use tokenization or encryption to prevent unauthorized access.

3.10.2 Invoice Generation

- **Automatic Invoice Creation:** The system should automatically generate invoices for complete transactions. For example, after a customer completes a purchase, the system can create an invoice with order details, payment information. The invoice can then be emailed to the customer or made available for download in the customer's account.
- **Invoice Customization:** The system allows administrators to customize the content and format of invoices. For example, administrators can add company branding, contact information, and terms and conditions to the invoice. This ensures that the invoice is professional and provides all necessary information to the customer.
- **Invoice History:** The system should keep a history of all invoices, allowing users to review past transactions. For example, users can search for invoices by order number or customer name to view details of a specific transaction. This helps with financial reconciliation and provides a record of all sales activity.

3.11 Inventory Management Enhancement

3.11.1 Automated Reordering

- **Real - Time Inventory Monitoring:** The system should continuously monitor inventory levels and automatically generate purchase orders when stock levels fall below a predefined threshold. For example, if the inventory level of a certain raw material drops below 50 units, the system can automatically create a purchase order to replenish the stock. This ensures that the company always has the necessary materials to meet production demands.
- **Supplier Integration:** The system should be integrated with supplier databases to automatically retrieve information on lead times and pricing. For example, when a purchase order is generated, the system can fetch the latest lead time and pricing information from the supplier's database. This helps in making informed decisions about inventory levels and reordering.

- **Reordering History:** The system should maintain a record of all reordering activities, including the date, quantity, and supplier. For example, a user can view the reordering history of a particular product to identify any trends or issues with the inventory management process. This helps to optimize inventory levels and improve procurement efficiency.

3.11.2 Supplier Management

- **Supplier Database:** The system should maintain a comprehensive supplier database that includes information such as contact details, product catalogs, and pricing. For example, a user can search for suppliers by name or product category to find the most suitable supplier for a particular raw material. This provides a centralized source of information for supplier management and enhances procurement efficiency.
- **Supplier Evaluation:** The system should allow users to evaluate and rate suppliers based on factors such as delivery time, product quality, and service. For example, after receiving a shipment from a supplier, a user can rate the supplier's performance and provide feedback. Optimize inventory levels and increase purchasing efficiency.
- **Supplier Communication:** The system should facilitate communication with suppliers, allowing users to send inquiries, requests for quotes, and purchase orders directly through the system. For example, a user can send a request for a quote to multiple suppliers and compare their responses to make an informed decision. This streamlines the procurement process and improves supplier relationships.

3.11.3 Stock Audits

- **Periodic Stock Audits:** The system should support periodic stock audits to ensure that physical inventory matches the system records. For example, a user can schedule a stock audit at the end of each month or quarter. The system can then generate a report comparing the physical inventory count with the recorded inventory levels, highlighting any discrepancies.
- **Discrepancy Resolution:** The system should provide tools to investigate and resolve discrepancies identified during stock audits. For example, if a discrepancy is found, the system can generate a task for the relevant department to investigate the cause and take corrective action. This helps to maintain the accuracy of inventory records and prevent potential losses.
- **Audit History:** The system should maintain a record of all stock audits, including the date, auditor, and results. For example, a user can view the audit history of a particular product to identify any recurring discrepancies or issues with the inventory management process. This helps to improve inventory accuracy and enhance overall inventory management.

3.12 Quality Control Monitoring

3.12.1 Quality Metrics Tracking

- **Quality Data Collection:** The system should collect and track quality metrics at each stage of production. For example, during the production process, the system can record data such as the number of defective units, the percentage of non - conforming products, and the root causes of quality issues. This provides a comprehensive view of the quality performance of the production process.
- **Quality Alerts:** The system should send alerts when quality metrics exceed predefined thresholds. For example, if the percentage of non - conforming products exceeds 5%, the system can send an alert to the quality control department, prompting them to take corrective action. This helps to identify and address quality issues in a timely manner, preventing the production of defective products.
- **Quality Reports:** The system should generate reports on quality metrics, providing insights into the quality performance of the production process. For example, a user can view a report that summarizes the quality data for a particular product or production line, highlighting any trends or areas for improvement. This helps to identify root causes of quality issues and implement corrective actions.

3.12.2 Non - Conformance Management

- **Non - Conformance Reporting:** The system should allow users to document and track non - conformance reports. For example, when a quality issue is identified, a user can create a non - conformance report in the system, detailing the issue, the affected products, and the corrective actions taken. This provides a centralized repository for non-conformance information and helps to manage the resolution process.
- **Corrective Action Tracking:** The system should track the progress of corrective actions for non-conformance reports. For example, after a corrective action is assigned, the system can send reminders to the responsible party and monitor the completion status. This ensures that corrective actions are implemented in a timely manner and that quality issues are resolved effectively.
- **Non - Conformance Analysis:** The system should analyze non - conformance data to identify root causes and trends. For example, the system can generate a report that summarizes the non-conformance data for a particular product or production line, highlighting any recurring issues or patterns. This helps to improve the quality of the production process and prevent future quality issues.

3.12.3 Supplier Quality Assessment

- **Supplier Quality Data Collection:** The system should collect and analyze quality data from suppliers. For example, the system can track the quality performance of suppliers based on factors such as the percentage of non-conforming products, the number of quality complaints, and the timeliness of deliveries. This provides a comprehensive view of the quality performance of suppliers.

- **Supplier Quality Scoring:** The system should assign a quality score to each supplier based on their performance. For example, the system can calculate a quality score for each supplier using a weighted formula that considers factors such as product quality, delivery time, and service. This helps to identify reliable suppliers and make informed decisions about sourcing.
- **Supplier Quality Improvement:** The system should facilitate communication with suppliers to address quality issues and improve quality performance. For example, the system can generate a report that highlights the quality issues of a particular supplier and provides recommendations for improvement. This helps to build a collaborative relationship with suppliers and enhance the overall quality of the supply chain.

3.13 Communication and Collaboration Platform

3.13.1 Centralized Communication

- **Meeting Scheduling:** The system should allow users to schedule meetings and invite participants from different departments. For example, a user can create a meeting event in the system, specifying the date, time, and location, and inviting relevant team members. The system can then send notifications to the participants and add the meeting to their calendars.
- **Document Sharing:** The system should provide a centralized platform for sharing documents related to projects or tasks. For example, a user can upload a document to the system and share it with specific team members or departments. The system can track the version history of the document and notify users of any changes or updates.
- **Project Progress Tracking:** The system should allow users to track the progress of projects or tasks in real - time. For example, a user can view the status of a project, including the completion percentage, the assigned tasks, and the responsible parties. This provides visibility into the project progress and helps to identify any delays or issues.

3.13.2 Instant Messaging

- **Real - Time Communication:** The system should support instant messaging, allowing users to communicate in real - time with colleagues or team members. For example, a user can send a message to another user or a group of users, and receive an immediate response. This enhances collaboration and reduces the need for email communication.
- **Message History:** The system should maintain a history of all instant messages, allowing users to review past conversations. For example, a user can search for messages by keyword or date to find specific information or discussions. This helps to keep track of important information and ensures that everyone is on the same page.
- **File Transfer:** The system should allow users to transfer files through instant messaging. For example, a user can send a document or an image to another user during a conversation. This facilitates the sharing of information and enhances collaboration.

3.14 Order Tracking

3.14.1 Real - Time Updates

- **Order Status Tracking:** The system should provide real-time updates on the status of orders, allowing users to view the current stage of the order fulfillment process. For example, a user can see whether an order is being processed, shipped, or delivered. This provides transparency into the order status and helps to manage customer expectations.
- **Customer Notifications:** The system should send automated notifications to customers regarding the status of their orders. For example, when an order is shipped, the system can send an email or SMS notification to the customer, providing the tracking number and estimated delivery date. This keeps customers informed and builds trust in the company's order fulfillment capabilities.
- **Order History:** The system should maintain a history of all orders, allowing users to review past transactions. For example, a user can search for orders by customer name or order number to view the details of a particular transaction. This helps with financial reconciliation and provides a record of all sales activities.

3.15 Customer Relationship Management (CRM)

3.15.1 Customer Segmentation

- **Segmentation Criteria:** The system should allow users to segment customers based on various criteria, such as demographics, purchase history, and behavior. For example, a user can create a customer segment for customers who have purchased a particular product in the last 30 days or for customers who have visited the company's website but have not made a purchase. This provides a more targeted approach to marketing and customer service.
- **Targeted Marketing:** The system should enable targeted marketing campaigns, allowing users to send personalized messages and offers to specific customer segments. For example, a user can send a promotional email to customers who have purchased a particular product, offering a discount on related products. This enhances customer engagement and increases the likelihood of repeat purchases.

4 Non-Functional Requirements

4.1 Security

- **Data Encryption:** The system should have a high level of security to protect user information and financial data. This includes using encryption technology to ensure the security of data transmission and storage. For example, the payment system should use SSL encryption to protect customer payment information during transmission.
- **Access Control:** The system should have strict access control measures to prevent unauthorized access. This includes implementing role-based access control (RBAC) and multi-factor authentication (MFA) to ensure that only authorized users can access sensitive information.
- **Security Audits:** The system should undergo regular security audits to identify and address vulnerabilities. This ensures that the system remains secure and compliant with industry standards.

4.2 Performance

- **Response Time:** The system should have good performance to ensure fast response times and high availability. This includes optimizing database queries and using caching mechanisms to reduce latency.
- **Scalability:** The system should be able to handle many user requests and transactions without experiencing significant delays or failures. This includes using load balancing and horizontal scaling to distribute traffic across multiple servers.
- **Reliability:** The system should be reliable and have a high degree of stability. This includes implementing redundancy and failover mechanisms to ensure that the system remains available even in the event of hardware failures.

4.3 Usability

- **User Interface:** The system should be easy to use and have a user-friendly interface. This includes designing intuitive workflows and providing clear and concise operation instructions and help with documentation.
- **Accessibility:** The system should be accessible to users with disabilities, complying with accessibility standards such as WCAG. This includes providing alternative text for images and ensuring that the system is navigable using keyboard shortcuts.
- **Training:** The system should provide training materials and tutorials to help users get up to speed quickly. This includes video tutorials, user guides, and interactive demos.

4.4 Reliability

- **Data Integrity:** The system should be reliable and have a high degree of stability. This includes implementing data validation and error handling mechanisms to ensure that data is accurate and consistent.

- **Backup and Recovery:** The system should have backup and recovery mechanisms to ensure the integrity and availability of data. This includes regular data backups and implementing disaster recovery plans to minimize downtime.
- **Monitoring:** The system should be monitored for performance and availability, with alerts sent to administrators in the event of issues. This ensures that problems are detected and resolved quickly.

4.5 Scalability

- **Horizontal Scaling:** The system should have good scalability to meet the changing needs of the company's business. This includes the ability to easily add new functions and modules without affecting the normal operation of the existing system. For example, as the company's business expands, the inventory management system should be able to easily add new inventory items and warehouses.
- **Vertical Scaling:** The system should support vertical scaling, allowing for the addition of more resources to a single server to handle increased load.
- **Modular Design:** The system should be designed with modular architecture, allowing for easy integration of new features and functionalities.

4.6 Interoperability

- **Standard Interfaces:** The system should be able to interoperate with other systems and platforms. This includes having standard interfaces and protocols to enable data exchange and integration with other systems. For example, the payment system should be able to integrate with various online payment platforms to provide customers with more payment options.
- **APIs:** The system should provide APIs to allow for integration with third-party systems and services. This includes RESTful APIs and webhooks to facilitate data exchange.
- **Data Formats:** The system should support common data formats, such as JSON and XML, to ensure compatibility with other systems.

4.7 Maintainability

- **Code Structure:** The system should be easy to maintain and upgrade. This includes having a clear and concise code structure and documentation to facilitate system maintenance and upgrade.
- **Version Control:** The system should use version control systems, such as Git, to manage code changes and collaborate effectively.
- **Continuous Integration/Continuous Deployment (CI/CD):** The system should support CI/CD pipelines to automate the build, test, and deployment processes. This ensures that updates and fixes are deployed quickly and reliably.

4.8 Compliance

- **Regulatory Compliance:** The system should comply with relevant laws and regulations, such as data protection laws and industry-specific regulations. This includes implementing necessary measures to ensure compliance and avoid legal risks.
- **Industry Standards:** The system should adhere to industry standards and best practices to ensure interoperability and compatibility with other systems and platforms.

4.9 Cost - Effectiveness

- **Development Cost:** The system should be developed within a reasonable budget, balancing functionality and cost. This includes optimizing development processes and using cost-effective technologies.
- **Operational Cost:** The system should have low operational costs, including server maintenance, software updates, and user support. This helps to reduce the overall cost of ownership and improve the return on investment.

4.10 Flexibility

- **Customization:** The system should be customizable to meet the specific needs of different users and departments. This includes allowing users to customize workflows, interfaces, and functionalities to suit their requirements.
- **Adaptability:** The system should be able to adapt to changing business needs and market conditions. This includes being able to quickly respond to new requirements and incorporate new technologies and trends.

5 Software and hardware

5.1.1 Computer



Companies purchase computers to meet the need for necessary tools at work and ensure that employees can complete their daily tasks efficiently and smoothly.

Item	Specification
Processor	Up to AMD Ryzen™ 9 5950X
Graphics Card	Up to NVIDIA® GeForce RTX™ 3070 Ti, 8GB GDDR6
Memory	Standard: 2×16GB UDIMM ARMOR DDR4-3200 Max: 128GB DDR4-3200, four UDIMM slots, dual-channel mode
Storage	2×3.5-inch HDD slots, 1×M.2 NVMe slot Max: Up to five drives
Motherboard	Chipset: AMD B550 Memory slots: Four DDR4 UDIMM slots Expansion slots: 1×PCIe 4.0×4 M.2 slot, 1×PCIe 3.0×2 M.2 slot
Power Supply	80 Plus Bronze certified 550W
Front Ports	2×USB-A (5Gbps), 1×USB-C (10Gbps), 1×3.5mm headphone jack
Cooling	Two 140mm fans (one intake, one exhaust)
Quantity	150
Price	10217(HKD)

Lenovo Legion T5 26AMR5, with its high-performance hardware, enterprise-level scalability, and reliable after-sales service, has become the ideal choice for corporate procurement. Whether for daily office work, professional design, or high-load computing, this model provides stable and efficient performance support.

5.1.2 Server



A server is a high-performance computer designed to provide data, resources, and services to other devices (such as clients), commonly used for networking, storage, and application hosting.

Item	Specification
Processor	Up to 2x 3rd Gen Intel® Xeon® Scalable processors (up to 40 cores per CPU)
Memory	Up to 2TB RDIMM or 4TB LRDIMM at 3200 MT/s (32x DDR4 DIMM slots)
Storage	Up to 24x 2.5" SAS/SATA/NVMe drives (max 368.64TB) Up to 2x rear 2.5" drives (max 30.72TB)
Power Supplies	800W, 1100W, 1400W, or 2400W (Platinum or Titanium rated)
Form Factor	2U rack server
Management	iDRAC9 with Lifecycle Controller, Quick Sync 2 wireless module
Operating Systems	Windows Server, VMware ESXi, Red Hat Enterprise Linux, SUSE Linux Enterprise Server
Security	TPM 2.0, Encrypted firmware, Secure boot,

	System lock (with iDRAC9 Enterprise)
PCIe Slots	Up to 8x PCIe 4.0 slots (up to 6x16)
Quantity	6
Price	53623(HKD)

It has powerful computing capabilities and expandability, supports virtualization and databases, and can ensure stable 7×24 operation to meet the core business needs of the enterprise.

5.1.3 Core Switch



A core switch is a high-performance device in a network, designed to connect multiple distributed networks and forward data at high speed, typically deployed at the core of a network.

Item	Specification
Ports	24x 10GE SFP+, 6x 40GE/100GE QSFP28, optional license upgrade to 6x 100GE QSFP28
Memory	4GB RAM
Flash	2GB physical storage
Backplane Bandwidth	2.56Tbps
Packet Forwarding Rate	1260Mpps
Power Supply	600W AC or 1000W DC
Dimensions	442.0mm × 420.0mm × 43.6mm
Weight	8.9kg
Operating Temperature	-5°C to +45°C (at altitudes up to 1800m)
Storage Temperature	-40°C to +70°C
Management Features	Supports stacking, VLAN support
Cooling	Air-cooled with intelligent speed adjustment
Quantity	2
Price	19304.28(HKD)

It provides a 10 - Gigabit backbone network connection, supports 10Gbps data transmission between departments, and has high bandwidth, low latency, and high reliability to ensure core data exchange in the enterprise network.

5.1.4 AccessSwitch



An access switch is a network device used to connect end devices (like computers and printers) to the network, providing basic data forwarding capabilities.

Item	Specification
Ports	8x Gigabit RJ45 PoE+ ports (IEEE 802.3af/at compliant) 1x Gigabit RJ45 port 1x Gigabit SFP port
PoE Capability	Total PoE budget: 63W Single port max PoE power: 30W
Switching Capacity	20Gbps
Dimensions	171×98×27mm
Power Supply	53.5VDC/1.22A
Operating Temperature	0°C to 40°C
Quantity	6
Price	3002.89(HKD)

It supports PoE power supply, which can directly power connected devices such as printers and IP phones, simplifying wiring and facilitating network deployment and management.

5.1.5 windows 11



Windows 11 is a modern operating system by Microsoft, offering a refreshed user interface, improved performance, and enhanced support for touch devices.

Reason

Windows 11 Enterprise Edition offers advanced security features, remote work support, and centralized management tools, ideal for improving organizational efficiency and safeguarding data.

Quantity	1
Price	free

5.1.6 Visual Studio 2019



Visual Studio is an Integrated Development Environment (IDE) developed by Microsoft, supporting multiple programming languages (e.g., C#, Python, JavaScript) and various application types (desktop, mobile, web, and cloud).

Reason

Choosing Visual Studio 2019 provides several benefits: it offers a stable and proven development environment, supports a wide range of languages and platforms, and integrates well with tools like Git. It's also a great choice for maintaining legacy projects or for developers who prefer its familiar interface over newer versions.

Quantity	1
Price	free

5.1.7 360Qihoo



Qihoo 360 is a Chinese internet security company founded in 2005, providing comprehensive cybersecurity solutions.

Reason

Choosing Qihoo 360 ensures robust cybersecurity solutions, leveraging advanced technology to protect against a wide range of online threats.

Quantity	1
Price	free

5.1.8 Ubuntu Server



Ubuntu Server is an efficient tool for building and managing server environments, suitable for a variety of scenarios such as web services, cloud computing, containerization, and database management.

Reason

Choose Ubuntu Server LTS version 22.04 LTS because it is free, stable, and has long-term support, which can provide enterprises with reliable performance and flexible scalability.

Quantity	1
Price	free

5.1.9 UPS



A device that provides backup power. It can continue to power equipment for a short period of time when the main power is interrupted to ensure the normal operation of important equipment (such as servers, computers, etc.), while also protecting the equipment from damage caused by voltage fluctuations or sudden power outages.

Parameter	Description
Capacity Range	30–800 kVA (modular scalability, single unit up to 800 kVA)
Input Voltage	380/400/415 V AC (ultra-wide range: 138–485 V AC, handles poor grid conditions)
Output Voltage	380/400/415 V AC, output PF 1.0 (supports inductive/capacitive loads)
Efficiency	Double conversion: up to 96%; ECO mode: 99%
Module Specifications	30kVA/2U (30–120kVA models); 50kVA/3U (200–800kVA models)
Battery Support	Lead-acid or lithium (SmartLi 3.0), 1–12 groups, 30–44 cells per group
Dimensions (W×D×H)	200–300kVA: 600×850×2000 mm; 800kVA: 2400×850×2000 mm (50% space-saving)
Operating Temperature	Standard: 0–40°C; Rugged: -40–70°C
Quantity	2
Price	96000(HKD)

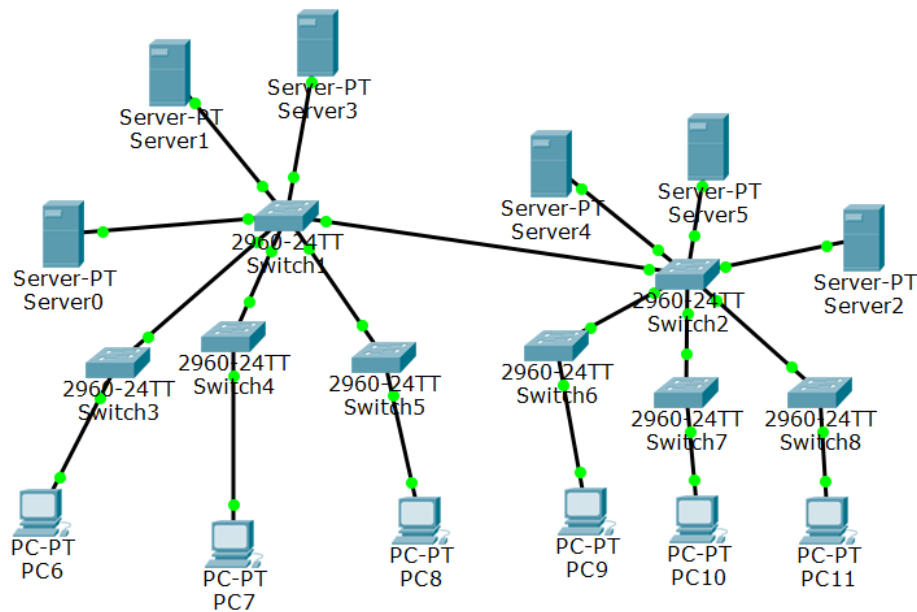
Reason

The Huawei UPS5000-E is the optimal choice for mission-critical infrastructure due to its modular hot-swappable design for seamless scalability, military-grade reliability under extreme grid conditions, 99% energy efficiency for cost savings, and AI-driven predictive maintenance ensuring uninterrupted operations.

5.1.10 Order Overview

Item	Price Range(HKD)	Quantity	Total Cost(HKD)
Lenovo Legion T5 26AMR	10217	33	141564.7
Dell PowerEdge R750	53623	6	321738
Huawei S6730 - H24X6C	19304.28	2	38608.56
TP - LINK TL - SG1210P	3002.89	6	18017.34
Windows 11	0	1	0
Visual Studio	0	1	0
360	64.35	33	2123.55
Ubuntu Server	0	1	0
HUAWEI UPS5000-E	96000	2	192000
Sum			2105037

6 Network topology diagram



This network topology diagram presents a structured network architecture. At the core lies Switch1 (2960-24TT), serving as the central hub to manage all network traffic. In the server area, Server0, Server1, and Server3 are directly connected to Switch1 via network cables, enabling efficient data transmission through the core network.

Switch2 (2960-24TT) plays a key role in the server segment. Server2, Server4, and Server5 first connect to Switch2, which then links to Switch1. This configuration distributes server connections reasonably, avoiding congestion on a single switch and enhancing server network reliability.

For departmental connections, each PC represents a specific department. Switch3 connects the department of PC6; Switch4 serves PC7's department; Switch5 links to PC8's department; Switch6 connects with PC9's department; Switch7 is responsible for PC10's department; and Switch8 handles PC11's department. Notably, Switch3 to Switch8 are all ultimately connected to the core Switch1, ensuring departmental network data flows through respective switches before converging to the core.

This topology offers distinct advantages. First, it enables centralized management. When issues arise in a department (e.g., PC6's department), maintenance teams can quickly troubleshoot by checking its dedicated switch (Switch3), improving efficiency. Second, it features strong scalability. If a department needs to add devices, adjustments can be made on its switch without disrupting the entire network, supporting flexible network expansion.

In summary, this network topology forms a clear, organized system. With Switch1 as the core, servers connect via Switch1 and Switch2, and each department's PC accesses the network through its dedicated switch, integrating into the core. This design ensures efficient server resource sharing and stable departmental network operation. It is a robust architecture for projects requiring reliable network support, balancing current usage and future development needs.

7 Benefits of a Centralized Computerized Management System

7.1 Data Security

The system boasts a secure login mechanism that is specifically designed to differentiate between various user groups, such as departments, employees, and customers. This meticulous differentiation ensures that each user can only access the information and functionalities that are relevant to their particular role. For example, an employee from the sales department will have access to sales-related data and tools, while a customer can only view and manage their own purchase history and account details. By doing so, it significantly reduces the risk of unauthorized access to sensitive information and potential data leaks, creating a more secure environment for all parties involved.

7.2 Inventory Management

The real - time inventory tracking feature of the system is highly beneficial. It constantly updates the status of stored products, providing users with up - to - the - minute information. This means that whether it's a warehouse manager checking the stock levels or a salesperson trying to confirm product availability for a customer, they can get accurate and timely data. As a result, businesses can avoid situations like stockouts, where they run out of products when customers want to make a purchase, or overstocking, which ties up valuable resources and storage space, leading to more efficient inventory management.

7.3 Logistics Visibility

The logistics query and order tracking systems play a crucial role in enhancing customer satisfaction. They offer real - time updates on the shipment and order status. Customers can easily track where their orders are, whether they are in transit, out for delivery, or have been delivered. This transparency not only makes customers feel more at ease but also increases their trust in the company's logistics capabilities. When customers know that they can rely on the company to provide accurate and timely information about their orders, they are more likely to have a positive view of the company and its services.

7.4 Customer Experience

The user-friendly online shopping cart and custom order system greatly improves customer experience. Customers can easily add or remove items from the shopping cart, change product quantities, and apply for any discounts or promotions. The custom order system allows customers to specify their unique requirements, such as special packaging or personalized information. This can improve user satisfaction and increase the user's sense of involvement in the product, as if they made it themselves.

7.5 E - commerce Trust

A secure payment system is essential to building customer trust in an e-commerce platform. The system supports multiple payment methods, providing customers with a variety of options to suit their preferences. At the same time, it takes strong measures to protect customers' financial information, and encryption technology must be used to ensure that credit card details and bank account information are not leaked. When customers are confident that their financial data is safe, they have a range of secure payment options, which is the most basic protection.

7.6 Communication and Collaboration

A centralized communication and collaboration platform is very beneficial to a company. It brings together employees from different departments, allowing them to communicate and share information more effectively. For example, the marketing department can easily collaborate with the product development team to discuss new product features and promotion strategies. This improved inter-departmental communication helps in the smooth implementation of projects, and when different departments can work together seamlessly, their communication efficiency can be increased.

7.7 Decision - making Insights

Data analytics is a powerful tool that provides valuable insights for decision - making. It analyzes various aspects of the business, such as production efficiency, sales trends, and customer preferences. For instance, by analyzing production data, the company can identify bottlenecks in the production process and take measures to improve efficiency. Sales trend analysis can help the company predict which products will be in high demand and adjust production and inventory accordingly. And understanding customer preferences allows the company to tailor its products and services to better meet customer needs. These insights enable more informed strategic decisions, leading the company in the right direction for growth and success.

7.8 Customer Relationships

The CRM system is an effective customer relationship management tool. It records all customer interactions like inquiries, complaints, and feedback, and offers after - service support to ensure timely, satisfactory issue - resolution. By handling these well, it strengthens customer relationships. When customers see their needs are met, they're more likely to have a positive bond with the company, keep doing business, and enable the company to improve services based on feedback.

8 Constraints and Limitations

8.1 Initial Cost and Investment:

8.1.1 High Implementation Cost:

Setting up a centralized system requires significant initial investment in hardware, software, and infrastructure.

8.1.2 Training Costs:

Employees need to be trained to use the new system effectively, which can incur additional costs and time.

8.2 Complexity:

8.2.1 System Complexity:

The system's complexity can make it difficult to manage and maintain, especially for organizations with limited IT resources.

8.2.2 Integration Challenges:

Integrating the new system with existing legacy systems can be challenging and may require custom development.

8.3 Data Security and Privacy:

8.3.1 Security Risks:

Centralizing data increases the risk of data breaches and cyber-attacks. Ensuring robust security measures is crucial but can be costly and complex.

8.3.2 Privacy Concerns:

Handling sensitive data requires strict compliance with data protection regulations, which can be challenging to manage.

8.4 Scalability:

8.4.1 Scalability Issues:

As the organization grows, the system must be able to scale accordingly. This can involve additional costs and technical challenges.

8.4.2 Performance Bottlenecks:

High volumes of data and transactions can lead to performance bottlenecks, affecting system efficiency.

8.5 Dependence on Technology:

8.5.1 System Downtime:

Any technical issues or system downtime can disrupt business operations, leading to potential losses.

8.5.2 Reliance on IT Support:

Continuous IT support is required to maintain and troubleshoot the system, which can be resource intensive.

8.6 User Resistance:

8.6.1 Resistance to Change:

Employees may resist adopting the new system due to unfamiliarity or fear of change, impacting the system's effectiveness.

8.6.2 Learning Curve:

There is a learning curve associated with using the new system, which can temporarily reduce productivity.

8.7 Customization and Flexibility:

8.7.1 Limited Customization:

Off-the-shelf solutions may not fully meet the specific needs of the organization, requiring additional customization.

8.7.2 Flexibility Constraints:

The system may lack the flexibility to adapt to unique or changing business processes without significant modifications.

8.8 Maintenance and Upgrades:

8.8.1 Ongoing Maintenance:

Regular maintenance is required to keep the system running smoothly, which can be time-consuming and costly.

8.8.2 Upgrade Challenges:

Upgrading the system to incorporate new features or technologies can be complex and may disrupt operations.

8.9 Compatibility Issues

8.9.1 Software Compatibility:

Newer versions of software or additional software modules integrated into the system may not be fully compatible with the existing software stack. This can lead to functionality failures, data transfer errors, or unexpected system behavior. For example, if a new analytics tool is added to the centralized system, it might not communicate properly with the database management system, causing data retrieval issues for reports and dashboards.

8.9.2 Hardware Compatibility:

As technology evolves, new hardware components may be needed to support system upgrades or increased workloads. However, these new hardware elements might not be fully compatible with the existing infrastructure. For instance, a high - performance graphics card installed to enhance certain visualization features in the system could cause power supply issues or conflicts with other hardware components, resulting in system instability.

8.10 Regulatory and Compliance Changes

8.10.1 Industry - Specific Regulations:

Industries are subject to evolving regulations. For example, in the financial sector, new regulations regarding data storage and reporting requirements may emerge. The centralized system may need to be updated to ensure compliance, which can involve significant development efforts, audits, and potential changes to business processes. Failure to adapt to these regulatory changes in a timely manner can result in legal penalties and damage to the organization's reputation.

8.10.2 International Compliance Variations:

If the organization operates across multiple countries, it must adhere to different compliance standards in each jurisdiction. For instance, data privacy laws vary widely between regions such as the European Union (with GDPR) and the United States. The centralized system must be configured to meet these diverse requirements, which can be a complex and resource-intensive task.

9 Actor Description

Actor	Production Manager
Description	Reviewing product specifications, timelines, and material requirements. Scheduling the manufacturing process to ensure it aligns with project deadlines. Allocating necessary resources such as machinery and labor to each stage of the process to maximize efficiency.

Actor	Quality Control Worker
Description	Checking the quality of materials from the previous step. Reporting the number of finished components and damaged materials at the end of the day. Ensuring that products meet quality standards before they are transferred to the next production line process.

Actor	Logistic Officer
Description	Organizing transportation logistics, selecting suitable carriers, and scheduling delivery times to meet customer demands. Preparing delivery notes, monitoring shipment status, managing delays or issues proactively, and verifying products against delivery notes upon arrival at customer locations.

Actor	Customer Service Representative
Description	Addressing customer inquiries or complaints through various channels such as phone or email. Gathering detailed information to understand the problem fully. Logging cases into a customer relationship record for tracking and resolution. Escalating technical issues to the technical support team for in-depth analysis and troubleshooting.

Actor	Inventory Control Specialist
Description	Checking stock levels after receiving requirement forms. Notifying procurement specialists to order more materials if stock is low. Coordinating with suppliers to ensure timely delivery of

	raw materials. Recording the movement of finished products and raw materials in and out of the warehouse.
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Actor	Procurement Specialist
Description	Sending purchase orders to raw material suppliers. Notifying inventory staff of estimated delivery dates provided by suppliers. Coordinating with suppliers to ensure timely delivery of raw materials. Negotiating with suppliers to resolve issues related to defective materials reported by the warehouse.

Actor	Finance Department
Description	Preparing invoices and sending them to customers for final payment. Managing accounts payable and receivable, ensuring timely payments and collections to maintain cash flow. Handling financial reporting, preparing accurate statements and reports for stakeholders.

Actor	Customer
Description	Requesting an existing product or a tailor-made product. Confirming order details, including product specifications, quantities, and special requirements. Reviewing prototypes and requesting adjustments. Confirming the final prototype before finalizing the order.

Actor	Project Coordinator
Description	Arranging meetings between customers and managers. Checking schedules with managers. Ensuring that the design is approved and prototypes meet all specifications and quality standards.

Actor	Engineer
Description	Assisting in building prototypes using 3D modeling. Conducting functionality and safety testing. Refining designs based on feedback from focus groups.

Actor	Production Officer
Description	Scheduling the manufacturing process according to product specifications. Ensuring that all necessary materials are procured and available on time. Attaching transfer forms before transferring components to the next production line process.

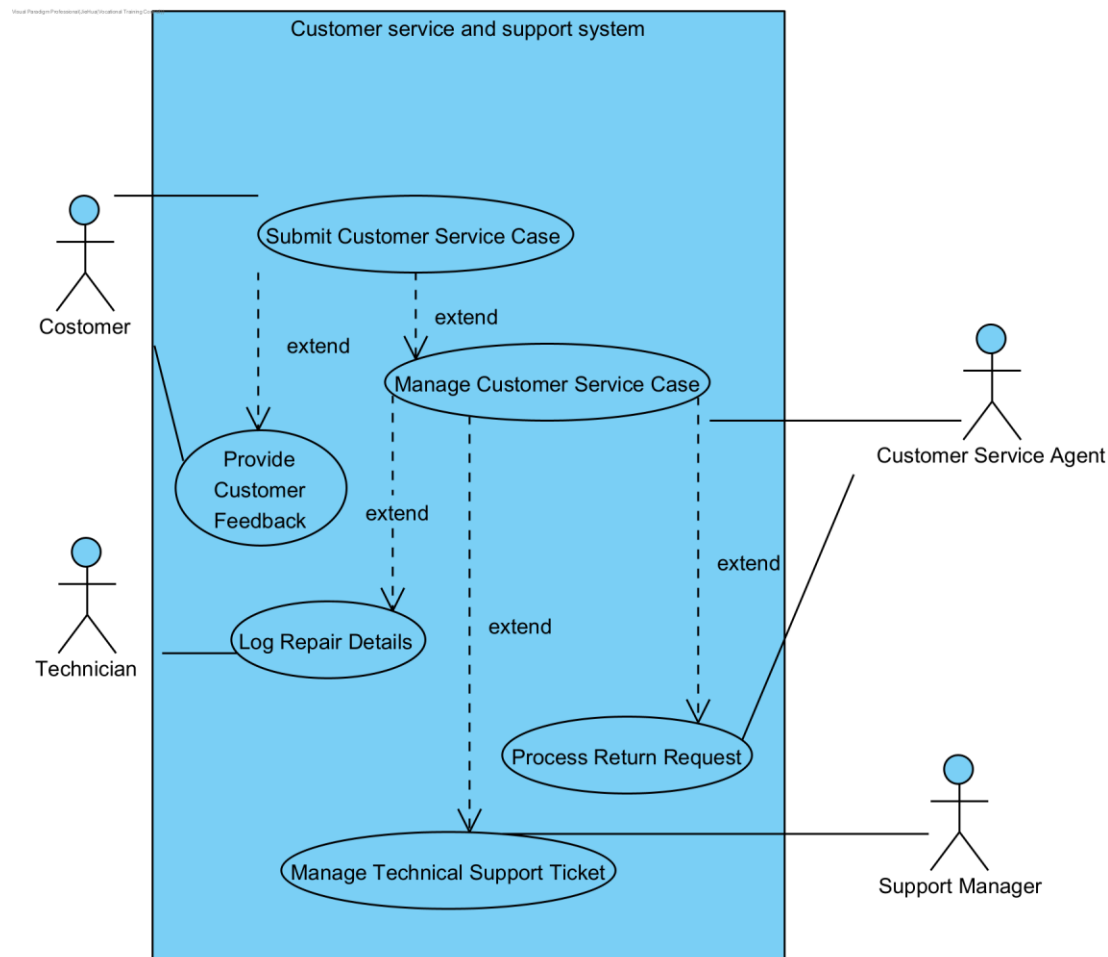
Actor	Warehouse Staff
Description	Sending raw materials from the warehouse to various production lines. Verifying items and quantities upon entry. Reporting defective materials to procurement staff.

Actor	Accountant
Description	Checking payments and notifying sales to pass order information to the production department. Preparing invoices and sending them to customers for final payment.

Actor	Product Manager
Description	Following up on tailor-made orders. Ensuring that prototypes meet all specifications and quality standards. Informing logistics to arrange delivery of completed products.

10 Use case diagram and description

10.1 Customer service and support system



Use case name:	Submit Customer Service Case
Use case ID:	UC-1100
Primary actor:	Customer
Secondary actor(s):	None
Brief description:	Customer submits a customer service case to report an issue or request support
Preconditions:	None
Flow of events:	1. The user logs into the system. 2. Customer navigates to the customer service section. 3. Customer enters case details, including caseID, customerID, and caseStatus. 4. Customer submits the service case. 5. The system validates the case information and saves it. 6. The system notifies the Customer that the service case has been successfully submitted.
Postconditions:	UC-1200

Alternative flows and exceptions:	In step 3, if the caseID already exists, the system notifies the Customer to use a different caseID. - In step 5, if required fields are missing, the system displays an error message.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can submit service cases.
Assumptions:	It is assumed that the Customer is familiar with the system and has received adequate training on how to use the customer service interface.
Issue:	The submission process may fail if the caseID is not unique or if the system encounters a validation error.

Use case name:	Manage Customer Service Case
Use case ID:	UC-1200
Primary actor:	Customer Service Agent
Secondary actor(s):	None
Brief description:	Customer Service Agent manages a customer service case to resolve the issue.
Preconditions:	UC-1100
Flow of events:	1. The user logs into the system. 2. Customer Service Agent navigates to the case management section. 3. Customer Service Agent selects a service case to manage. 4. Customer Service Agent updates the case details, including resolutionTime and caseStatus. 5. Customer Service Agent submits the updated case. 6. The system saves the updated case information. 7. The system notifies the Customer Service Agent that the case has been successfully updated.
Postconditions:	UC-1400
Alternative flows and exceptions:	In step 4, if the case cannot be resolved, the Customer Service Agent escalates the case to technical support.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can manage service cases.
Assumptions:	It is assumed that the Customer Service Agent is familiar with the system and has received adequate training on how to use the case management interface.
Issue:	The case management process may be delayed if the case details are incomplete or if escalation is required.

Use case name:	Provide Customer Feedback
Use case ID:	UC-1300
Primary actor:	Customer
Secondary actor(s):	None
Brief description:	Customer provides feedback on a resolved customer service case.
Preconditions:	UC-1100
Flow of events:	1. The user logs into the system. 2. Customer navigates to the feedback section. 3. Customer selects a resolved service case. 4. Customer enters feedback details, including satisfactionScore. 5. Customer submits the feedback. 6. The system saves the feedback information. 7. The system notifies the Customer that the feedback has been successfully submitted.
Postconditions:	None.
Alternative flows and exceptions:	In step 3, if the case is not yet resolved, the system notifies the Customer to wait until the case is closed.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can provide feedback.
Assumptions:	It is assumed that the Customer is familiar with the system and has received adequate training on how to use the feedback interface.
Issue:	The feedback submission may fail if the case is not yet resolved or if the system encounters a validation error.

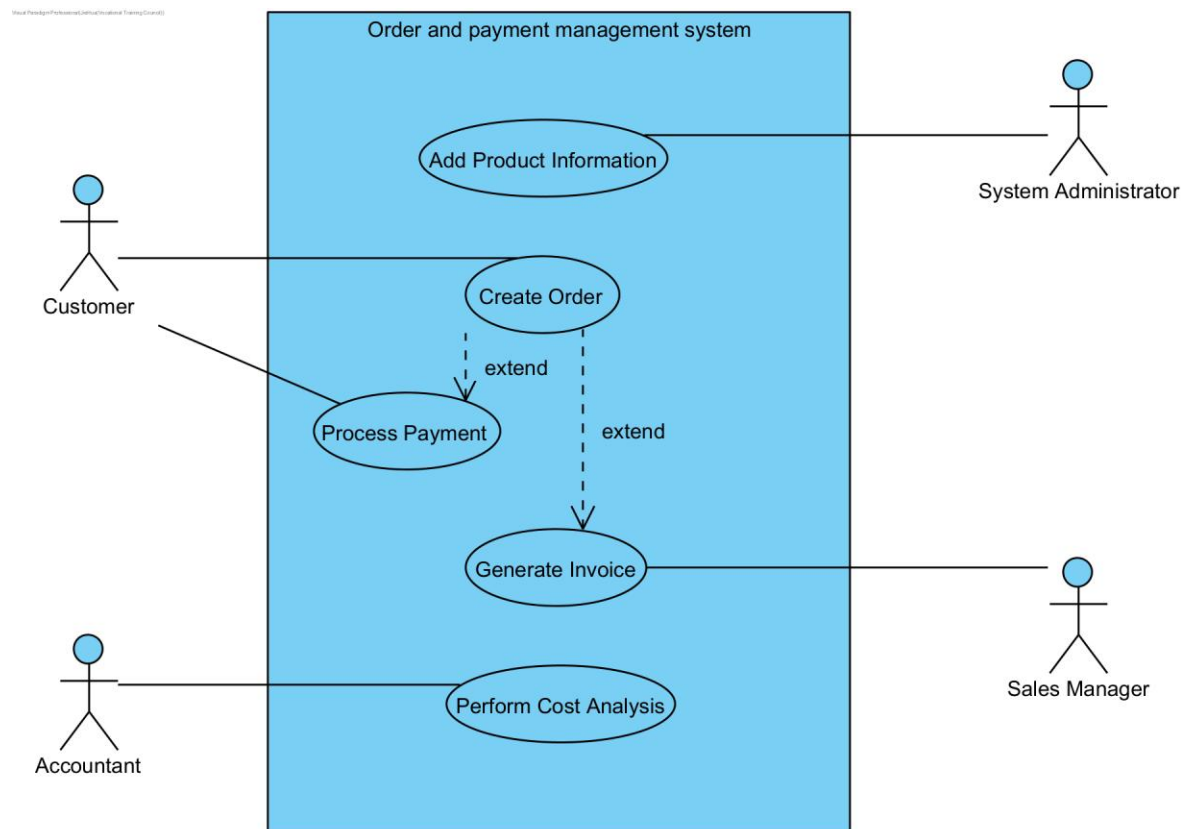
Use case name:	Log Repair Details
Use case ID:	UC-1400
Primary actor:	Technician
Secondary actor(s):	None
Brief description:	Technician logs repair details for a customer service case.
Preconditions:	UC-1200
Flow of events:	1. The user logs into the system. 2. Technician navigates to the repair log section. 3. Technician selects a service case to log repairs for. 4. Technician enters repair details, including logID, repairDetails, and repairDate. 5. Technician submits the repair log. 6. The system saves the repair log information. 7. The system notifies the Technician that the repair log has been successfully recorded.

Postconditions:	None.
Alternative flows and exceptions:	In step 4, if the repair details are incomplete, the system prompts the Technician to provide additional information.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can log repair details.
Assumptions:	It is assumed that the Technician is familiar with the system and has received adequate training on how to use the repair log interface.
Issue:	The repair logging process may be delayed if the service case details are incomplete or if the system encounters a validation error.

Use case name:	Process Return Request
Use case ID:	UC-1500
Primary actor:	Customer Service Agent
Secondary actor(s):	None
Brief description:	Customer Service Agent processes a return request for a customer service case.
Preconditions:	UC-1200
Flow of events:	1. The user logs into the system. 2. Customer Service Agent navigates to the return request section. 3. Customer Service Agent selects a service case with a return request. 4. Customer Service Agent enters return details, including return Reason and refund Amount. 5. Customer Service Agent submits the return request. 6. The system saves the return request information. 7. The system notifies the Customer Service Agent that the return request has been successfully processed.
Postconditions:	None.
Alternative flows and exceptions:	In step 4, if the return reason is invalid, the system prompts the Customer Service Agent to provide a valid reason.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can process return requests.
Assumptions:	It is assumed that the Customer Service Agent is familiar with the system and has received adequate training on how to use the return request interface.
Issue:	The return request process may be delayed if the return reason is invalid or if the system encounters a validation error.

Use case name:	Manage Technical Support Ticket
Use case ID:	UC-1600
Primary actor:	Support Manager
Secondary actor(s):	None
Brief description:	Support Manager manages a technical support ticket for a customer service case.
Preconditions:	UC-6200
Flow of events:	1. The user logs into the system. 2. Support Manager navigates to the technical support section. 3. Support Manager selects a service case with a technical support ticket. 4. Support Manager enters ticket details, including severity Level. 5. Support Manager submits the ticket. 6. The system saves the ticket information. 7. The system notifies the Support Manager that the technical support ticket has been successfully managed.
Postconditions:	None.
Alternative flows and exceptions:	In step 4, if the severity level is critical, the system escalates the ticket to a higher priority.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can manage technical support tickets.
Assumptions:	It is assumed that the Support Manager is familiar with the system and has received adequate training on how to use the technical support interface.
Issue:	The ticket management process may be delayed if the case details are incomplete or if escalation is required.

10.2 Order and payment management system



Use case name:	Add Product Information
Use case ID:	UC-2100
Primary actor:	System Administrator
Secondary actor(s):	None
Brief description:	System Administrator adds product information to the system.
Preconditions:	None
Flow of events:	1. The user logs into the system. 2. The System Administrator navigates to the product management section. 3. The System Administrator enters product details, including productID, name, specifications, safetyCertification, and creationDate. 4. The System Administrator submits the product information. 5. The system validates the input and saves the product information. 6. The system displays a confirmation message indicating that the product has been successfully added.
Postconditions:	UC-2200

Alternative flows and exceptions:	In step 3, if a product with the same productID already exists, the system notifies the user and prompts them to enter a different productID. In step 5, if required fields are missing, the system displays an error message.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can add product information.
Assumptions:	It is assumed that the System Administrator is familiar with the system and has received adequate training on how to use the product management interface.
Issue:	The product addition may fail if the system encounters a database error or if the productID is not unique.

Use case name:	Create Order
Use case ID:	UC-2200
Primary actor:	Customer
Secondary actor(s):	None
Brief description:	Customer creates an order for products.
Preconditions:	UC-2100
Flow of events:	1. The user logs into the system. 2. The Customer navigates to the order creation section. 3. The Customer selects products to include in the order. 4. The Customer enters order details, including deliveryTerms, orderDate, and customerID. 5. The Customer submits the order. 6. The system validates the order and saves it. 7. The system notifies the Customer that the order has been successfully created. 8. Postconditions: UC-2300, UC-2400
Postconditions:	None.
Alternative flows and exceptions:	In step 3, if no products are available, the system notifies the Customer to wait until products are added. In step 6, if required fields are missing, the system displays an error message.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can create orders.
Assumptions:	It is assumed that the Customer is familiar with the system and has received adequate training on how to use the order creation interface.
Issue:	The order creation may fail if the selected products are out of stock or if the system encounters a validation error.

Use case name:	Process Payment
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Use case ID:	UC-2300
Primary actor:	Customer
Secondary actor(s):	None
Brief description:	Customer processes payment for an order.
Preconditions:	UC-5200
Flow of events:	1. The user logs into the system. 2. The Customer navigates to the payment section. 3. The Customer selects the order to pay for. 4. The Customer enters payment details, including amount, paymentDate, and selects a payment gateway. 5. The Customer submits the payment. 6. The system processes the payment through the selected PaymentGateway and retrieves an approvalCode. 7. The system notifies the Customer that the payment has been successfully processed.
Postconditions:	None.
Alternative flows and exceptions:	In step 6, if the payment fails (e.g., due to insufficient funds), the system notifies the Customer to retry with a different payment method.
Non-behavior requirements:	The system must ensure secure payment processing and protect user data.
Assumptions:	It is assumed that the Customer has a valid payment method and is familiar with the payment interface.
Issue:	The payment process may fail if the payment gateway is down or if the Customer enters incorrect payment details.

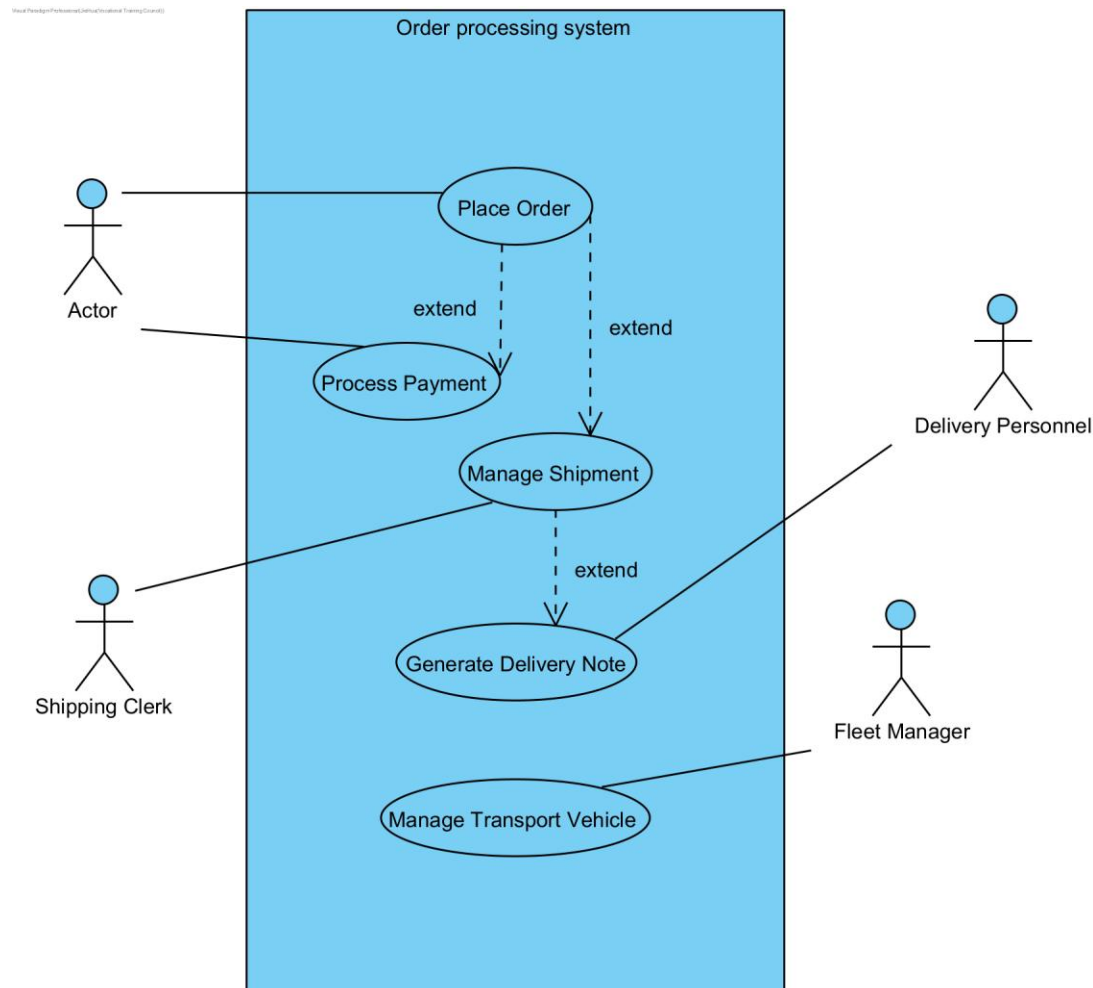
Use case name:	Generate Invoice
Use case ID:	UC-2400
Primary actor:	Sales Manager
Secondary actor(s):	None
Brief description:	Sales Manager generates an invoice for an order.
Preconditions:	UC-5200
Flow of events:	1. The user logs into the system. 2. The Sales Manager navigates to the invoice generation section. 3. The Sales Manager selects an order to generate an invoice for. 4. The Sales Manager enters invoice details, including invoiceID, invoiceAmount, taxDetails, and paymentDueDate.

	5. The Sales Manager submits the invoice. 6. The system validates the invoice information and saves it. 7. The system notifies the Sales Manager that the invoice has been successfully generated.
Postconditions:	UC-2500
Alternative flows and exceptions:	In step 4, if the order has not been paid, the system prompts the Sales Manager to ensure payment is completed first. In step 6, if required fields are missing, the system displays an error message.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can generate invoices.
Assumptions:	It is assumed that the Sales Manager is familiar with the system and has received adequate training on how to use the invoice generation interface.
Issue:	The invoice generation may fail if the order is not fully paid or if the system encounters a validation error.

Use case name:	Perform Cost Analysis
Use case ID:	UC-2500
Primary actor:	Accountant
Secondary actor(s):	None
Brief description:	Accountant performs cost analysis based on invoices.
Preconditions:	UC-5400
Flow of events:	1. The user logs into the system. 2. The accountant navigates the cost analysis section. 3. The accountant selects an invoice to analyze. 4. The accountant enters cost details, including materialCost and laborCost. 5. The accountant submits the cost analysis. 6. The system saves the cost analysis information. 7. The system notifies the accountant that the cost analysis has been successfully completed.
Postconditions:	None.
Alternative flows and exceptions:	In step 3, if no invoices are available, the system notifies the Accountant to wait until an invoice is generated. In step 4, if cost details are incomplete, the system prompts the accountant to provide additional information.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can perform cost analysis.
Assumptions:	It is assumed that the Accountant is familiar with the system and has received adequate training on how to use the cost

	analysis interface.
Issue:	The cost analysis may be delayed if the invoice data is incomplete or if the system encounters a validation error.

10.3 Order processing system.



Use case name:	Place Order
Use case ID:	UC-3100
Primary actor:	Customer
Secondary actor(s):	None
Brief description:	Customers place orders in the system to purchase products
Preconditions:	None
Flow of events:	1. The user logs into the system. 2. The customer navigates to the order creation page. 3. The customer selects the products to order. 4. The customer enters the order details, including deliveryTerms, orderDate, and customerID. 5. The customer submits the order.

	6. The system validates the order information and saves it. 7. The system notifies the customer that the order has been successfully submitted.
Postconditions:	UC-3200, UC-3300
Alternative flows and exceptions:	In step 3, if the product is out of stock, the system notifies the Customer to wait for the product to be restocked. In step 6, if the required fields are missing, the system displays an error message and asks the Customer to complete the information.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can place orders.
Assumptions:	It is assumed that Customer is familiar with the system operation and has been trained on how to use the order interface.
Issue:	Order creation may fail if the selected product is out of stock or if there is a system validation error.

Use case name:	Process Payment
Use case ID:	UC-3200
Primary actor:	Customer
Secondary actor(s):	None
Brief description:	Customers pay for order processing.
Preconditions:	UC-3100
Flow of events:	1. The user logs into the system. 2. The customer navigates to the payment page. 3. The customer selects the order to be paid. 4. The customer enters the payment details, including amount and paymentDate. 5. The customer submits the payment. 6. The system processes the payment and saves the payment information. 7. The system notifies the customer that the payment has been successfully processed.
Postconditions:	None.
Alternative flows and exceptions:	In step 5, if the payment fails (for example, insufficient balance), the system notifies the Customer to try again using another payment method.
Non-behavior requirements:	The system must ensure that the payment process is secure and protect user data.
Assumptions:	It is assumed that the Customer has a valid payment method and is familiar with the payment interface.
Issue:	The payment process may fail if the payment gateway is unavailable, or the Customer enters incorrect payment

	information.
Use case name:	Manage Shipment
Use case ID:	UC-3300
Primary actor:	Shipping Clerk
Secondary actor(s):	None
Brief description:	The shipper manages the shipping information for the order.
Preconditions:	UC-3100
Flow of events:	1. The user logs into the system. 2. The Shipping Clerk navigates to the shipment management page. 3. The Shipping Clerk selects the order to be managed. 4. The Shipping Clerk enters the shipment details, including trackingNumber, carrierInfo, shipmentDate, and deliveryStatus. 5. The Shipping Clerk submits the shipment information. 6. The system verifies the shipment details and saves them. 7. The system notifies the Shipping Clerk that the shipment has been successfully managed.
Postconditions:	UC-3400
Alternative flows and exceptions:	In step 4, if the shipment details are incomplete, the Shipping Clerk is prompted to enter the missing information.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can manage shipments.
Assumptions:	It is assumed that the Shipping Clerk is already familiar with the system and has been trained on the Shipping Management Interface.
Issue:	If the order has not been confirmed or the shipping details are incorrect, the shipping management process may be delayed.

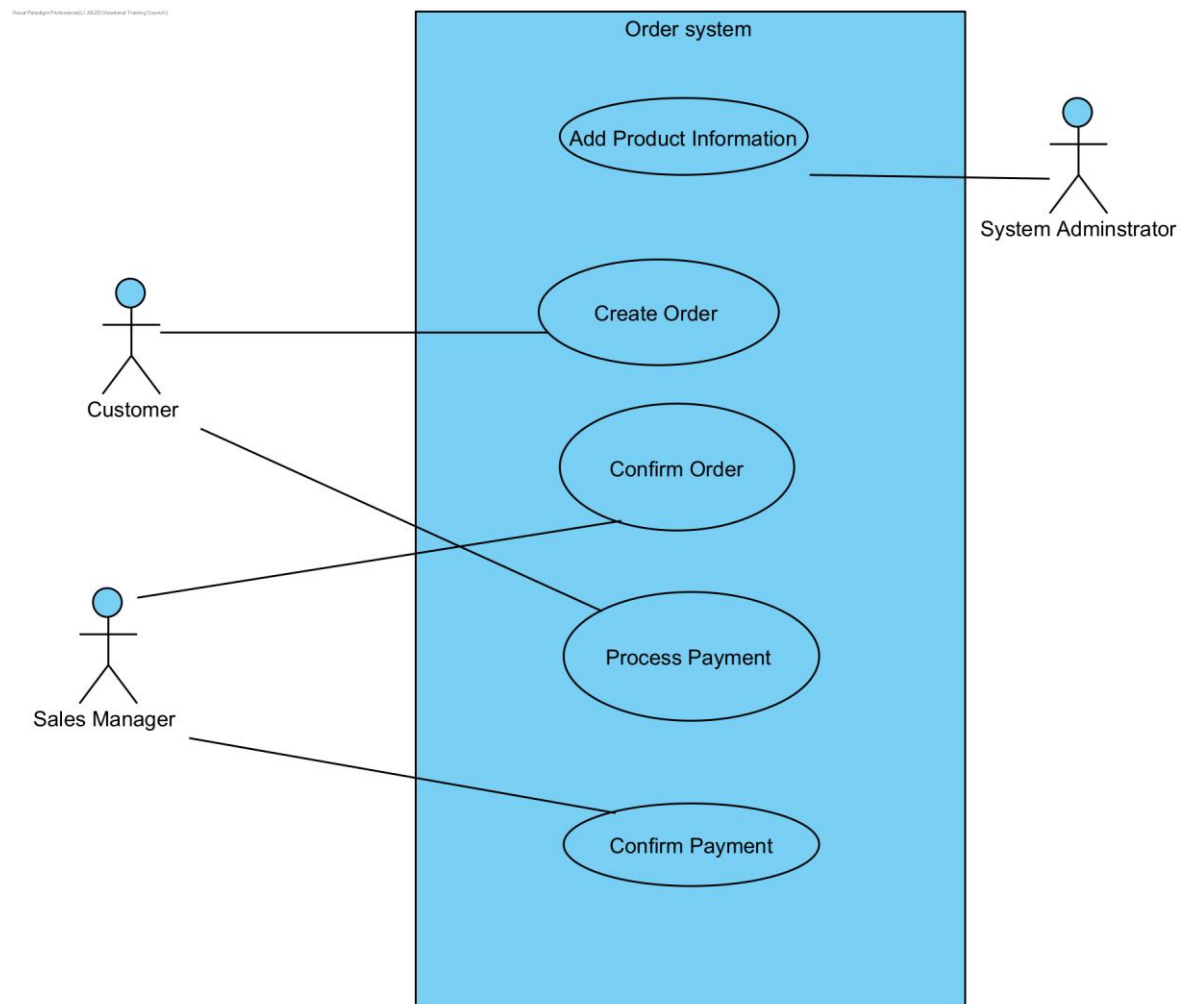
Use case name:	Generate Delivery Note
Use case ID:	UC-3400
Primary actor:	Delivery Personnel
Secondary actor(s):	None
Brief description:	The delivery person generates a delivery note for the shipment.
Preconditions:	UC-3300
Flow of events:	1. The user logs into the system. 2. The Delivery Personnel navigates to the delivery note generation page.

	3. The Delivery Personnel selects the shipment for which the delivery note needs to be generated. 4. The Delivery Personnel enters the delivery note details, including noteID and recipientSignature. 5. The Delivery Personnel submits the delivery note. 6. The system verifies the delivery of note information and saves it. 7. The system notifies the Delivery Personnel that the delivery note has been successfully generated.
Postconditions:	None.
Alternative flows and exceptions:	In step 4, if the recipient signature is missing, the Delivery Personnel is prompted to obtain a signature.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can generate delivery orders.
Assumptions:	It is assumed that the Delivery Personnel is familiar with the system and have been trained on the delivery note generation interface.
Issue:	The delivery note generation process may be delayed if the shipment has not yet been delivered, or the consignee's signature cannot be obtained.

Use case name:	Manage Transport Vehicle
Use case ID:	UC-3500
Primary actor:	Fleet Manager
Secondary actor(s):	None
Brief description:	Fleet managers manage transport vehicle information.
Preconditions:	None
Flow of events:	1. User logs into the system. 2. Fleet Manager navigates to the transport vehicle management page. 3. Fleet Manager enters vehicle details, including vehicleID and loadCapacity. 4. Fleet Manager submits the vehicle information. 5. System validates vehicle details and saves. 6. System notifies Fleet Manager that the transport vehicle has been successfully managed.
Postconditions:	None
Alternative flows and exceptions:	In step 3, if the vehicleID already exists, the system tells Fleet Manager to enter a different vehicleID. In step 5, if a required field is missing, the system displays an error message.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can manage transport vehicles.

Assumptions:	It is assumed that the Fleet Manager is familiar with the system and has been trained on the transport vehicle management interface.
Issue:	The transport vehicle management process may fail if there is an error in system validation or the vehicleID is not unique.

10.4 Order system



Use case name:	Add Product Information
Use case ID:	UC-4100
Primary actor:	System Administrator
Secondary actor(s):	None
Brief description:	System Administrator adds product information to the system.
Preconditions:	None
Flow of events:	1. The user logs into the system.

	2. The System Administrator navigates to the product management section. 3. The System Administrator enters product details, including productID, name, specifications, safetyCertification, and creationDate. 4. The System Administrator submits the product information. 5. The system validates the input and saves the product information. 6. The system displays a confirmation message indicating that the product has been successfully added.
Postconditions:	UC-4200
Alternative flows and exceptions:	In step 3, if a product with the same productID already exists, the system notifies the user and prompts them to enter a different productID. - In step 5, if required fields are missing, the system displays an error message.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can add product information.
Assumptions:	It is assumed that the System Administrator is familiar with the system and has received adequate training on how to use the product management interface.
Issue:	The product addition may fail if the system encounters a database error or if the productID is not unique.

Use case name:	Create Order
Use case ID:	UC-4200
Primary actor:	Customer
Secondary actor(s):	None
Brief description:	Customer creates an order for products.
Preconditions:	UC-4100
Flow of events:	1. The user logs into the system. 2. The Customer navigates to the order creation section. 3. The Customer selects products to include in the order. 4. The Customer enters order details, including deliveryTerms, orderDate, and customerID. 5. The Customer submits the order. 6. The system validates the order and saves it. 7. The system notifies the Customer that the order has been successfully created.
Postconditions:	UC-4300, UC-4400
Alternative flows and exceptions:	In step 3, if no products are available, the system notifies the Customer to wait until products are added. - In step 6, if required fields are missing, the system displays an error

	message.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can create orders.
Assumptions:	It is assumed that the Customer is familiar with the system and has received adequate training on how to use the order creation interface.
Issue:	The order creation may fail if the selected products are out of stock or if the system encounters a validation error.

Use case name:	Confirm Order
Use case ID:	UC-4300
Primary actor:	Sales Manager
Secondary actor(s):	None
Brief description:	The Sales Manager confirms the customer's order.
Preconditions:	UC-4200
Flow of events:	1. The user logs into the system. 2. The Sales Manager navigates to the order confirmation section. 3. The Sales Manager selects an order to confirm from a list of submitted orders. 4. The Sales Manager reviews the order details, including delivery terms, orderDate, and associated products. 5. The Sales Manager clicks the "Confirm" button to approve the order. 6. The system updates the order status to "Confirmed". 7. The system notifies the Customer of the order confirmation.
Postconditions:	None
Alternative flows and exceptions:	In step 4, if the order details are incomplete, the Sales Manager requests revisions from the Customer.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can confirm orders.
Assumptions:	It is assumed that the Sales Manager is familiar with the system and has received adequate training on how to use the order confirmation interface.
Issue:	The confirmation process may be delayed if the order lacks necessary details or if the products are unavailable.

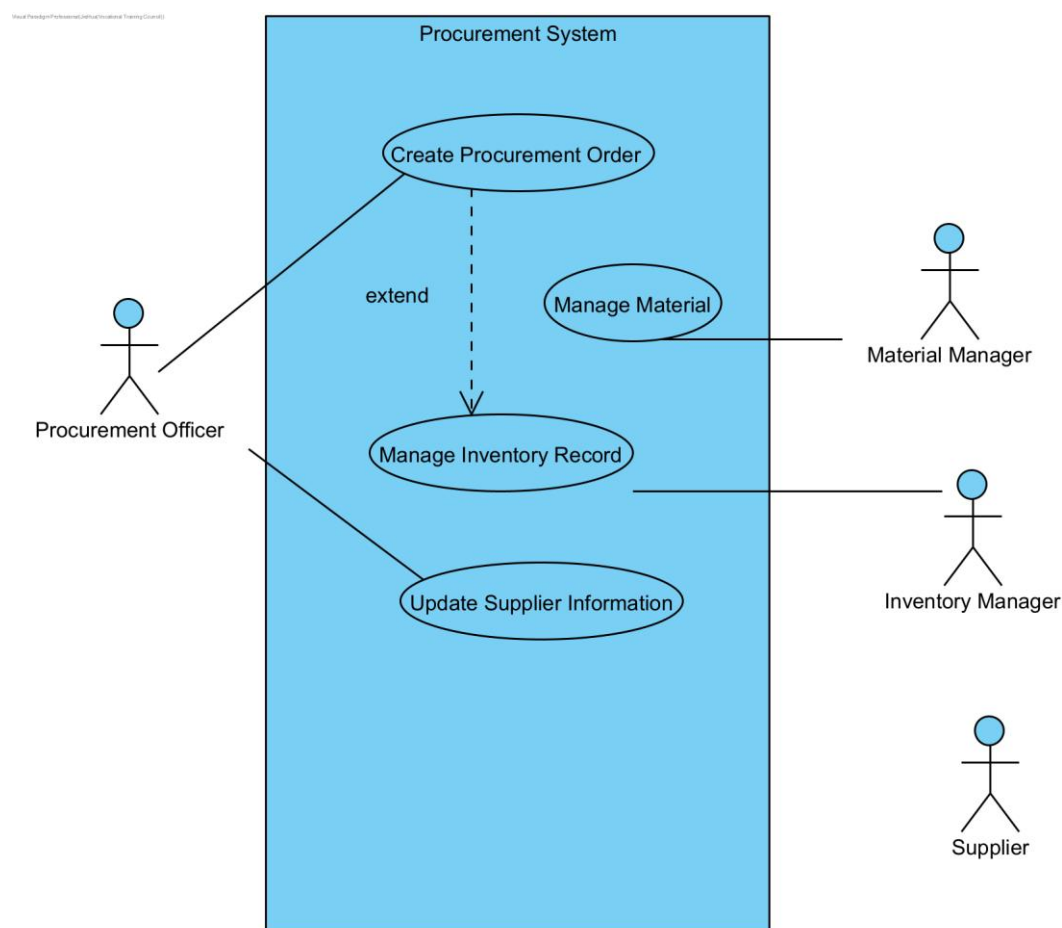
Use case name:	Process Payment
Use case ID:	UC-4400
Primary actor:	Customer
Secondary actor(s):	None

Brief description:	Customer processes payment for an order.
Preconditions:	UC-4200
Flow of events:	1. The user logs into the system. 2. The Customer navigates to the payment section. 3. The Customer selects the order to pay for. 4. The Customer enters payment details, including amount and paymentDate, and selects a payment gateway. 5. The Customer submits the payment. 6. The system processes the payment through the selected PaymentGateway and retrieves an approval Code. 7. The system notifies the Customer that the payment has been successfully processed.
Postconditions:	UC-4500
Alternative flows and exceptions:	In step 6, if the payment fails (e.g., due to insufficient funds), the system notifies the Customer to retry with a different payment method.
Non-behavior requirements:	The system must ensure secure payment processing and protect user data.
Assumptions:	It is assumed that the Customer has a valid payment method and is familiar with the payment interface.
Issue:	The payment process may fail if the payment gateway is down or if the Customer enters incorrect payment details.

Use case name:	Confirm Payment
Use case ID:	UC-4500
Primary actor:	Sales Manager
Secondary actor(s):	None
Brief description:	Sales Manager confirms the payment for an order.
Preconditions:	UC-4400
Flow of events:	1. The user logs into the system. 2. The Sales Manager navigates to the payment confirmation section. 3. The Sales Manager selects payment to confirm from a list of processed payments. 4. The Sales Manager reviews the payment details, including amount, paymentDate, and approvalCode. 5. The Sales Manager clicks the "Confirm" button to approve the payment. 6. The system updates the payment status to "Confirmed". 7. The system notifies the Customer of the payment confirmation.
Postconditions:	None.

Alternative flows and exceptions:	In step 4, if the payment details are incorrect or the approvalCode is invalid, the Sales Manager requests the Customer to reprocess the payment.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can confirm payments.
Assumptions:	It is assumed that the Sales Manager is familiar with the system and has received adequate training on how to use the payment confirmation interface.
Issue:	The confirmation process may be delayed if the payment gateway fails to provide a valid approvalCode.

10.5 Procurement system.



Use case name:	Create Procurement Order
Use case ID:	UC-5100
Primary actor:	Procurement Officer
Secondary actor(s):	None
Brief description:	The buyer creates a purchase order to order materials.
Preconditions:	None

Flow of events:	1. The user logs into the system. 2. The Procurement Officer navigates the purchase order creation page. 3. The Procurement Officer enters the purchase order details, including PO_Number, supplierID, and orderDate. 4. The Procurement Officer selects the materials to be ordered. 5. The Procurement Officer submits the purchase order. 6. The system validates the order information and saves it. 7. The system notifies the Procurement Officer that the purchase order has been successfully created.
Postconditions:	UC-5300
Alternative flows and exceptions:	In step 3, if the supplierID does not exist, the system prompts the Procurement Officer to select a valid supplier. In step 6, if the required fields are missing, the system displays an error message and asks the Procurement Officer to complete the information.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can create purchase orders.
Assumptions:	It is assumed that the Procurement Officer is familiar with the system operation and has been trained in how to use the purchase order interface.
Issue:	Purchase order creation may fail if the supplier information is incomplete, or the material is unavailable.

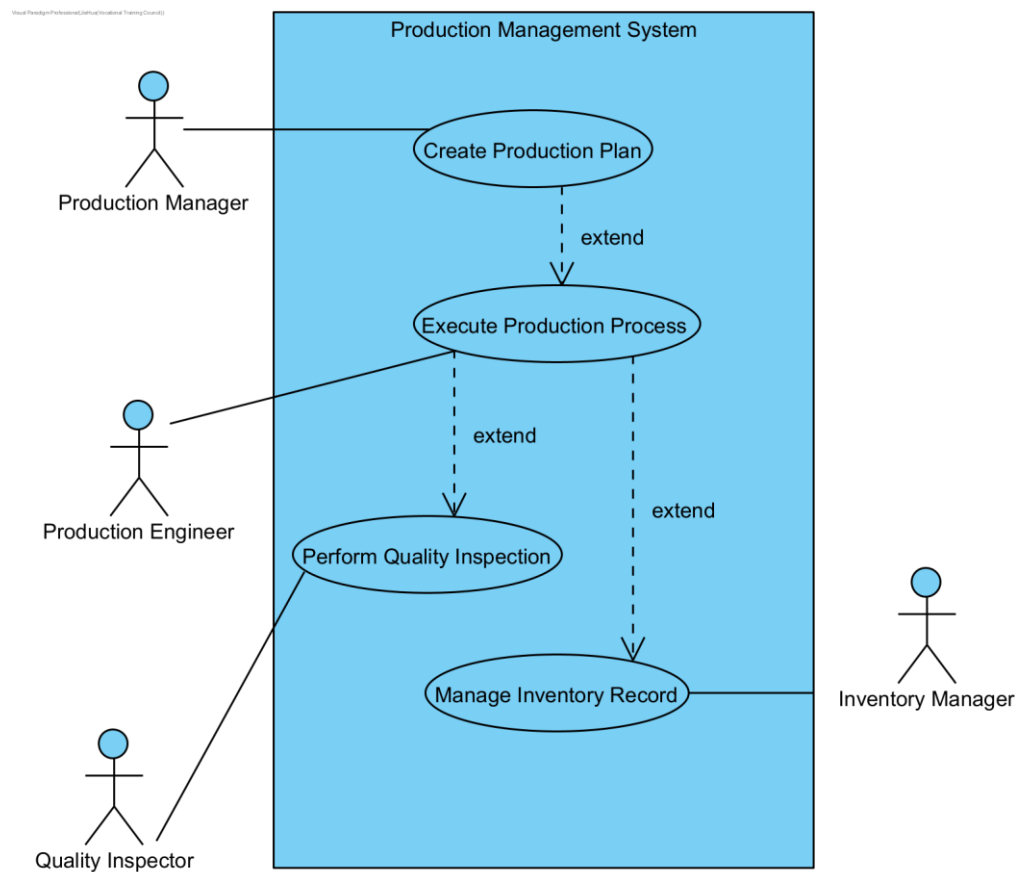
Use case name:	Manage Material
Use case ID:	UC-5200
Primary actor:	Material Manager
Secondary actor(s):	None
Brief description:	The material manager manages material information.
Preconditions:	None
Flow of events:	1. User logs into the system. 2. Material Manager navigates to the material management page. 3. Material Manager enters material details, including material_Number, reorderLevel, and quantity. 4. Material Manager submits the material information. 5. The system validates the material information and saves it. 6. The system notifies Material Manager that the material information has been successfully managed.
Postconditions:	None
Alternative flows	In step 3, if the material_Number already exists, the system

and exceptions:	prompts Material Manager to enter a different material_Number. In step 5, if a required field is missing, the system displays an error message.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can manage material information.
Assumptions:	It is assumed that the Material Manager is familiar with the system and has been trained on the material management interface.
Issue:	The material management process may fail if material numbers are duplicated or if system validation errors occur.

Use case name:	Manage Inventory Record
Use case ID:	UC-5300
Primary actor:	Inventory Manager
Secondary actor(s):	None
Brief description:	Inventory clerks manage inventory records to track material inventory.
Preconditions:	UC-5100
Flow of events:	1. The user logs into the system. 2. The Inventory Manager navigates to the Inventory Management page. 3. The Inventory Manager selects the material associated with the purchase order. 4. The Inventory Manager enters the inventory record details, including stockMovementType, transactionTime, and quantity. 5. The Inventory Manager submits the inventory record. 6. The system validates the inventory record information and saves it. 7. The system notifies the Inventory Manager that the inventory record has been successfully updated.
Postconditions:	None
Alternative flows and exceptions:	In step 4, if the material inventory is below reordering Level, the system notifies the Inventory Manager to reorder the material. In step 6, if the required fields are missing, the system displays an error message.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can manage inventory records.
Assumptions:	It is assumed that the Inventory Manager is familiar with the system and has been trained on the Inventory Management Interface.

Issue:	The inventory management process can fail if material data is incorrect or if system validation goes awry.
Use case name:	Update Supplier Information
Use case ID:	UC-5400
Primary actor:	Procurement Officer
Secondary actor(s):	None
Brief description:	The buyer updates the supplier's information.
Preconditions:	None
Flow of events:	1. The user logs into the system. 2. The Procurement Officer navigates the supplier management page. 3. The Procurement Officer selects the supplier to be updated. 4. The Procurement Officer enters the supplier details, including reliabilityRating and contactInfo. 5. The Procurement Officer submits the update. 6. The system validates the supplier's information and saves it. 7. The system notifies the Procurement Officer that the supplier information has been successfully updated.
Postconditions:	None
Alternative flows and exceptions:	In step 3, if the supplier does not exist, the system prompts the Procurement Officer to add a new supplier. In step 6, if the mandatory fields are missing, the system displays an error message.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can update supplier information.
Assumptions:	It is assumed that the Procurement Officer is familiar with the system and has been trained on the supplier management interface.
Issue:	The update process may fail if the supplier information is incomplete or if there are errors in system verification.

10.6 Production management system



Use case name:	Create Production Plan
Use case ID:	UC-6100
Primary actor:	Production Manager
Secondary actor(s):	None
Brief description:	Production Manager creates a production plan for an order.
Preconditions:	None
Flow of events:	1. The user logs into the system. 2. The Production Manager navigates to the production planning section. 3. The Production Manager selects an order to create a production plan for. 4. The Production Manager defines the plan details, including resourceAllocation, startDate, endDate, and status. 5. The Production Manager submits the production plan. 6. The system validates the plan and saves it. 7. The system notifies the Production Manager that the production plan has been successfully created.

Postconditions:	UC-6200.
Alternative flows and exceptions:	In step 3, if no orders are available, the system notifies the Production Manager to wait until an order is created. In step 6, if required fields are missing, the system displays an error message.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can create production plans.
Assumptions:	It is assumed that the Production Manager is familiar with the system and has received adequate training on how to use the production planning interface.
Issue:	The production plan creation may fail if the system encounters a validation error or if resources are overallocated.

Use case name:	Execute Production Process
Use case ID:	UC-6200
Primary actor:	Production Engineer
Secondary actor(s):	None
Brief description:	Production Engineer executes a production process as part of a production plan.
Preconditions:	UC-100
Flow of events:	1. The user logs into the system. 2. The Production Engineer navigates to the production process execution section. 3. The Production Engineer selects a production plan to work on. 4. The Production Engineer defines the process details, including processID and sequengceNumber. 5. The Production Engineer initiates the production process. 6. The system saves the production process details. 7. The system notifies the Production Engineer that the production process has been successfully initiated.
Postconditions:	UC-6300, UC-6400
Alternative flows and exceptions:	In step 3, if no production plans are available, the system notifies the Production Engineer to wait until a plan is created. In step 6, if required fields are missing, the system displays an error message.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can execute production processes.
Assumptions:	It is assumed that the Production Engineer is familiar with

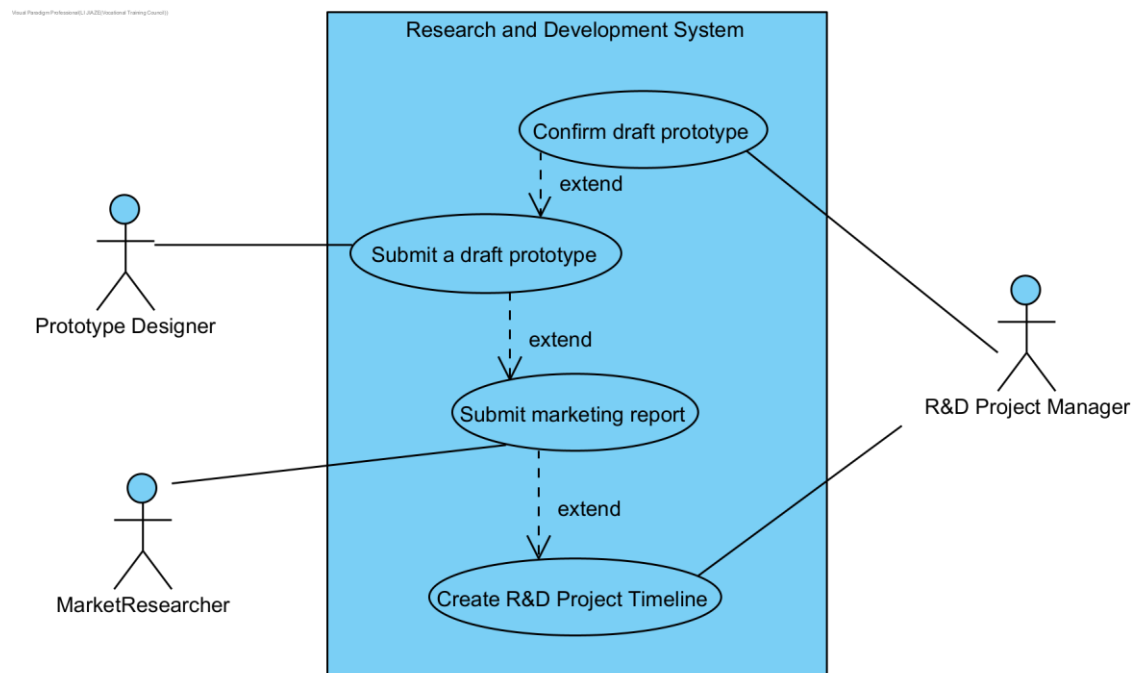
	the system and has received adequate training on how to use the production process execution interface.
Issue:	The production process execution may be delayed if required materials are unavailable or if the sequenceNumber conflicts with other processes.

Use case name:	Perform Quality Inspection
Use case ID:	UC-6300
Primary actor:	Quality Inspector
Secondary actor(s):	None
Brief description:	Quality Inspector performs a quality inspection on a production process.
Preconditions:	UC-3200
Flow of events:	1. The user logs into the system. 2. The Quality Inspector navigates to the quality inspection section. 3. The Quality Inspector selects a production process to inspect. 4. The Quality Inspector enters inspection details, including inspectionID, defectTypes, and inspectionDate. 5. The Quality Inspector submits the inspection report. 6. The system saves the inspection details. 7. The system notifies the Quality Inspector that the inspection has been successfully recorded.
Postconditions:	None.
Alternative flows and exceptions:	In step 4, if critical defects are found, the Quality Inspector flags the production process for review by the Production Engineer.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can perform quality inspections.
Assumptions:	It is assumed that the Quality Inspector is familiar with the system and has received adequate training on how to use the quality inspection interface.
Issue:	The quality inspection process may be delayed if the production process is not yet complete or if the inspection tools are unavailable.

Use case name:	Manage Inventory Record
Use case ID:	UC-6400
Primary actor:	Inventory Manager
Secondary actor(s):	None
Brief description:	Inventory Manager manages inventory records for a

	production process.
Preconditions:	UC-6200
Flow of events:	1. The user logs into the system. 2. The Inventory Manager navigates to the inventory management section. 3. The Inventory Manager selects a production process to manage inventory for. 4. The Inventory Manager enters inventory details, including stockMovementType, transactionTime, quantity, and associated material. 5. The Inventory Manager submits the inventory record. 6. The system validates the record and saves it. 7. The system notifies the Inventory Manager that the inventory record has been successfully updated.
Postconditions:	None.
Alternative flows and exceptions:	In step 4, if the material stock is below the orderLevel, the system notifies the Inventory Manager to reorder materials. In step 6, if required fields are missing, the system displays an error message.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can manage inventory records.
Assumptions:	It is assumed that the Inventory Manager is familiar with the system and has received adequate training on how to use the inventory management interface.
Issue:	The inventory management process may fail if the material data is incorrect or if the system encounters a validation error.

10.7 Research & development system



Use case name:	Confirm draft prototype
Use case ID:	UC-6400
Primary actor:	R&D Project Manager
Secondary actor(s):	None
Brief description:	R&D Project Manager confirm draft prototype.
Preconditions:	UC-7300
Flow of events:	1. The user logs into the system. 2. The Manager navigates to the section for reviewing draft prototypes. 3. The Manager selects the draft prototype to be confirmed from a list of submitted prototypes. 4. The Manager reviews the prototype details, including design specifications, functionality, and any attached supporting documents. 5. Once satisfied with the prototype, the Manager clicks the "Confirm" button to approve the draft prototype.
Postconditions:	None.
Alternative flows and exceptions:	In step 3, Instead of confirming the prototype, the Manager requests revisions from the Prototype Designer.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can confirm prototypes.
Assumptions:	It is assumed that the R&D Project Manager is familiar with the system and has received adequate training on how to use

	the prototype confirmation interface.
Issue:	The confirmation process may be hindered if the draft prototype lacks necessary details or supporting documents.

Use case name:	Submit a draft prototype
Use case ID:	UC-6300
Primary actor:	Prototype Designer
Secondary actor(s):	None
Brief description:	Prototype Designer submit a draft prototype.
Preconditions:	Study the marketing report
Flow of events:	1. The user logs into the system. 2. The user navigates to the prototype submission section. 3. The user selects the draft prototype to be submitted from their list of prototypes. 4. The user reviews the prototype details to ensure all necessary information is included. 5. The user makes any final adjustments or edits to the prototype. 6. The user attaches any supporting documents or files that are required for the submission. 7. The system notifies the user that the draft prototype has been successfully submitted.
Postconditions:	UC-6400
Alternative flows and exceptions:	In step 3, if the draft prototype already exists, the system notifies the user and prompts them to enter a different draft prototype. In step 6, if the user does not attaches any supporting documents or files. The system validates the submission and detects that some required fields are incomplete.
Non-behavior requirements:	The system must be able to handle multiple prototype submissions simultaneously without significant delay. The system must ensure that only authenticated and authorized users can submit prototypes.
Assumptions:	It is assumed that the Prototype Designer is familiar with the system and has received adequate training on how to use the prototype submission interface.
Issue:	The Designer might encounter issues while uploading supporting documents, such as file size limits being exceeded or unsupported file formats.

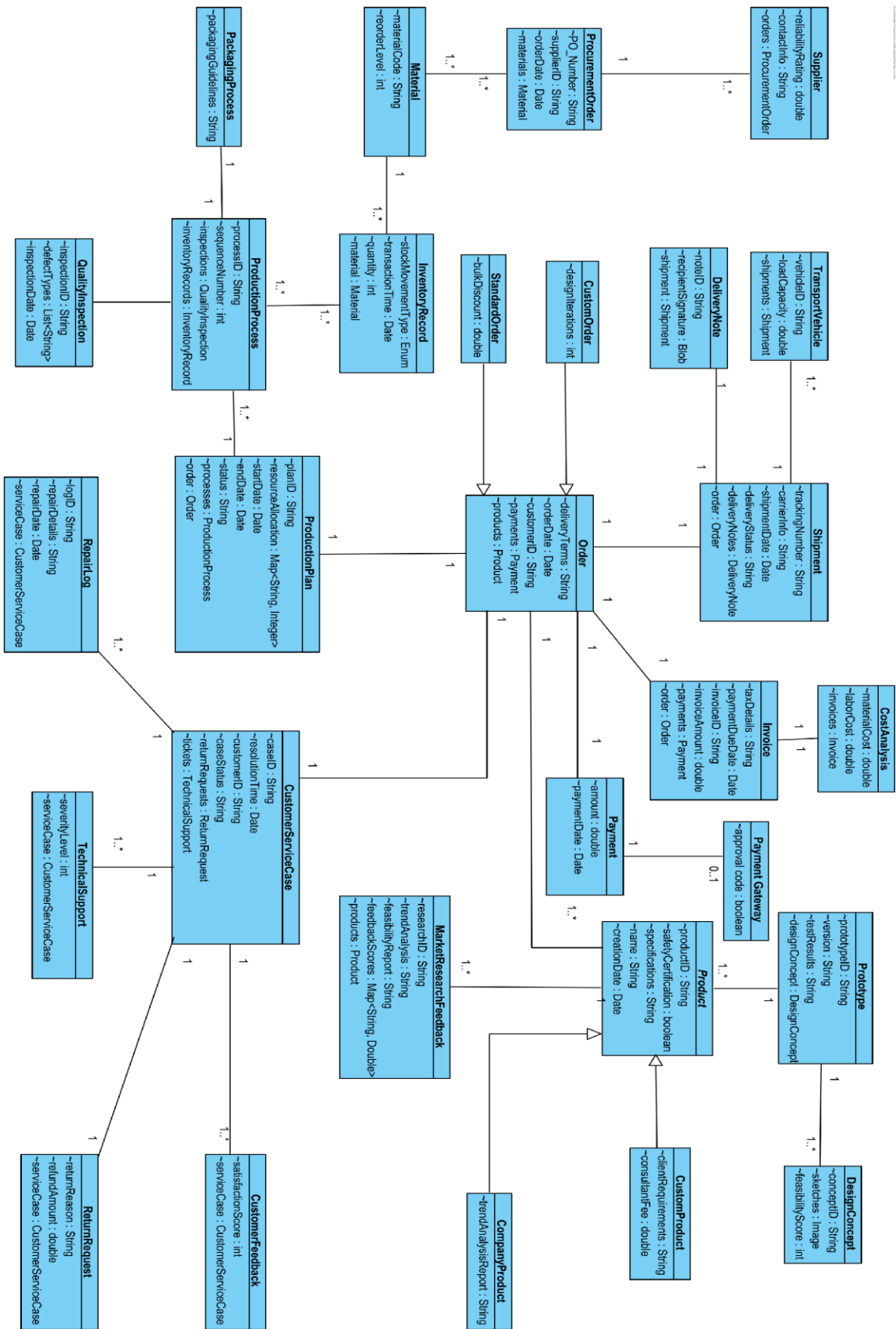
Use case name:	Submit marketing report
Use case ID:	UC-6200
Primary actor:	MarketResearcher

Secondary actor(s):	None
Brief description:	MarketResearcher submit marketing report.
Preconditions:	UC-6100
Flow of events:	1. The user logs into the system. 2. The user navigates to the report submission section. 3. The Researcher selects the marketing report to be submitted from their list of reports. 4. The Researcher attaches any supporting documents or files that are required for the submission. 5. The Researcher clicks the "Submit" button to submit the marketing report. 6. The system validates the submission to ensure all required fields are completed. 7. The system displays a confirmation message indicating that the report has been successfully submitted.
Postconditions:	UC-6300
Alternative flows and exceptions:	In step 3, The system detects that a report with the same name or version already exists. In step 6, The system validates the submission and detects that some required fields are incomplete.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can submit reports. The report submission interface must be intuitive and user-friendly, allowing researchers to easily navigate and complete the submission process.
Assumptions:	It is assumed that the Market Researcher is familiar with the system and has received adequate training on how to use the report submission interface.
Issue:	The report submission may be rejected if required fields are not filled out completely or accurately.

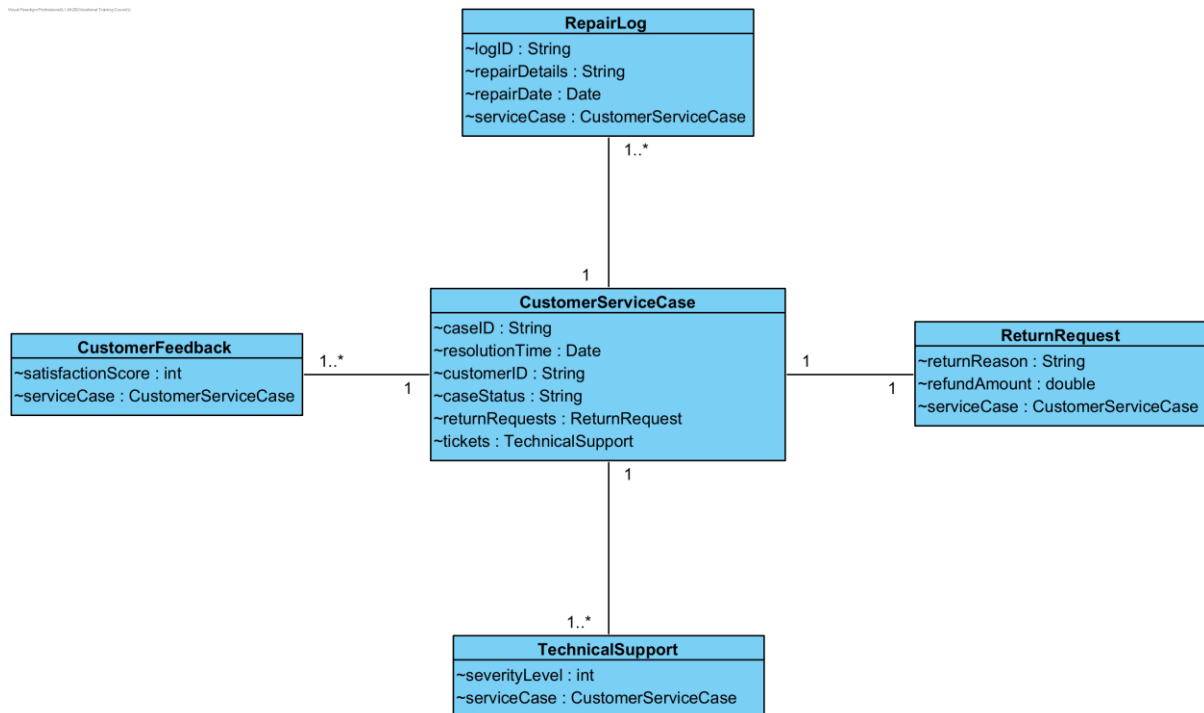
Use case name:	Create R&D Project Timeline
Use case ID:	UC-6100
Primary actor:	R&D Project Manager
Secondary actor(s):	None
Brief description:	R&D Project Manager create R&D Project Timeline.
Preconditions:	None
Flow of events:	1. The user logs into the system. 2. The user navigates to the section for creating a new project timeline. 3. The user defines key milestones for the project,

	including start and end dates for each milestone. 4. The Manager submits the project timeline for approval from relevant stakeholders or higher management. 5. The system notifies stakeholders of the new timeline submission.
Postconditions:	None.
Alternative flows and exceptions:	In step 3, The system detects that a timeline for the selected project already exists. In step 3, The system detects that some required milestone information is missing or incomplete.
Non-behavior requirements:	The system must ensure that only authenticated and authorized users can create and submit project timelines.
Assumptions:	It is assumed that the R&D Project Manager is familiar with the system and has received adequate training on how to use the timeline creation interface.
Issue:	There might be conflicts in resource allocation, such as overbooking of personnel, which could delay the project timeline.

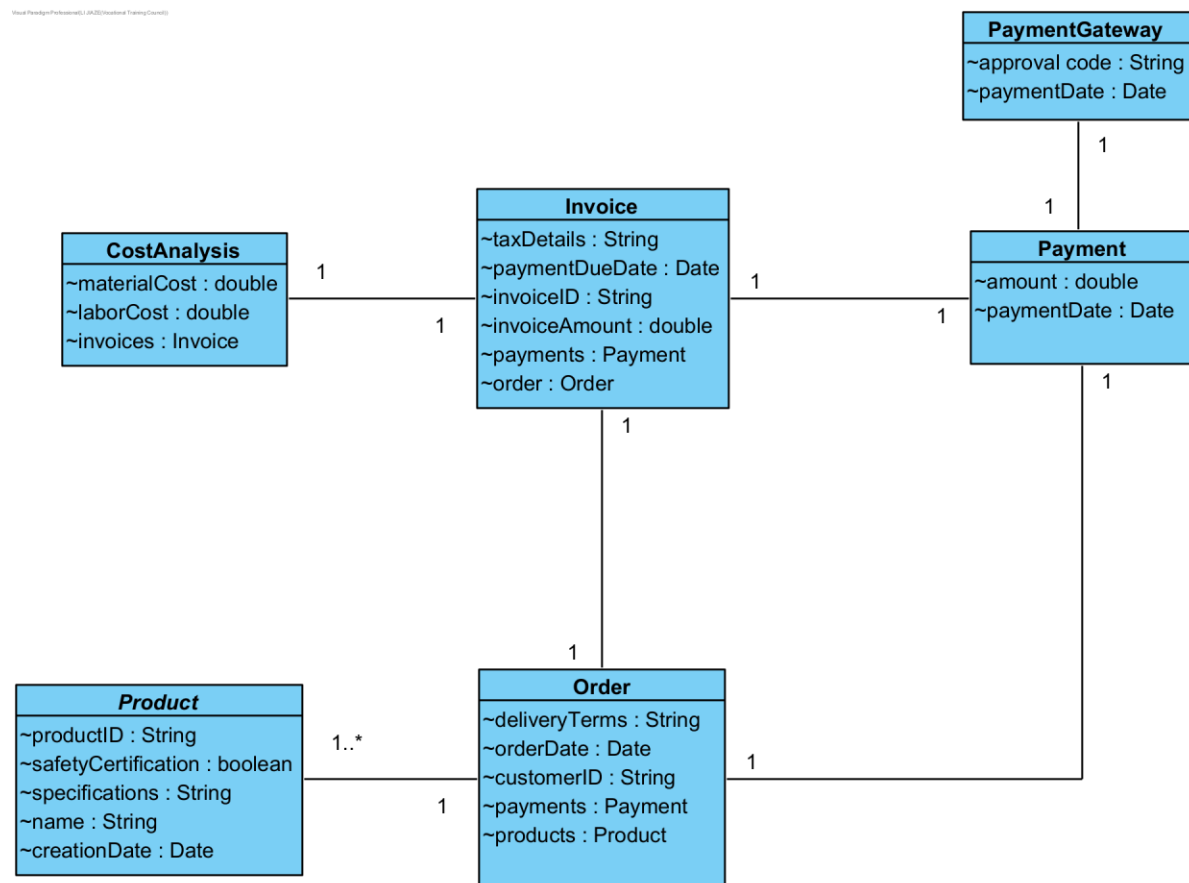
11 Class Diagram



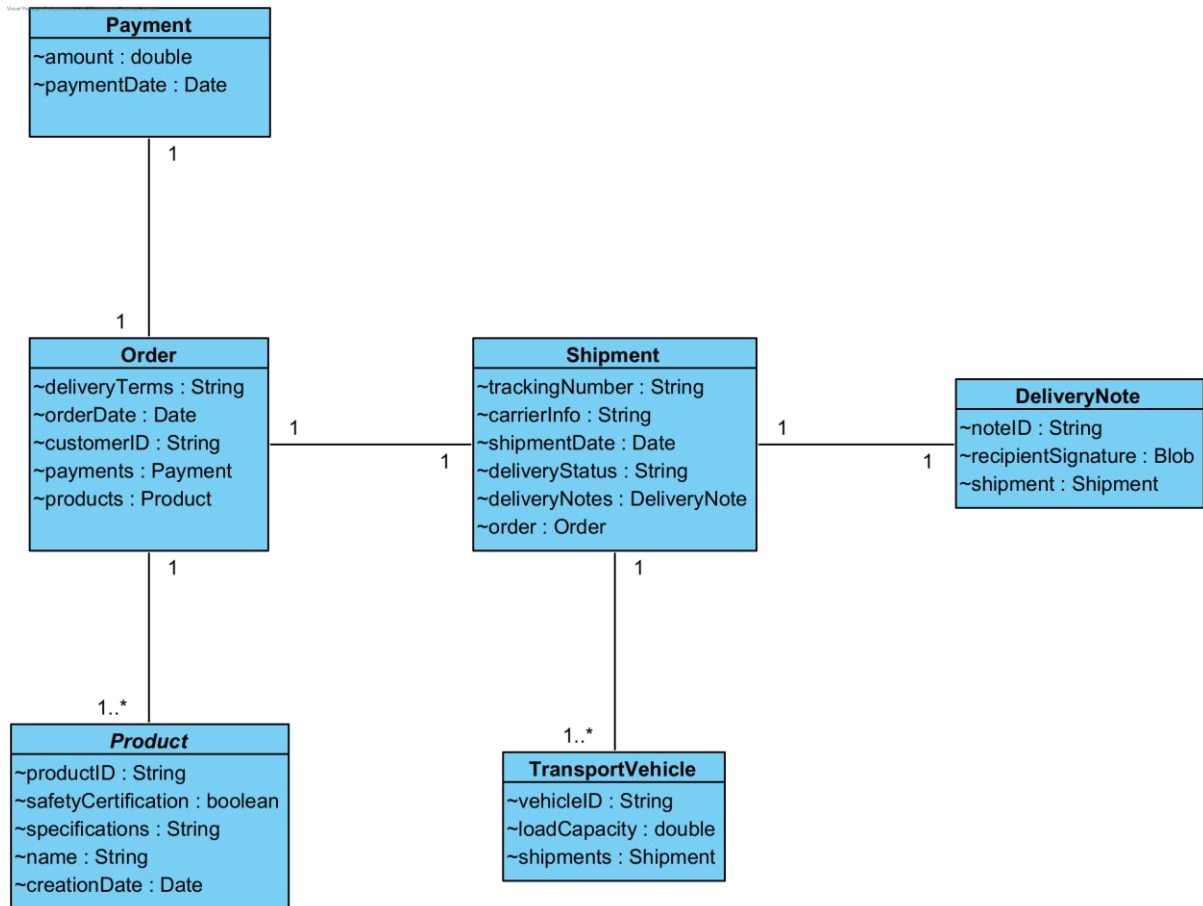
11.1 Customer service and support system



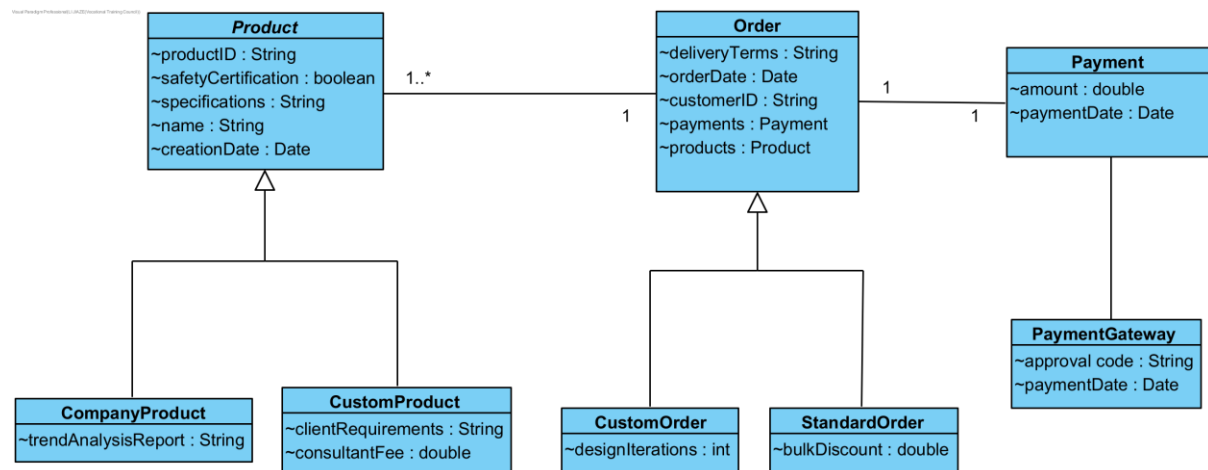
11.2 Order and payment management system



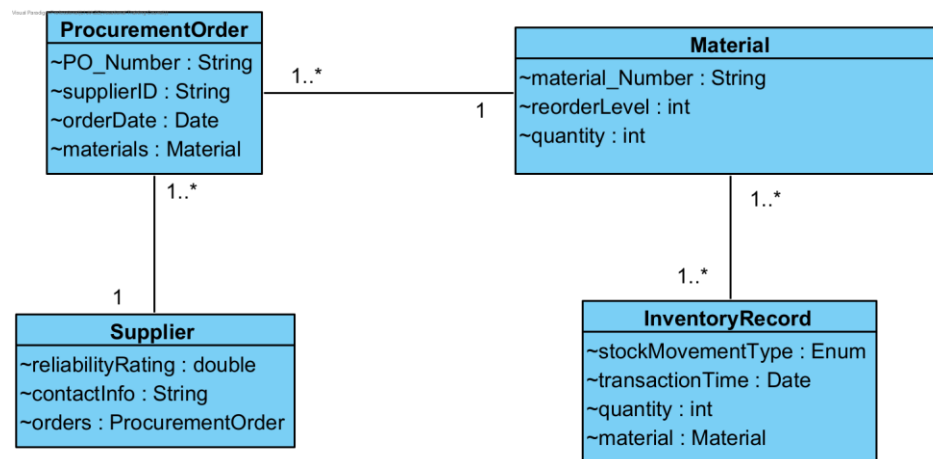
11.3 Order processing system



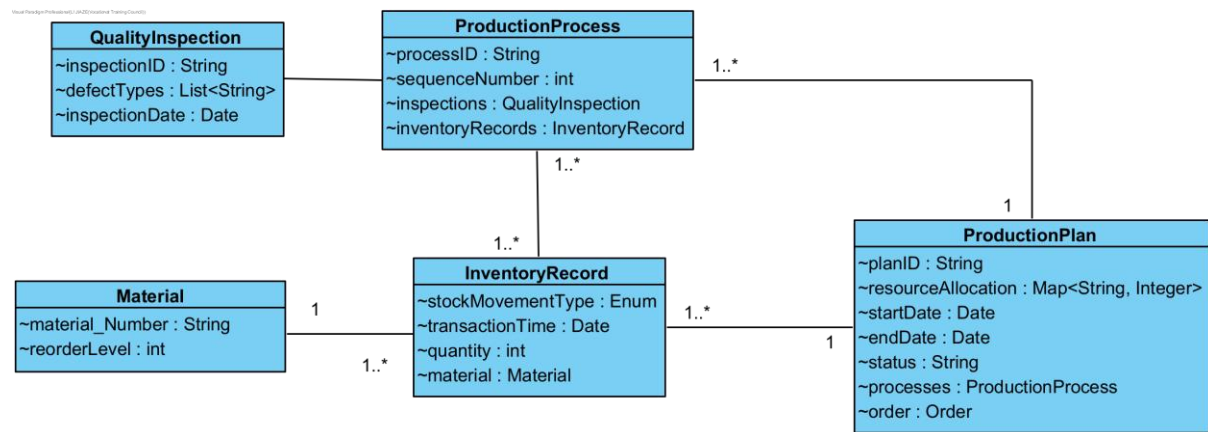
11.4 Order system



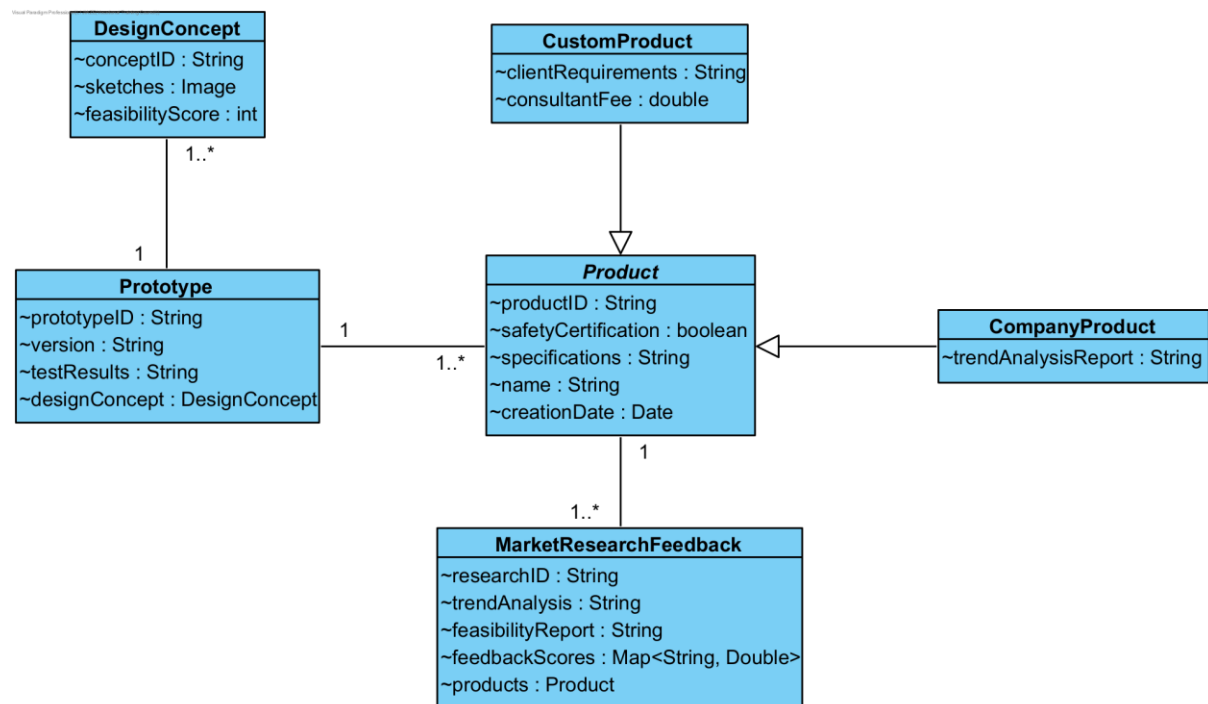
11.5 Procurement system



11.6 Production Management System



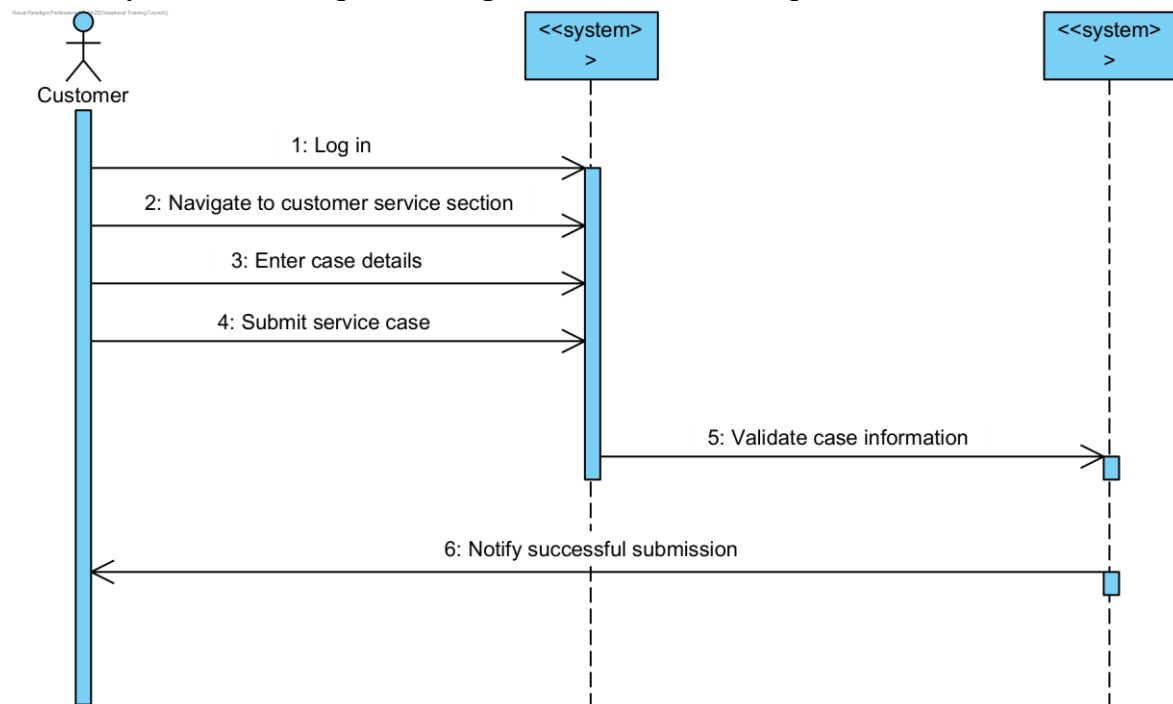
11.7 Research & Development system



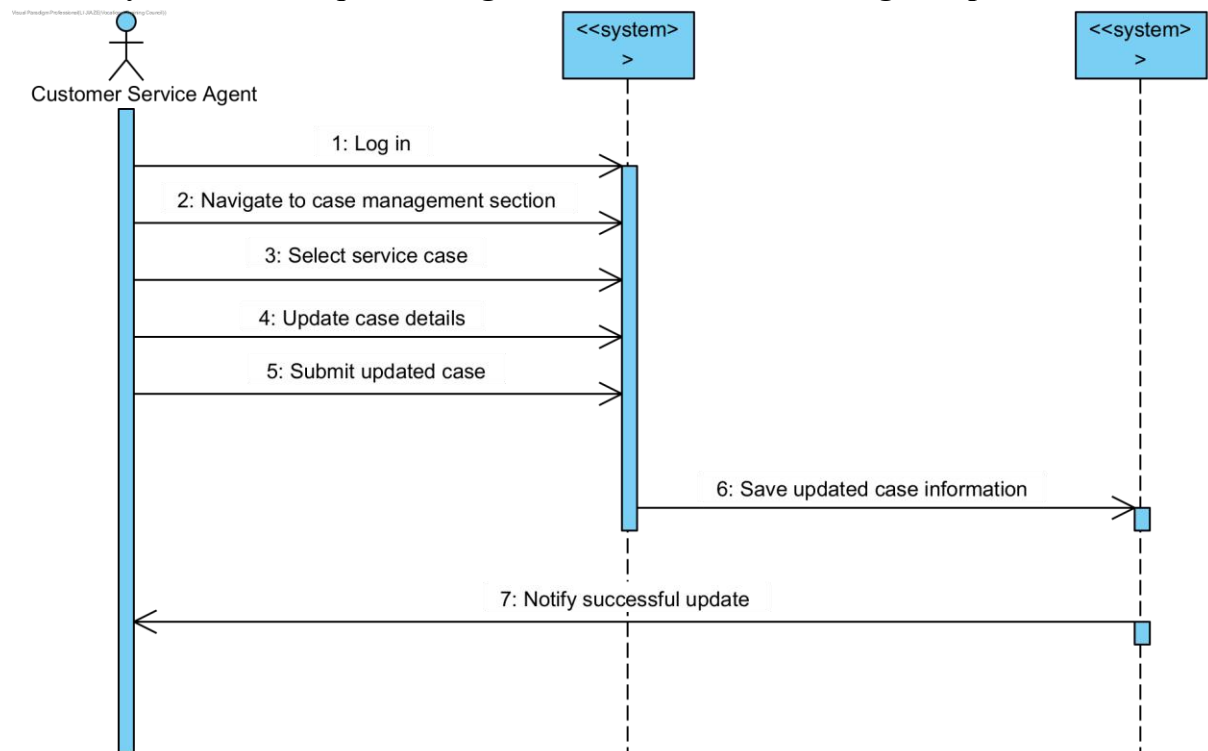
12 Sequence Diagram

12.1 System Level Sequence Diagram for Customer service and support

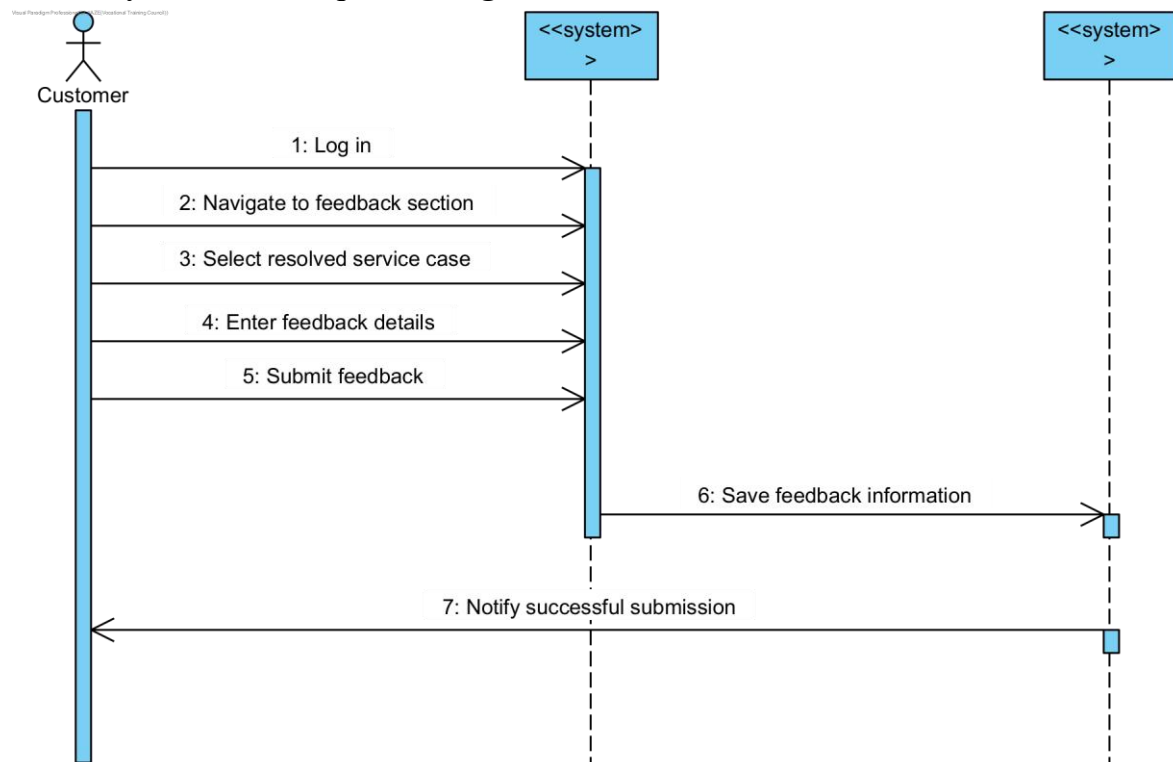
12.1.1 System Level Sequence Diagram for Customer request service



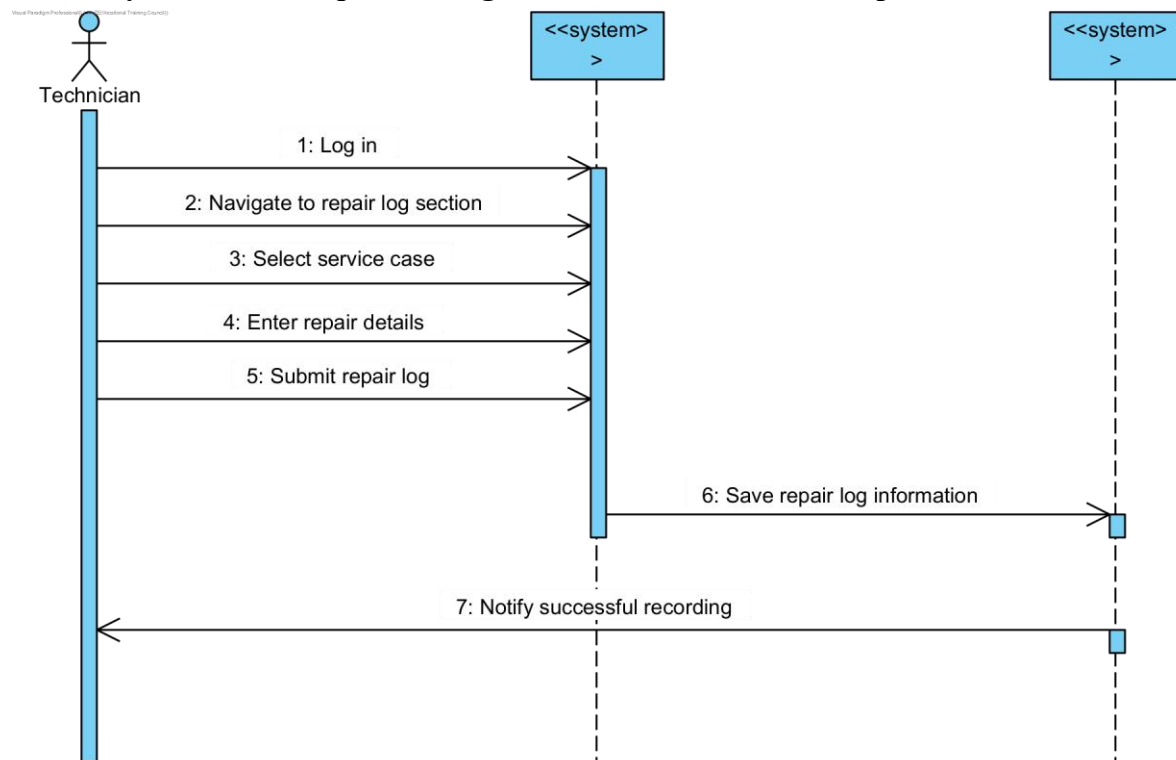
12.1.2 System Level Sequence Diagram for Customer service Agent Update case details



12.1.3 System Level Sequence Diagram for Customer receive feedback

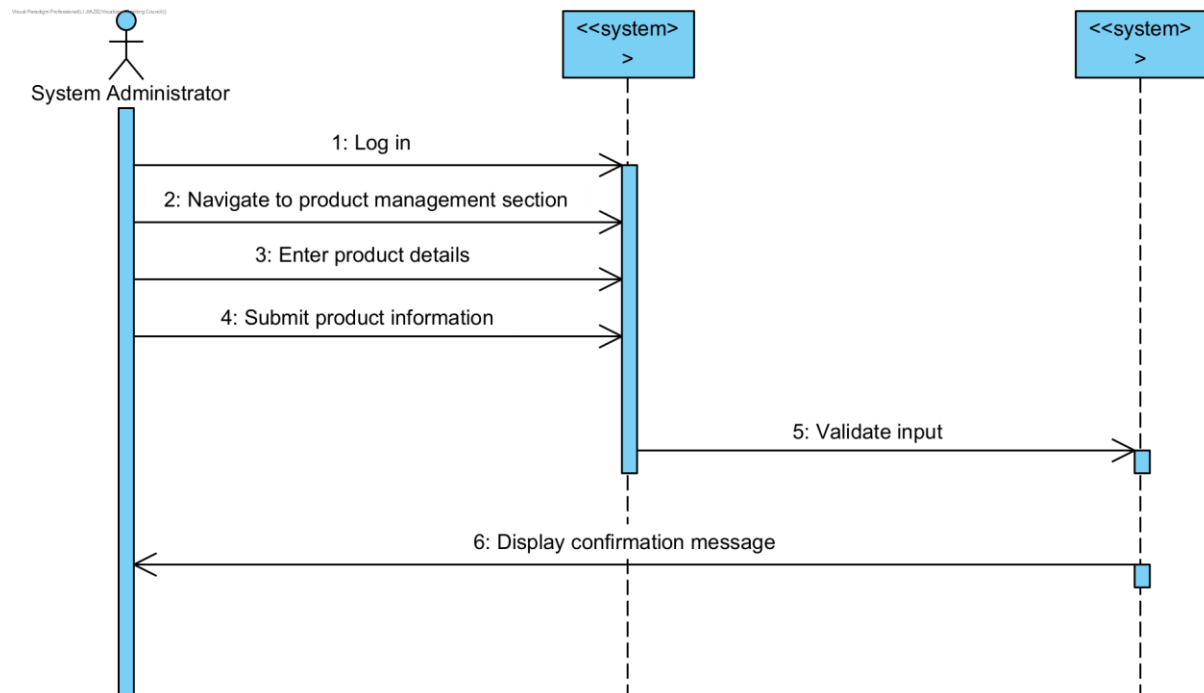


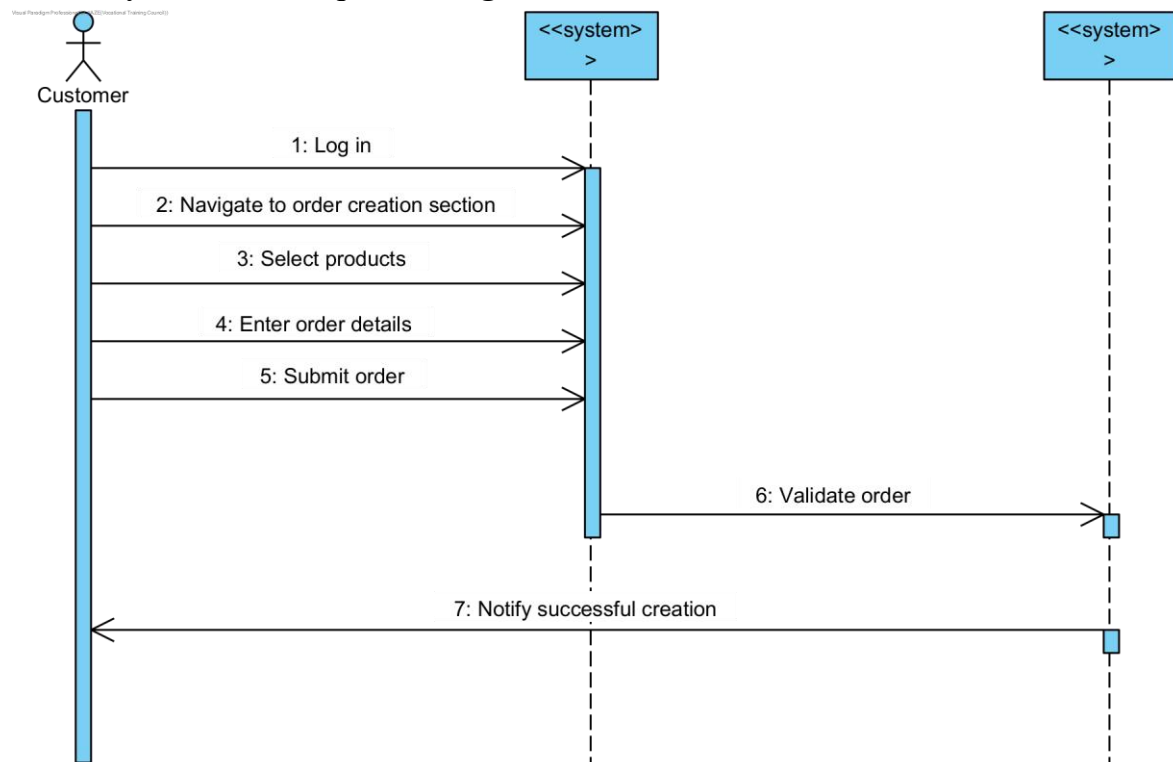
12.1.4 System Level Sequence Diagram for Technician enter repair details



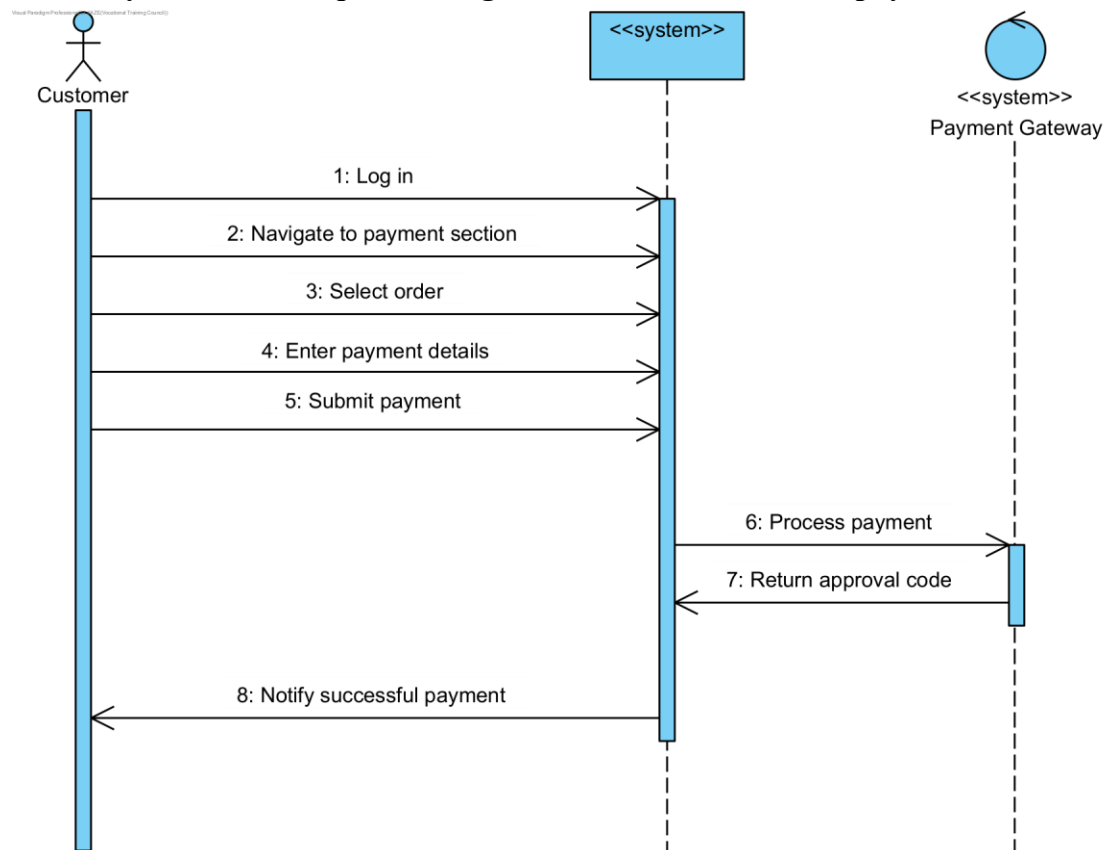
12.2 System Level Sequence Diagram for Order and payment management

12.2.1 System Level Sequence Diagram for System Administrator enter product details

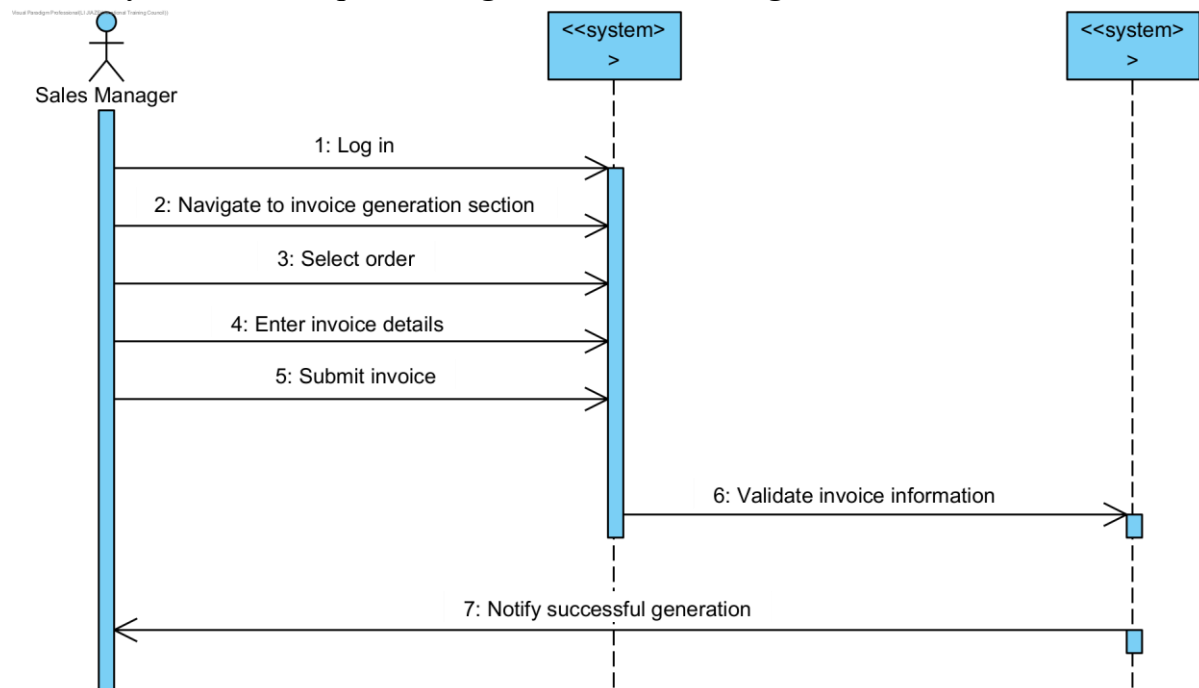


12.2.2 System Level Sequence Diagram for Customer submit order

12.2.3 System Level Sequence Diagram for Customer submit payment

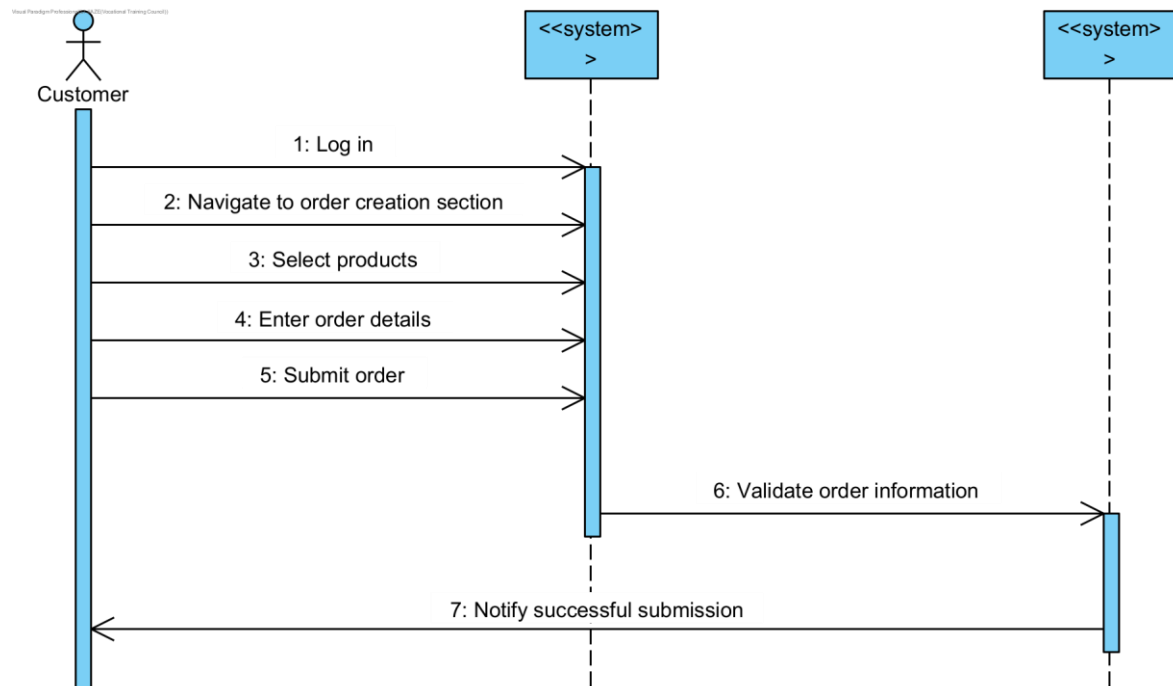


12.2.4 System Level Sequence Diagram for Sales Manager submit invoice

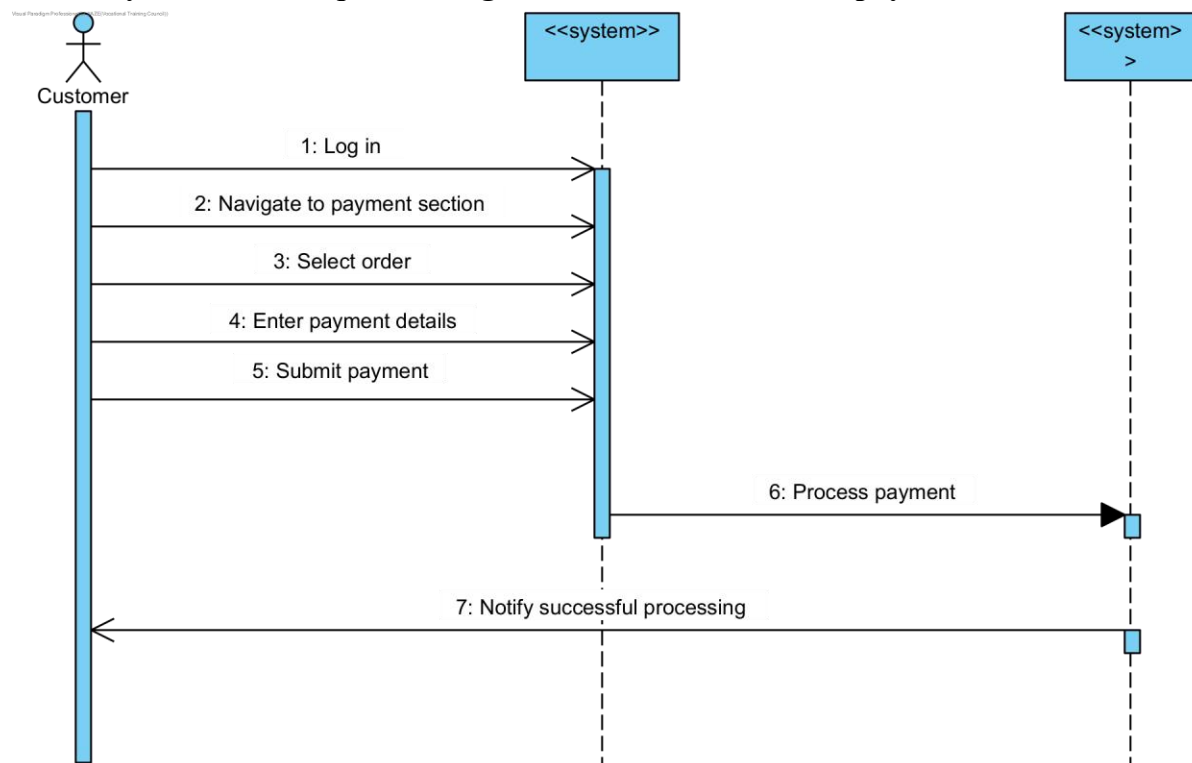


12.3 System Level Sequence Diagram for Order processing

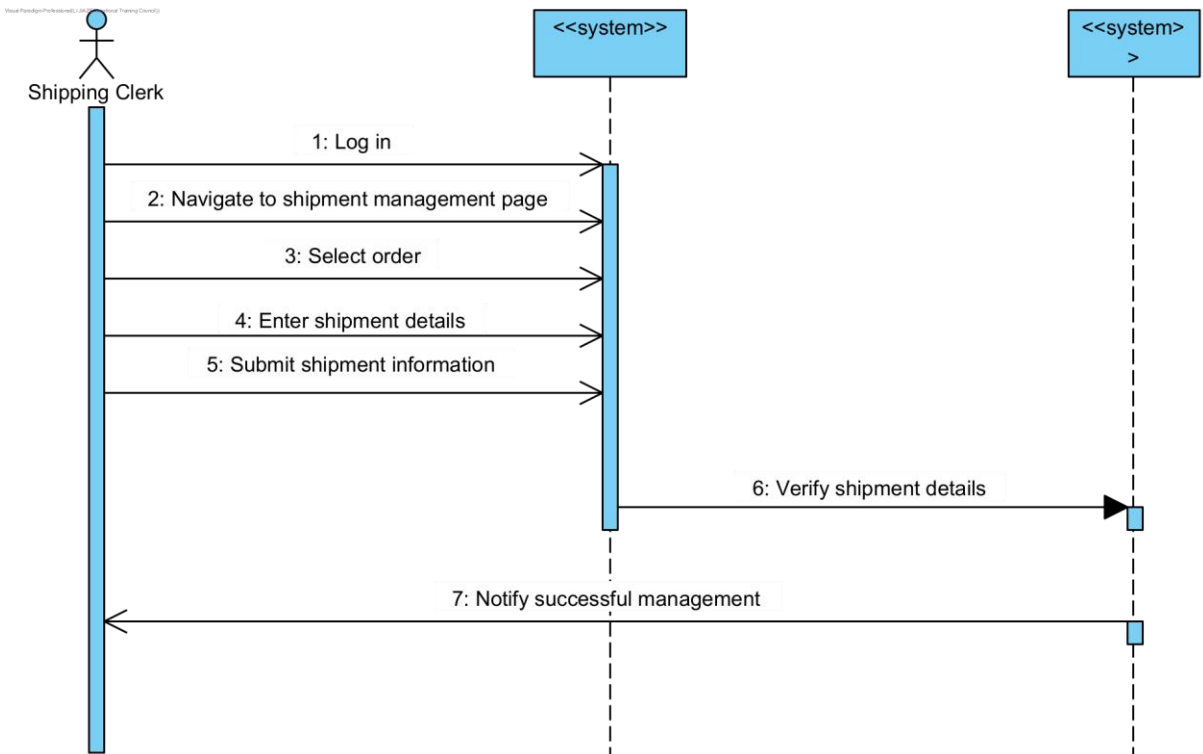
12.3.1 System Level Sequence Diagram for Customer submit order



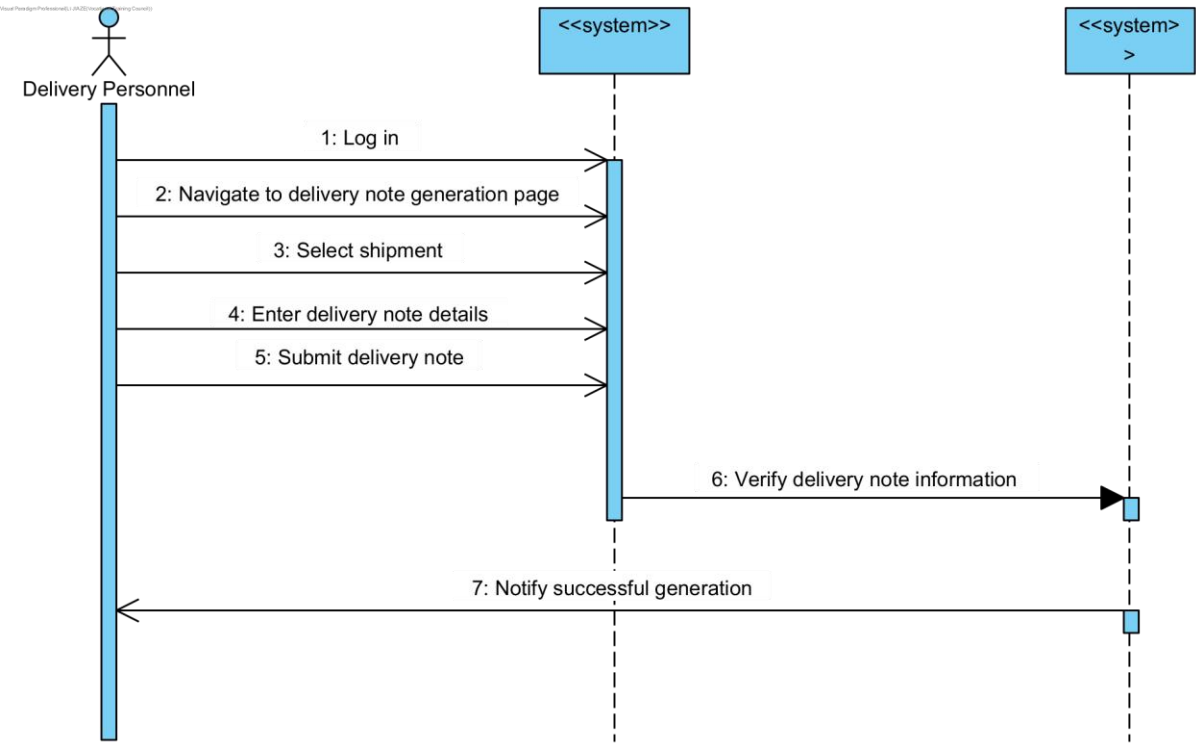
12.3.2 System Level Sequence Diagram for Customer submit payment



12.3.3 System Level Sequence Diagram for Shipping Clerk

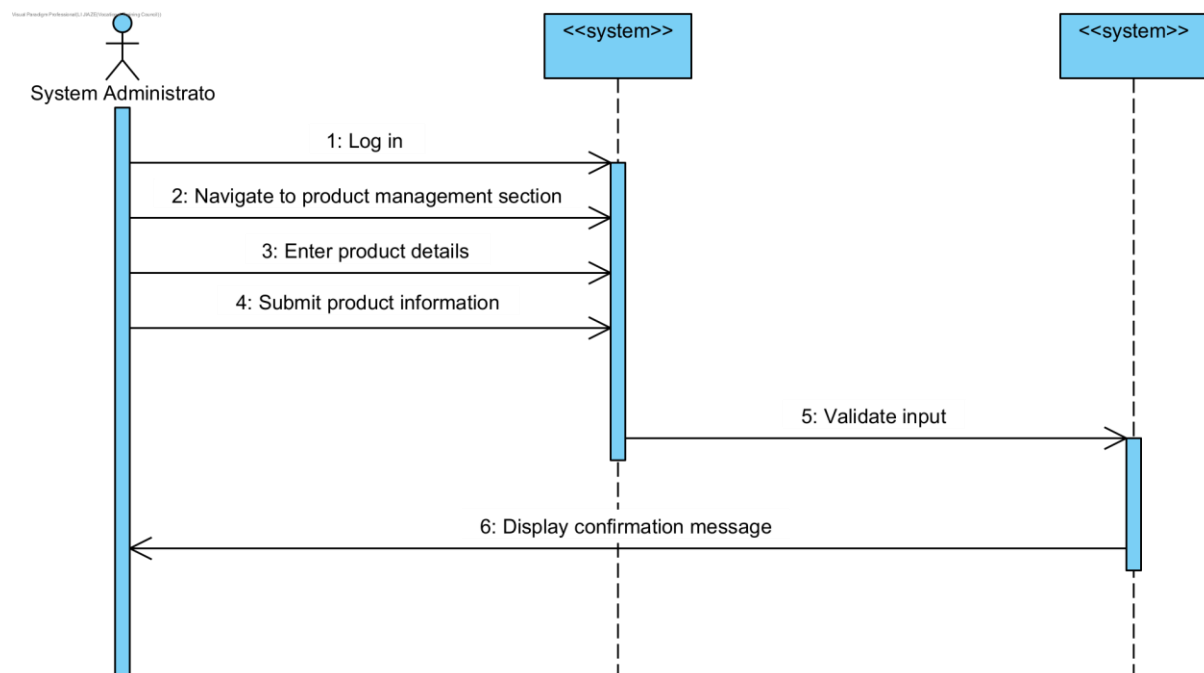


12.3.4 System Level Sequence Diagram for Delivery Personnel

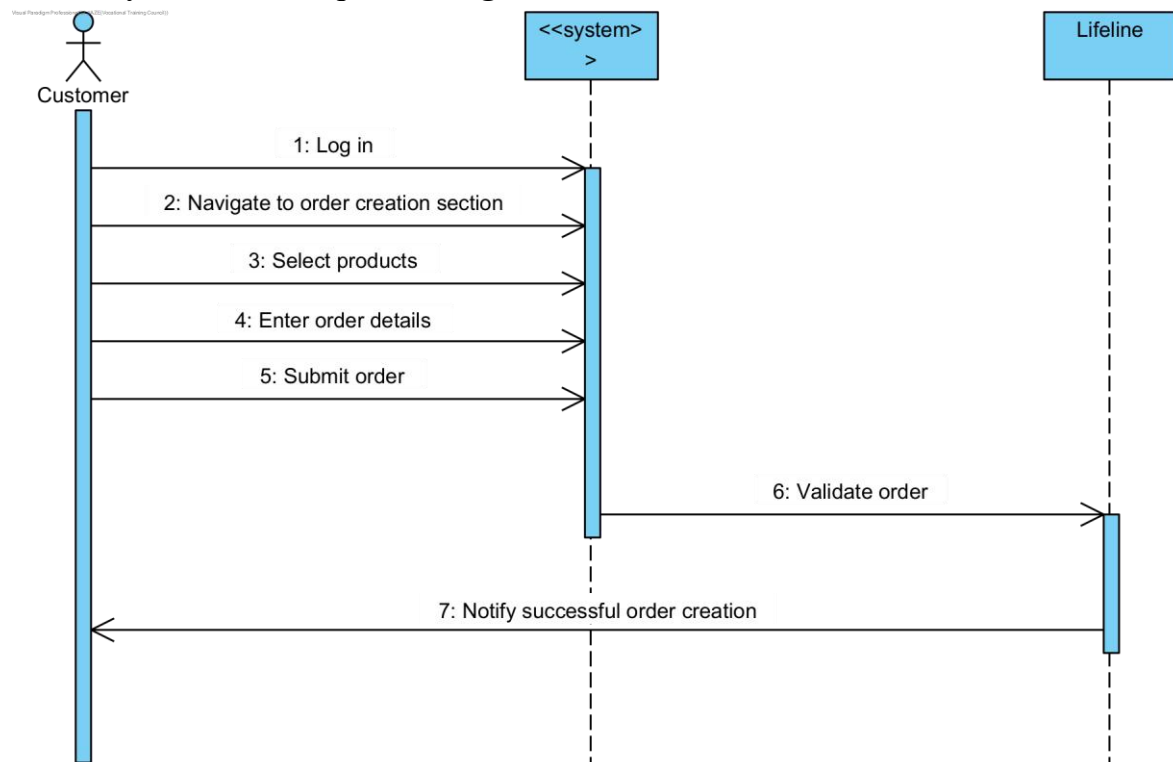


12.4 System Level Sequence Diagram for Order system

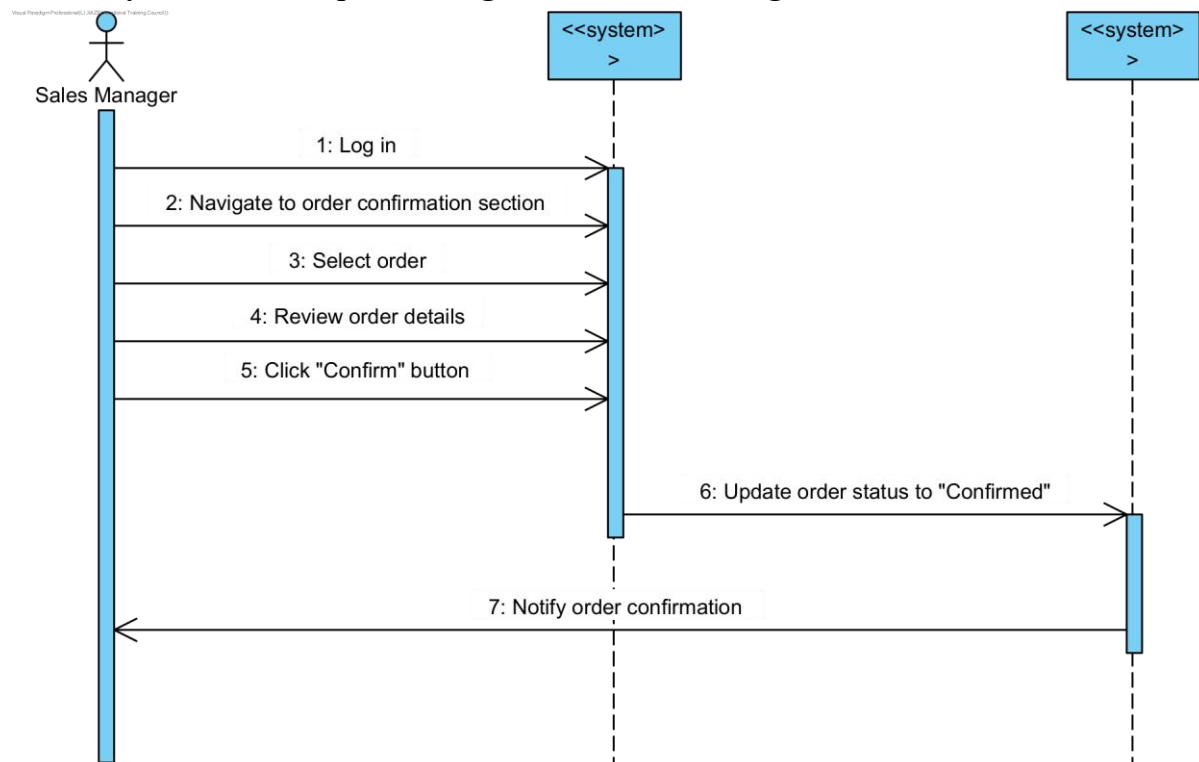
12.4.1 System Level Sequence Diagram for System Administrator update product details



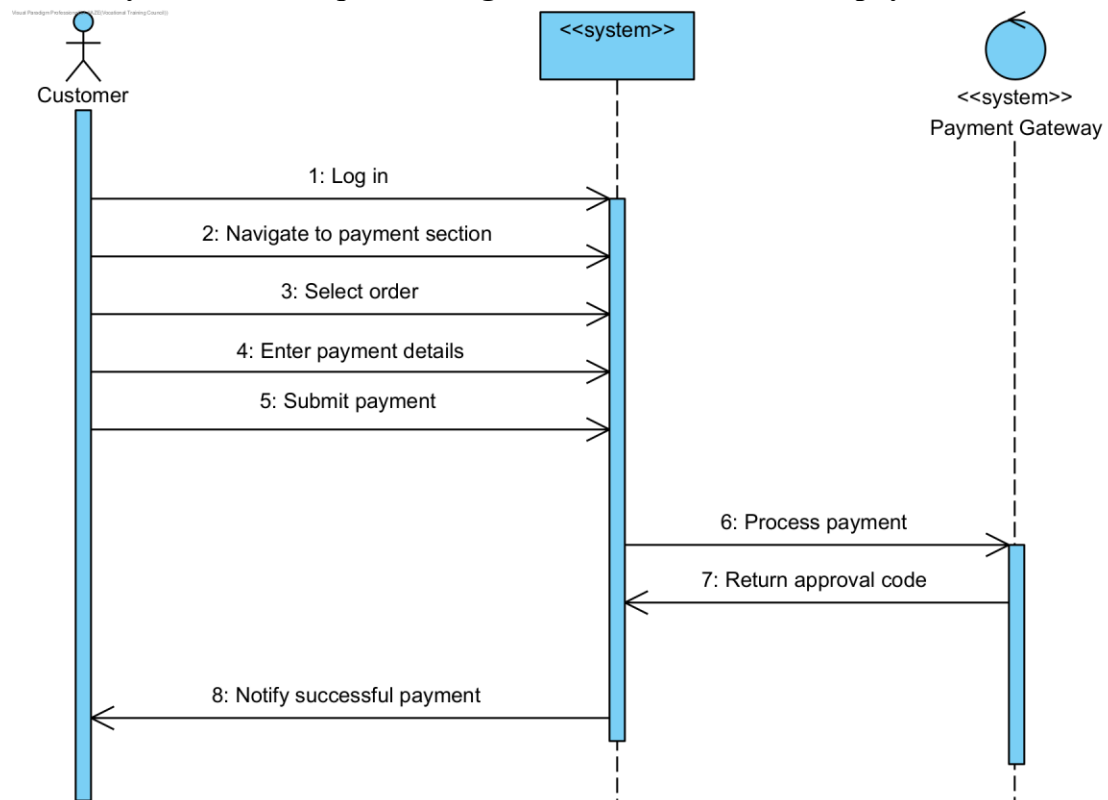
12.4.2 System Level Sequence Diagram for Customer submit order



12.4.3 System Level Sequence Diagram for Sales Manager confirm Order

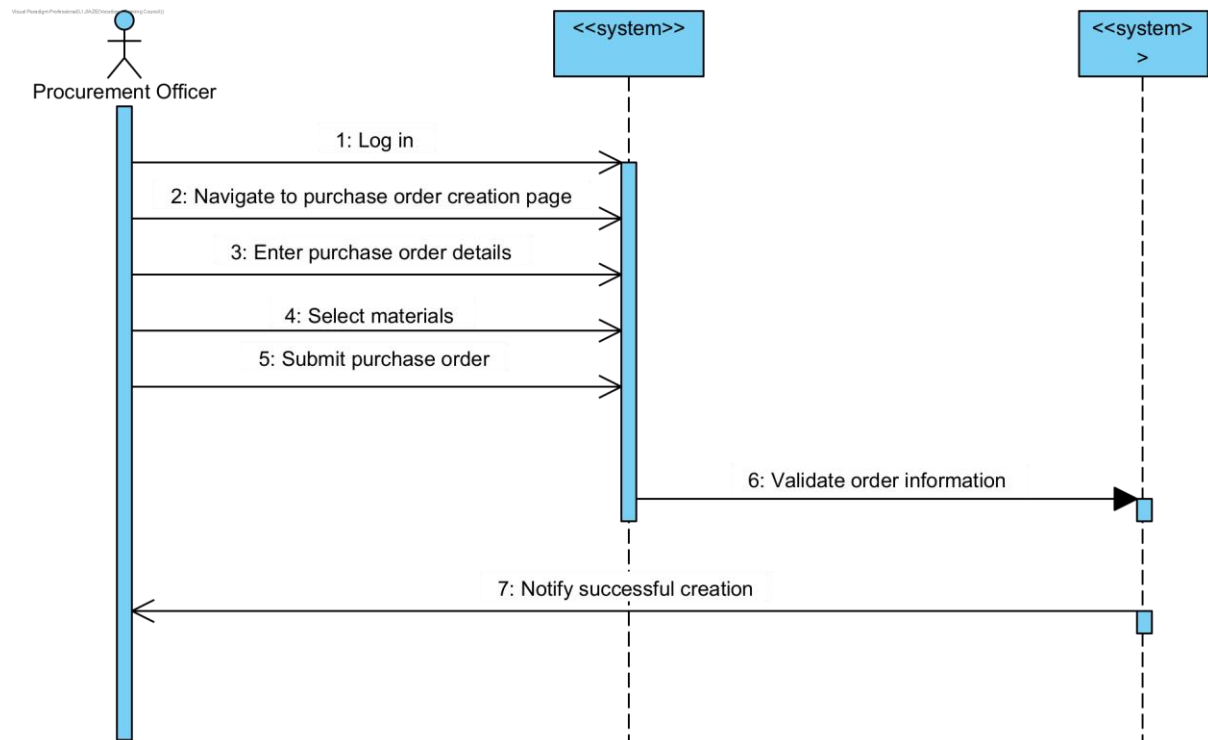


12.4.4 System Level Sequence Diagram for Customer submit payment

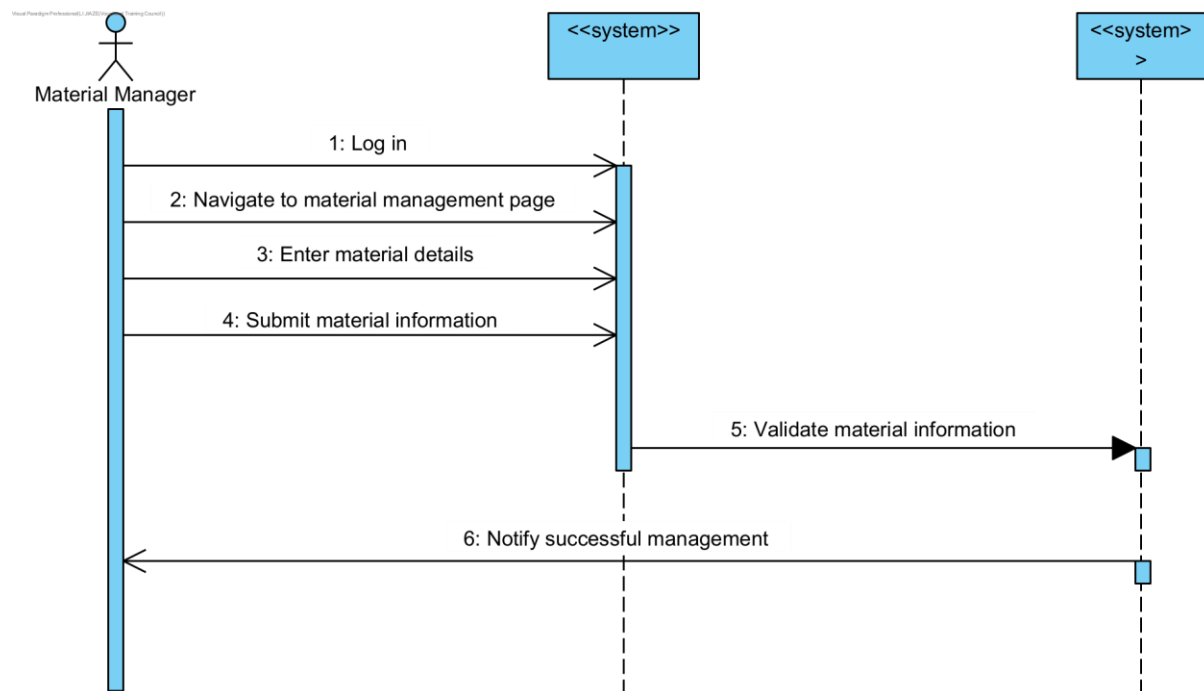


12.5 System Level Sequence Diagram for Procurement system

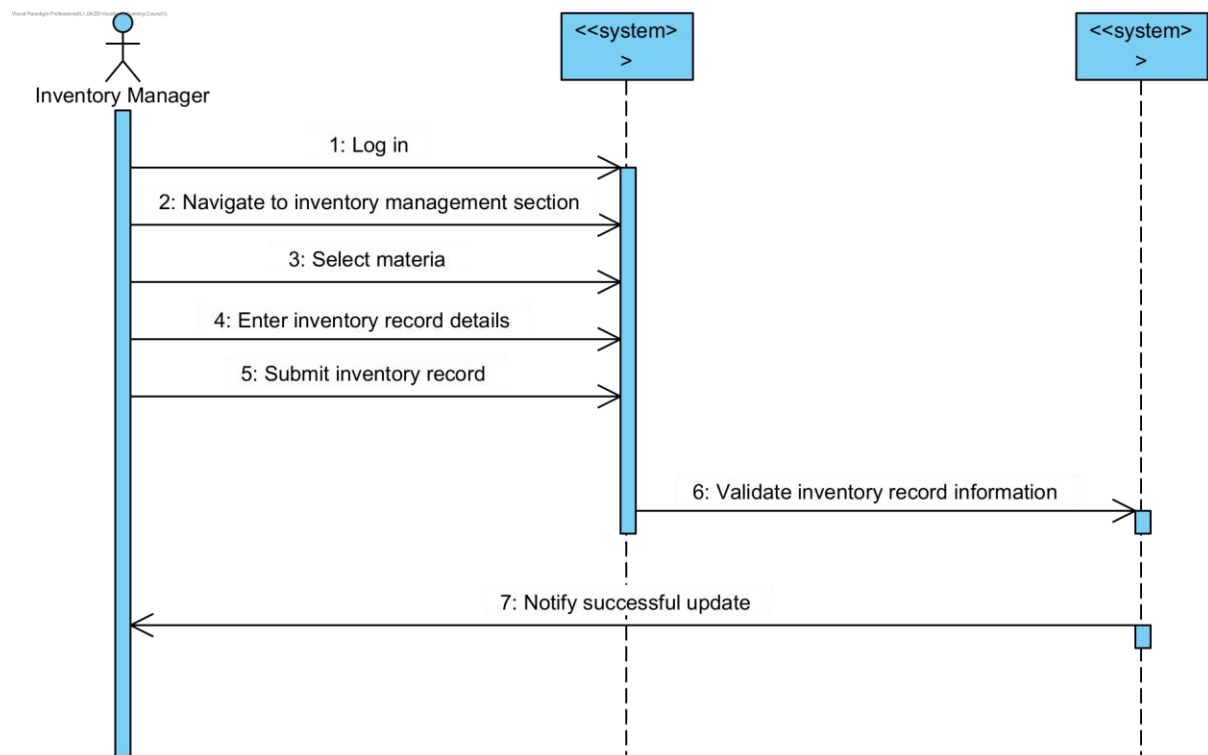
12.5.1 System Level Sequence Diagram for Procurement Officer submit purchase order



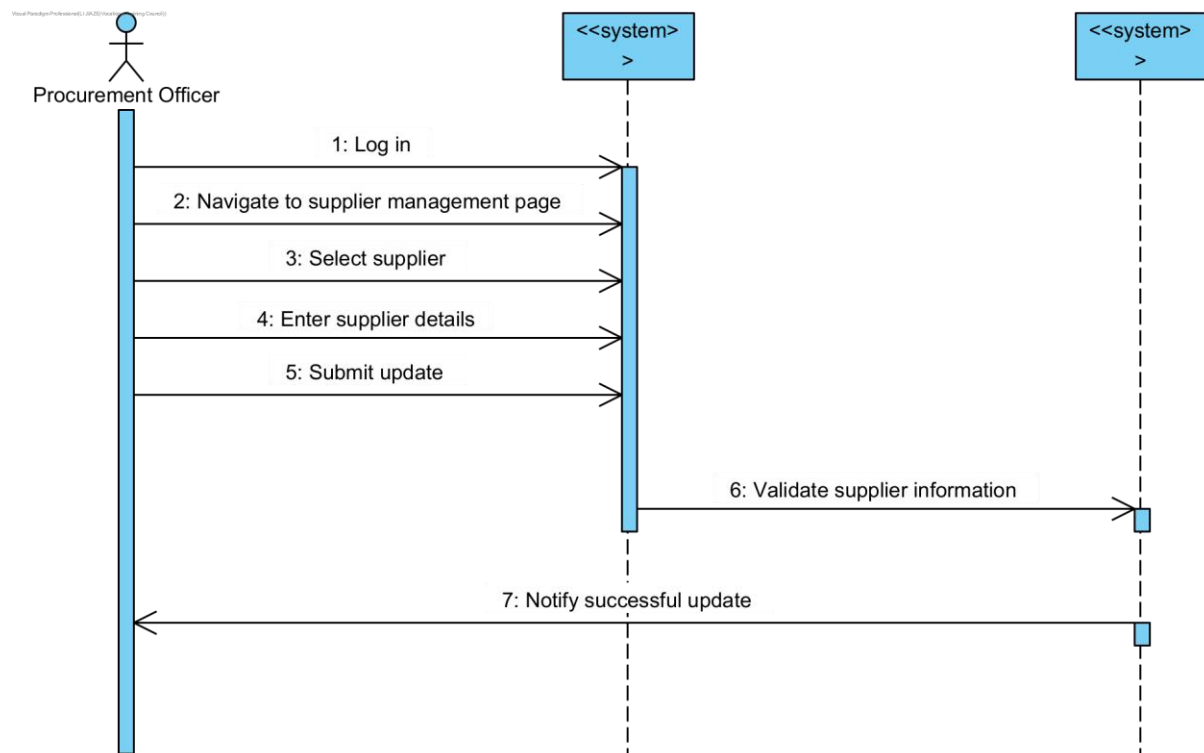
12.5.2 System Level Sequence Diagram for Material Manager update material information



12.5.3 System Level Sequence Diagram for Inventory Manager update inventory record

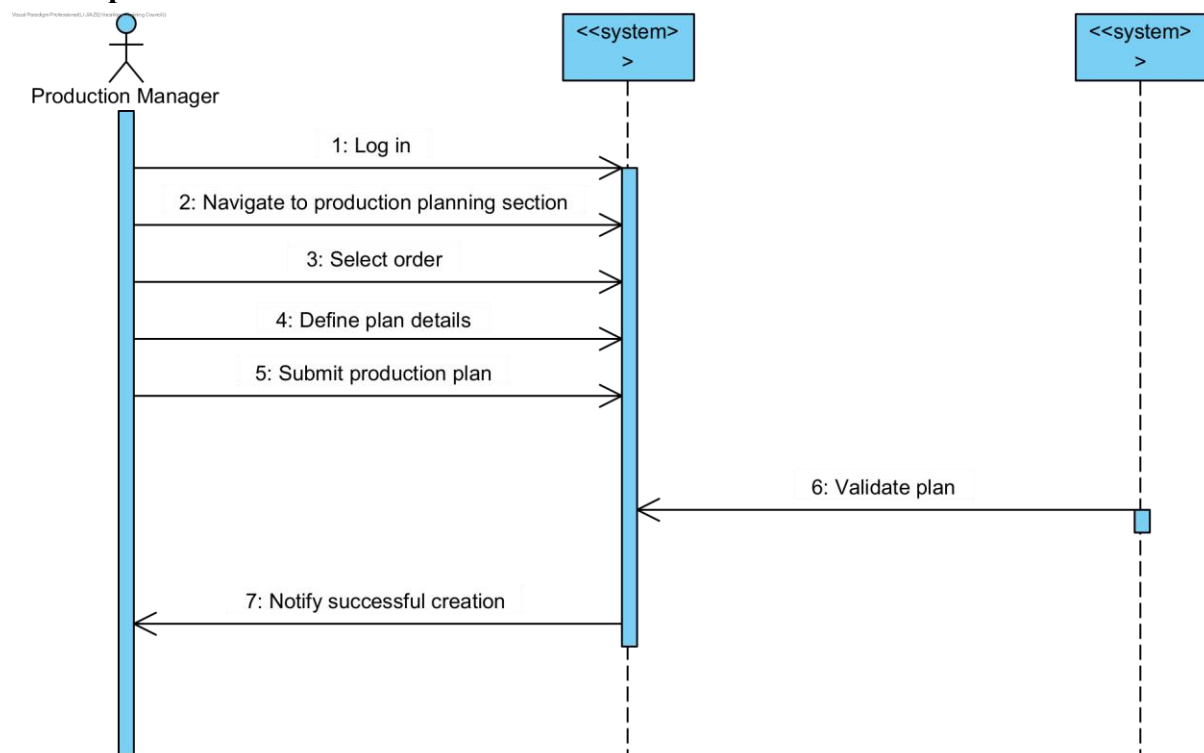


12.5.4 System Level Sequence Diagram for Procurement Officer manage supplier information

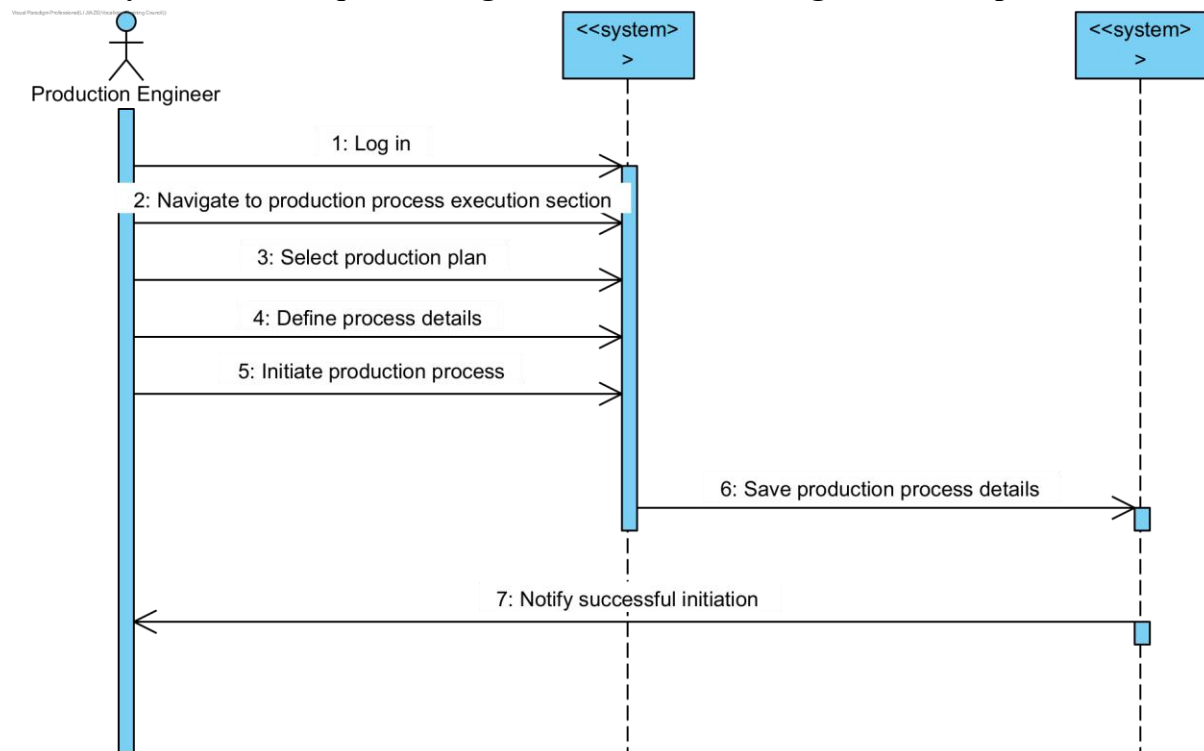


12.6 System Level Sequence Diagram for Production Management

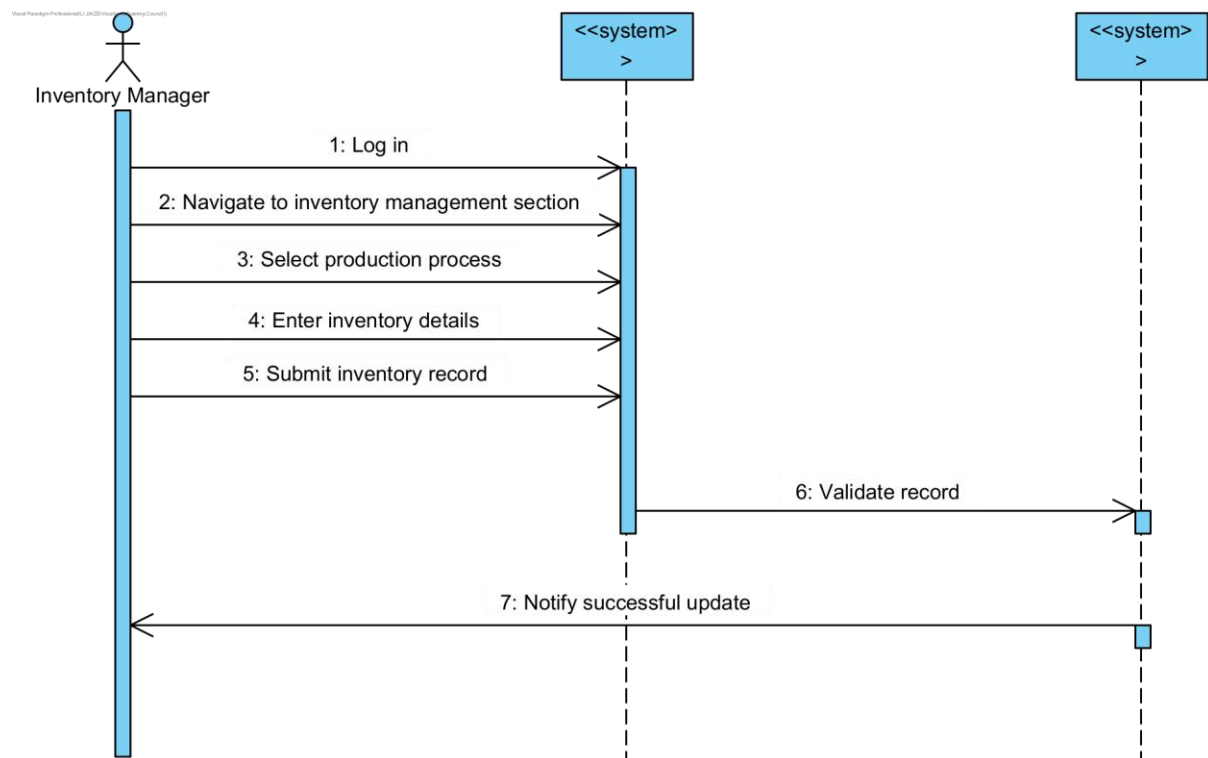
12.6.1 System Level Sequence Diagram for Production Manager submit production plan

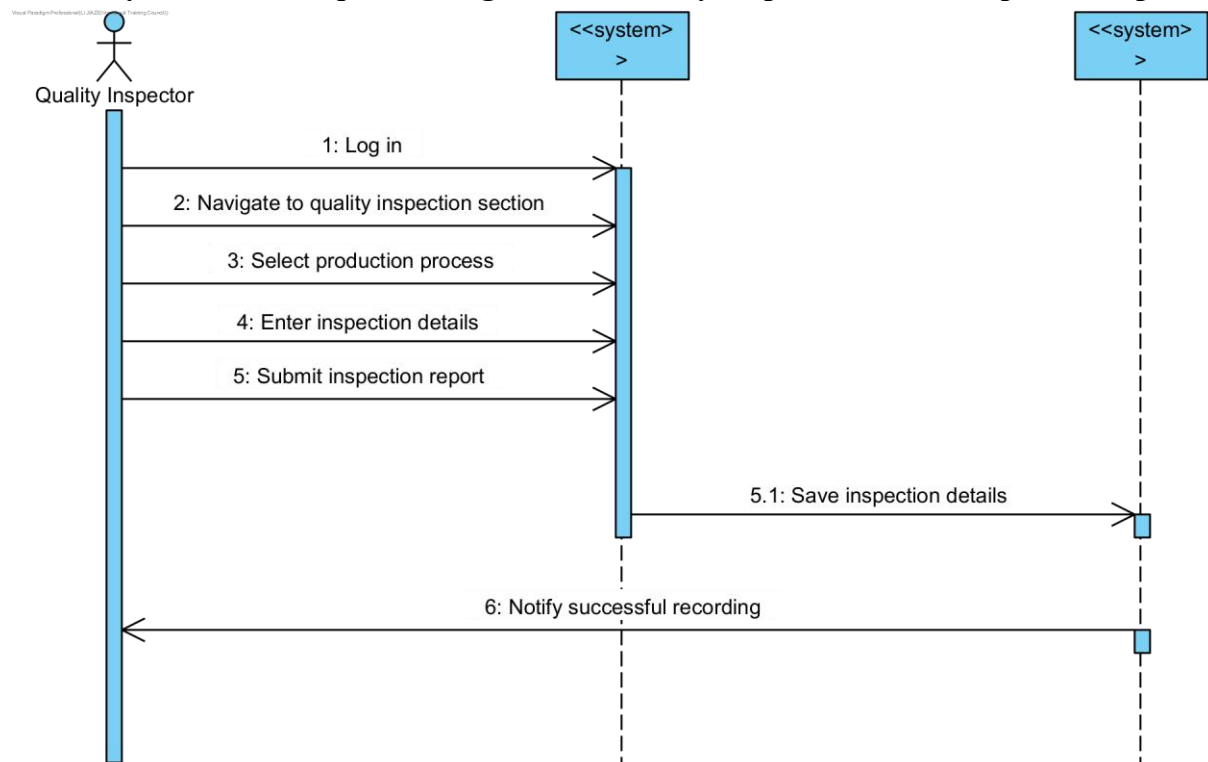


12.6.2 System Level Sequence Diagram for Production Engineer define process details



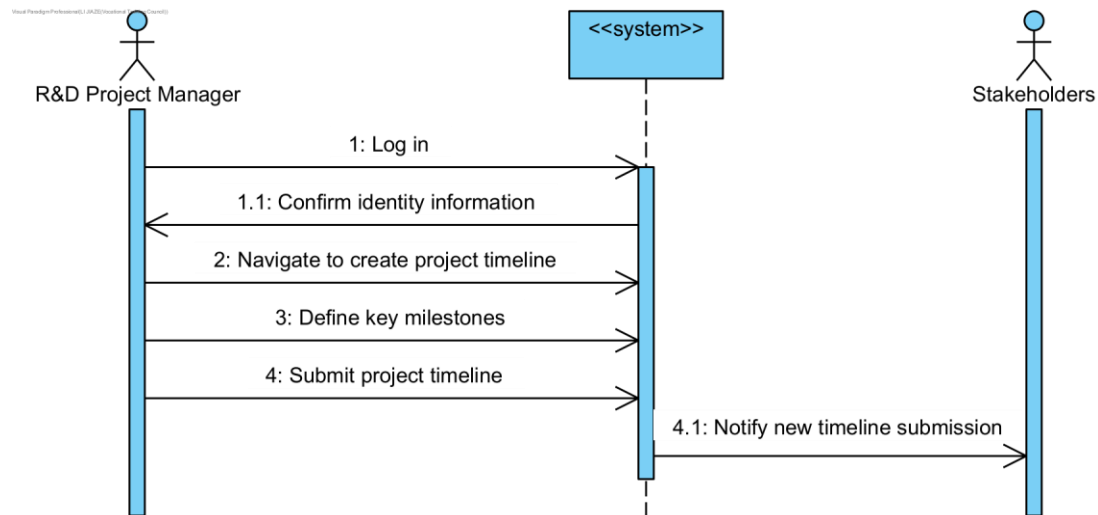
12.6.3 System Level Sequence Diagram for Inventory Manager update inventory record



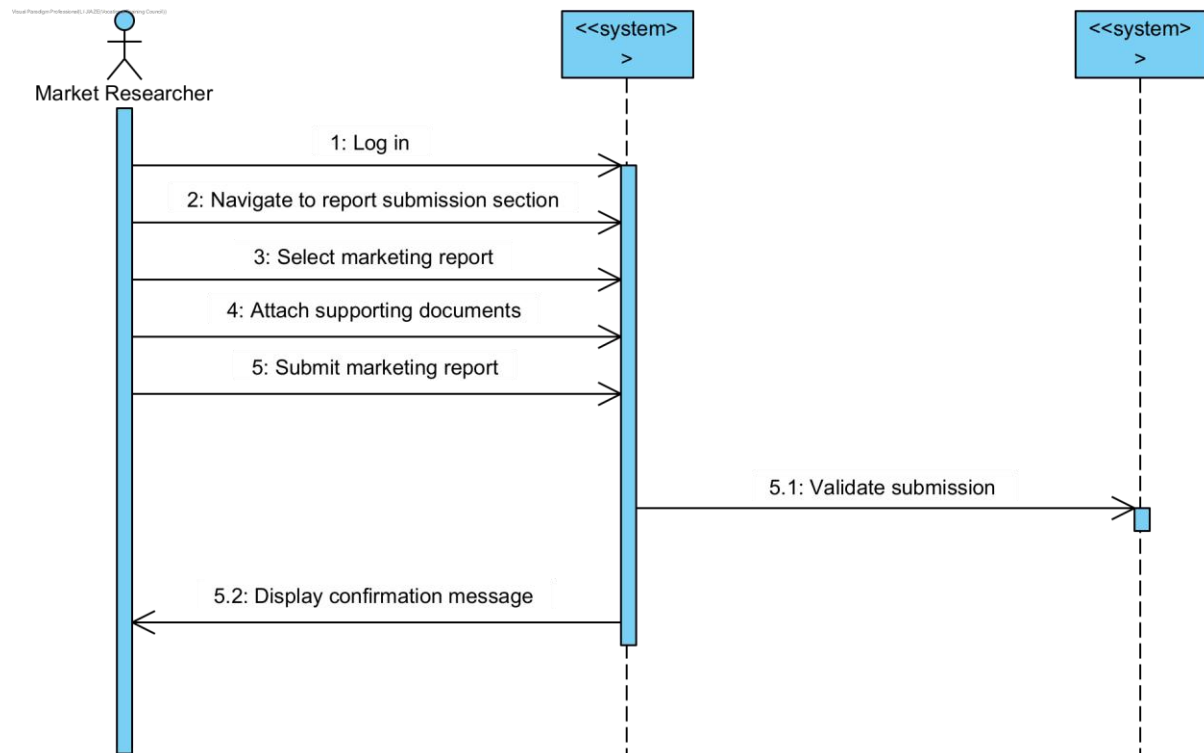
12.6.4 System Level Sequence Diagram for Quality Inspector submit inspection report

12.7 System Level Sequence Diagram for Research & Development

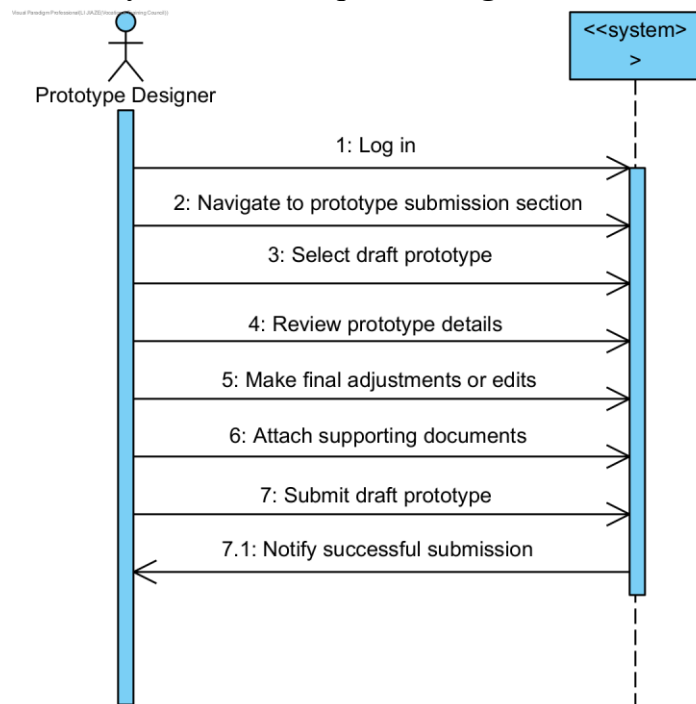
12.7.1 System Level Sequence Diagram for Project Manager submit project timeline



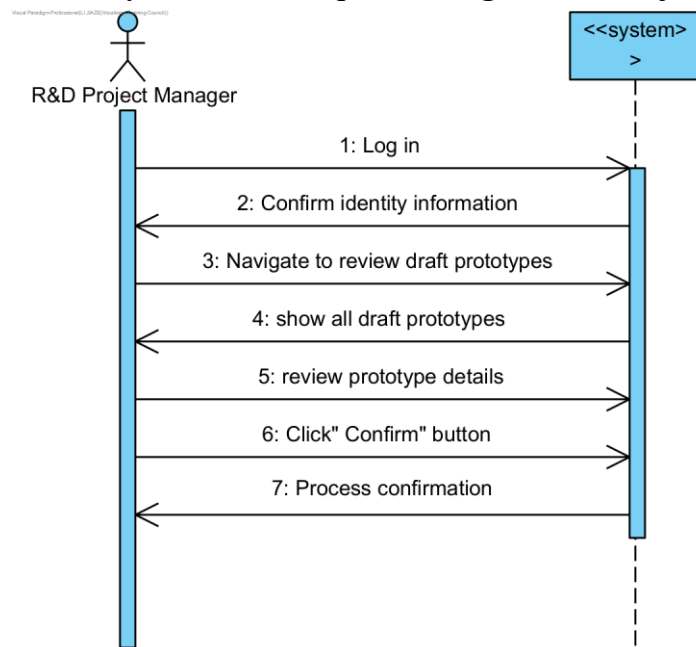
12.7.2 System Level Sequence Diagram for Market Researcher submit marketing report



12.7.3 System Level Sequence Diagram for Prototype Designer submit draft prototype



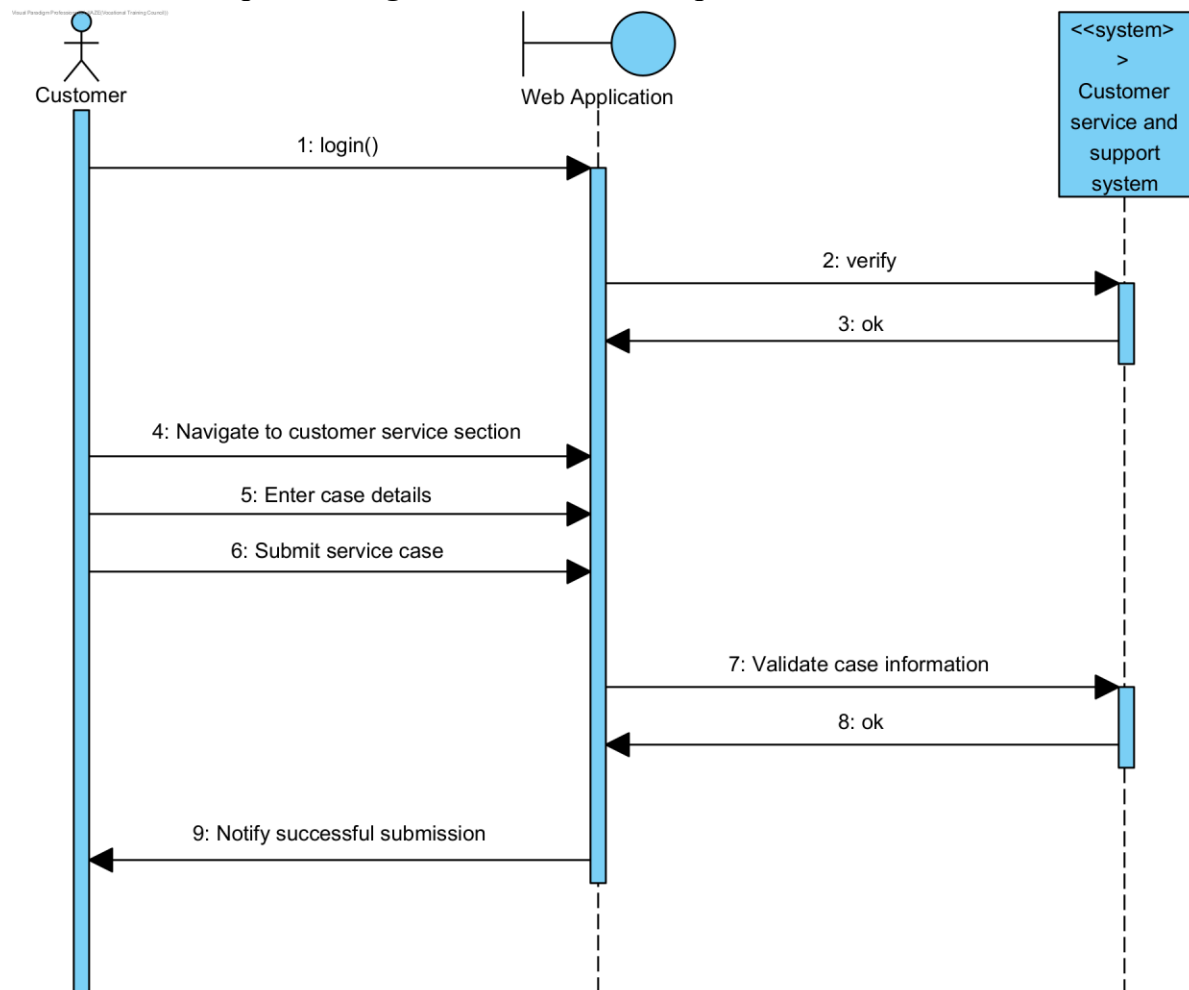
12.7.4 System Level Sequence Diagram for Project Manager production approval

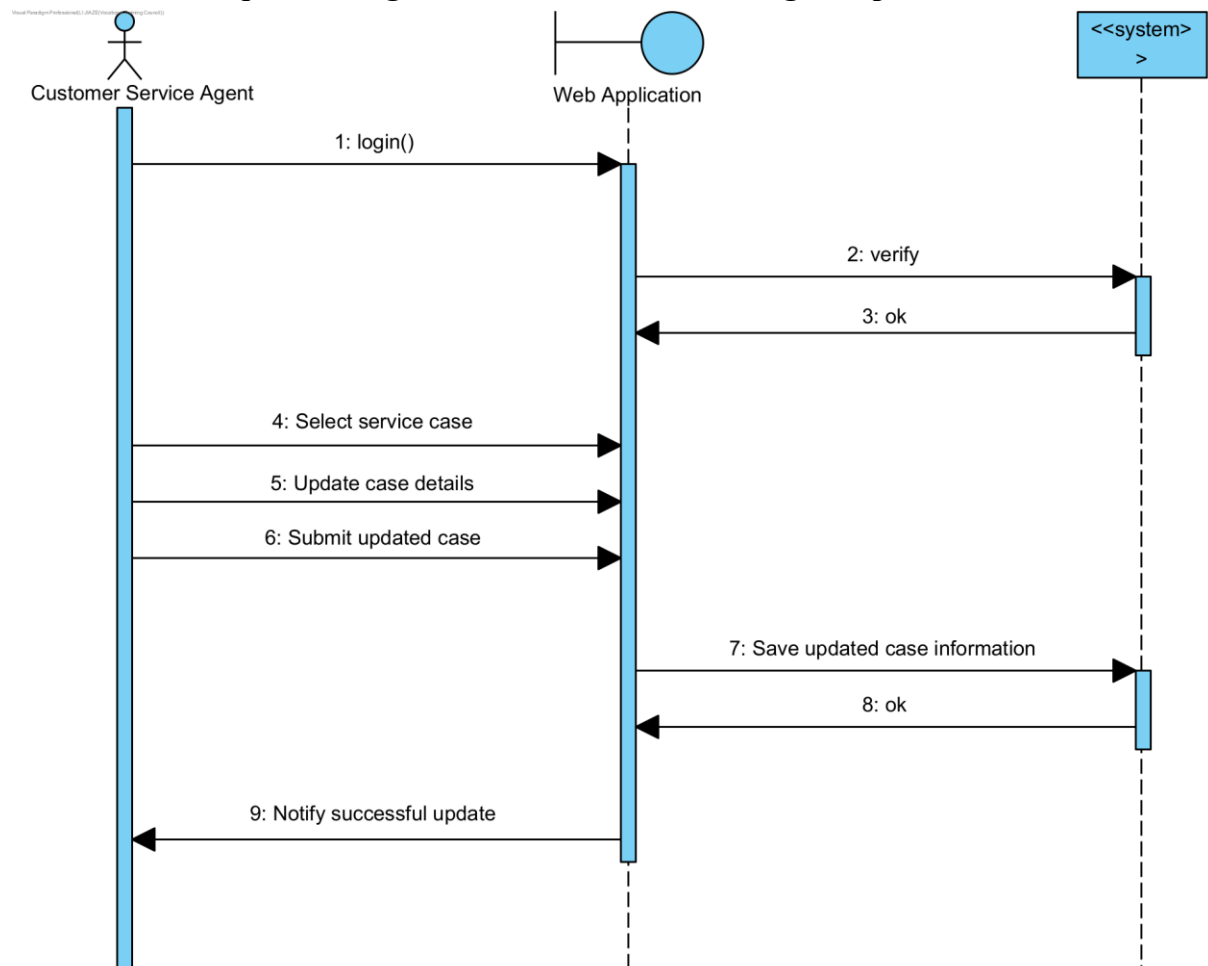


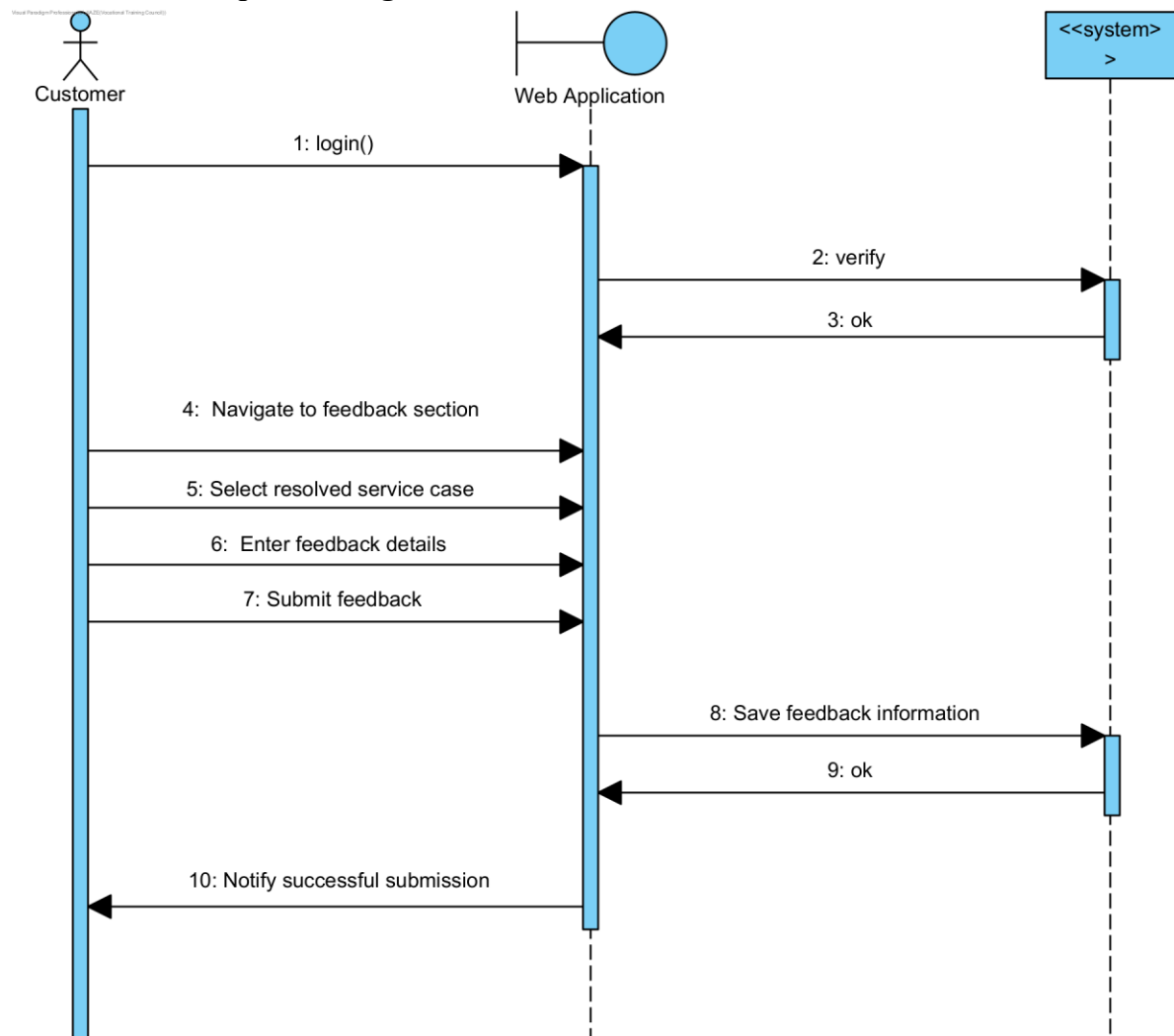
13 3-Tier sequence diagrams

13.1 3-Tier Sequence Diagram for Customer service and support

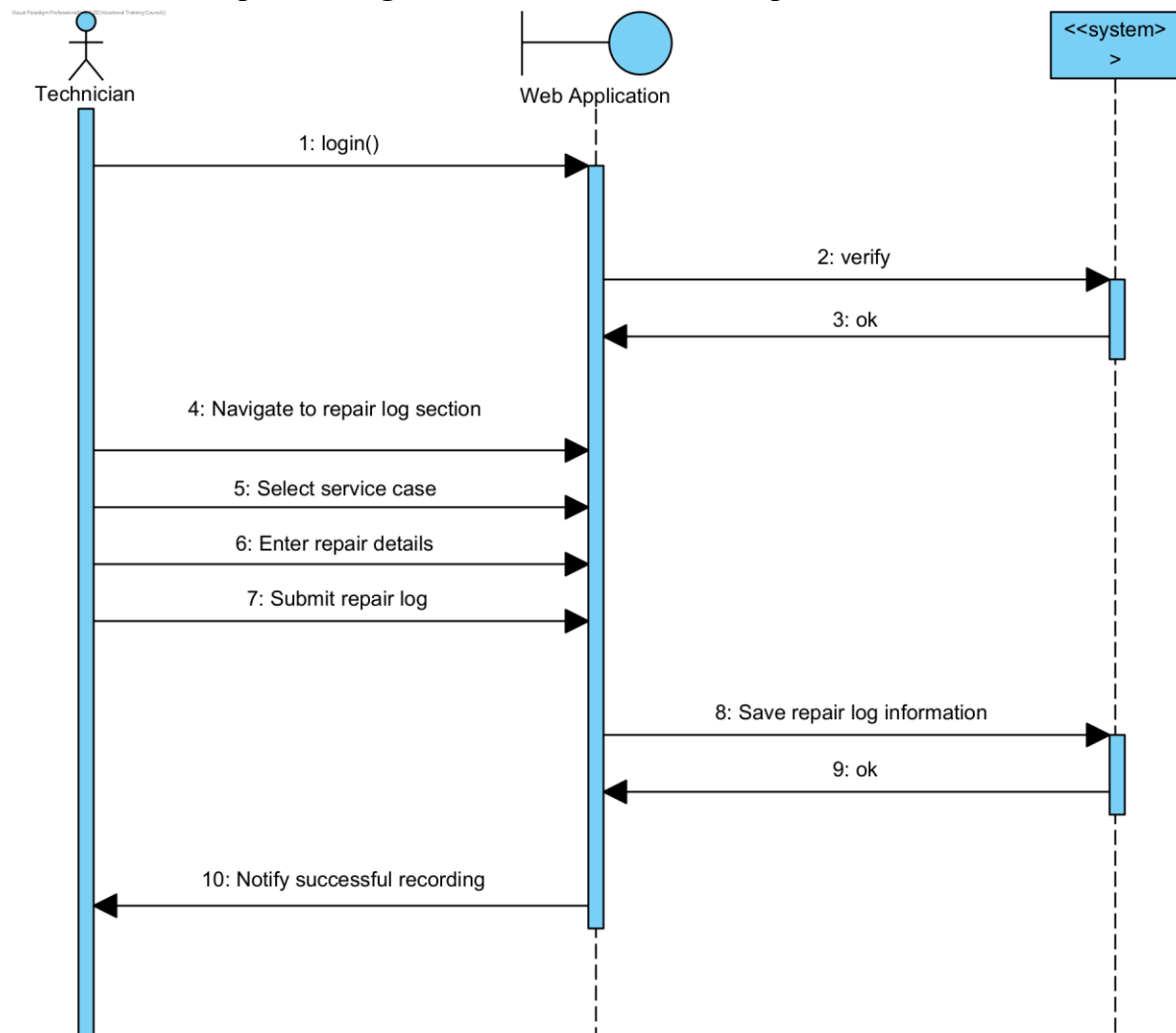
13.1.1 3-Tier Sequence Diagram for Customer request service



13.1.2 3-Tier Sequence Diagram for Customer service Agent Update case details

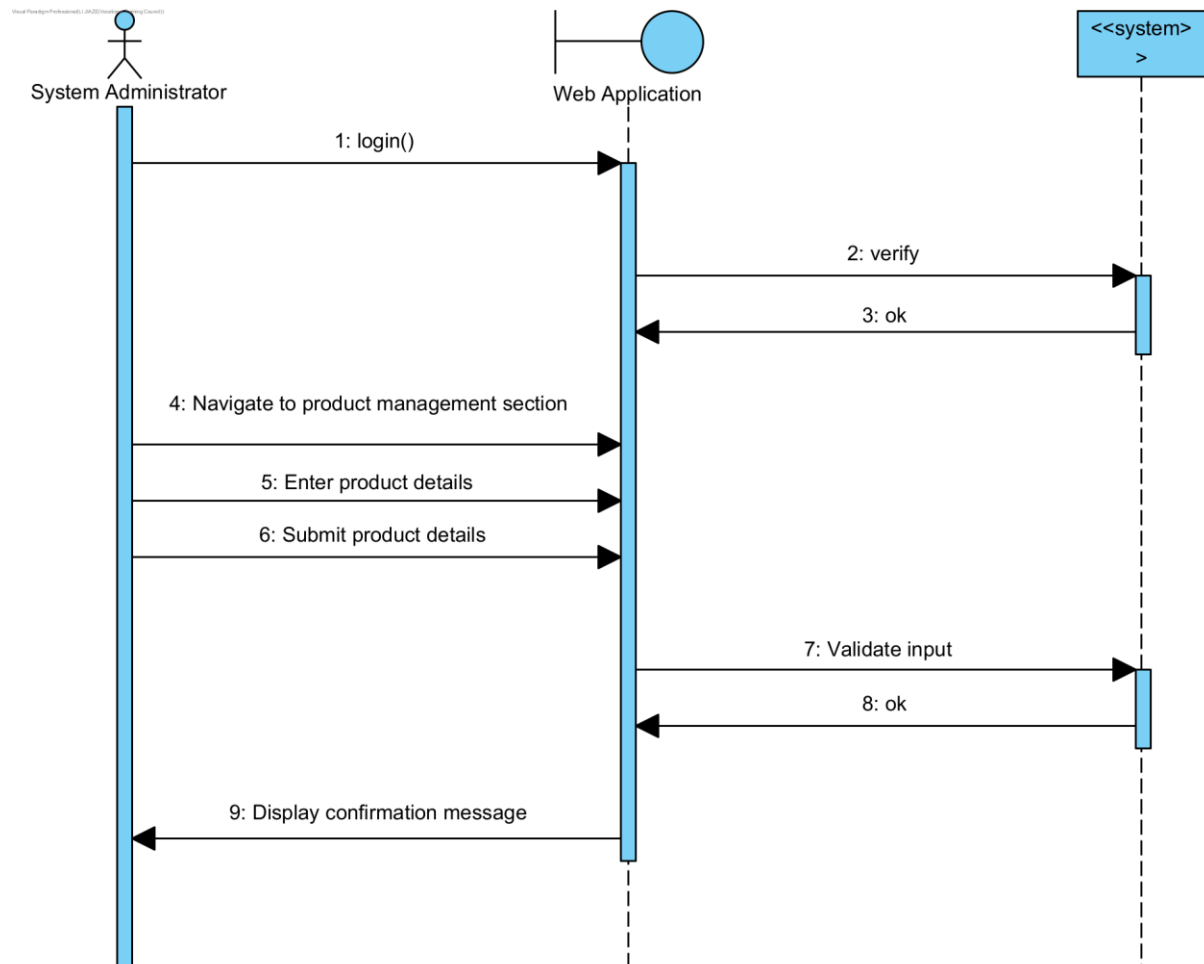
13.1.3 3-Tier Sequence Diagram for Customer receive feedback

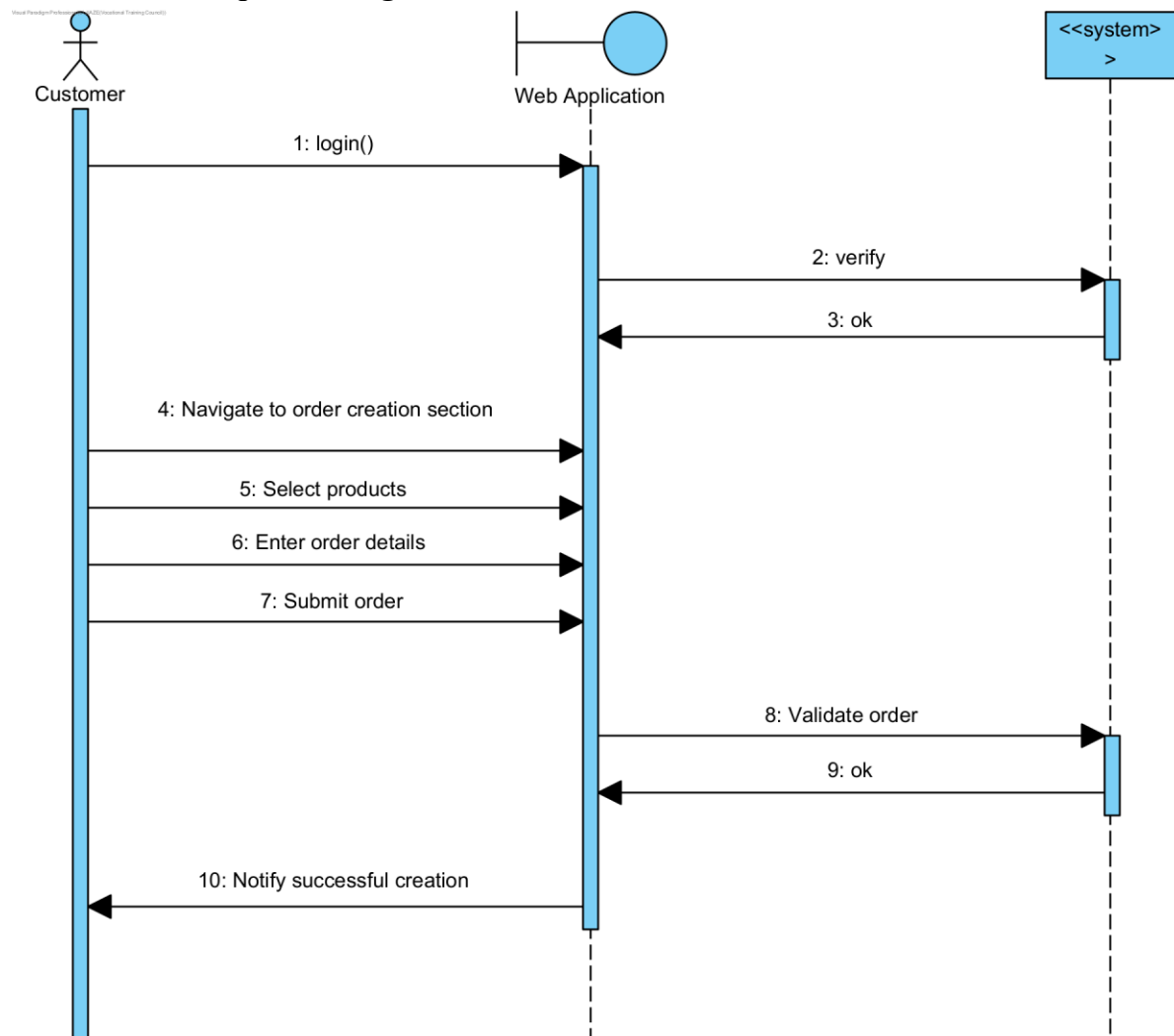
13.1.4 3-Tier Sequence Diagram for Technician enter repair details

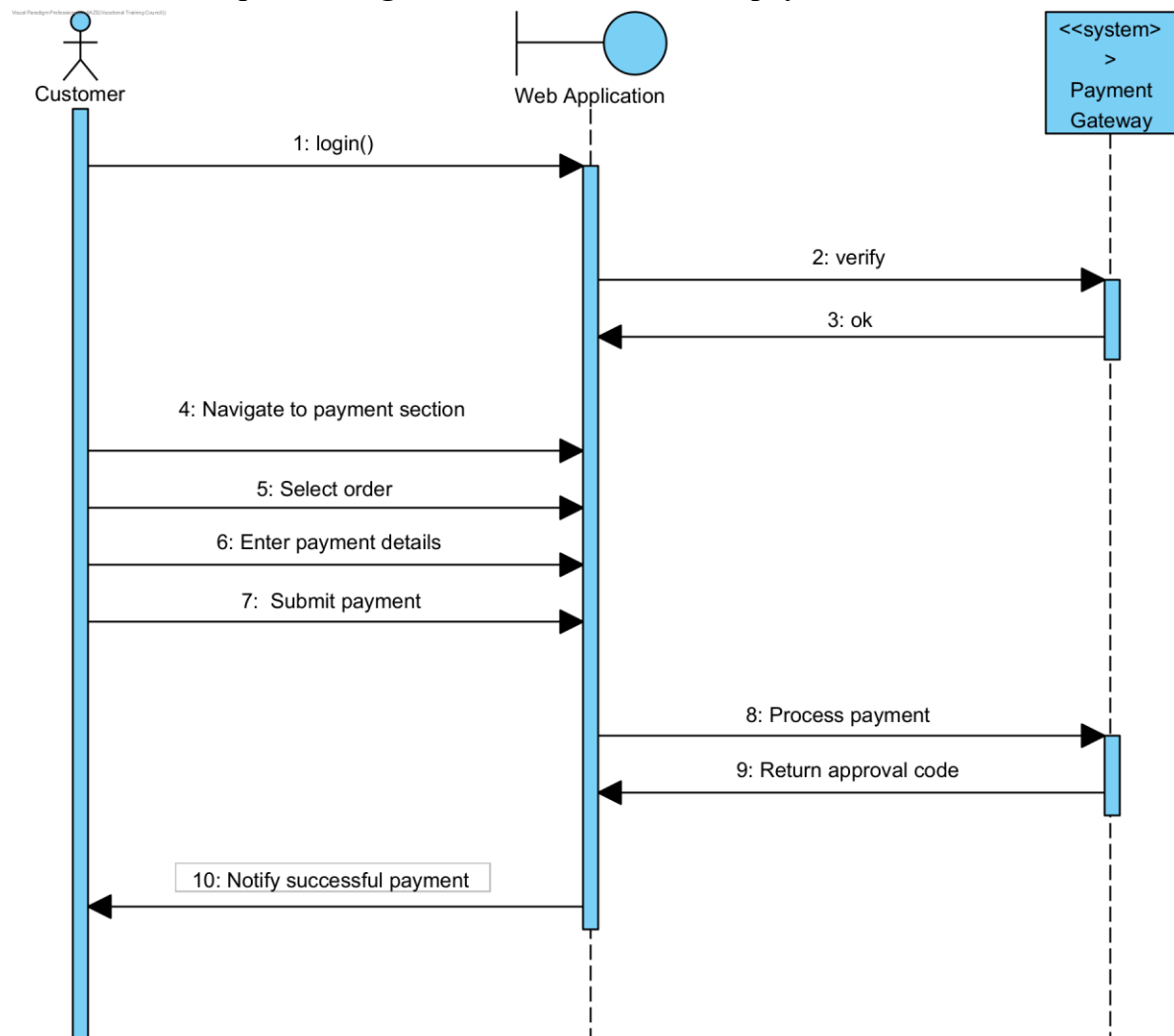


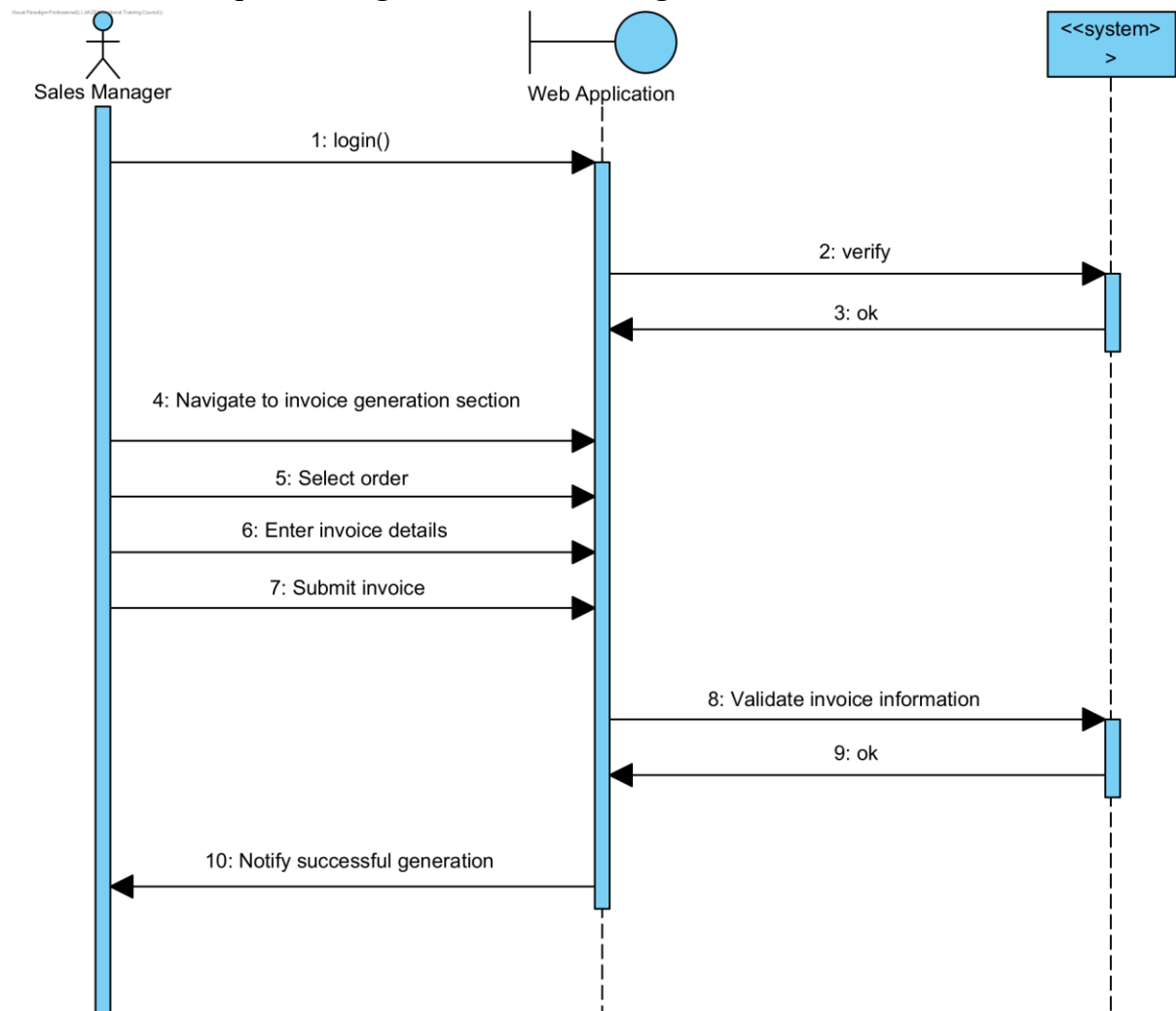
13.2 3-Tier Sequence Diagram for Order and payment management

13.2.1 3-Tier Sequence Diagram for System Administrator enter product details



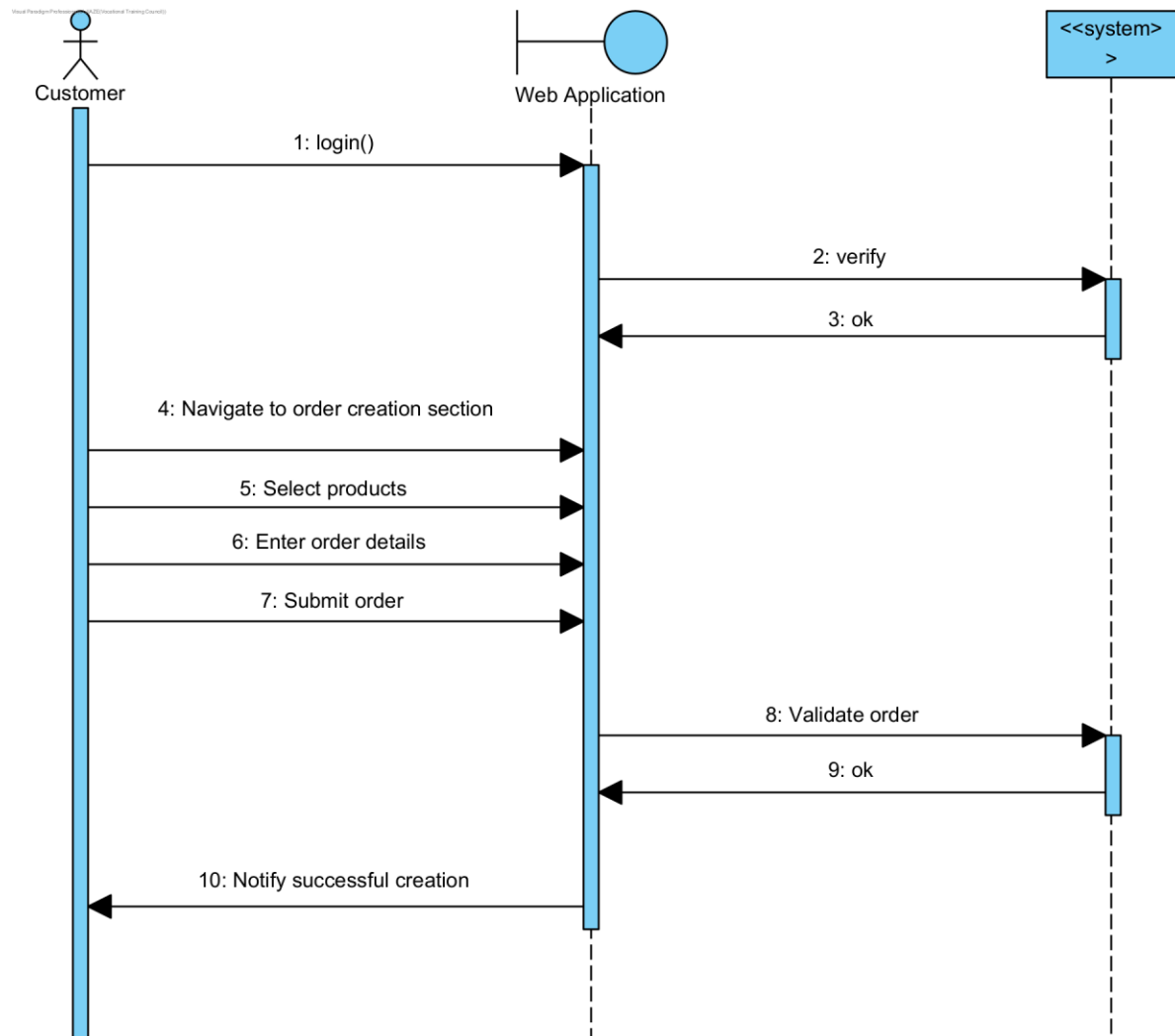
13.2.2 3-Tier Sequence Diagram for Customer submit order

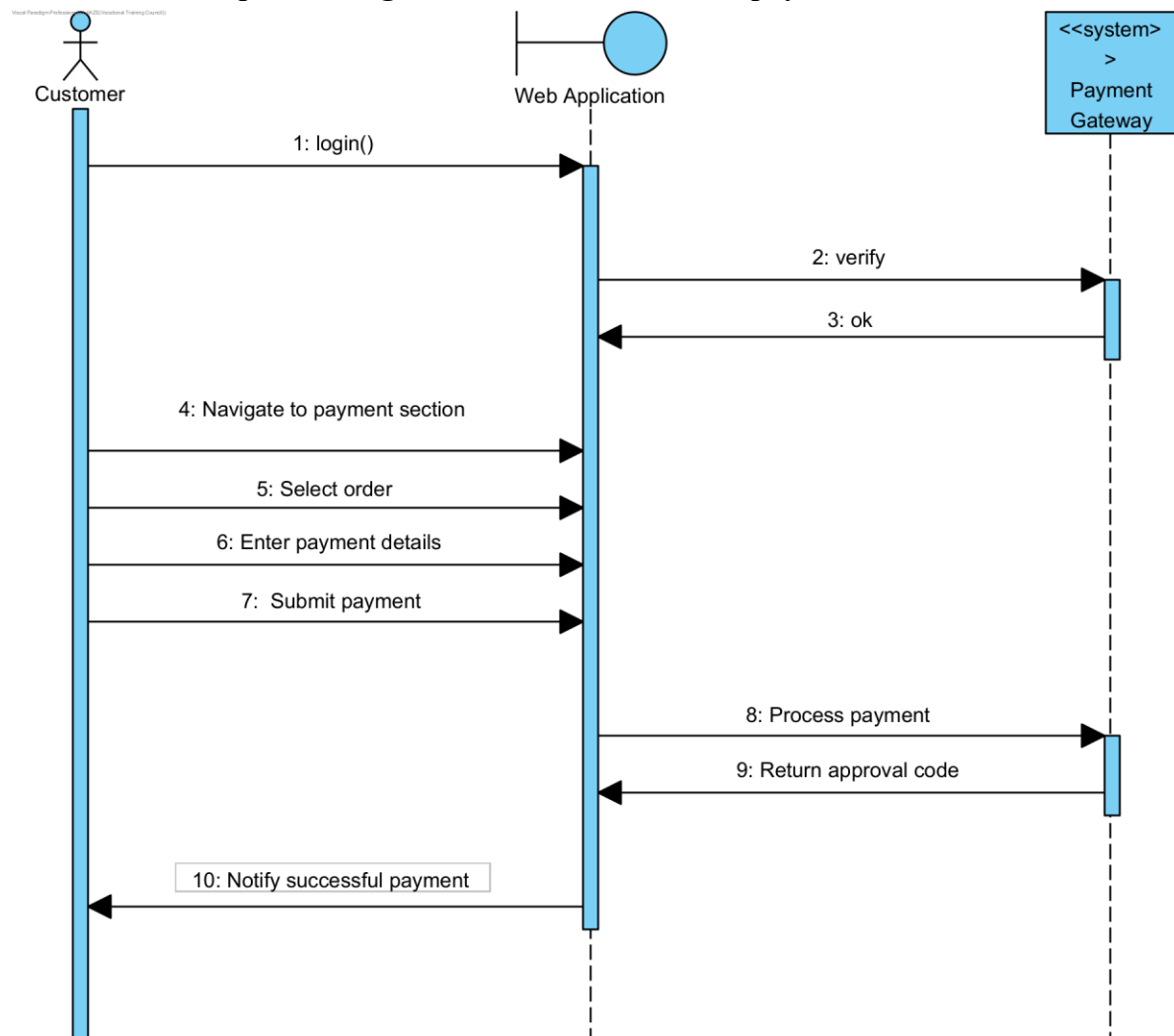
13.2.3 3-Tier Sequence Diagram for Customer submit payment

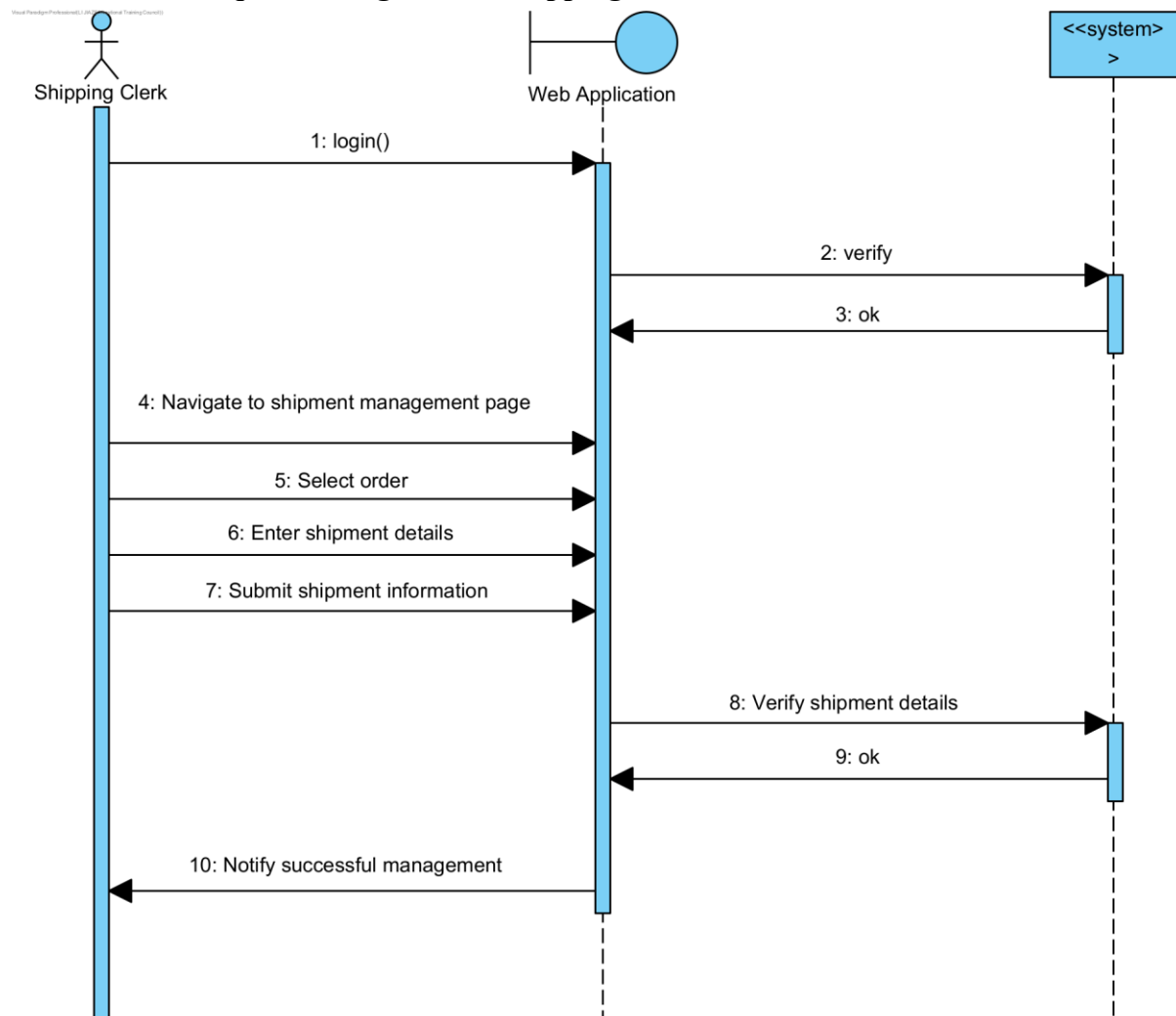
13.2.4 3-Tier Sequence Diagram for Sales Manager submit invoice

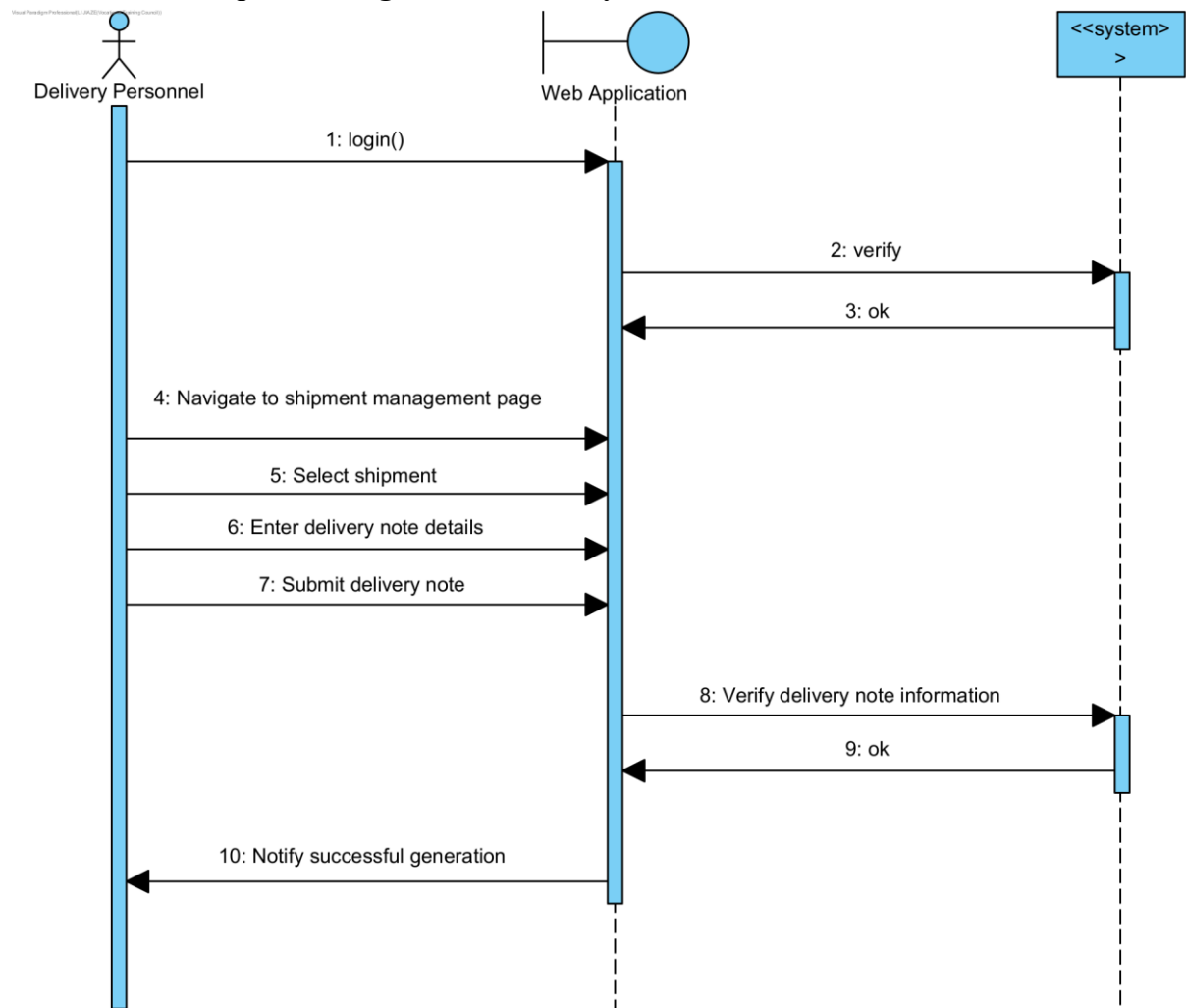
13.3 3-Tier Sequence Diagram for Order processing

13.3.1 3-Tier Sequence Diagram for Customer submit order



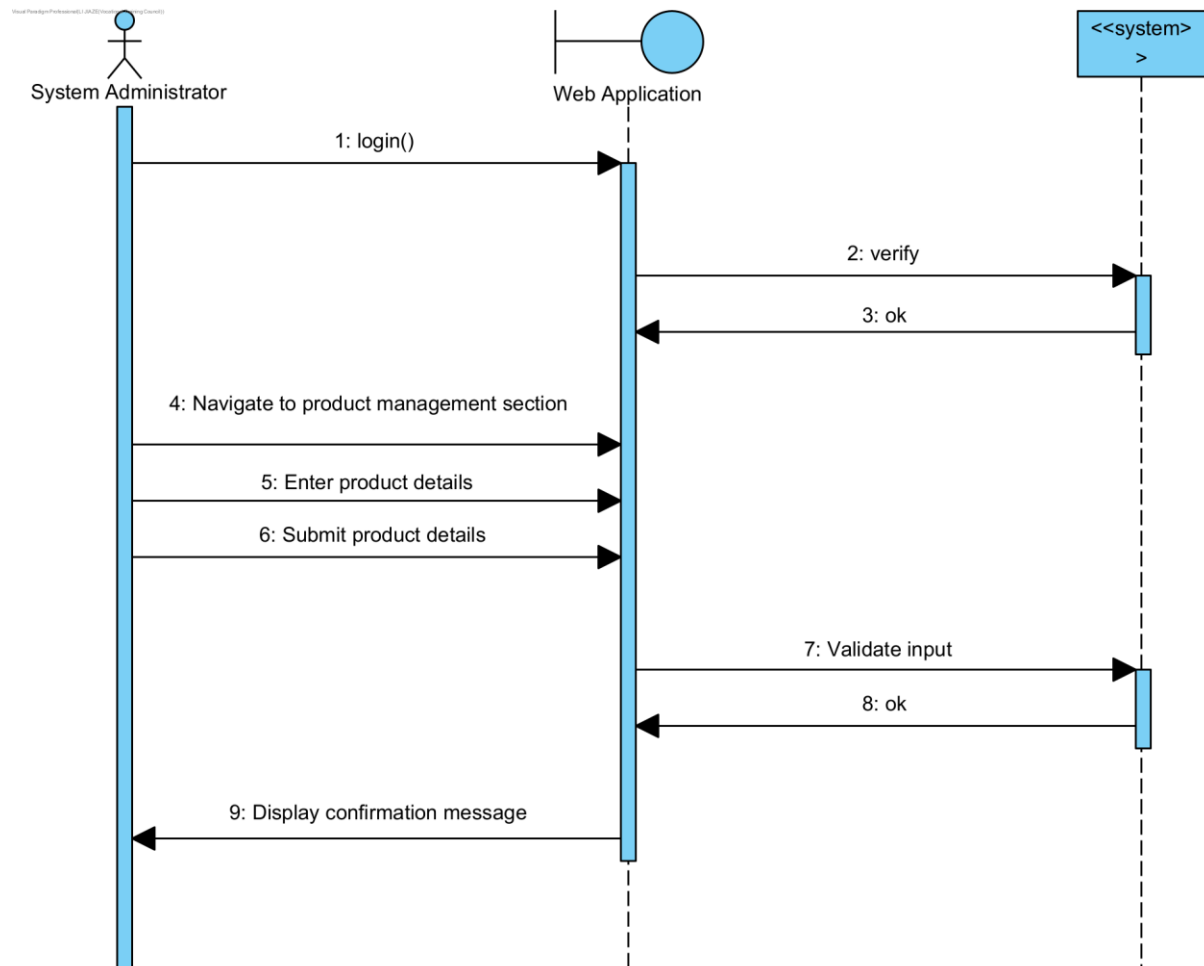
13.3.2 3-Tier Sequence Diagram for Customer submit payment

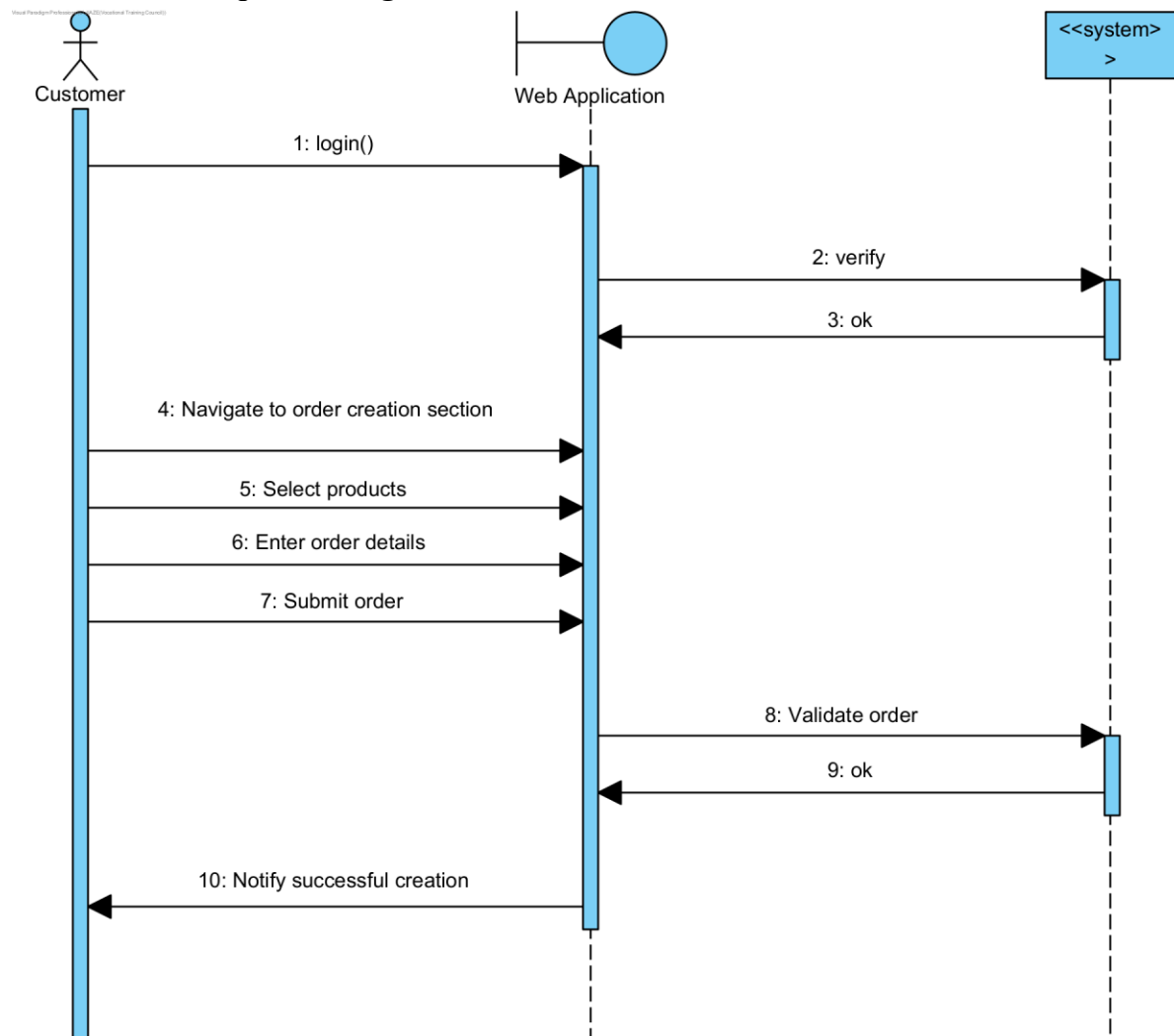
13.3.3 3-Tier Sequence Diagram for Shipping Clerk

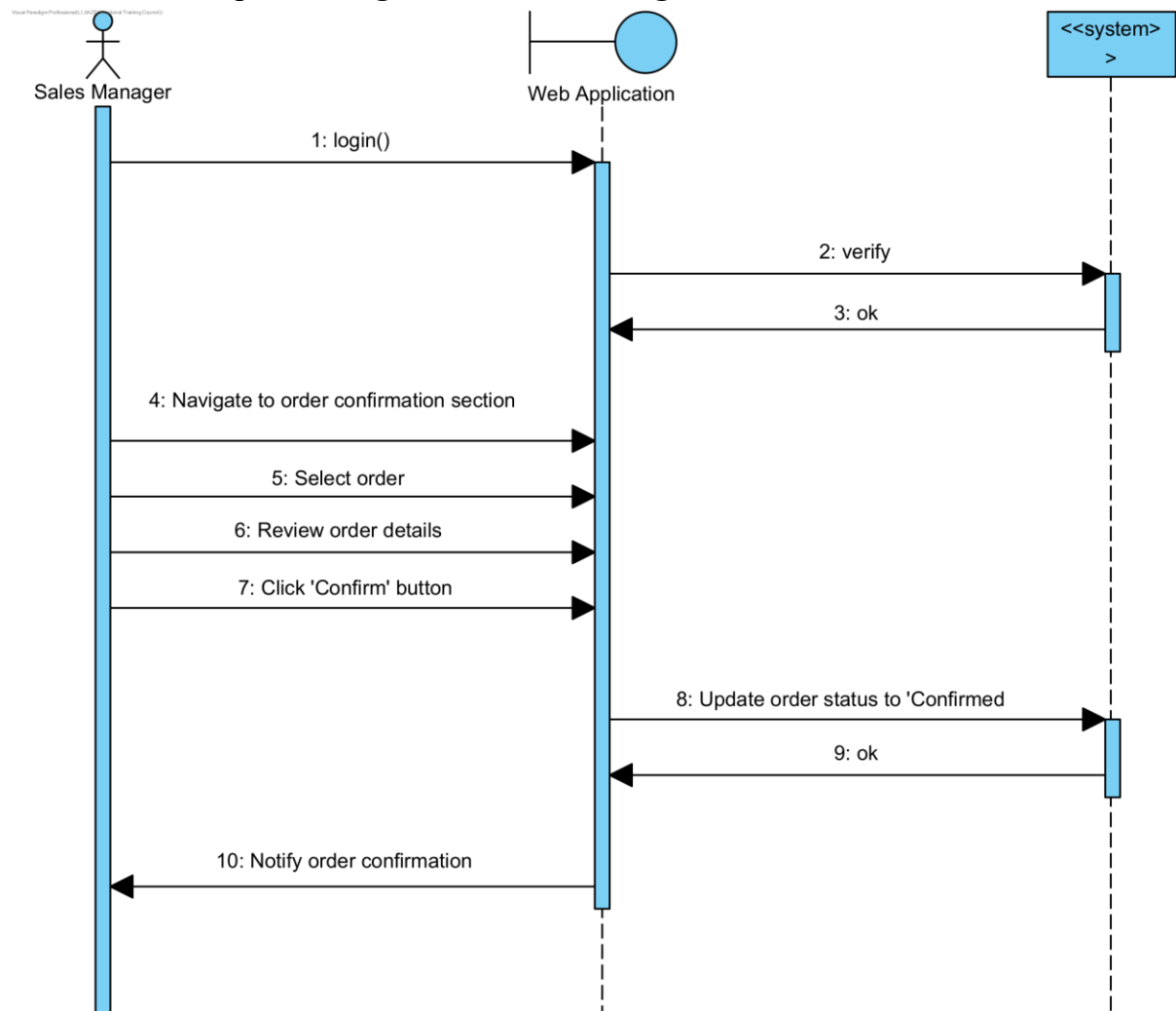
13.3.4 3-Tier Sequence Diagram for Delivery Personnel

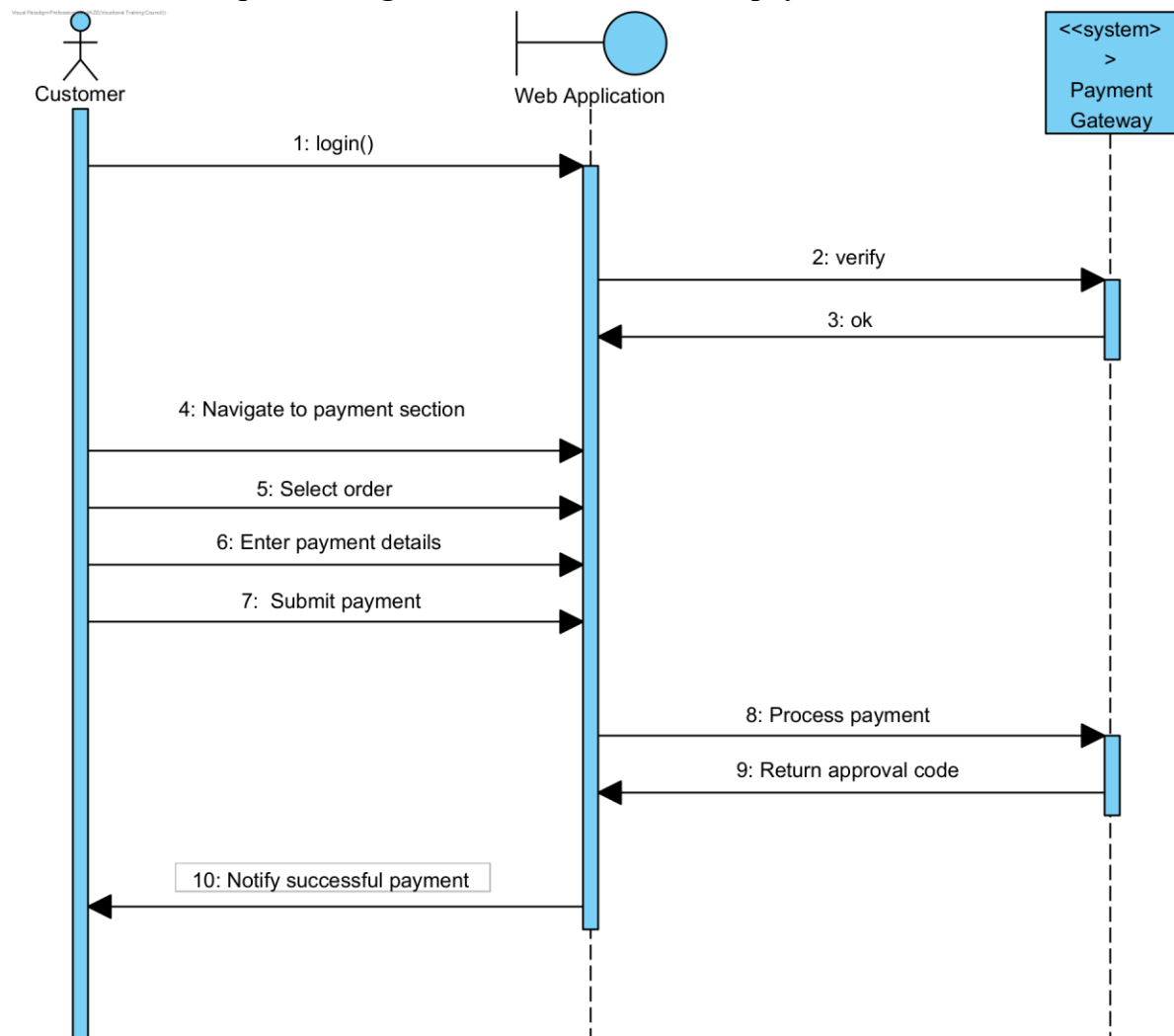
13.4 3-Tier Sequence Diagram for Order system

13.4.1 3-Tier Sequence Diagram for System Administrator update product details



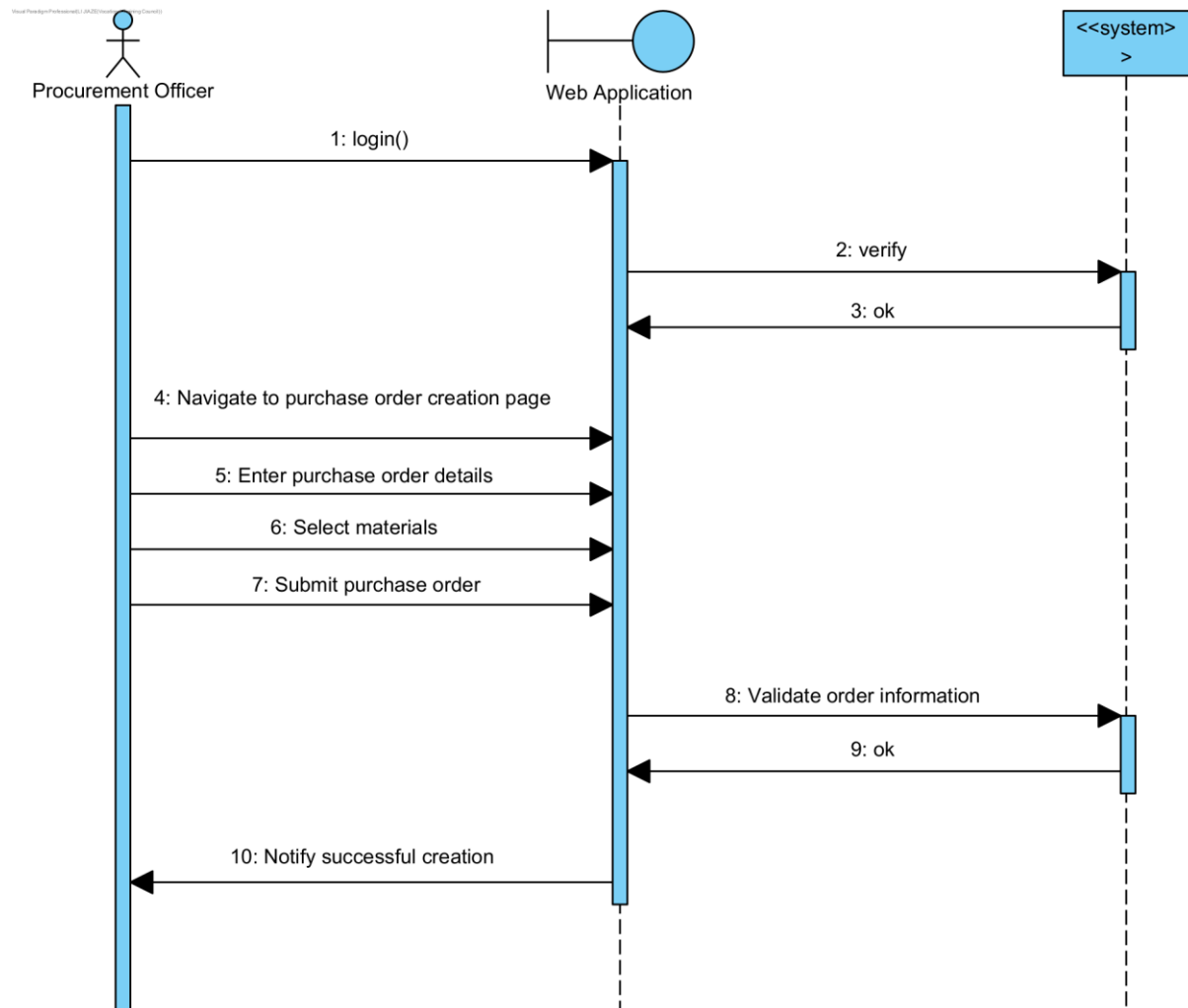
13.4.2 3-Tier Sequence Diagram for Customer submit order

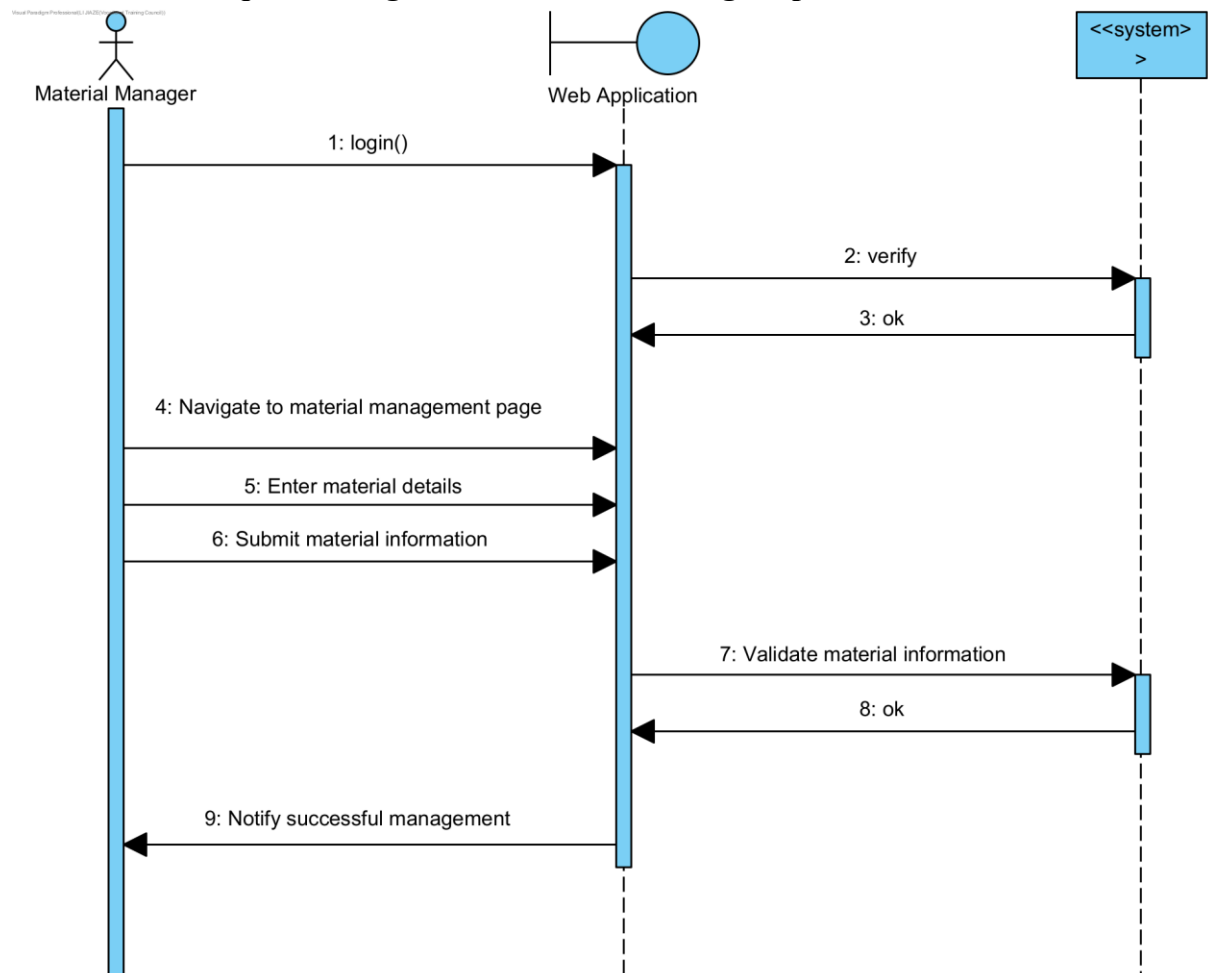
13.4.3 3-Tier Sequence Diagram for Sales Manager confirm Order

13.4.4 3-Tier Sequence Diagram for Customer submit payment

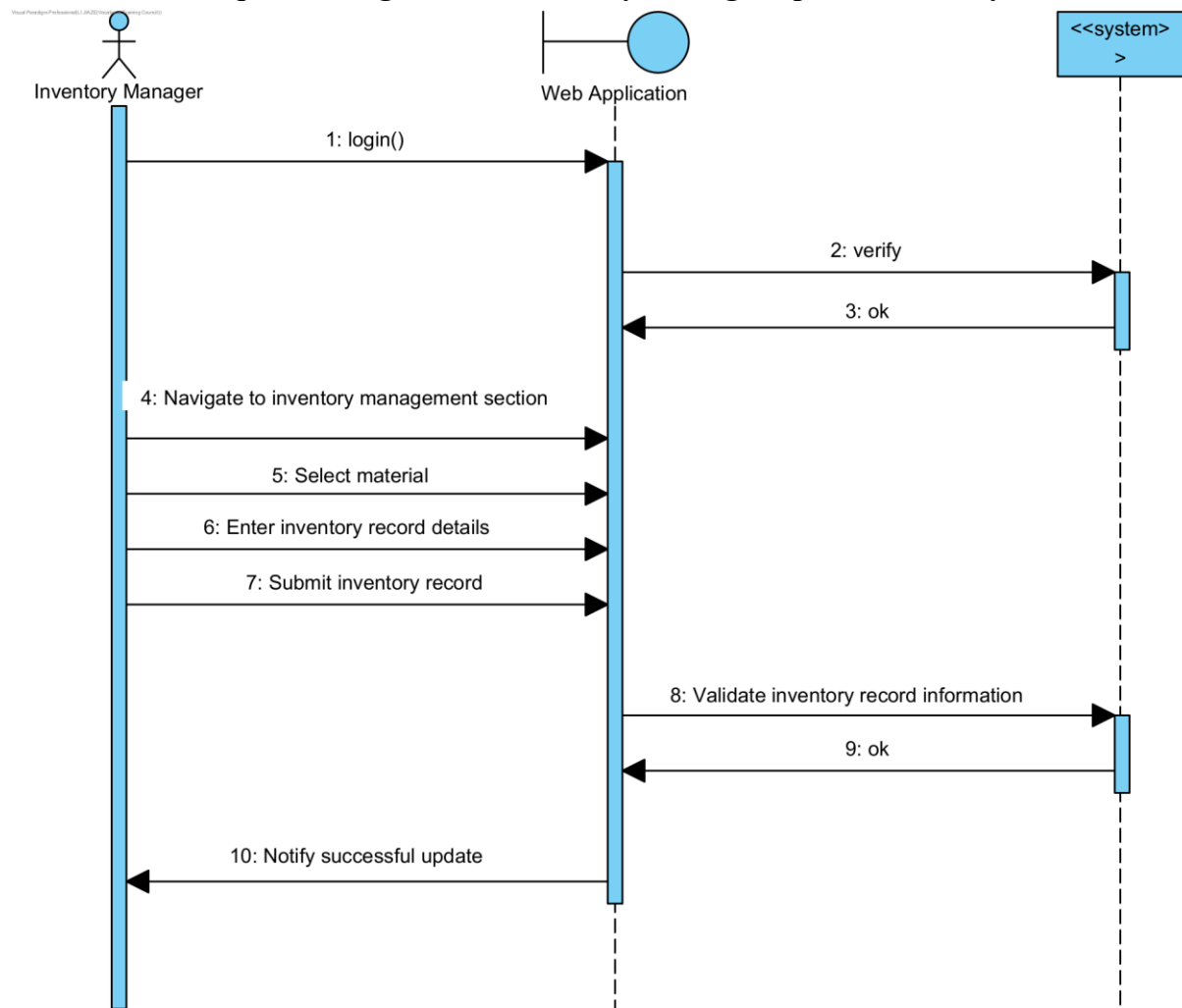
13.5 3-Tier Sequence Diagram for Procurement system

13.5.1 3-Tier Sequence Diagram for Procurement Officer submit purchase order

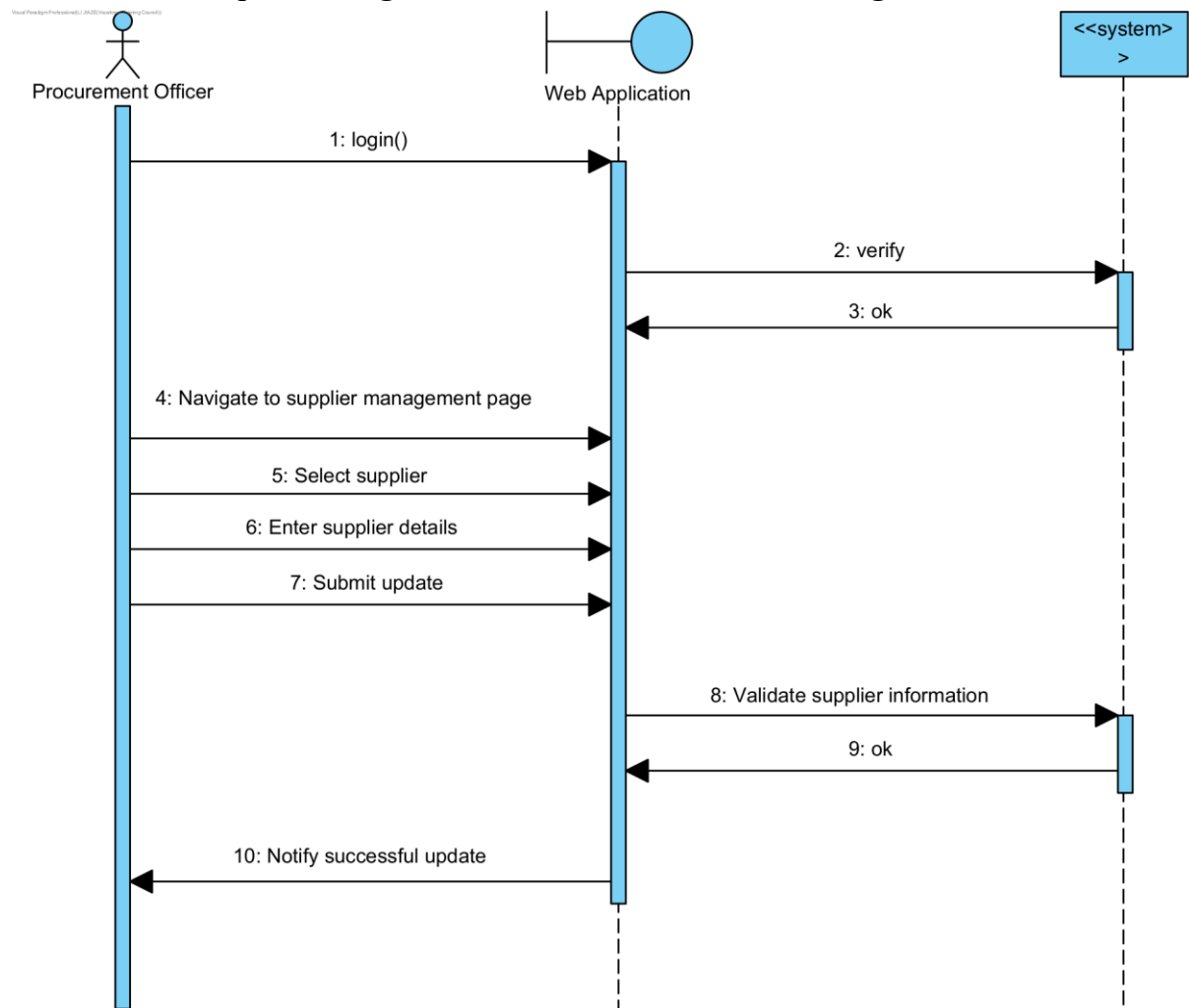


13.5.2 3-Tier Sequence Diagram for Material Manager update material information

13.5.3 3-Tier Sequence Diagram for Inventory Manager update inventory record

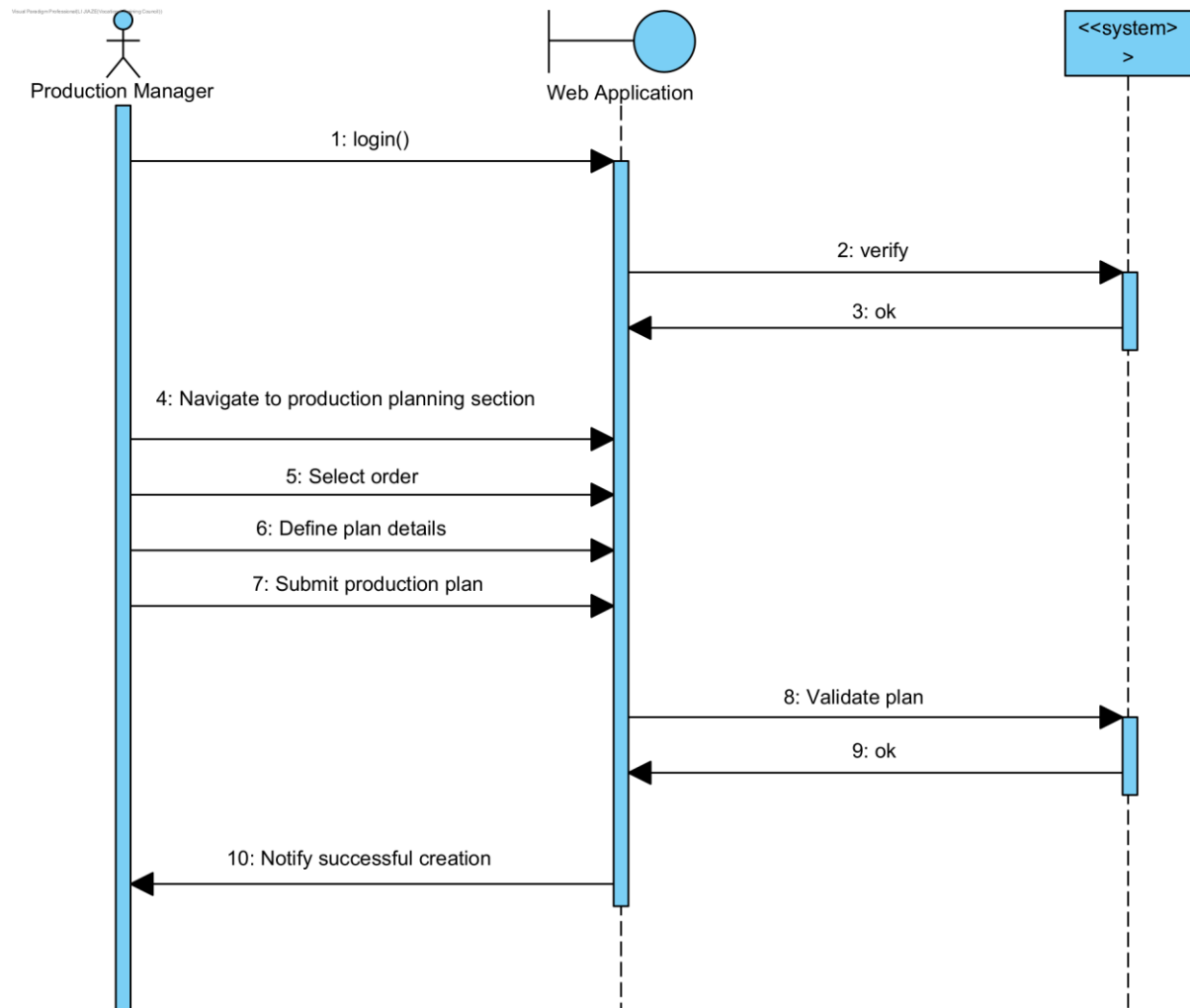


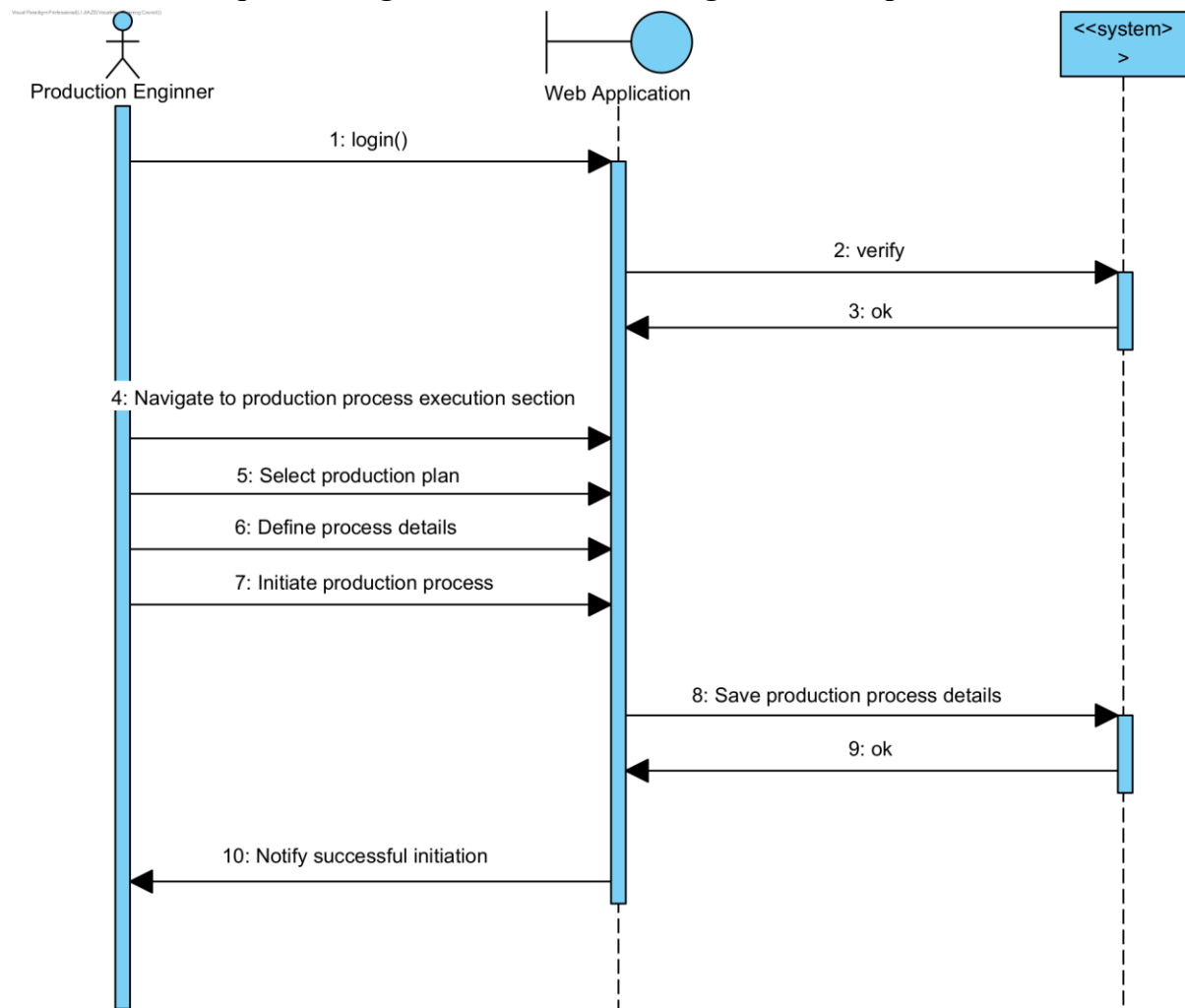
13.5.4 3-Tier Sequence Diagram for Procurement Officer managerial information

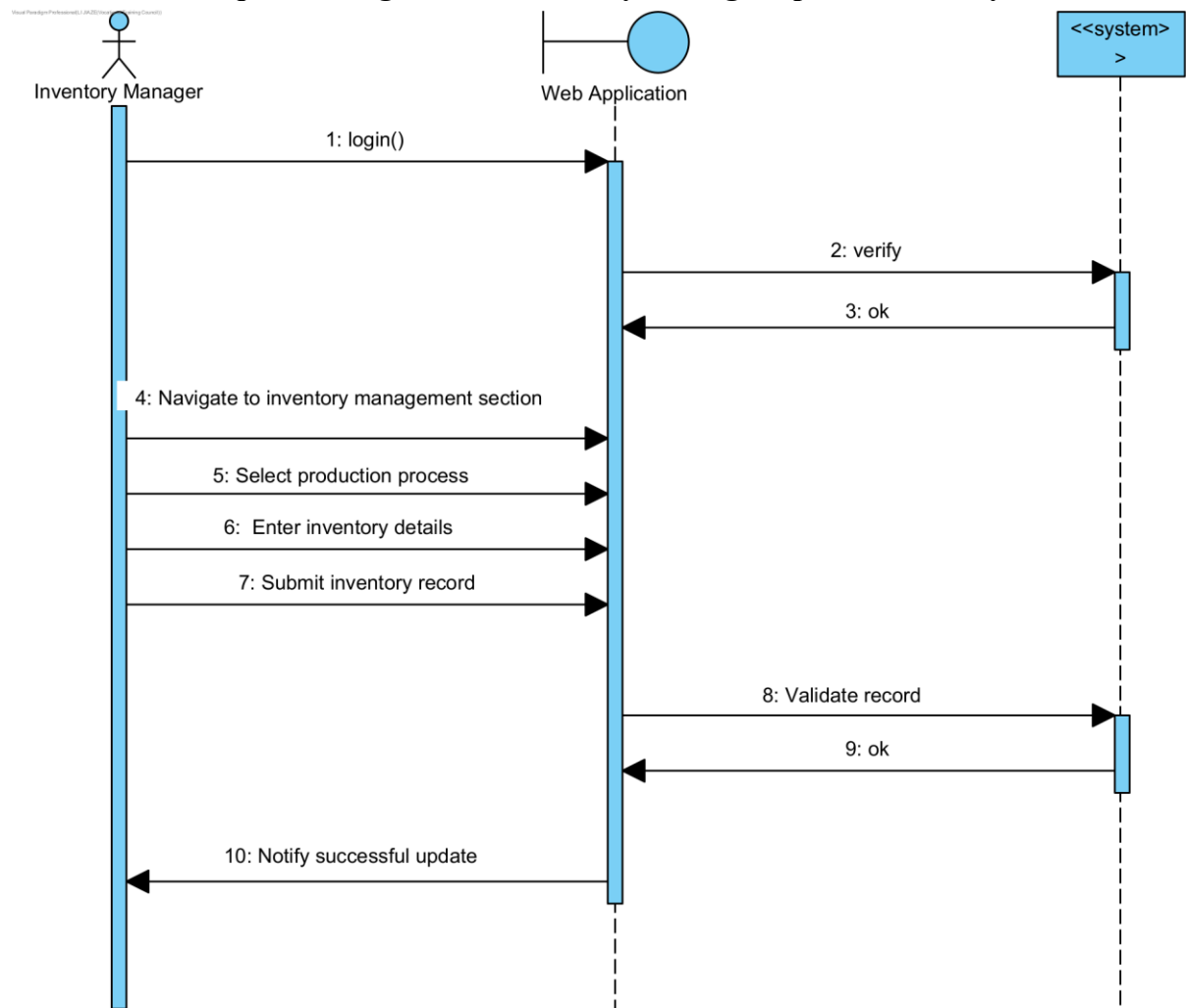


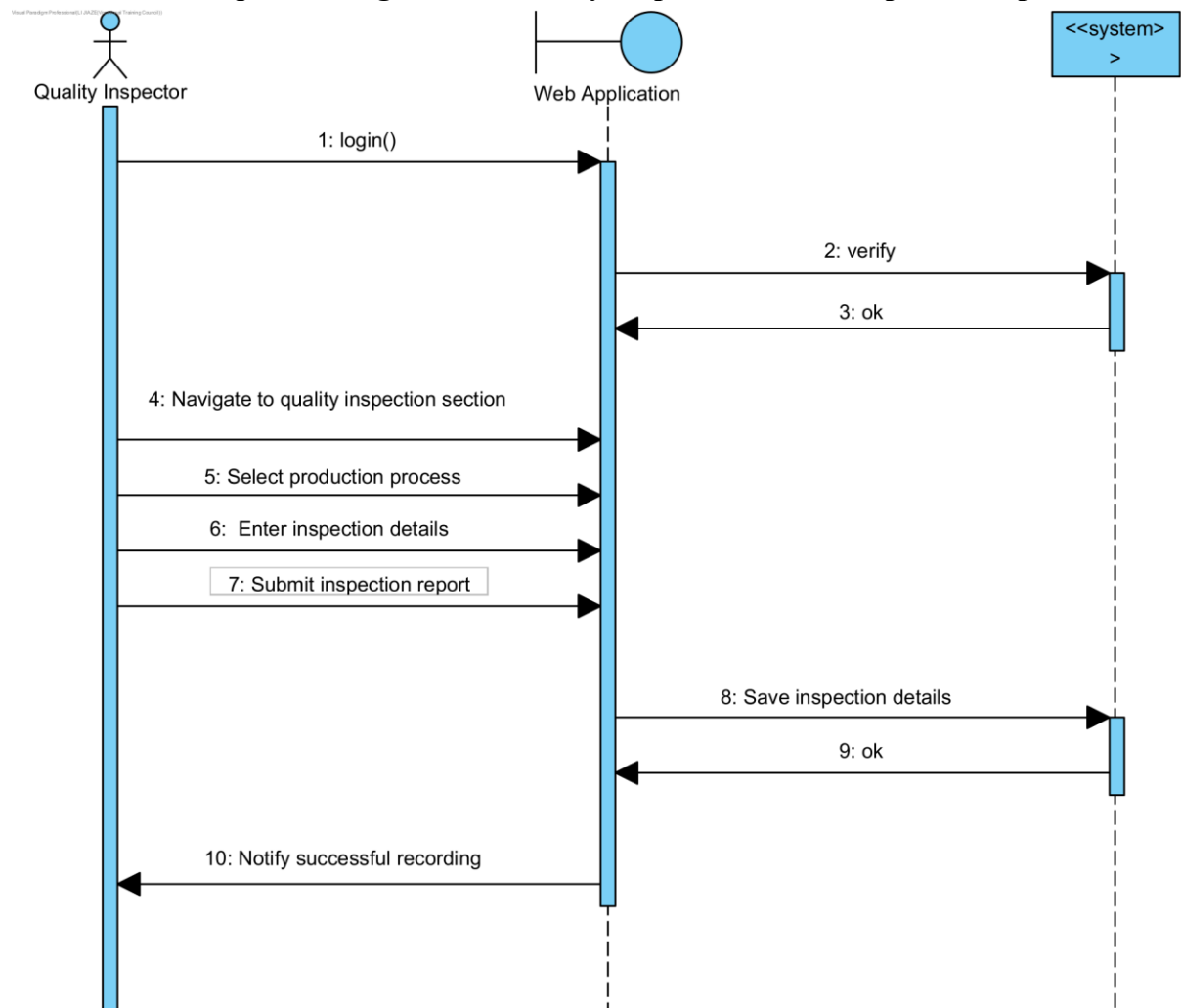
13.6 3-Tier Sequence Diagram for Production Management

13.6.1 3-Tier Sequence Diagram for Production Manager submit production plan



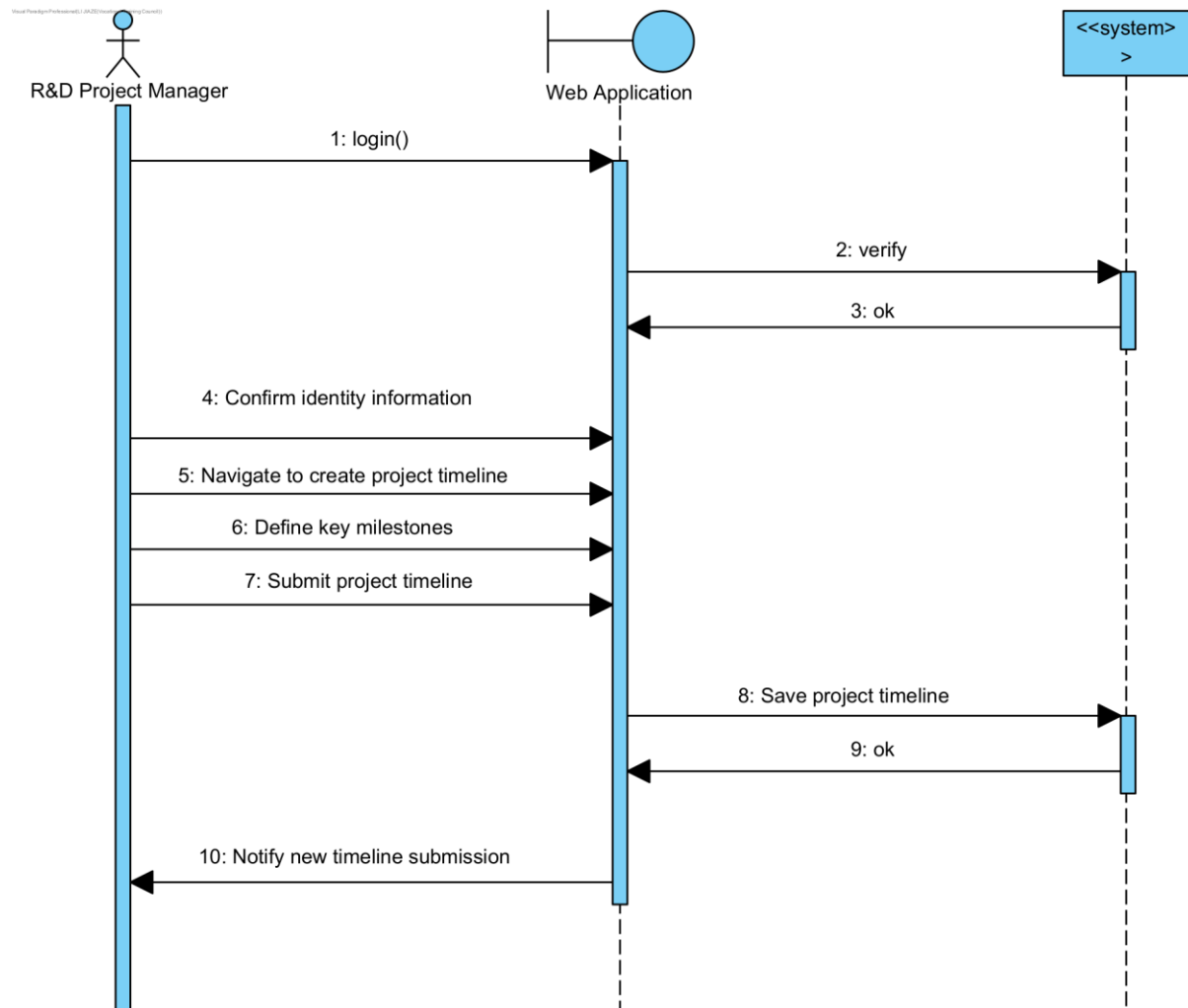
13.6.2 3-Tier Sequence Diagram for Production Engineer define process details

13.6.3 3-Tier Sequence Diagram for Inventory Manager update inventory record

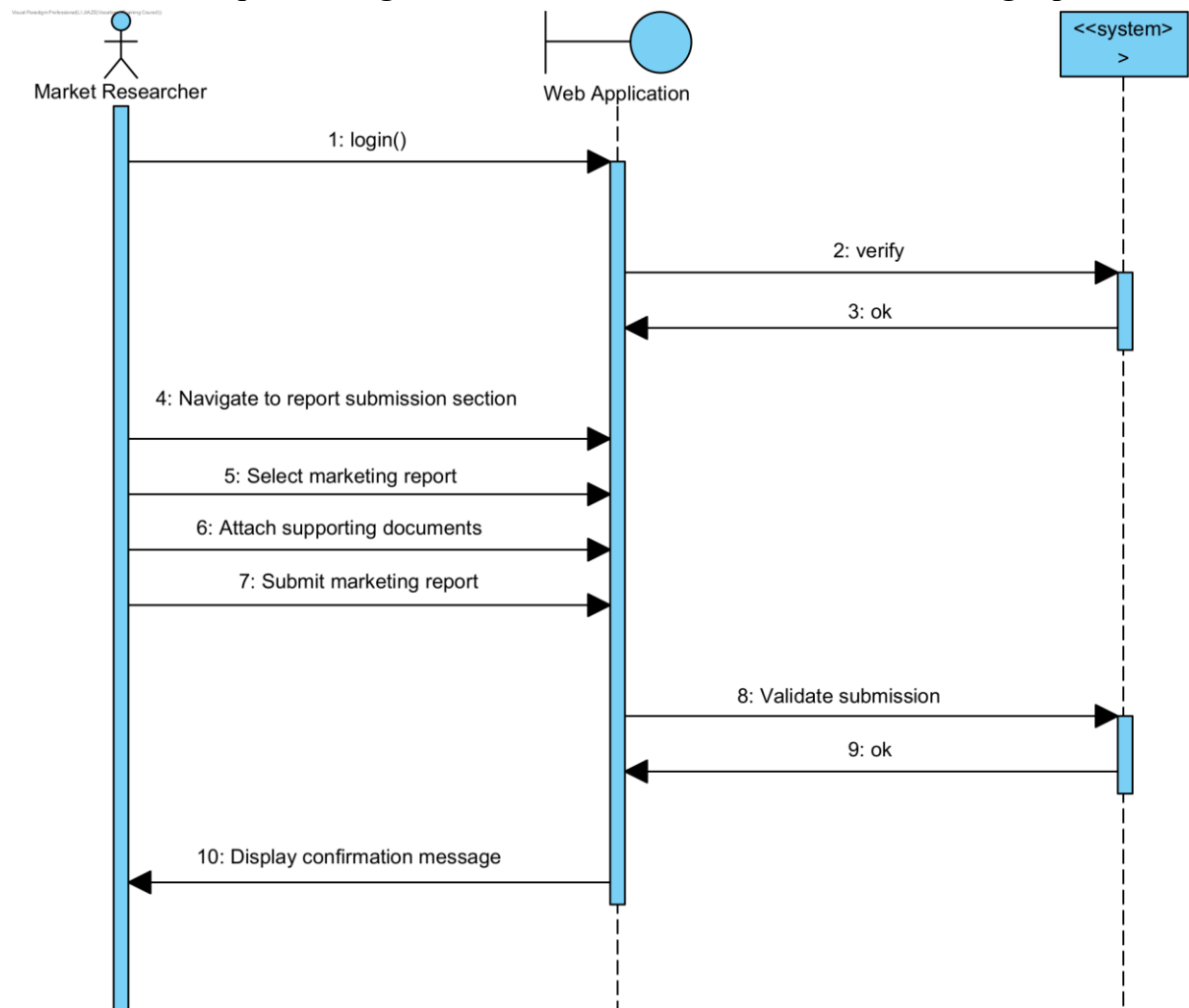
13.6.4 3-Tier Sequence Diagram for Quality Inspector submit inspection report

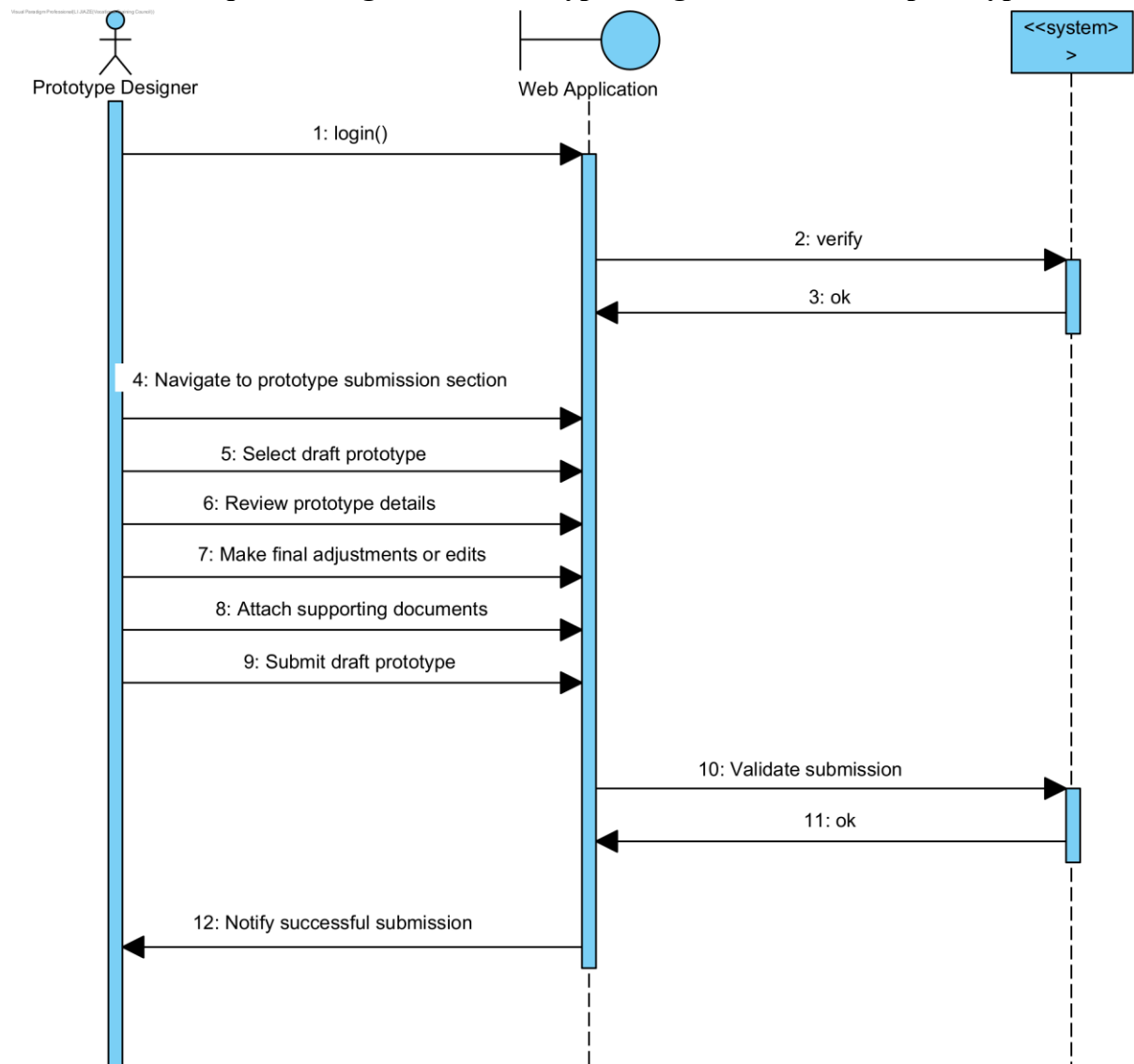
13.7 3-Tier Sequence Diagram for Research & Development

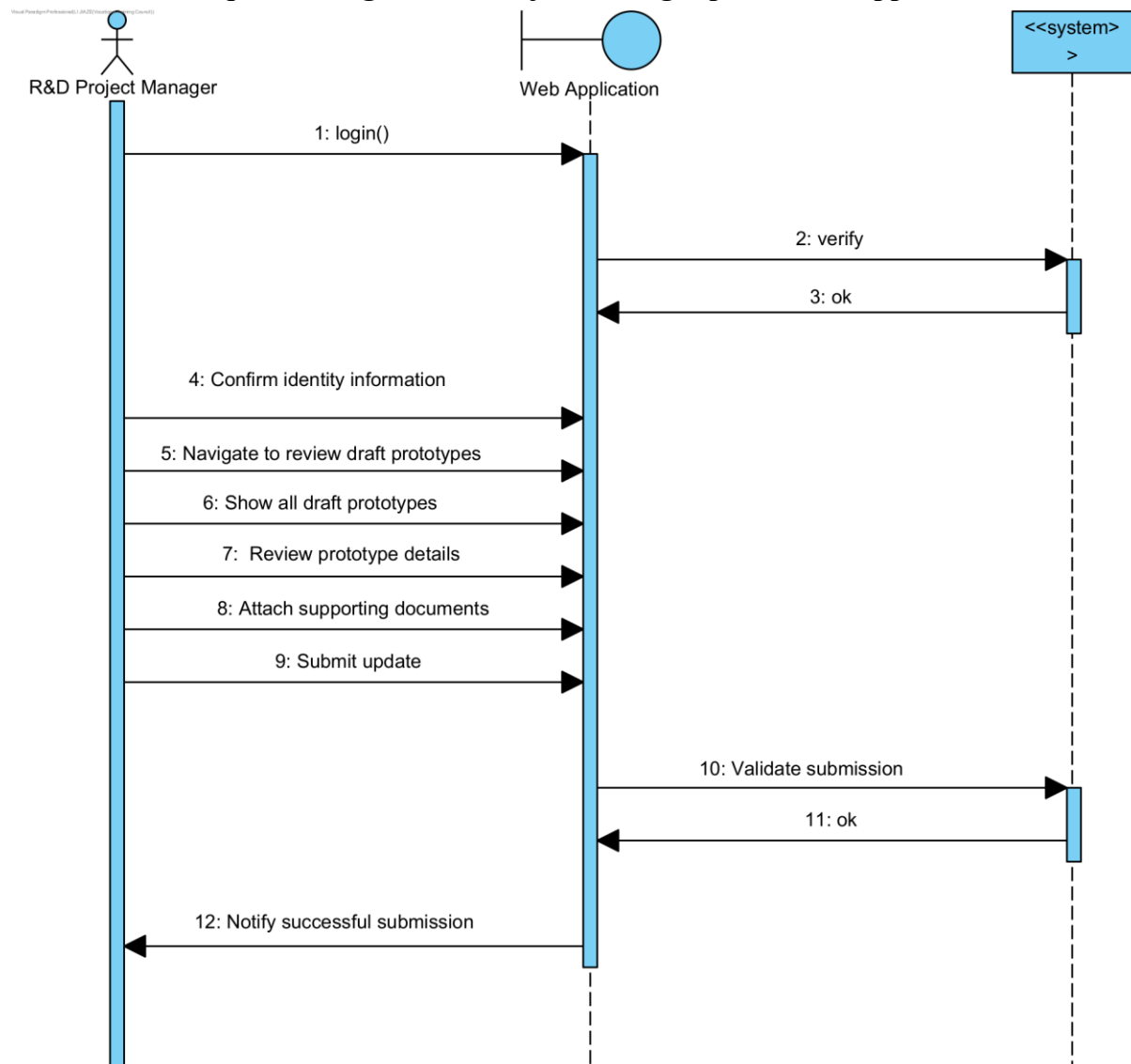
13.7.1 3-Tier Sequence Diagram for Project Manager submit project timeline



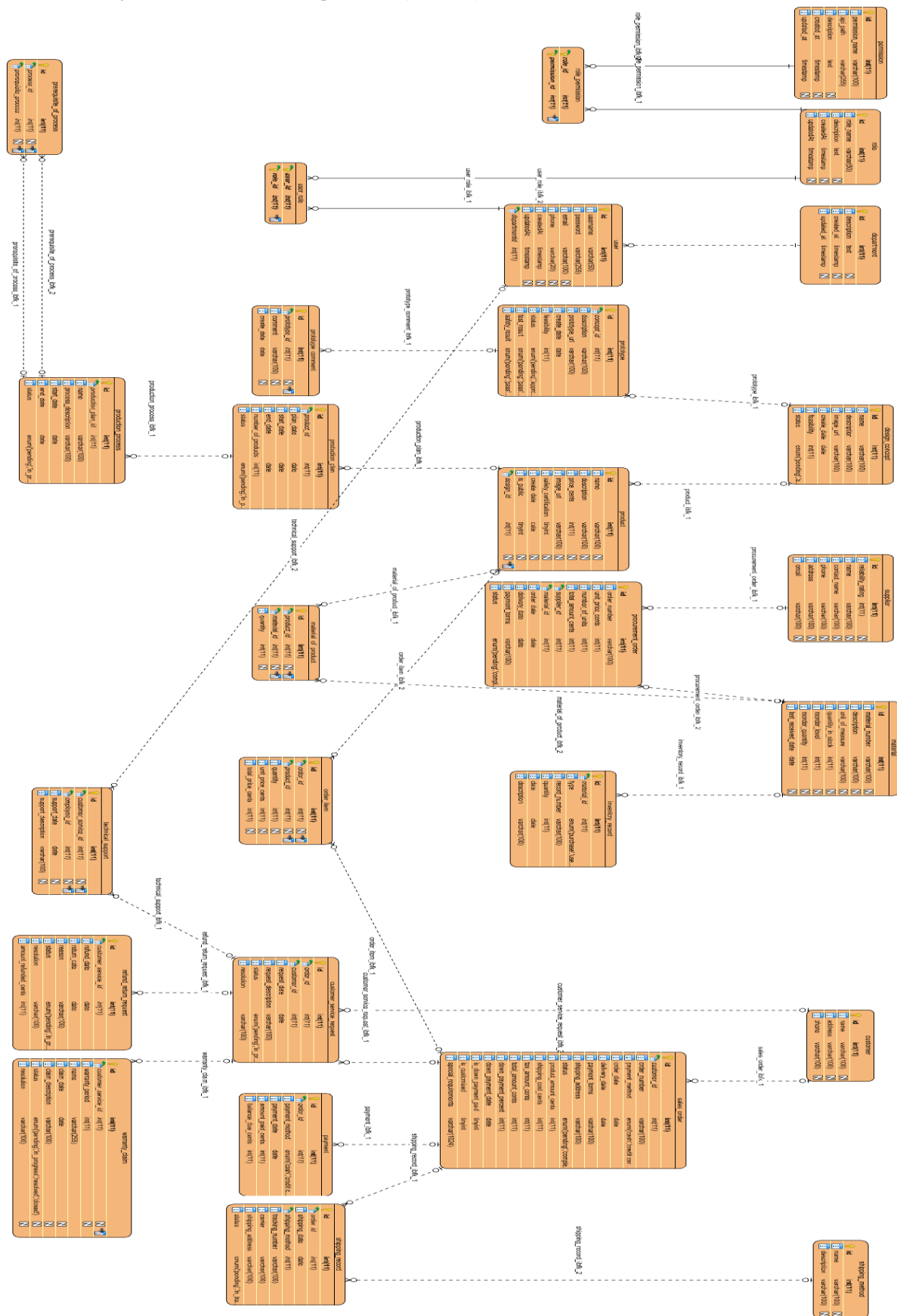
13.7.2 3-Tier Sequence Diagram for Market Researcher submit marketing report



13.7.3 3-Tier Sequence Diagram for Prototype Designer submit draft prototype

13.7.4 3-Tier Sequence Diagram for Project Manager production approval

14 Entity Relation Diagram (ERD)



14.1 department

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Department ID, primary key, auto-incremented (unique identifier for each department)
name	VARCHAR	255	-	-	Department name (required field, cannot be NULL)
description	TEXT	65535	-	-	Department description (optional field, can store detailed information about department)
created_at	TIMESTAMP	4	'1970-01-01'	'2038-01-19'	Creation timestamp, defaults to current time when record is inserted
updated_at	TIMESTAMP	4	'1970-01-01'	'2038-01-19'	Last update timestamp, automatically updates to current time when record is modified

14.2 user

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	User ID, primary key, auto-incremented
username	VARCHAR	50	-	-	Username, unique identifier
password	VARCHAR	255	-	-	User password, encrypted storage
email	VARCHAR	100	-	-	User email (optional)
phone	VARCHAR	20	-	-	User phone number (optional)
createdAt	TIMESTAMP	4	'1970-01-01'	'2038-01-19'	Creation timestamp, defaults to current time
updatedAt	TIMESTAMP	4	'1970-01-01'	'2038-01-19'	Last update timestamp, auto-updates

14.3 role

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Role ID, primary key, auto-incremented
role_name	VARCHAR	50	-	-	Role name (e.g., admin, user)
description	TEXT	65535	-	-	Role description (optional)
createdAt	TIMESTAMP	4	'1970-01-01'	'2038-01-19'	Creation timestamp, defaults to current time
updatedAt	TIMESTAMP	4	'1970-01-01'	'2038-01-19'	Last update timestamp, auto-updates

14.4 permission

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Permission ID, primary key, auto-incremented
permission_name	VARCHAR	100	-	-	Permission name
api_path	VARCHAR	255	-	-	Corresponding API path
description	TEXT	65535	-	-	Permission description (optional)
created_at	TIMESTAMP	4	'1970-01-01'	'2038-01-19'	Creation timestamp
updated_at	TIMESTAMP	4	'1970-01-01'	'2038-01-19'	Last update timestamp

14.5 user_role

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
userId	INT	4	1	2147483647	User ID (references user.id)
roleId	INT	4	1	2147483647	Role ID (references role.id)

14.6 role_permission

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
role_id	INT	4	1	2147483647	Role ID (references role.id)
permission_id	INT	4	1	2147483647	Permission ID (references permission.id)

14.7 supplier

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented supplier ID
reliability_rating	INT	4	-2147483648	2147483647	Supplier reliability rating score
name	VARCHAR	100	-	-	Supplier company name
contact_name	VARCHAR	100	-	-	Primary contact person name
phone	VARCHAR	100	-	-	Contact phone number
address	VARCHAR	100	-	-	Physical business address
email	VARCHAR	100	-	-	Business email address

14.8 material

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented material ID
material_number	VARCHAR	100	-	-	Unique identifier/number for the material
description	VARCHAR	100	-	-	Brief description of the material
unit_of_measure	VARCHAR	100	-	-	Measurement unit (e.g., kg, liter, piece)
quantity_in_stock	INT	4	0	2147483647	Current available quantity in inventory
reorder_level	INT	4	0	2147483647	Minimum stock level triggering reorder
reorder_quantity	INT	4	1	2147483647	Standard quantity to order when replenishing
last_received_date	DATE	3	-	-	Most recent date when material was received

14.9 procurement_order

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented order ID
order_number	VARCHAR	100	-	-	Unique identifier for the procurement order
unit_price_cents	INT	4	0	2147483647	Price per unit in cents
number_of_units	INT	4	1	2147483647	Quantity of units ordered
total_amount_cents	INT	4	0	2147483647	Total order amount in cents
supplier_id	INT	4	1	2147483647	Reference to supplier (foreign key)
material_id	INT	4	1	2147483647	Reference to material (foreign key)
order_date	DATE	3	-	-	Date when order was placed
delivery_date	DATE	3	-	-	Expected/actual delivery date
payment_terms	VARCHAR	100	-	-	Terms and conditions for payment
status	ENUM	1	-	-	Order status (pending/completed/cancelled)

14.10 inventory_record

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented record ID
material_id	INT	4	1	2147483647	Reference to material (foreign key)
type	ENUM	1	-	-	Transaction type (purchase/used)
record_number	VARCHAR	100	-	-	Unique identifier for the inventory record
quantity	INT	4	1	2147483647	Quantity of material involved
date	DATE	3	-	-	Date of inventory transaction
description	VARCHAR	100	-	-	Additional notes about the transaction

14.11 design_concept

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented concept ID
name	VARCHAR	100	-	-	Name of the design concept
description	VARCHAR	100	-	-	Brief description of the concept
image_url	VARCHAR	100	-	-	URL linking to concept image/visual
create_date	DATE	3	-	-	Date when concept was created
feasibility	INT	4	0	100	Feasibility rating (0-100 scale)
status	ENUM	1	-	-	Review status (pending/approved/rejected)

14.12 prototype

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented prototype ID
concept_id	INT	4	1	2147483647	Reference to design concept (foreign key)
description	VARCHAR	100	-	-	Brief description of the prototype
prototype_url	VARCHAR	100	-	-	URL linking to prototype documentation
create_date	DATE	3	-	-	Date when prototype was created
feasibility	INT	4	0	100	Feasibility rating (0-100 scale)
status	ENUM	1	-	-	Review status (pending/approved/rejected)
test_result	ENUM	1	-	-	Functional test result status
safety_result	ENUM	1	-	-	Safety evaluation result status

14.13 prototype_comment

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented comment ID
prototype_id	INT	4	1	2147483647	Reference to prototype (foreign key)
comment	VARCHAR	100	-	-	Feedback comment about the prototype
create_date	DATE	3	-	-	Date when comment was created

14.14 product

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented product ID
name	VARCHAR	100	-	-	Name of the product
description	VARCHAR	100	-	-	Brief description of the product
price_cents	INT	4	0	2147483647	Product price in cents
image_url	VARCHAR	100	-	-	URL linking to product image
safety_certification	BOOLEAN	1	-	-	Whether product has safety certification
create_date	DATE	3	-	-	Date when product was created
is_public	BOOLEAN	1	-	-	Whether product is publicly visible
design_id	INT	4	1	2147483647	Reference to design concept

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
					(foreign key)

14.15 material_of_product

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented record ID
product_id	INT	4	1	2147483647	Reference to product (foreign key)
material_id	INT	4	1	2147483647	Reference to material (foreign key)
quantity	INT	4	1	2147483647	Quantity of material used in product

14.16 production_plan

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented plan ID
product_id	INT	4	1	2147483647	Reference to product (foreign key)
plan_date	DATE	3	-	-	Date when plan was created
start_date	DATE	3	-	-	Scheduled start date of production
end_date	DATE	3	-	-	Scheduled completion date of production
number_of_products	INT	4	1	2147483647	Target quantity of products to manufacture
status	ENUM	1	-	-	Current status of production plan

14.17 production_process

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented process ID
production_plan_id	INT	4	1	2147483647	Reference to production plan (foreign key)
name	VARCHAR	100	-	-	Name of the production process
process_description	VARCHAR	100	-	-	Description of the process
start_date	DATE	3	-	-	Scheduled start date of process
end_date	DATE	3	-	-	Scheduled completion date of process
status	ENUM	1	-	-	Current status of production process

14.18 prerequisite_of_process

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented record ID
process_id	INT	4	1	2147483647	Reference to main process (foreign key)
prerequisite_process	INT	4	1	2147483647	Reference to prerequisite process (foreign key)

14.19 customer

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented customer ID
name	VARCHAR	100	-	-	Customer name
address	VARCHAR	100	-	-	Customer physical address
phone	VARCHAR	100	-	-	Customer contact phone number

14.20 sales_order

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented order ID
customer_id	INT	4	1	2147483647	Reference to customer (foreign key)
order_number	VARCHAR	100	-	-	Unique order identifier
payment_method	ENUM	1	-	-	Payment method used
order_date	DATE	3	-	-	Date order was placed
delivery_date	DATE	3	-	-	Expected/actual delivery date
payment_terms	VARCHAR	100	-	-	Terms and conditions for payment
shipping_address	VARCHAR	100	-	-	Delivery destination address
status	ENUM	1	-	-	Current order status
product_amount_cents	INT	4	0	2147483647	Subtotal for products in cents
shipping_cost_cents	INT	4	0	2147483647	Shipping cost in cents
tax_amount_cents	INT	4	0	2147483647	Tax amount in cents

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
total_amount_cents	INT	4	0	2147483647	Grand total amount in cents
down_payment_percent	INT	4	0	100	Percentage of down payment required
down_payment_date	DATE	3	-	-	Due date for down payment
is_down_payment_paid	BOOLEAN	1	-	-	Whether down payment was received
is_customized	BOOLEAN	1	-	-	Whether order has custom specifications
special_requirements	VARCHAR	1024	-	-	Customization details and special requests

14.21 order_item

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented item ID
order_id	INT	4	1	2147483647	Reference to sales order (foreign key)
product_id	INT	4	1	2147483647	Reference to product (foreign key)
quantity	INT	4	1	2147483647	Number of units ordered
unit_price_cents	INT	4	0	2147483647	Price per unit in cents at time of order
total_price_cents	INT	4	0	2147483647	Extended price (quantity × unit price)

14.22 payment

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented payment ID
order_id	INT	4	1	2147483647	Reference to sales order (foreign key)
payment_method	ENUM	1	-	-	Method used for payment
payment_date	DATE	3	-	-	Date payment was received
amount_paid_cents	INT	4	0	2147483647	Amount paid in cents
balance_due_cents	INT	4	0	2147483647	Remaining balance due in cents

14.23 shipping_method

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented shipping method ID
name	VARCHAR	100	-	-	Name of the shipping method
description	VARCHAR	100	-	-	Description of the shipping method

14.24 shipping_record

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented shipping ID
order_id	INT	4	1	2147483647	Reference to sales order (foreign key)
shipping_date	DATE	3	-	-	Date shipment was sent
shipping_method	INT	4	1	2147483647	Reference to shipping method (foreign key)
tracking_number	VARCHAR	100	-	-	Carrier's tracking identifier
carrier	VARCHAR	100	-	-	Name of shipping carrier
shipping_address	VARCHAR	100	-	-	Destination delivery address
status	ENUM	1	-	-	Current shipping status

14.25 customer_service_request

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented request ID
order_id	INT	4	1	2147483647	Reference to sales order (foreign key)
customer_id	INT	4	1	2147483647	Reference to customer (foreign key)
request_date	DATE	3	-	-	Date request was submitted
request_description	VARCHAR	100	-	-	Description of the service request
status	ENUM	1	-	-	Current status of the request
resolution	VARCHAR	100	-	-	Resolution details when request is closed

14.26 technical_support

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented support ID
customer_service_id	INT	4	1	2147483647	Reference to service request (foreign key)
employee_id	INT	4	1	2147483647	Reference to employee/user (foreign key)
support_date	DATE	3	-	-	Date technical support was provided
support_description	VARCHAR	100	-	-	Description of technical support provided

14.27 warranty_claim

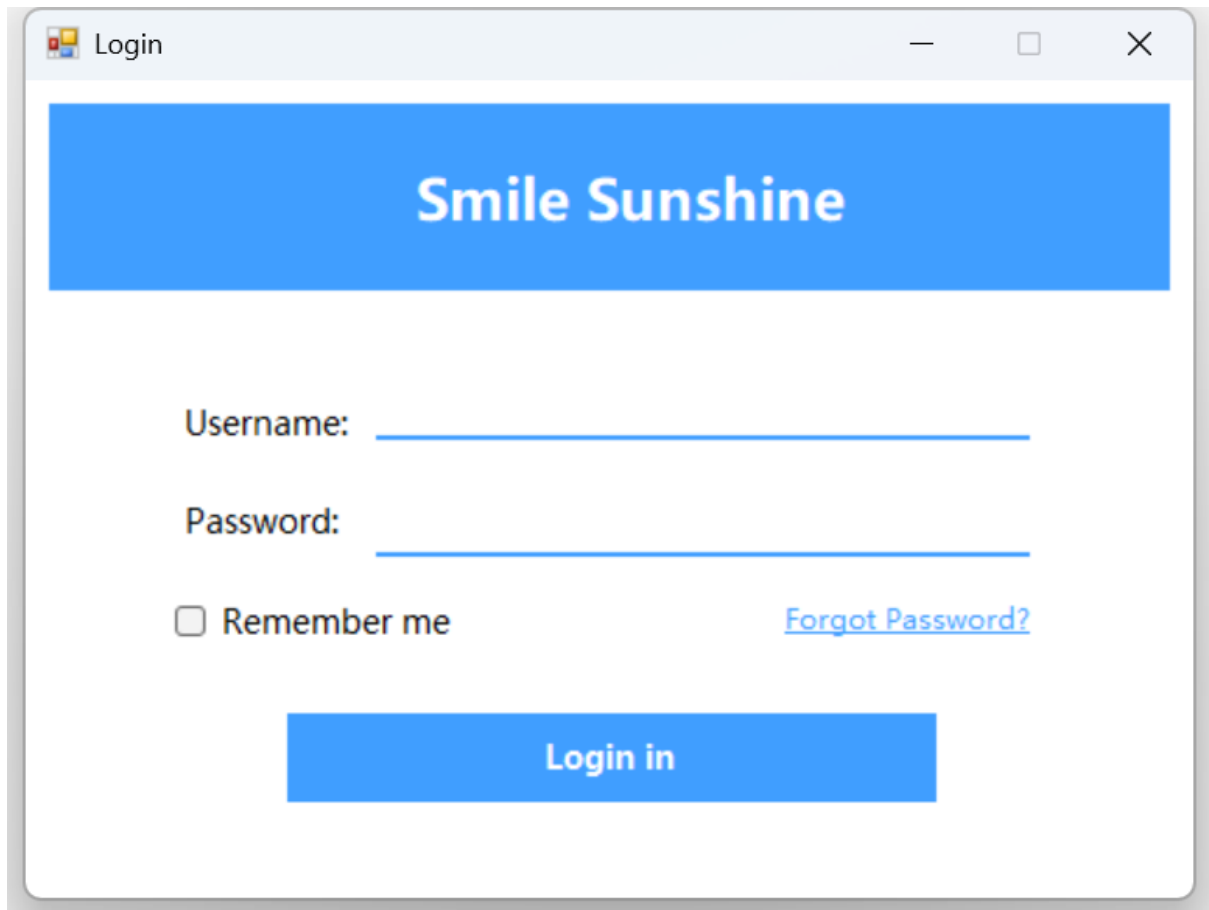
Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented claim ID
customer_service_id	INT	4	1	2147483647	Reference to service request (foreign key)
warranty_period	INT	4	0	2147483647	Warranty duration in days
claim_date	DATE	3	-	-	Date warranty claim was submitted
claim_description	VARCHAR	100	-	-	Description of the warranty issue
status	ENUM	1	-	-	Current status of the claim
resolution	VARCHAR	100	-	-	Resolution details when claim is closed

14.28 refund_return_request

Data Item Name	Data Type	Data Length	Minimum Value	Maximum Value	Description of Data Item
id	INT	4	1	2147483647	Primary key, auto-incremented request ID
customer_service_id	INT	4	1	2147483647	Reference to service request (foreign key)
refund_date	DATE	3	-	-	Date refund was processed
return_date	DATE	3	-	-	Date item was returned
reason	VARCHAR	100	-	-	Reason for return/refund
status	ENUM	1	-	-	Current status of the request
resolution	VARCHAR	100	-	-	Resolution details when request is closed
amount_refunded_cents	INT	4	0	2147483647	Amount refunded to customer in cents

15 User Interface and Report Design

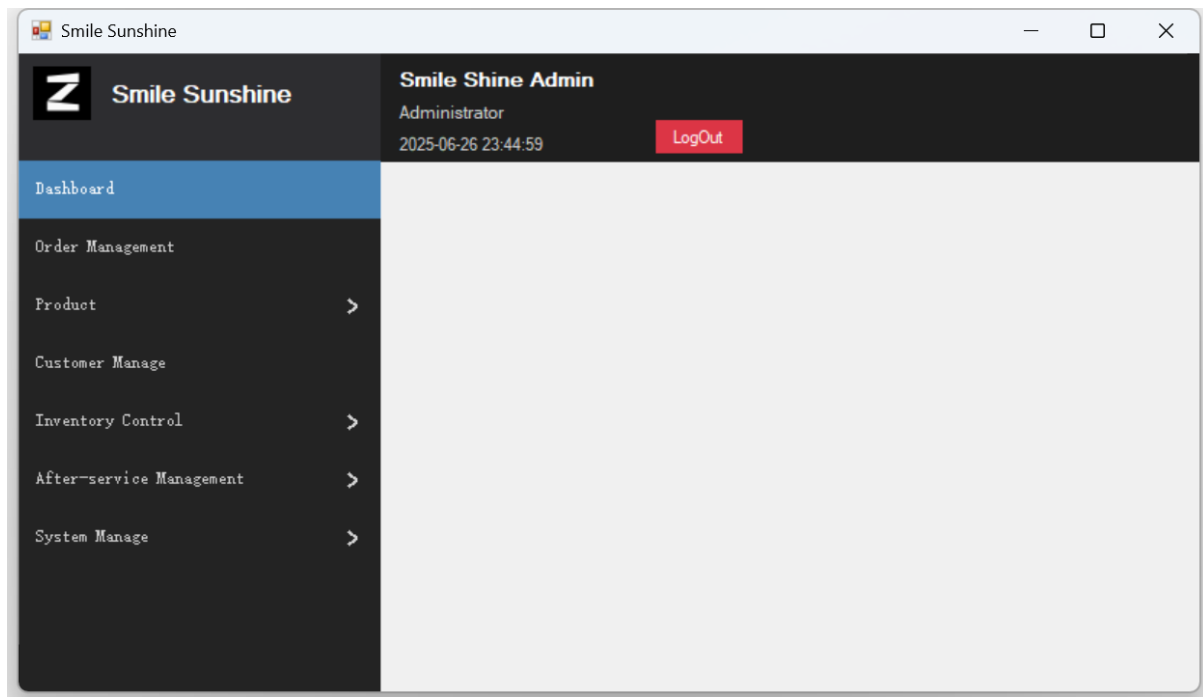
15.1 Login Page



The image shows a web browser window with the title "Login". The main content area has a blue header with the text "Smile Sunshine". Below the header, there are two input fields: "Username:" and "Password:". Below the "Password:" field, there is a checkbox labeled "Remember me" and a blue link labeled "Forgot Password?". At the bottom, there is a large blue button labeled "Login in".

This is a login interface. Users can enter their username and password in the corresponding input fields. There's a "Remember me" checkbox to let the system remember the login info for future convenience. If users forget their password, they can click the "Forgot Password?" link to retrieve it. Finally, clicking the "Login in" button submits the credentials to access the relevant system or application.

15.2 Dashboard Page



This is the main interface of the “Smile Sunshine” system after logging in as an administrator.

On the left, there’s a navigation menu with options like Dashboard (for overview), Order Management (to handle orders), Product (manage products), Customer Manage (deal with customers), Inventory Control (monitor stock), After - service Management (manage after - sales), and System Manage (configure system settings).

At the top right, it shows the admin role “Smile Shine Admin”, the access time “2025 - 06 - 26 23:44:59”, and a “LogOut” button to exit the system. The right - hand area is likely for displaying detailed content based on the selected menu item.

15.3 Order Management Page

The screenshot displays the 'Smile Sunshine Admin' interface. The left sidebar contains a navigation menu with 'Order Management' selected. The top header shows the user is an Administrator, logged in on 2025-06-26 at 23:49:28, with 'Edit' and 'Export PDF' buttons. The main content area features an 'Order List' table and an 'Order Item' section.

OrderID	OrderNumber	CustomerId	PaymentMethod	OrderDate	DeliveryDate	Status	TotalAmountCent	IsCustomized	ShippingAddress
1	08202506260001	1	cash	2025-06-26	2025-07-03	pending	\$272.46	F	18 Queen's Road Central, Central, Hong Kong Island
2	08202506260002	2	credit card	2025-06-26	2025-07-03	pending	\$1,472.07	N	123 Nathan Road, Tsim Sha Tsui, Kowloon
3	08202506260004	6	bank transfer	2025-06-26	2025-07-03	pending	\$478.17	Y	555 Lockhart Road, Causeway Bay, Hong Kong Island

Below the table, the 'Order Item' section displays three items:

- ToyRobot**: Quantity: 6, Total: \$59.94, Unit Price: \$9.99. Description: Cute robot toy with interactive features.
- ToyBear**: Quantity: 6, Total: \$59.94, Unit Price: \$9.99. Description: A soft, cuddly plush bear for endless play and comfort.
- ToySailboat**: Quantity: 9, Total: \$116.91, Unit Price: \$12.99. Description: A mini toy sailboat for imaginative play.

This is the “Order Management” section of the “Smile Sunshine” system.

On the left, the navigation menu shows “Order Management” is selected. The main area has an “Order List” with details like OrderID, OrderNumber, CustomerId, PaymentMethod, OrderDate, DeliveryDate, Status, TotalAmountCent, IsCustomized, and ShippingAddress.

There are “Edit” and “Export PDF” buttons for order operations. Below, the “Order Item” section displays purchased items (ToyRobot, ToyBear, ToySailboat) with their descriptions, quantities, unit prices, and total amounts. It helps admins view and manage order - related info comprehensively.

15.4 Edit Order Page

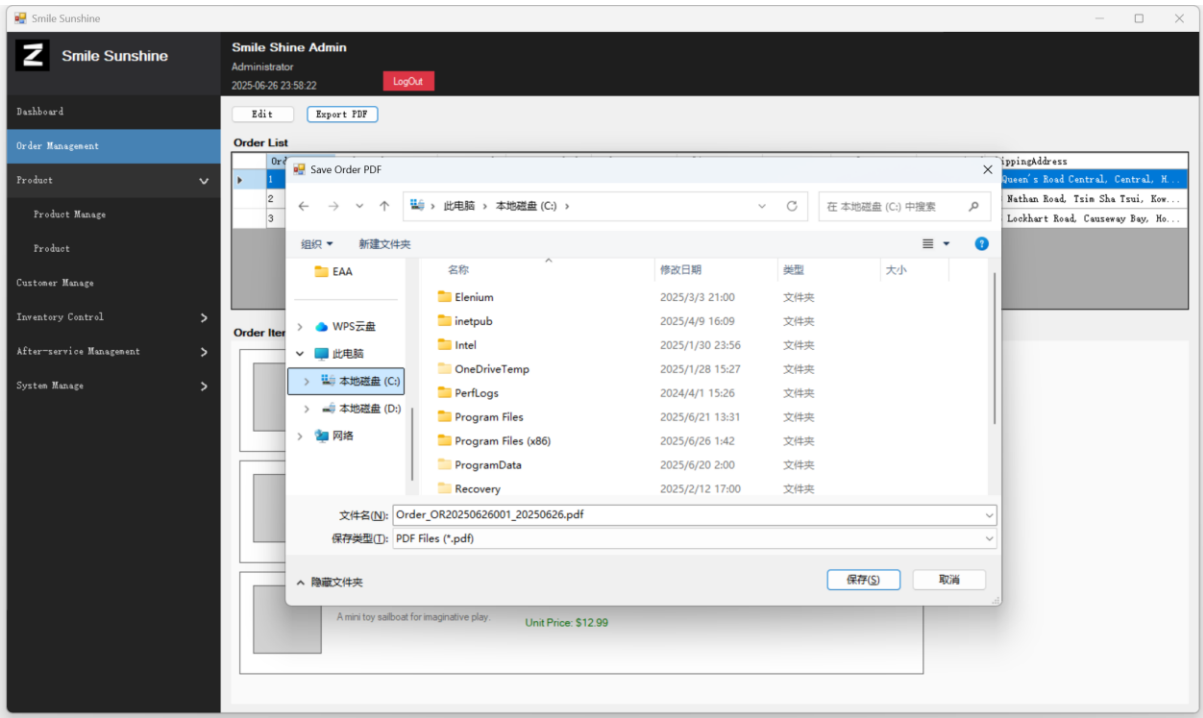
The screenshot shows the 'Edit Order' window in the Smile Sunshine Admin system. The window is titled 'Edit Order - OR20250626001'. It contains the following fields and controls:

- Order List:** A table with columns 'Order ID' and 'Order Number'. The first row is selected, showing '1' and 'OR20250626001'.
- Order Details:**
 - Order Number: OR20250626001
 - Customer ID: 1
 - Payment Methods: (dropdown menu)
 - Order Date: 2025年 6月26日
 - Delivery Date: 2025年 7月 3日
 - Status: (dropdown menu)
 - Shipping address: 18 Queen's Road Central, Central, Hong Kong Island
- Order Item:**
 - Total amount (in dollars): 272.46
 - ☒ Custom Orders
 - Special Requests: (text box)
- Buttons:** 'Save' and 'Cancel' buttons at the bottom.

The background shows the 'Smile Sunshine Admin' dashboard with a sidebar menu containing 'Dashboard', 'Order Management', 'Product', 'Customer Manage', 'Inventory Control', 'After-service Management', and 'System Manage'. The 'Order Management' section is active, showing a list of orders.

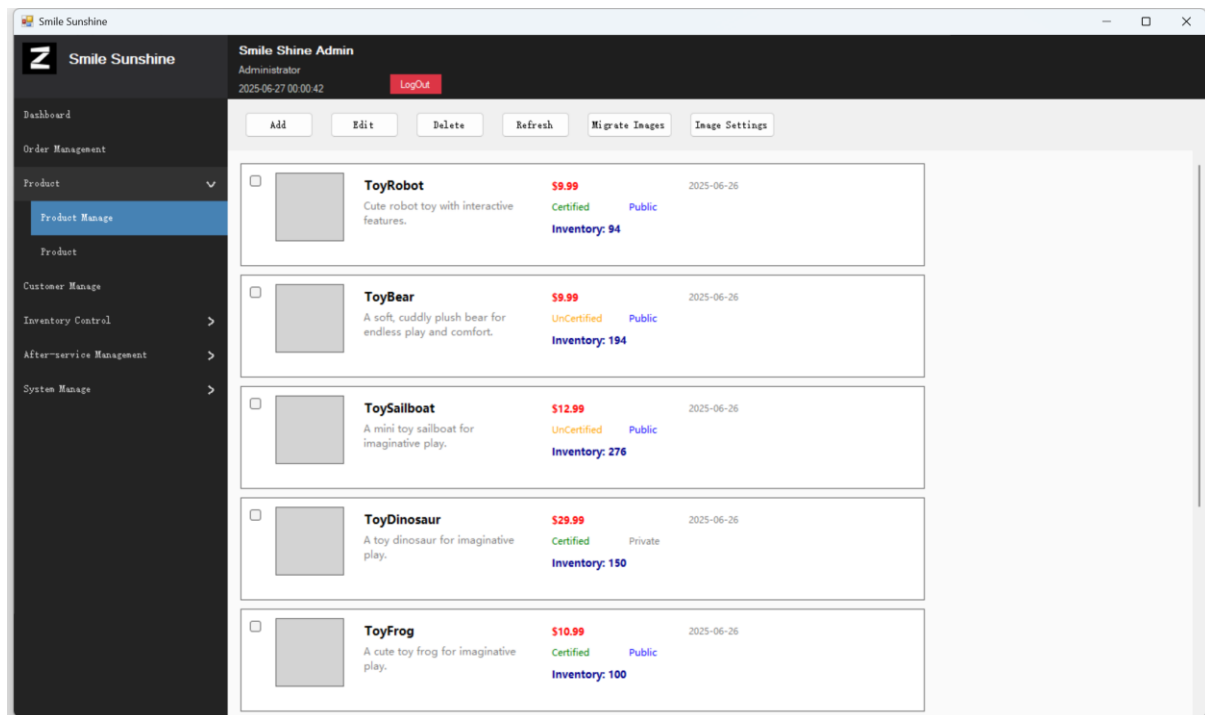
This is the “Edit Order” window triggered by clicking “Edit” on the order management page. It displays detailed information of order “OR20250626001”, including fields like order number, customer ID, payment method, order date, delivery date, status, shipping address, total amount, a checkbox for “custom order”, and a text box for “special requests”. There are “Save” and “Cancel” buttons to confirm or discard changes, allowing administrators to modify order details as needed.

15.5 Save Order PDF Page



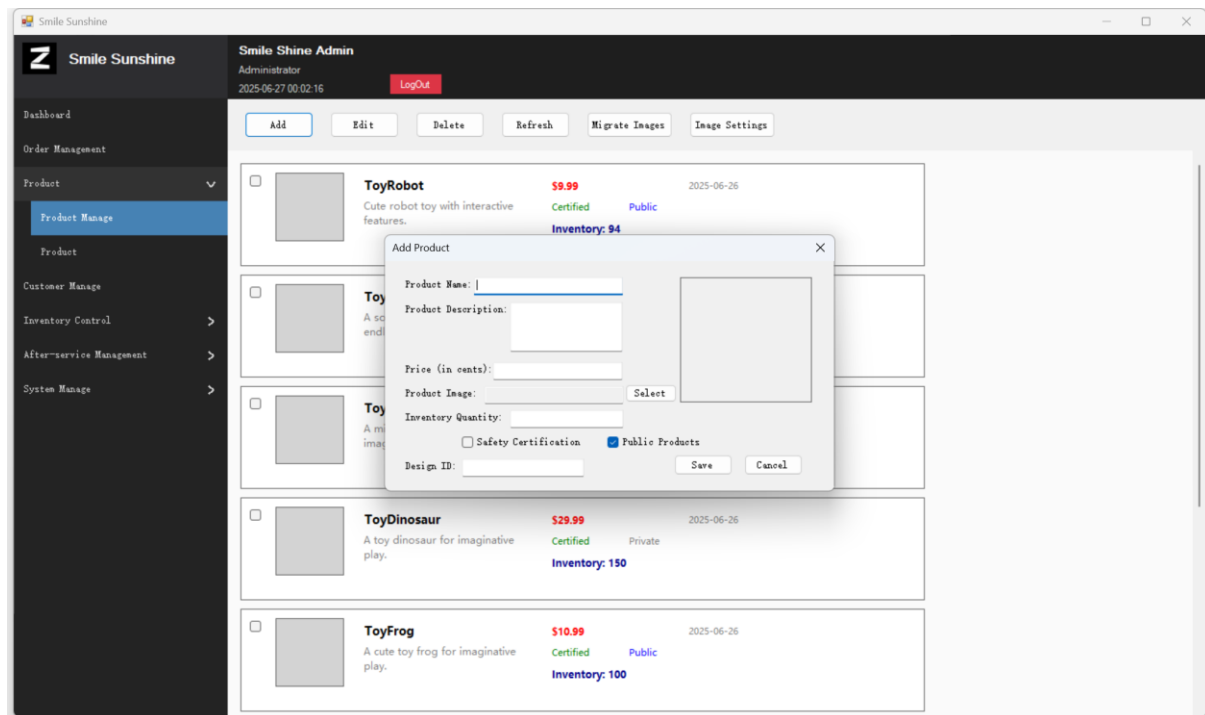
This is the “Save Order PDF” dialog box that pops up after clicking “Export PDF” in the order management interface. It allows the user to choose a location on the local disk (currently showing the C: drive) to save the PDF file of the order. The file is named “Order_OR20250626001_20250626.pdf” by default and is set to be saved as a PDF format file. The user can click “Save” to confirm the save or “Cancel” to abort the operation.

15.6 Product Manage Page



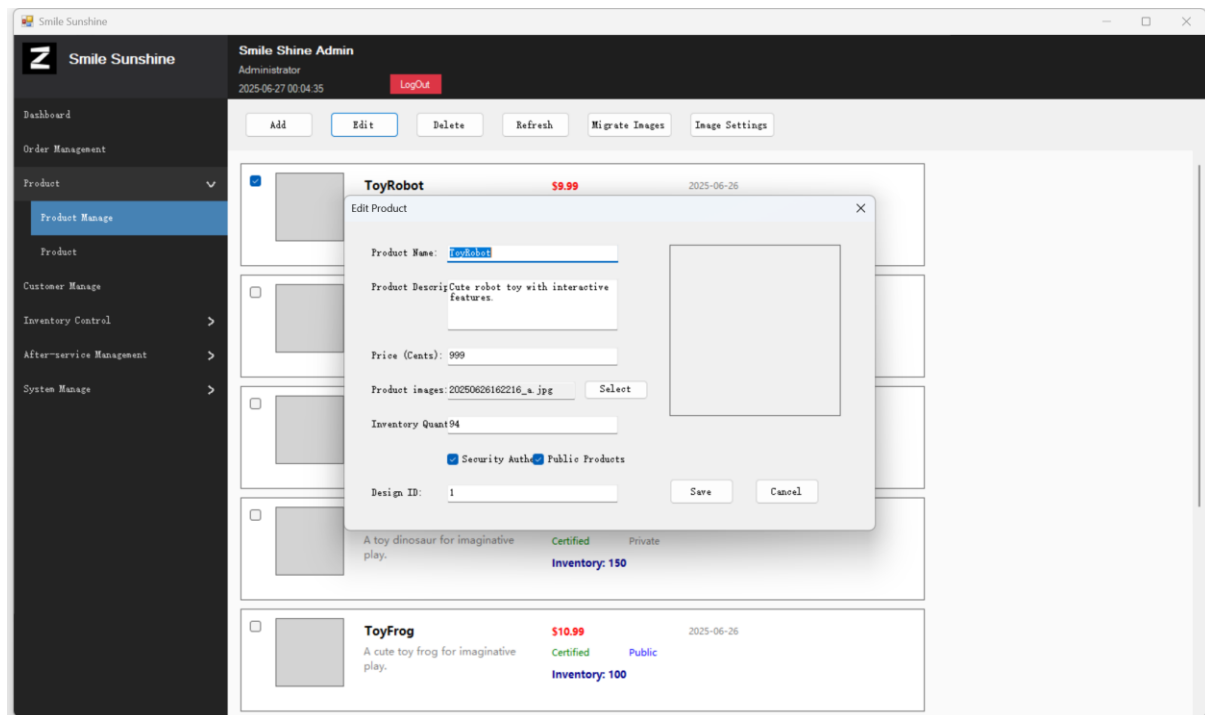
This is the “Product Manage” section of the “Smile Sunshine” system. The left navigation menu shows “Product” expanded, with “Product Manage” selected. The main area lists products like ToyRobot, ToyBear, etc. Each product entry displays details: name, price, description, certification status (Certified/UnCertified), visibility (Public/Private), inventory quantity, and date (2025 - 06 - 26). At the top, there are action buttons: Add (create new product), Edit (modify selected product), Delete (remove product), Refresh (update list), Migrate Images (handle product images), and Image Settings (configure image - related settings). It’s a centralized place for admins to manage product catalog info.

15.7 Add Product Page



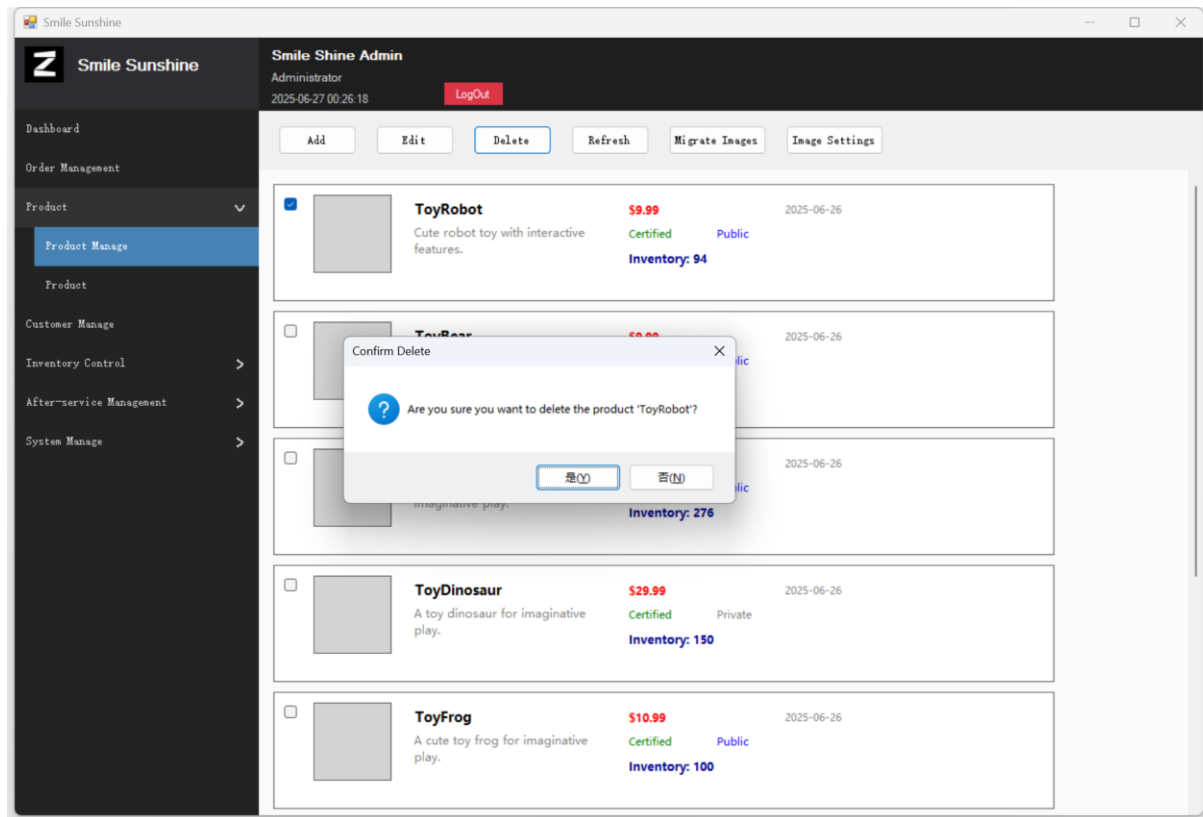
This is the “Add Product” pop - up window that appears after clicking the “Add” button in the “Product Manage” section. It provides input fields for creating a new product, including Product Name, Product Description, Price (in cents), Product Image (with a “Select” button for uploading), Inventory Quantity, and options for Safety Certification and Public Products. There's also a Design ID field. Admins can fill in these details and click “Save” to add the new product or “Cancel” to abandon the operation.

15.8 Edit Product Page



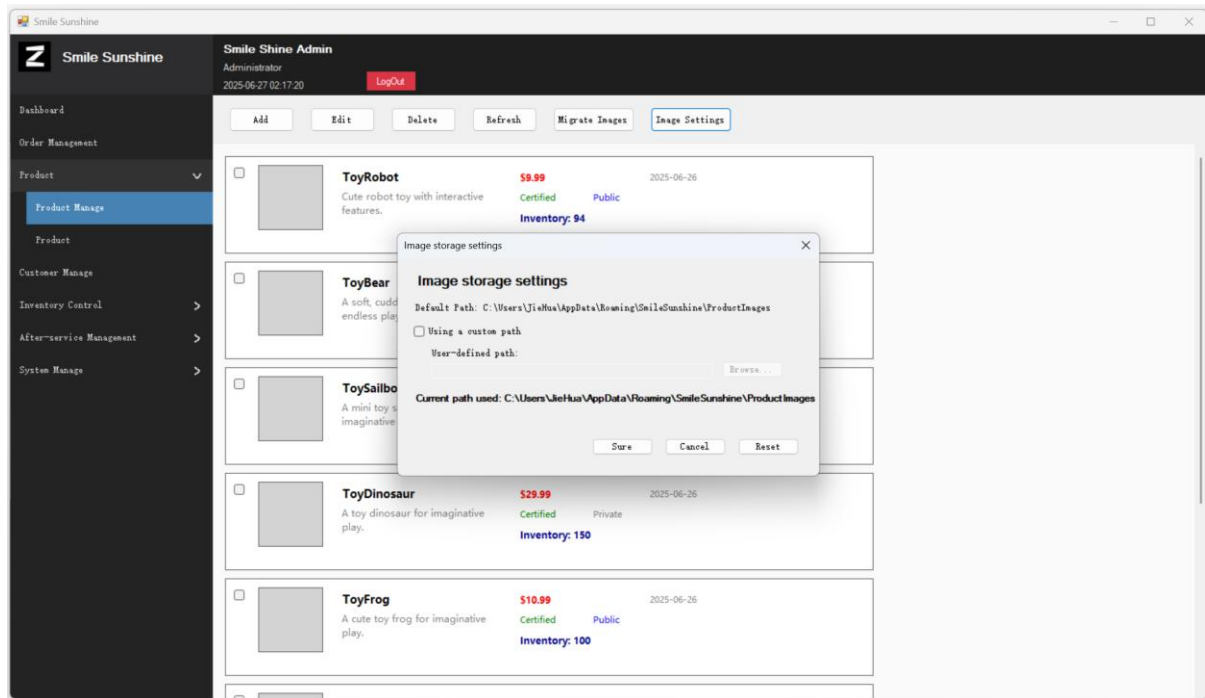
This is the “Edit Product” pop - up window triggered by clicking “Edit” in the “Product Manage” section. It shows details of “ToyRobot” for modification, including fields like Product Name, Product Description, Price (in cents), Product image (with a “Select” button to change), Inventory Quantity, checkboxes for Security Authentication and Public Products, and Design ID. Admins can update these details and click “Save” to apply changes or “Cancel” to discard them.

15.9 Delete Product Page



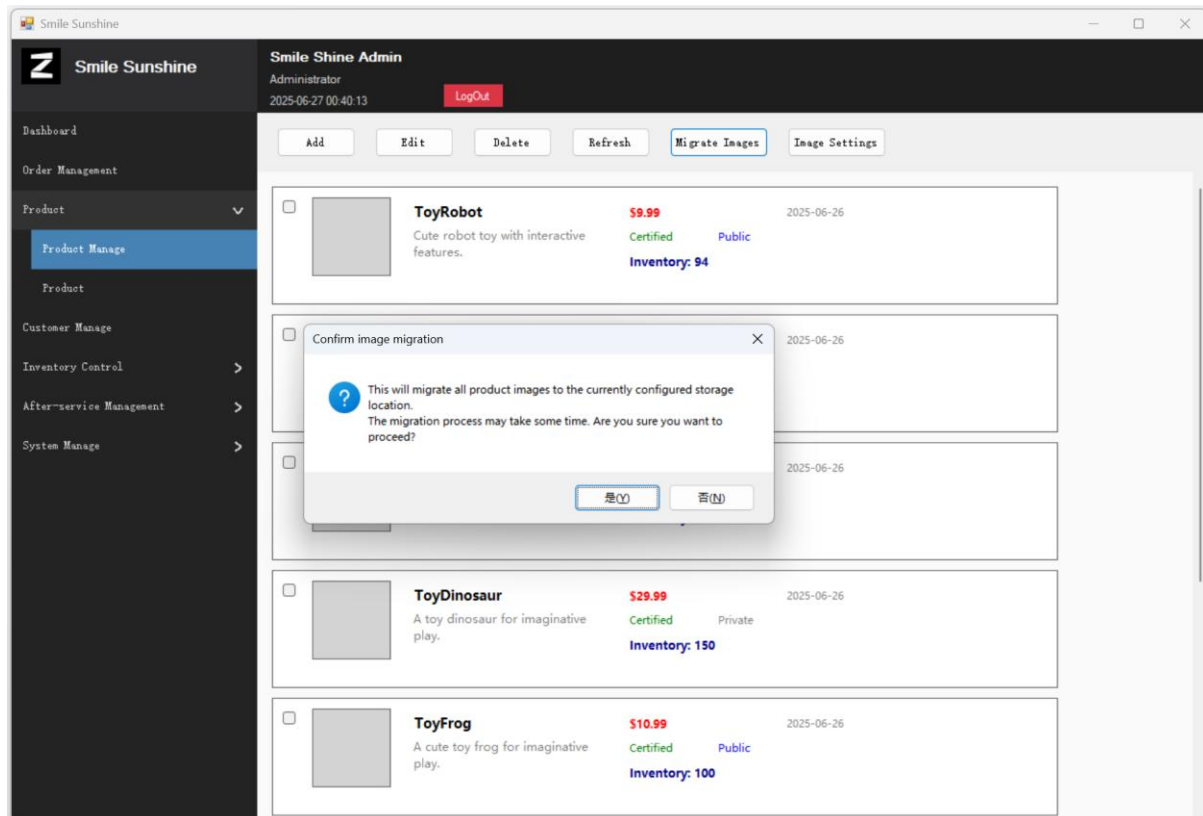
After clicking "Delete", a confirmation dialog box titled "Confirm Delete" pops up. It asks "Are you sure you want to delete the product ToyRobot?" with two options: "(Y)" (Yes) to confirm the deletion and "(N)" (No) to cancel the deletion. This prevents accidental removal of the product "ToyRobot" from the "Product Manage" list.

15.10 Edit Product Page



This is the “Image Settings” pop - up in the “Product Manage” section of the “Smile Sunshine” system. It shows the “Default Path” for storing product images as “C:\Users\lieHu\AppData\Roaming\SmileSunshine\ProductImages”. There's an option “Use Custom Path” which is not checked. If checked, users can set a “Custom Path” via the “Browse” button. The “Current path used” is the same as the default path. Buttons “Confirm”, “Cancel”, and “Reset” are for confirming, canceling, or resetting the settings.

15.11 Confirm Product Image Migration Page



This is a “Confirm image migration” dialog box in the “Product Manage” section of the “Smile Sunshine” system. It asks whether to migrate all product images to the currently configured storage location, warning that the process may take time. Options are “YES(Y)” (Yes) to proceed and “NO(N)” (No) to cancel.

15.12 Customer Manage Page

Smile Sunshine Admin
Administrator
2025-06-27 00:41:54 [Logout](#)

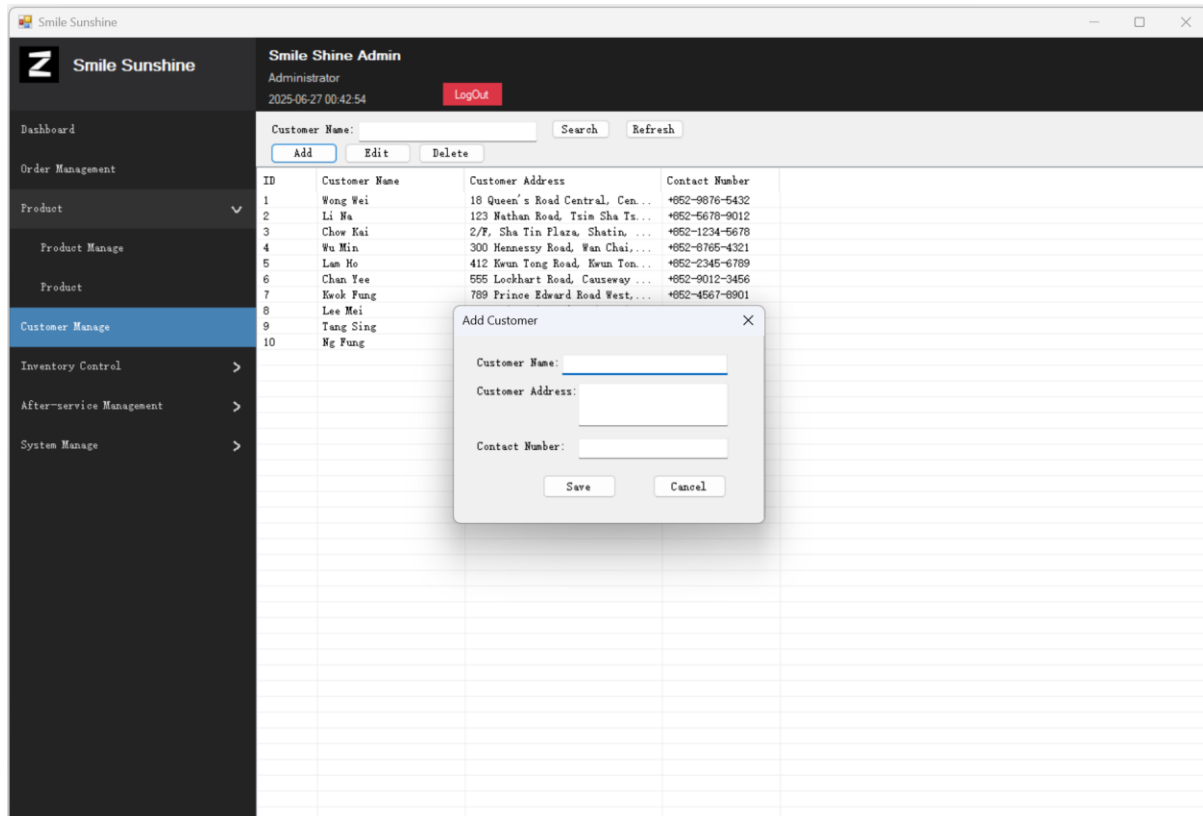
Customer Name: [Search](#) [Refresh](#)

[Add](#) [Edit](#) [Delete](#)

ID	Customer Name	Customer Address	Contact Number
1	Wong Wei	18 Queen's Road Central, Cen...	+652-9876-5432
2	Li Na	123 Nathan Road, Tsim Sha Ts...	+652-5678-9012
3	Chow Kai	2/F, Sha Tin Plaza, Shatin, ...	+652-1234-5678
4	Wu Min	300 Hennessy Road, Wan Chai,...	+652-8765-4321
5	Lau Ho	412 Kwun Tong Road, Kwun Ton...	+652-2345-6789
6	Chan Yee	555 Lockhart Road, Causeway ...	+652-9012-3456
7	Kwok Fung	789 Prince Edward Road West,...	+652-4567-8901
8	Lee Mei	321 Tsing Yi Road, Tsing Yi,...	+652-6789-0123
9	Tang Sing	987 Lai Chi Kok Road, Lai Ch...	+652-3456-7890
10	Ng Fung	654 Jordan Road, Yau Ma Tei,...	+652-7890-1234

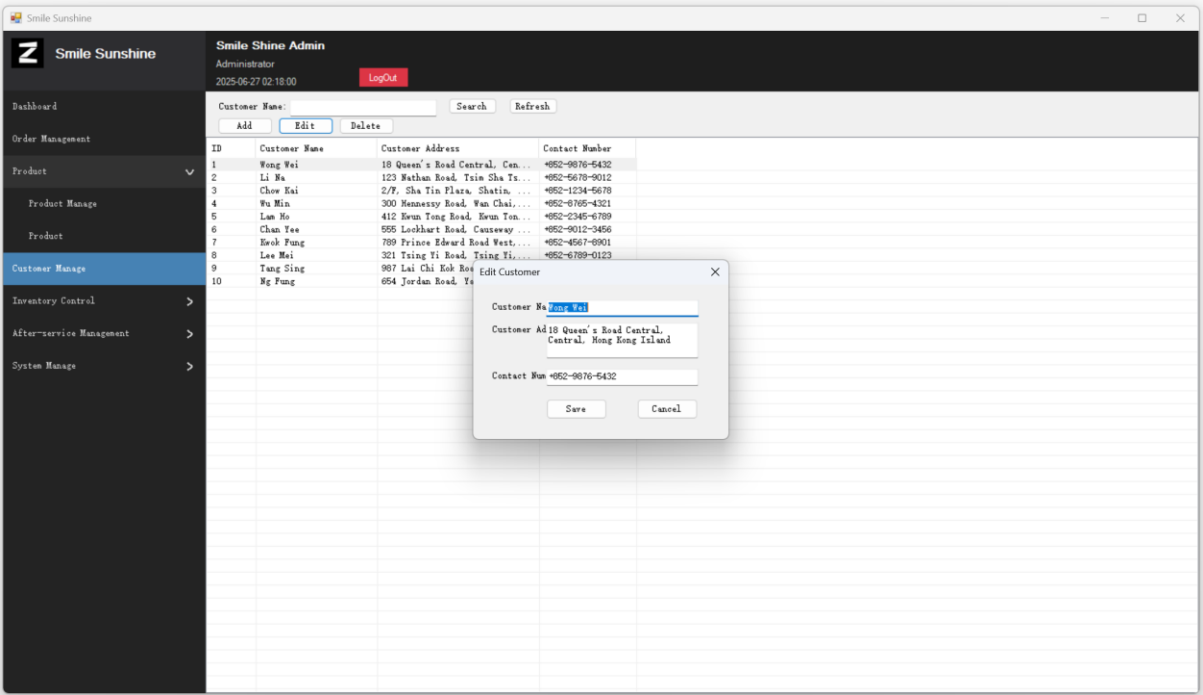
This is the “Customer Manage” section of the “Smile Sunshine” system. The left navigation menu shows “Customer Manage” is selected. The main area lists customer information with columns: ID, Customer Name (like Wong Wai, Li Na), Customer Address, and Contact Number. There are action buttons: Add (create new customer), Edit (modify selected customer), Delete (remove customer), Search (find customer by name), and Refresh (update list). It's used for administrators to manage customer data.

15.13 Customer Add Customer Page



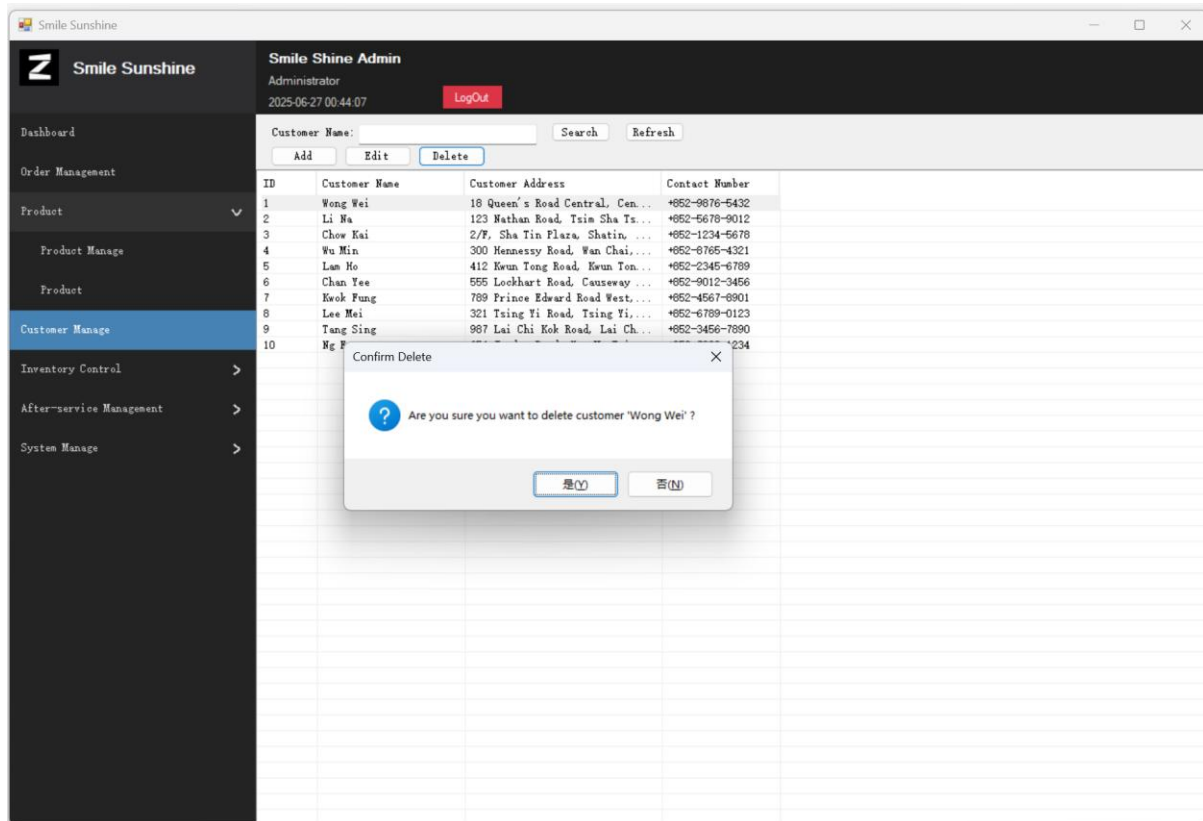
This is the “Add Customer” pop - up window in the “Customer Manage” section of the “Smile Sunshine” system. It has input fields for “Customer Name”, “Customer Address”, and “Contact Number”, along with “Save” and “Cancel” buttons. Used for administrators to add new customer information to the list showing existing customers like Wong Wai, Li Na, etc.

15.14 Customer Edit Customer Page

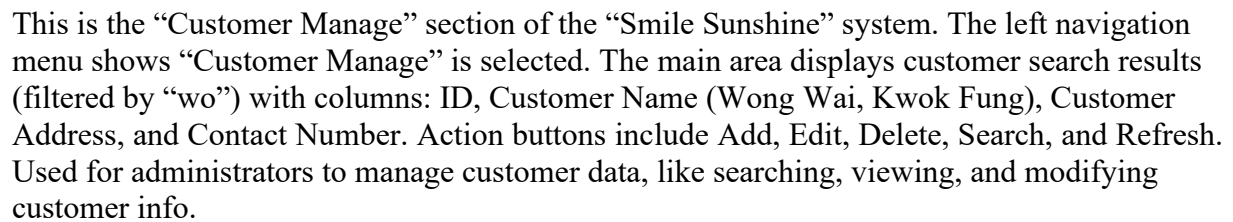


This is the “Edit Customer” pop - up window in the “Customer Manage” section of the “Smile Sunshine” system. It shows the customer info of “Wong Wai” for editing, including fields like customer name, address, and contact number. There are “Save” and “Cancel” buttons to confirm or discard changes. Used for administrators to modify existing customer data in the list.

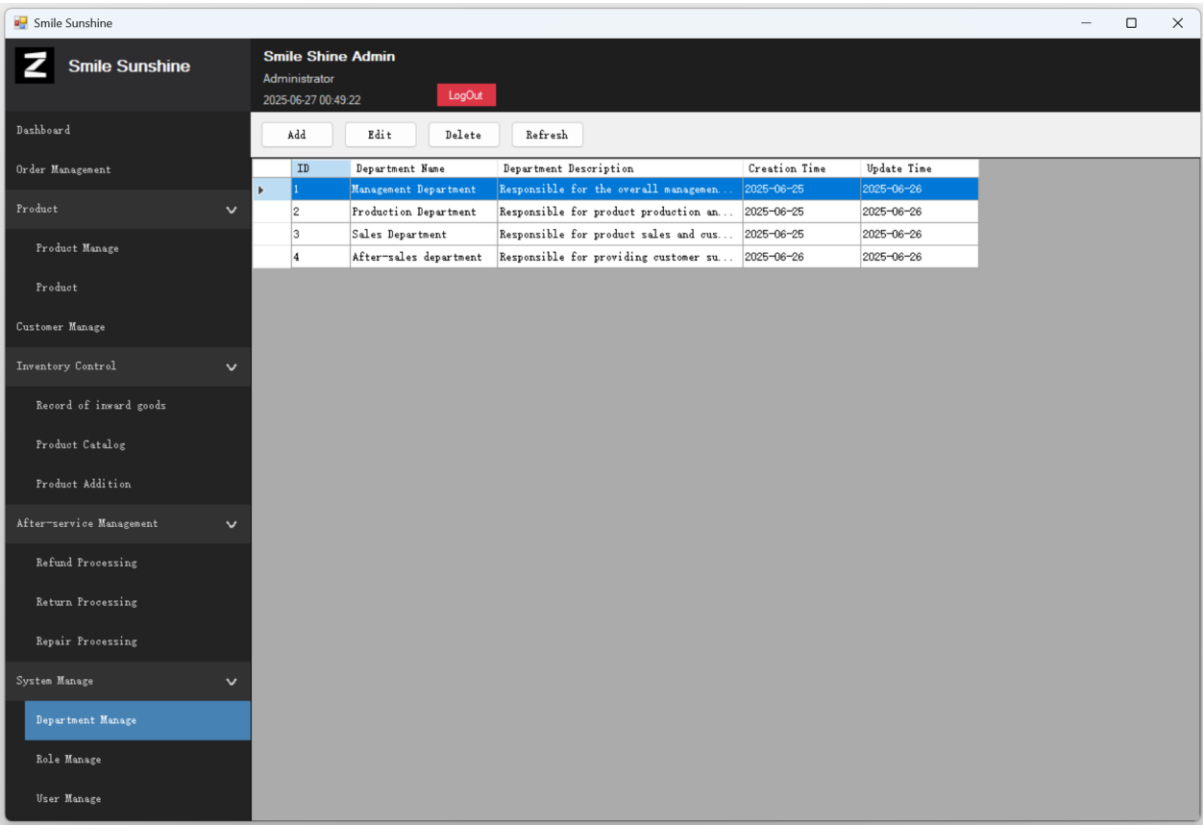
15.15 Customer Confirm Delete Page



This is a “Confirm Delete” dialog box in the “Customer Manage” section of the “Smile Sunshine” system. It asks whether to delete the customer “Wong Wai”, with options “YES(Y)” (Yes) to confirm deletion and “NO(N)” (No) to cancel.

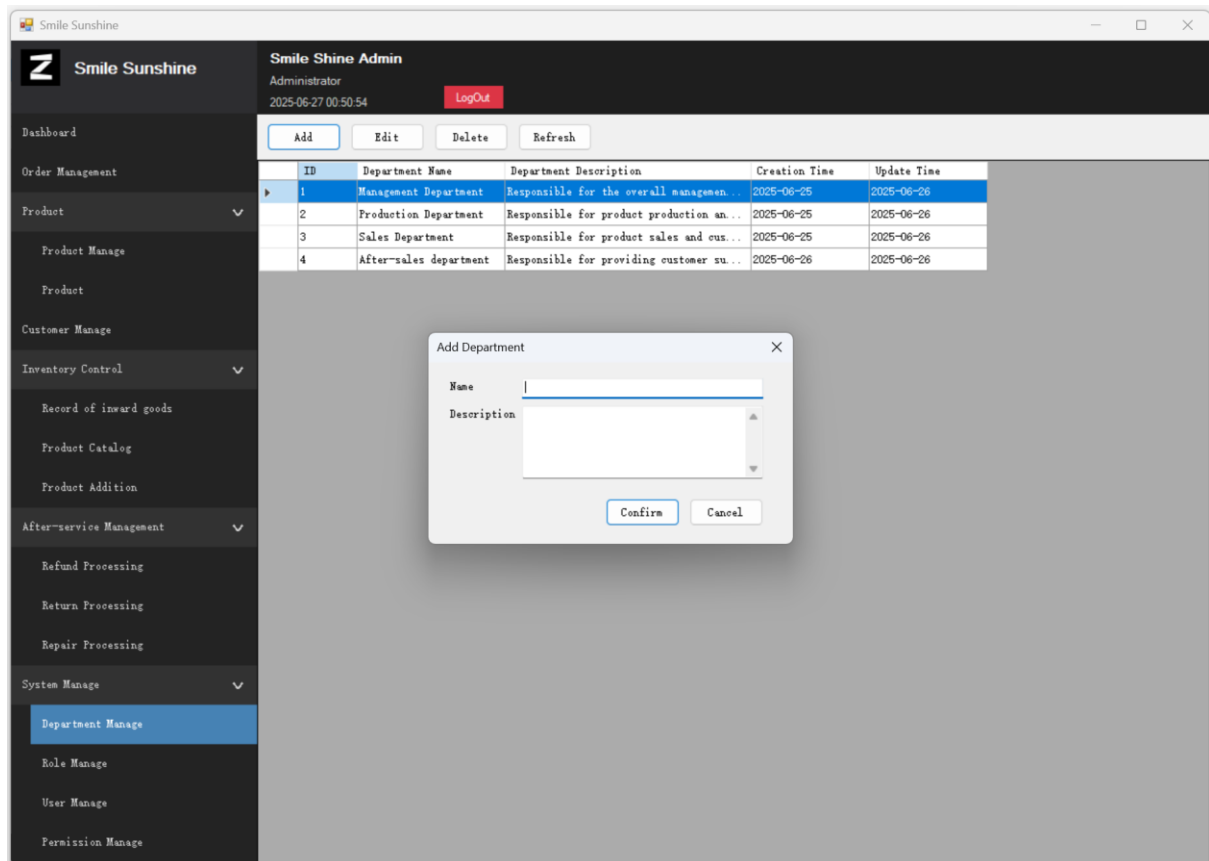


15.17 Department Manage Page



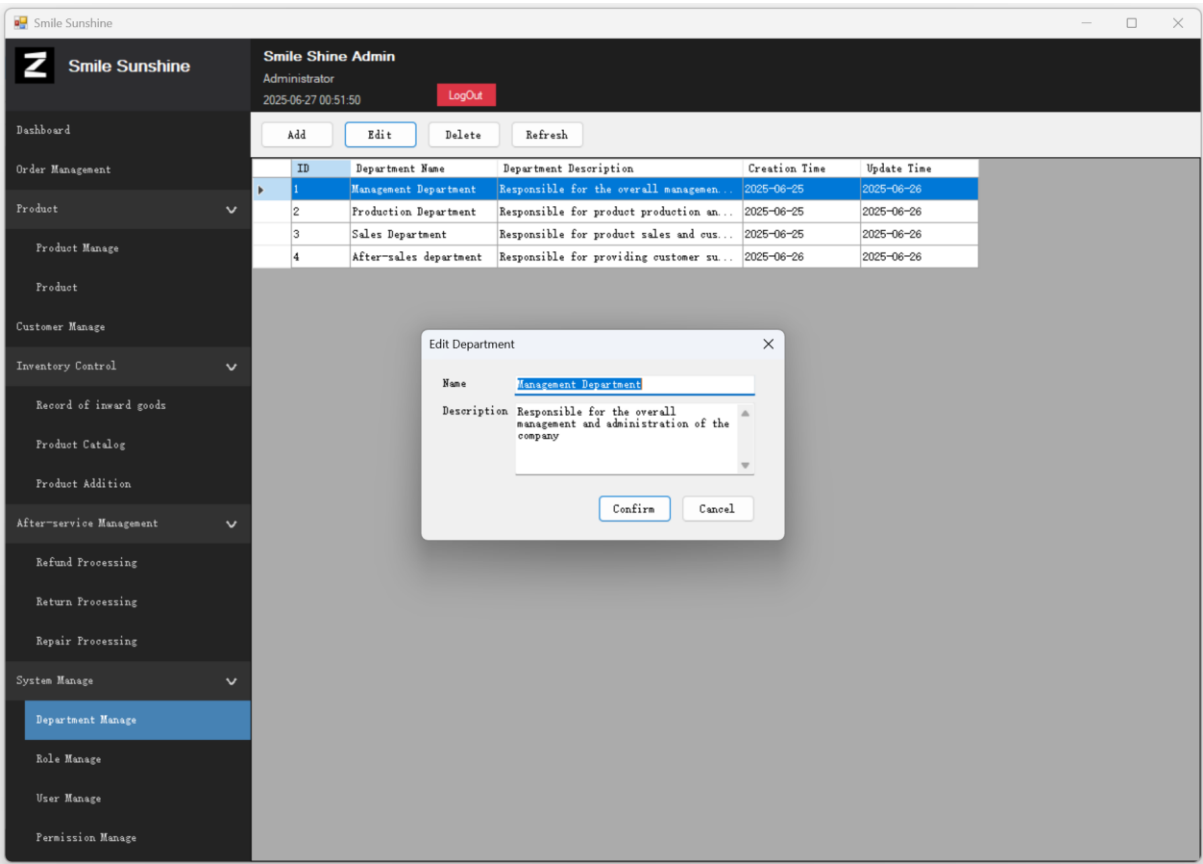
This is the “Department Manage” section of the “Smile Sunshine” system. The left navigation menu shows “Department Manage” under “System Manage” is selected. The main area lists departments with columns: ID, Department Name (Management Department, Production Department, etc.), Department Description, Creation Time, and Update Time. Action buttons include Add, Edit, Delete, and Refresh. Used for administrators to manage company departments and their info.

15.18 Department Add Department Page



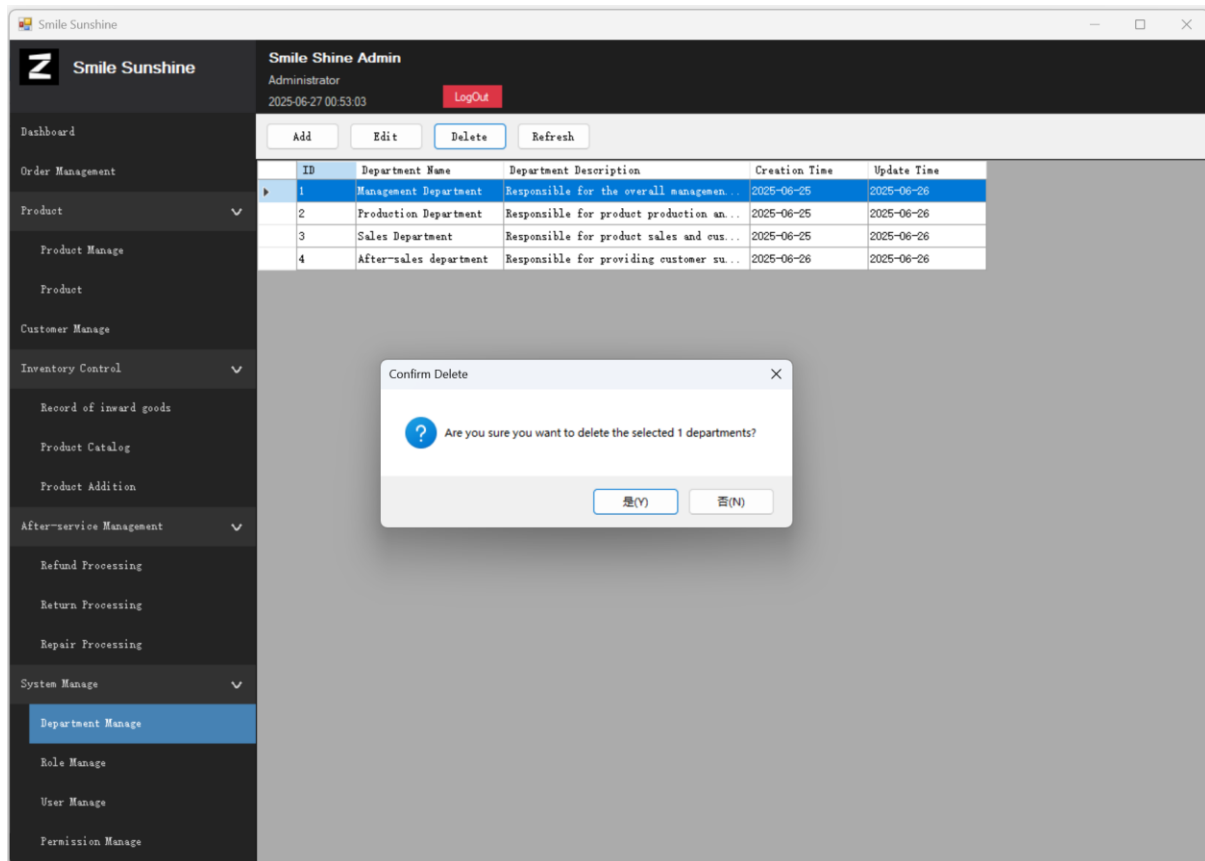
This is the “Add Department” pop - up window in the “Department Manage” section of the “Smile Sunshine” system. It has input fields for “Name” and “Description” of a new department, along with “Confirm” and “Cancel” buttons. Used for administrators to add new departments to the list showing existing ones like Management Department, Production Department, etc.

15.19 Department Edit Department Page



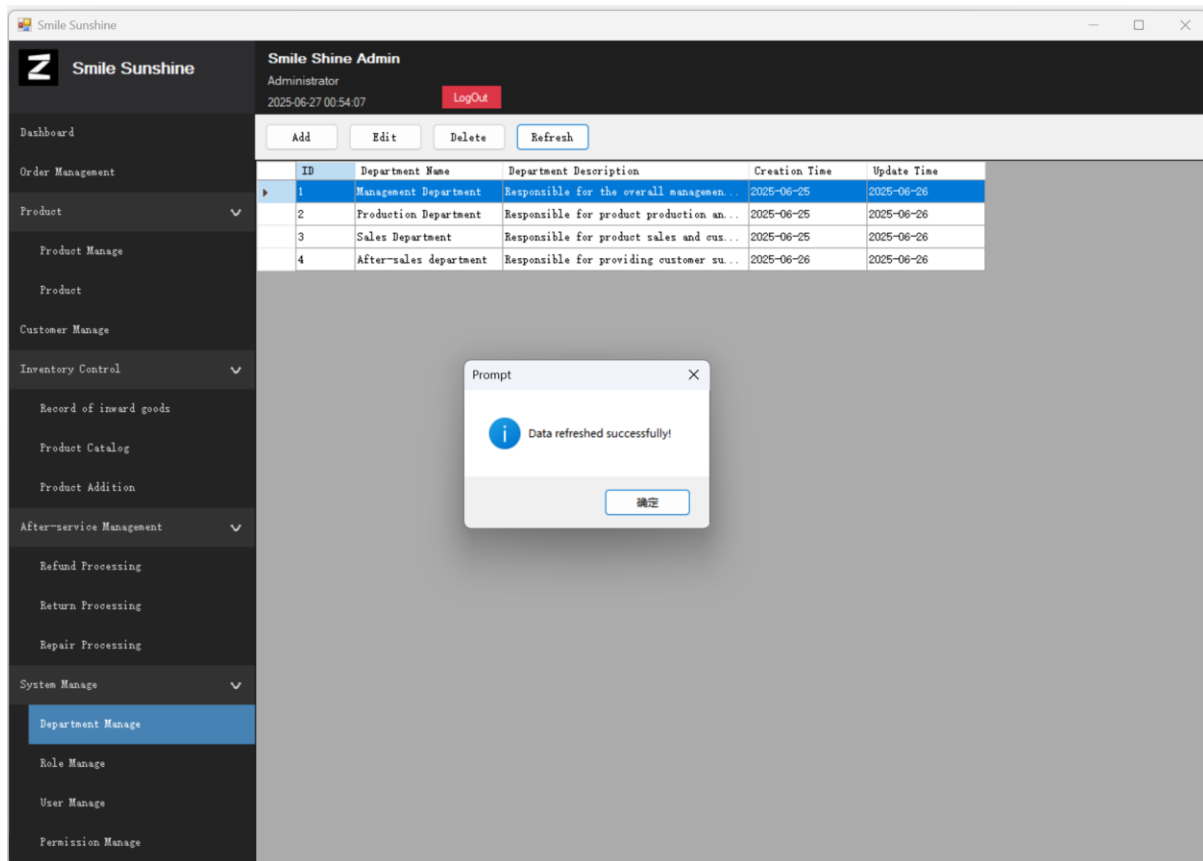
This is the “Edit Department” pop - up window in the “Department Manage” section of the “Smile Sunshine” system. It shows the info of the “Management Department” for editing, including fields for “Name” and “Description”. There are “Confirm” and “Cancel” buttons to save or discard changes. Used for administrators to modify department details in the list.

15.20 Department Confirm Delete Page



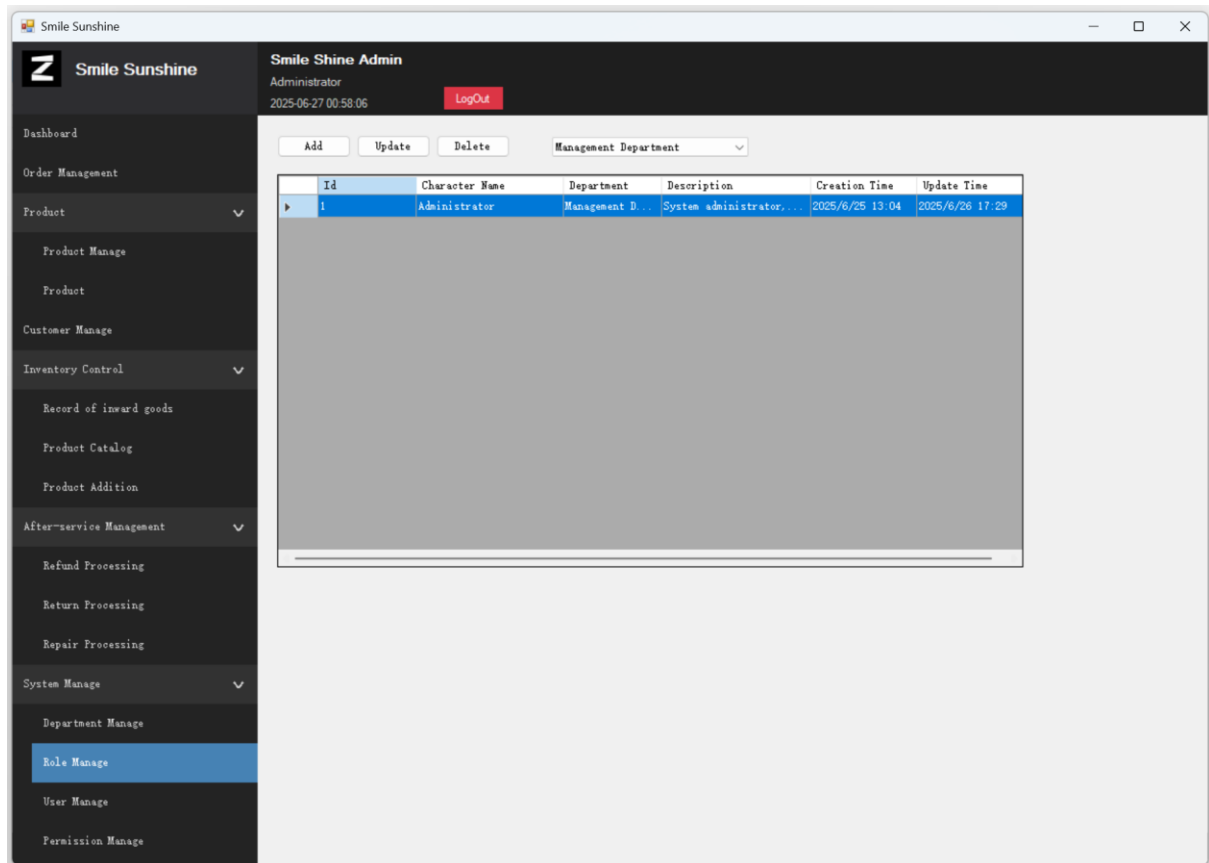
This is a “Confirm Delete” dialog box in the “Department Manage” section of the “Smile Sunshine” system. It asks whether to delete the selected department (Management Department), with options “YES(Y)” (Yes) to confirm deletion and “NO(N)” (No) to cancel.

15.21 Department Refresh Page



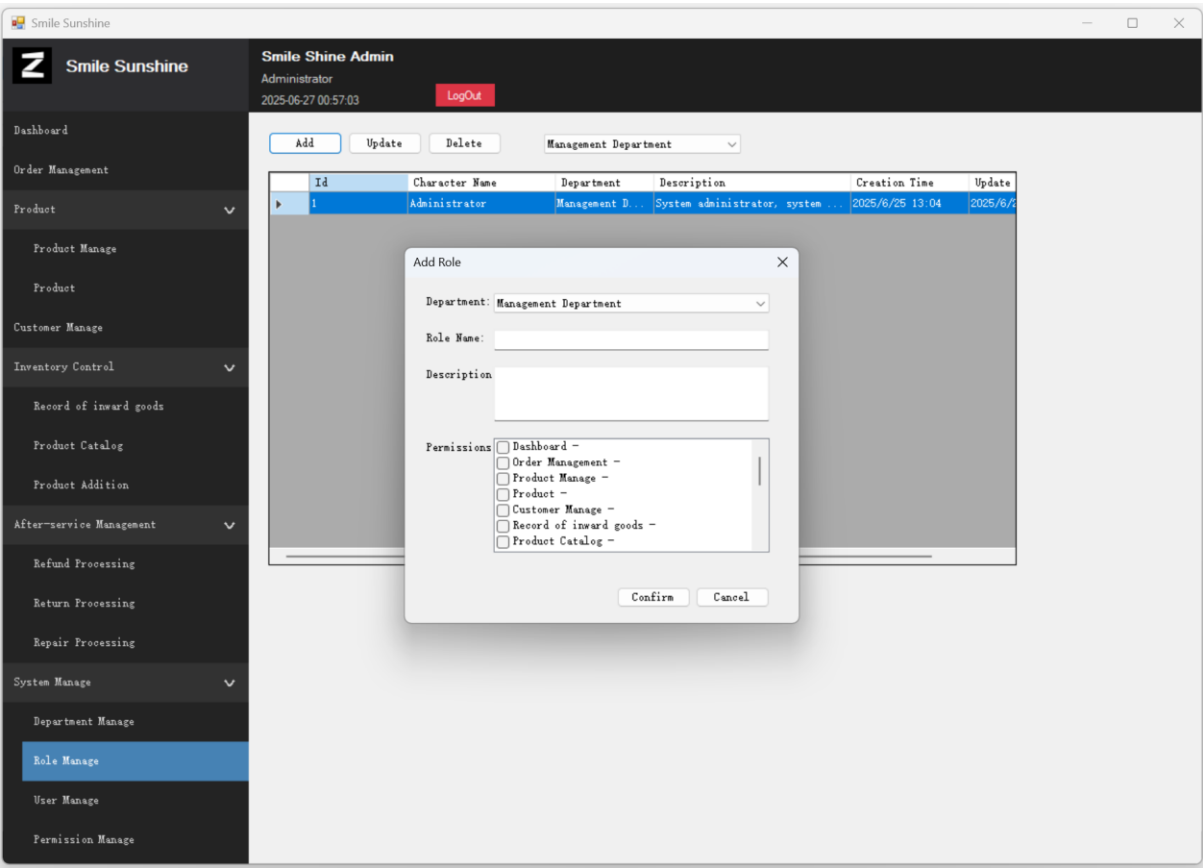
This is a “Prompt” window in the “Department Manage” section of the “Smile Sunshine” system. It shows the message “Data refreshed successfully!” indicating that the data in the department management list has been updated. There's a “Ensure (Confirm)” button to close the prompt.

15.22 Role Manage Page



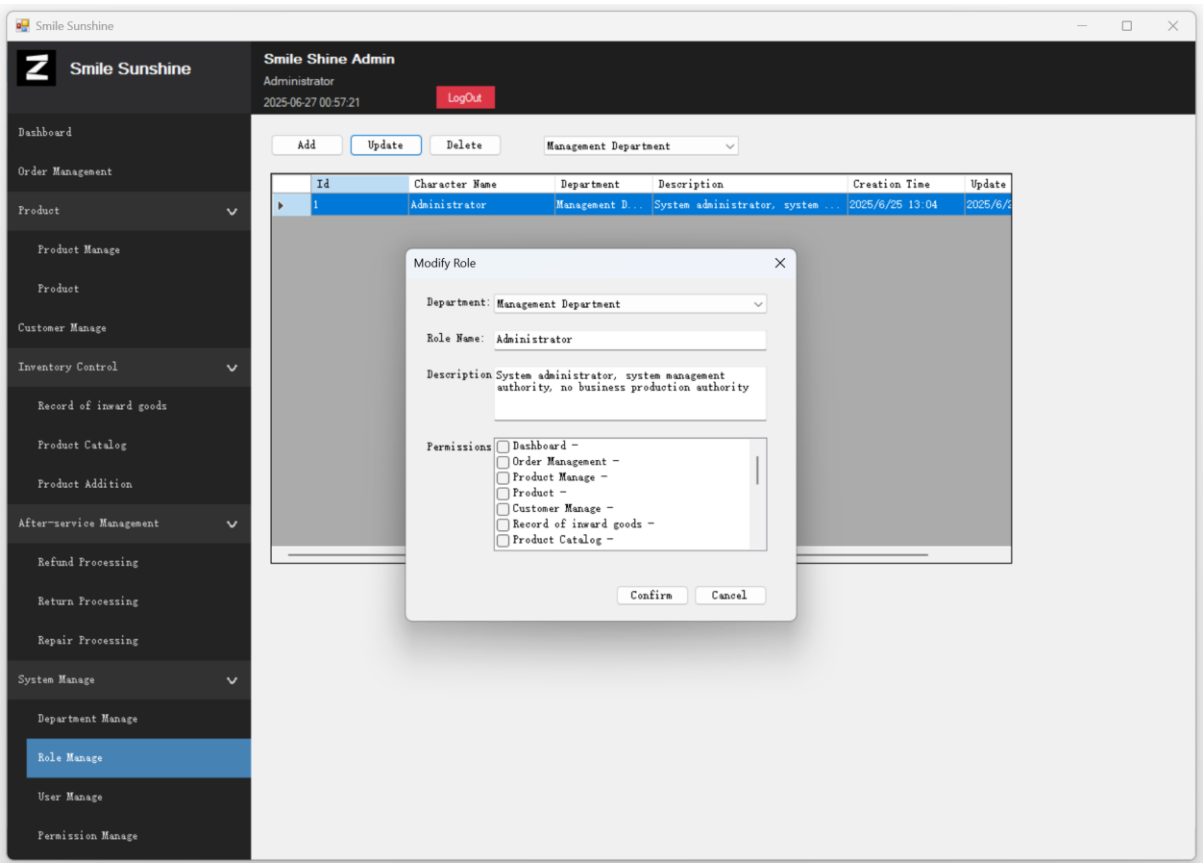
This is the “Role Manage” section of the “Smile Sunshine” system. The left navigation menu shows “Role Manage” under “System Manage” is selected. The main area lists the “Administrator” role with details: Id 1, Character Name “Administrator”, Department “Management D...”, Description “System administrators...”, Creation Time “2025/6/25 13:04”, Update Time “2025/6/26 17:29”. Action buttons include Add, Update, Delete, and a dropdown for selecting departments (currently “Management Department”). Used for administrators to manage system roles.

15.23 Role Add Role Page



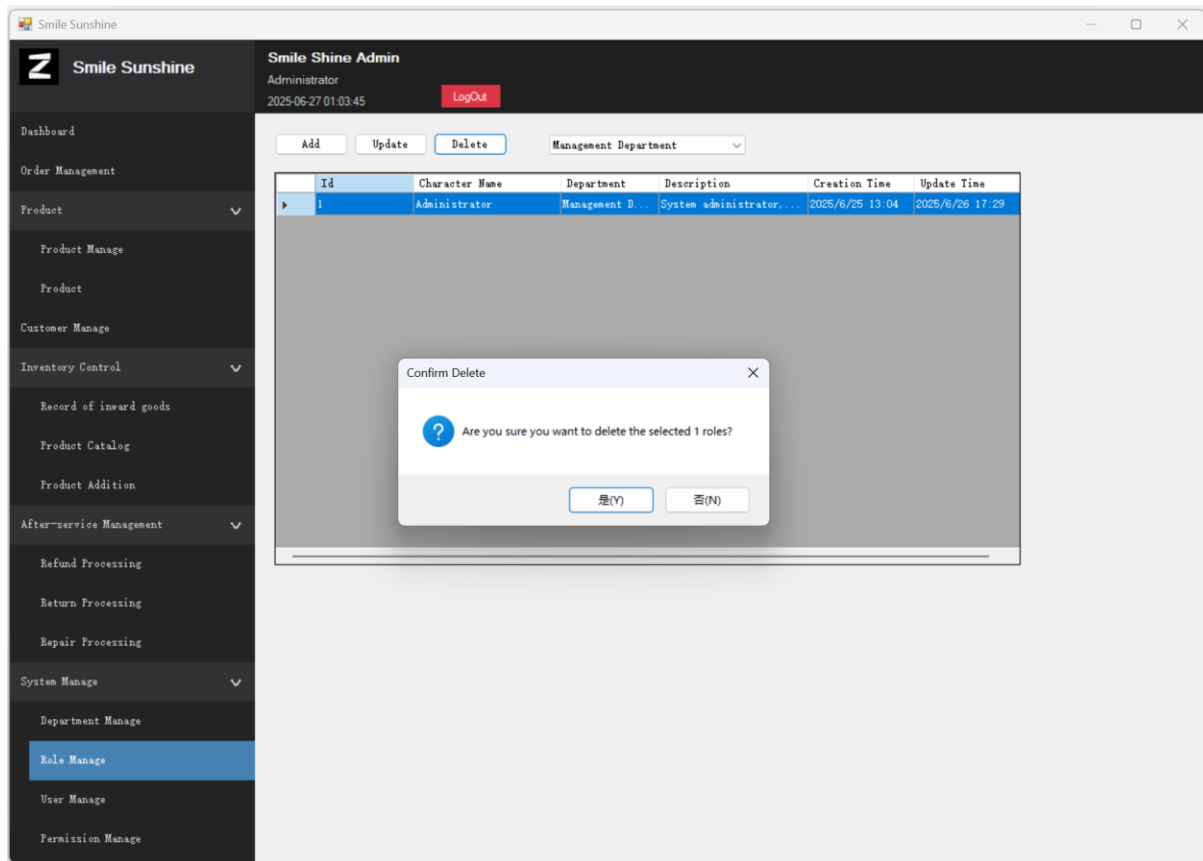
This is the “Add Role” pop - up window in the “Role Manage” section of the “Smile Sunshine” system. It has fields for selecting a “Department” (Management Department), entering a “Role Name” and “Description”, and setting “Permissions” (like Dashboard, Order Management). There are “Confirm” and “Cancel” buttons. Used for administrators to create new system roles with defined permissions.

15.24 Role Modify Role Page



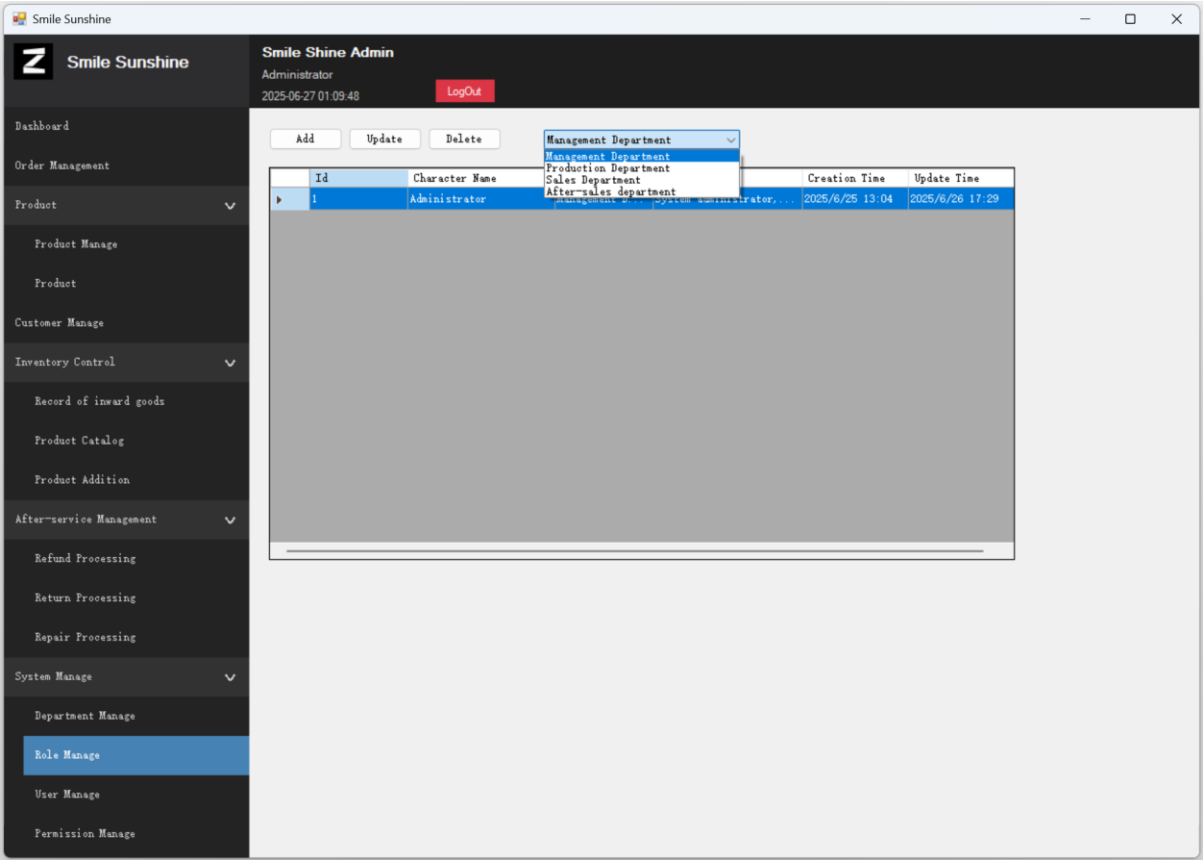
This is the “Modify Role” pop - up window in the “Role Manage” section of the “Smile Sunshine” system. It shows details of the “Administrator” role for modification, including fields for “Department” (Management Department), “Role Name”, “Description”, and “Permissions” (like Dashboard, Order Management). There are “Confirm” and “Cancel” buttons. Used for administrators to update role details and permissions.

15.25 Role Confirm Delete Page



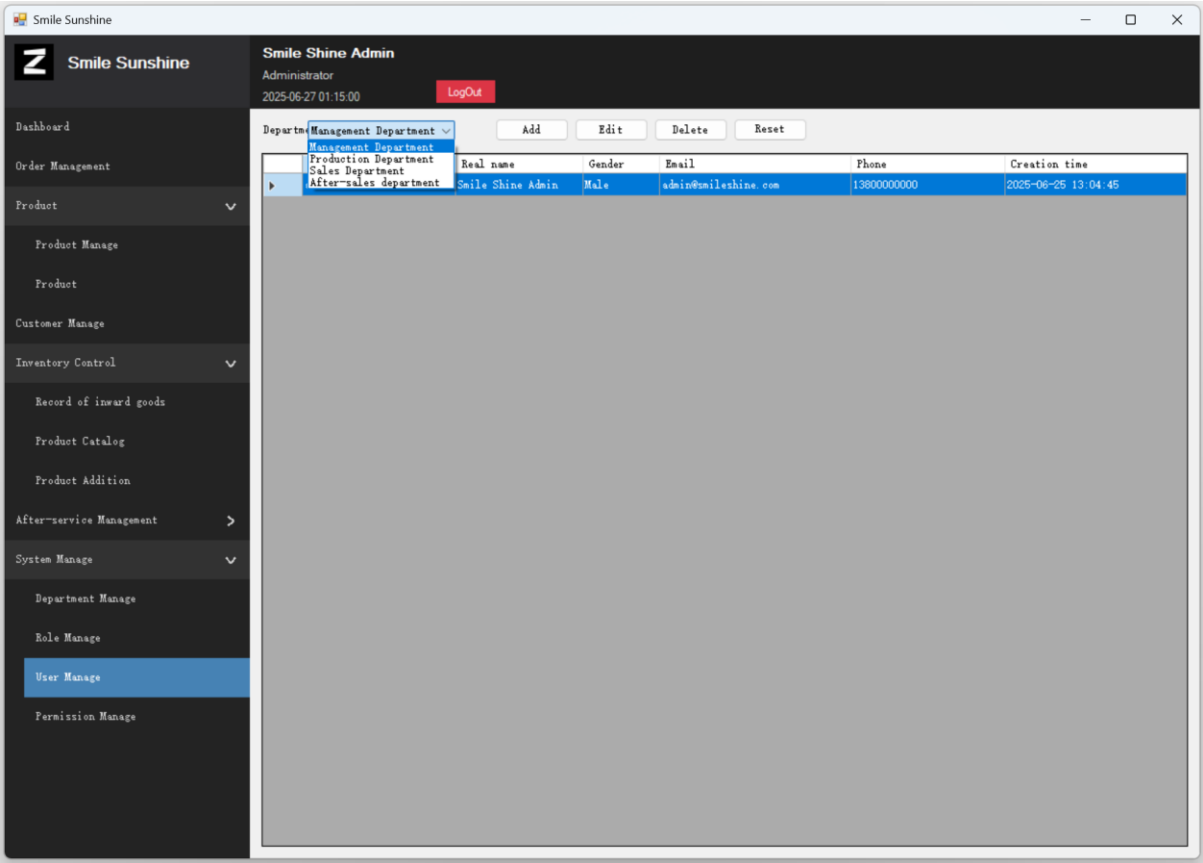
This is a “Confirm Delete” dialog box in the “Role Manage” section of the “Smile Sunshine” system. It asks whether to delete the selected “Administrator” role, with options “YES(Y)” (Yes) to confirm deletion and “NO(N)” (No) to cancel.

15.26 Role Of Department Navigation Menu Page



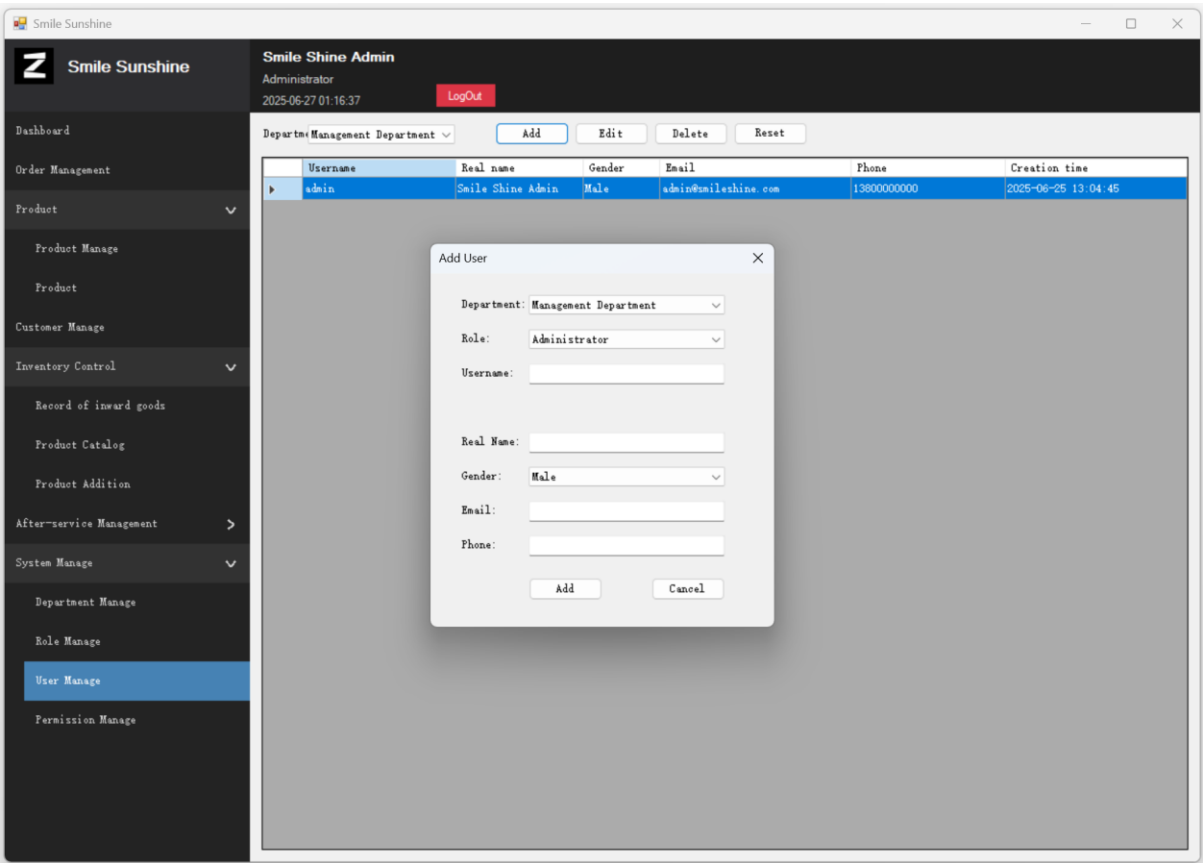
This is the “Role Manage” section of the “Smile Sunshine” system. The left navigation menu shows “Role Manage” under “System Manage” is selected. The main area has a dropdown for selecting departments (showing options like Management Department, Production Department).

15.27 User Manage Page



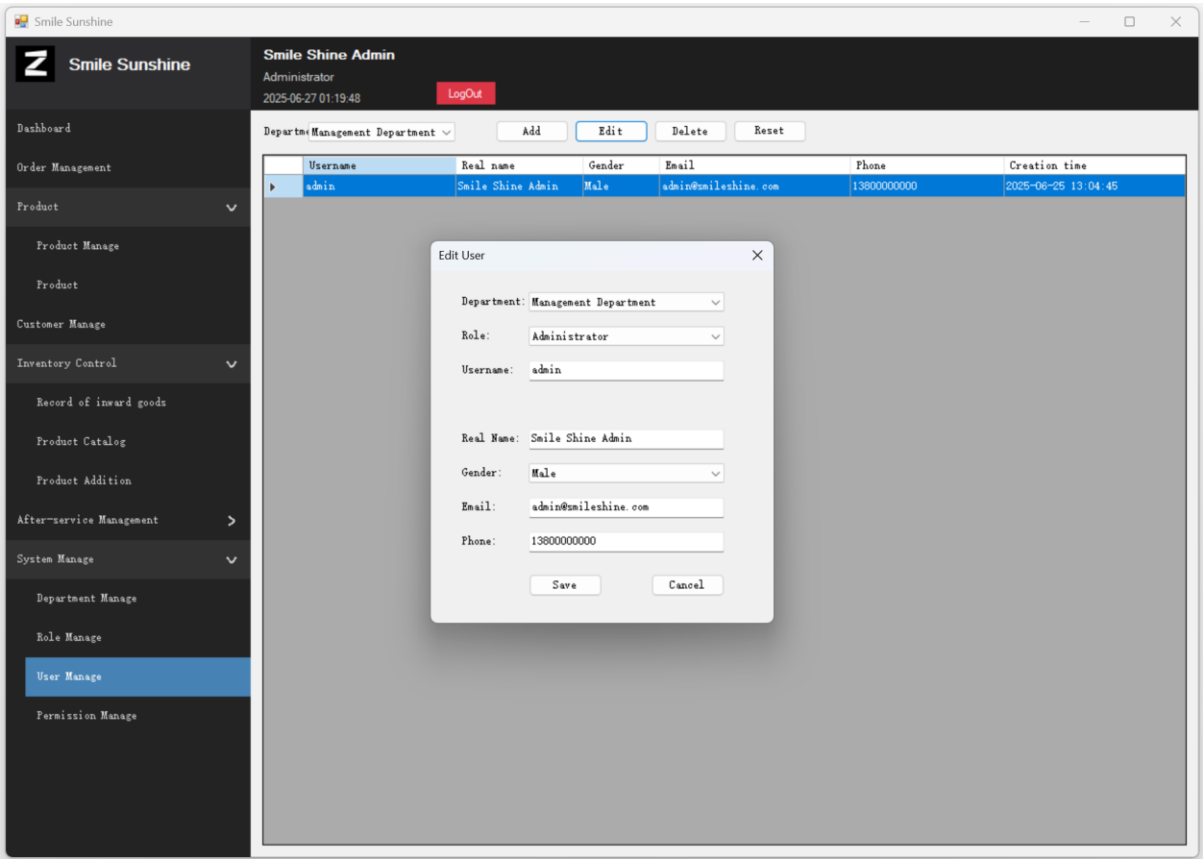
This is the “User Manage” section of the “Smile Sunshine” system. The left navigation menu shows “User Manage” under “System Manage” is selected. The main area has a dropdown for selecting departments (Management Department, Production Department, etc.) and lists a user “Smile Shine Admin” with details: Real name, Gender (Male), Email (admin@smileshine.com), Phone (13800000000), Creation time (2025-06-25 13:04:45). Action buttons include Add, Edit, Delete, Reset. Used for administrators to manage system user accounts.

15.28 Add User Page



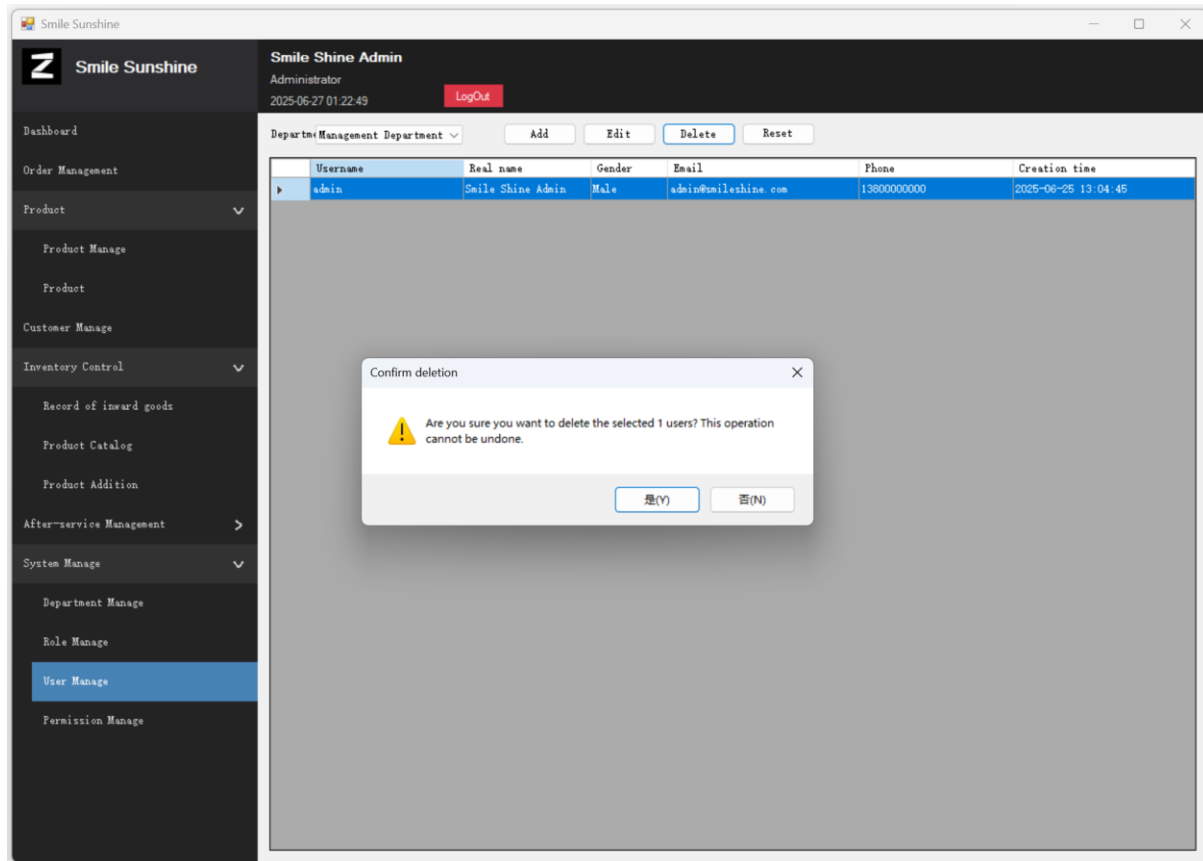
This is the “Add User” pop - up window in the “User Manage” section of the “Smile Sunshine” system. It has fields for selecting “Department” (Management Department), “Role” (Administrator), and entering “Username”, “Real Name”, “Gender” (Male), “Email”, “Phone”. There are “Add” and “Cancel” buttons. Used for administrators to create new system user accounts.

15.29 Edit User Page



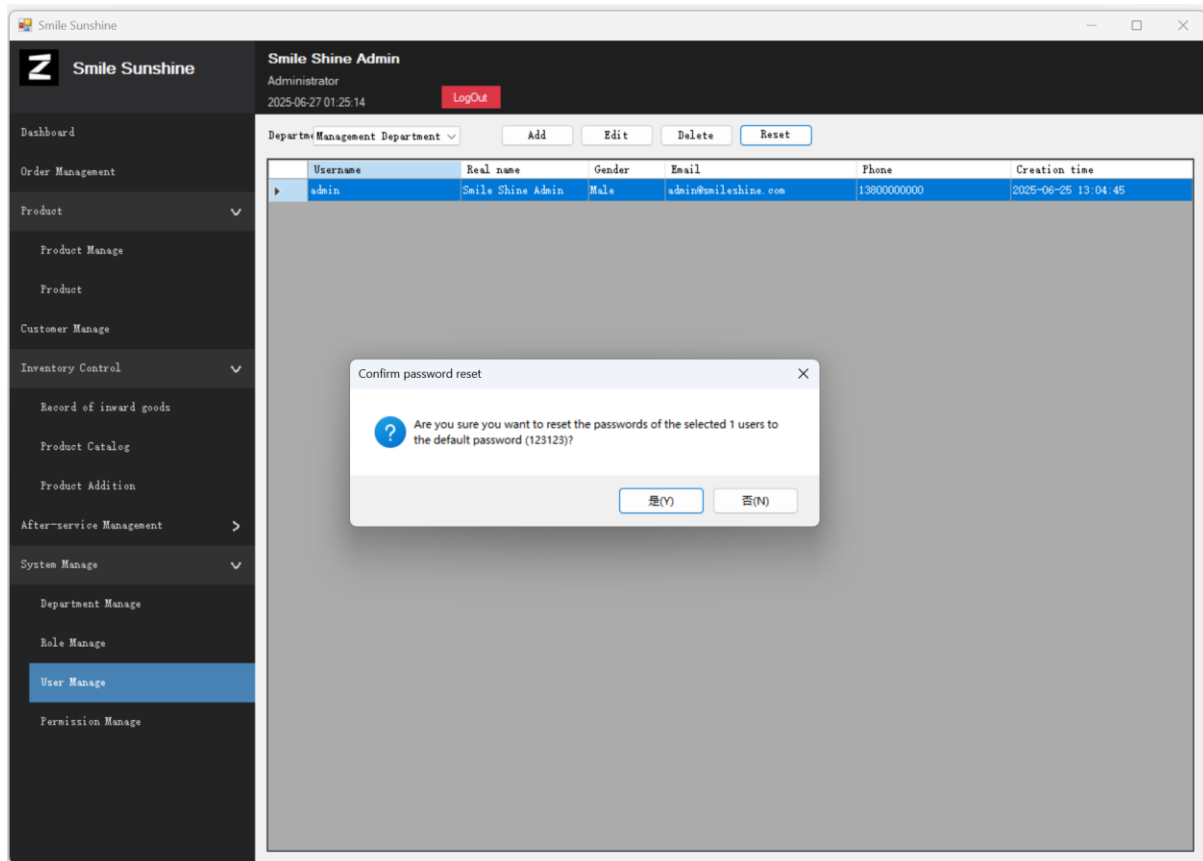
This is the “Edit User” pop - up window in the “User Manage” section of the “Smile Sunshine” system. It shows details of the user “admin” for editing, including fields for “Department” (Management Department), “Role” (Administrator), “Username”, “Real name” (Smile Shine Admin), “Gender” (Male), “Email” (admin@smileshine.com), “Phone” (13800000000). There are “Save” and “Cancel” buttons. Used for administrators to update user account details.

15.30 Confirm deletion Page



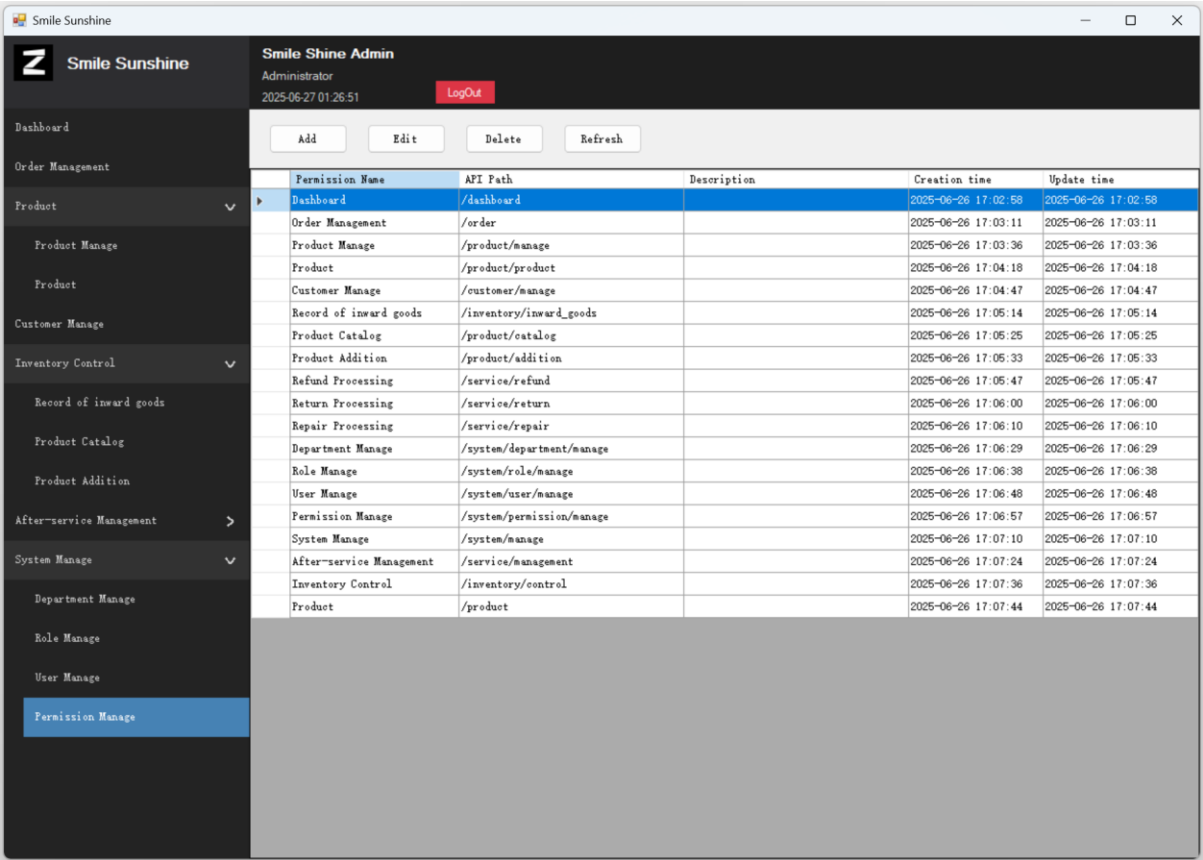
This is a “Confirm deletion” dialog box in the “User Manage” section of the “Smile Sunshine” system. It asks whether to delete the selected user “admin”, warning that the operation cannot be undone. Options are “YES(Y)” (Yes) to confirm deletion and “NO(N)” (No) to cancel.

15.31 Confirm Password Reset Page



This is a “Confirm password reset” dialog box in the “User Manage” section of the “Smile Sunshine” system. It asks whether to reset the password of the selected user “admin” to the default password (123123). Options are “YES(Y)” (Yes) to confirm reset and “NO(N)” (No) to cancel.

15.32 Permission Manage Page



This is the “Permission Manage” section of the “Smile Sunshine” system. The left navigation menu shows “Permission Manage” under “System Manage” is selected. The main area lists permissions with columns: Permission Name (Dashboard, Order Management, etc.), API Path, Description, Creation time, Update time. Action buttons include Add, Edit, Delete, Refresh. Used for administrators to manage system permissions.

16 TeseCase

Function Description	The user logs into the system by entering a username and password.
Case ID	TC _001
Brief Description	Verify that a user can successfully log in using the correct username and password.
Detailed Steps	1. Open the login page. 2. Enter a valid username. 3. Enter a valid password. 4. Click the "Login" button.
Priority	High
Preconditions	The user is registered and has a valid username and password.
Expected Result	The system redirects to the homepage, displaying a successful login message.
Actual Result	success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	User logs into the system by entering username and password
Case ID	TC_002
Brief Description	Verify that system displays error when attempting to login with non-existent username.
Detailed Steps	1. Open login page 2. Enter invalid username 3. Enter any password 4. Click "Login" button
Priority	High
Preconditions	None
Expected Result	System displays "Username does not exist" or similar error message.
Actual Result	success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	User logs into the system by entering username and password
Case ID	TC_003
Brief Description	Verify login fails when username is correct but password is incorrect
Detailed Steps	1. Open login page 2. Enter valid username 3. Enter incorrect password 4. Click "Login" button
Priority	High
Preconditions	User is registered
Expected Result	System displays "Incorrect password" or similar error message
Actual Result	success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	User logs into the system by entering username and password
Case ID	TC_004
Brief Description	Verify login fails when both username and password fields are empty
Detailed Steps	1. Open login page 2. Leave username field blank 3. Leave password field blank 4. Click "Login" button
Priority	Medium
Preconditions	None
Expected Result	System displays "Username and password cannot be empty" error message
Actual Result	success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	User logs into the system by entering username and password
Case ID	TC_005
Brief Description	Verify login fails when username field is empty
Detailed Steps	1. Open login page 2. Leave username field blank 3. Enter valid password 4. Click "Login" button
Priority	Medium
Preconditions	None
Expected Result	System displays "Username cannot be empty" error message
Actual Result	success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	Users can view order list in the order management page
Case ID	TC _006
Brief Description	Verify users can properly view order list after login
Detailed Steps	1. Open system main interface 2. Enter valid username on login page 3. Enter valid password 4. Click "Login" button 5. After login, click "Order Management" in main menu 6. Access order management page 7. Check order list section
Priority	High
Preconditions	User is registered Order data exists in system
Expected Result	Order list displays all order information correctly
Actual Result	success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	Users can edit order information in the order management page
Case ID	TC _007
Brief Description	Verify users can edit selected orders after login
Detailed Steps	1. Open system main interface 2. Enter valid username on login page 3. Enter valid password 4. Click "Login" button 5. After login, click "Order Management" in main menu 6. Access order management page 7. Select an order from the list 8. Click "Edit" button 9. Modify order information in edit dialog 10. Click "Save" button
Priority	High
Preconditions	1. User is registered with edit permissions 2. Order data exists in system
Expected Result	1. Order information updates successfully 2. Updated data displays in order list
Actual Result	success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	Users can export orders as PDF files
Case ID	TC _008
Brief Description	Verify users can export order list as PDF after login
Detailed Steps	1. Open system main interface 2. Enter valid username on login page 3. Enter valid password 4. Click "Login" button 5. After login, click "Order Management" in main menu 6. Access order management page 7. Click "Export PDF" button 8. Select save path and confirm
Priority	Medium
Preconditions	1. User is registered 2. Order data exists in system
Expected Result	System generates and downloads PDF file containing order information
Actual Result	success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	Users can view detailed product information of orders
Case ID	TC _009
Brief Description	Verify users can view order product details after login
Detailed Steps	1. Open system main interface 2. Enter valid username on login page 3. Enter valid password 4. Click "Login" button 5. After login, click "Order Management" in main menu 6. Access order management page 7. Select an order from the list 8. Check product details section on the right
Priority	Medium
Preconditions	1. User is registered 2. Orders contain product details in system
Expected Result	Right panel correctly displays product details of selected order
Actual Result	success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	Order management page should display proper prompt when no orders exist
Case ID	TC_010
Brief Description	Verify the page shows user-friendly prompt when empty after login
Detailed Steps	1. Open system main interface 2. Enter valid username on login page 3. Enter valid password 4. Click "Login" button 5. After login, click "Order Management" in main menu 6. Access order management page 7. Confirm no order data exists in system
Priority	Low
Preconditions	1. User is registered 2. No order data in system
Expected Result	Page displays "No orders available" or similar prompt message
Actual Result	success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	Users can view product list in the product management page
Case ID	TC_011
Brief Description	Verify users can properly view product list after login
Detailed Steps	1. Open system main interface 2. Enter valid username on login page 3. Enter valid password 4. Click "Login" button 5. After login, click "Product Management" in main menu 6. Access product management page 7. Check product list section
Priority	High
Preconditions	1. User is registered 2. Product data exists in system
Expected Result	Product list displays all product information correctly
Actual Result	success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	Users can add new products in the product management page
Case ID	TC_012
Brief Description	Verify users can successfully add new products after login
Detailed Steps	1. Open system main interface 2. Enter valid username on login page 3. Enter valid password 4. Click "Login" button 5. After login, click "Product Management" in main menu 6. Access product management page 7. Click "Add Product" button 8. Enter product information (name, price, inventory etc.) in popup form 9. Click "Save" button
Priority	High
Preconditions	1. User is registered 2. User has product creation permissions
Expected Result	1. Product is successfully created 2. New product appears in product list
Actual Result	success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	Users can edit product information in the product management page
Case ID	TC_013
Brief Description	Verify users can edit selected products after login
Detailed Steps	1. Open system main interface 2. Enter valid username on login page 3. Enter valid password 4. Click "Login" button 5. After login, click "Product Management" in main menu 6. Access product management page 7. Select a product from the list 8. Click "Edit" button 9. Modify product information in edit dialog 10. Click "Save" button
Priority	High
Preconditions	1. User is registered with edit permissions 2. Product data exists in system
Expected Result	1. Product information updates successfully 2. Updated data displays in product list
Actual Result	success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	Users can delete products in the product management page
Case ID	TC_014
Brief Description	Verify users can delete selected products after login
Detailed Steps	1. Open system main interface 2. Enter valid username on login page 3. Enter valid password 4. Click "Login" button 5. After login, click "Product Management" in main menu 6. Access product management page 7. Select a product from the list 8. Click "Delete" button 9. Click "Confirm" in the popup dialog
Priority	Medium
Preconditions	1. User is registered with delete permissions 2. Product data exists in system
Expected Result	1. Product is successfully deleted 2. Product no longer appears in product list
Actual Result	success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	Product management page should display proper prompt when no products exist
Case ID	TC_015
Brief Description	Verify the page shows user-friendly prompt when empty after login
Detailed Steps	1. Open system main interface 2. Enter valid username on login page 3. Enter valid password 4. Click "Login" button 5. After login, click "Product Management" in main menu 6. Access product management page 7. Confirm no product data exists in system
Priority	Low
Preconditions	1. User is registered 2. No product data in system
Expected Result	Page displays "No products available" or similar prompt message
Actual Result	success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator adds a new permission to the system.
Case ID	TC_016
Brief Description	Verify that an administrator can successfully add a new permission.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "System Manage" section. 3. Click on "Permission Manage". 4. Click the "Add" button. 5. Enter a valid Permission Name. 6. Enter a valid API Path. 7. Provide a Description for the permission. 8. Click "Save".
Priority	High
Preconditions	The user is logged in as an administrator with the necessary privileges to add permissions.
Expected Result	The new permission should be added successfully and appear in the permission list with the correct details.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator edits an existing permission in the system.
Case ID	TC_017
Brief Description	Verify that an administrator can successfully edit an existing permission.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "System Manage" section. 3. Click on "Permission Manage". 4. Select existing permission from the list. 5. Click the "Edit" button. 6. Modify the Permission Name or API Path. 7. Update the Description if necessary. 8. Click "Save".
Priority	High
Preconditions	The user is logged in as an administrator with the necessary privileges to edit permissions, and there is at least one permission to edit.
Expected Result	The selected permission details should be updated successfully and reflect the changes in the permission list.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator deletes a permission from the system.
Case ID	TC_018
Brief Description	Verify that an administrator can successfully delete a permission.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "System Manage" section. 3. Click on "Permission Manage". 4. Select a permission from the list. 5. Click the "Delete" button. 6. Confirm the deletion.
Priority	High
Preconditions	The user is logged in as an administrator with the necessary privileges to delete permissions, and there is at least one permission to delete.
Expected Result	The selected permission should be deleted successfully and no longer appear in the permission list.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator refreshes the permission list.
Case ID	TC_019
Brief Description	Verify that the permission list can be refreshed successfully.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "System Manage" section. 3. Click on "Permission Manage". 4. Click the "Refresh" button.
Priority	Medium
Preconditions	The user is logged in as an administrator.
Expected Result	The permission list should be refreshed and display the most current data.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The system displays the permission list correctly with all necessary columns and data.
Case ID	TC_020
Brief Description	Verify that the permission list is displayed correctly with all columns and data.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "System Manage" section. 3. Click on "Permission Manage". 4. Verify that the permission list is displayed with the following columns: Permission Name, API Path, Description, Creation time, Update time. 5. Check that each column contains the appropriate data for each permission.
Priority	Medium
Preconditions	The user is logged in as an administrator.
Expected Result	The permission list should be displayed correctly with all columns and appropriate data for each permission.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator attempts to edit a permission that does not exist.
Case ID	TC_021
Brief Description	Verify that the system handles the attempt to edit a non-existent permission correctly.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "System Manage" section. 3. Click on "Permission Manage". 4. Attempt to select a non-existent permission from the list. 5. Click the "Edit" button.
Priority	Medium
Preconditions	The user is logged in as an administrator and there are no permissions to edit.
Expected Result	The system should display an error message indicating that the selected permission does not exist or is not found.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator attempts to delete a permission that does not exist.
Case ID	TC_022
Brief Description	Verify that the system handles the attempt to delete a non-existent permission correctly.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "System Manage" section. 3. Click on "Permission Manage". 4. Attempt to select a non-existent permission from the list. 5. Click the "Delete" button. 6. Confirm the deletion.
Priority	Medium
Preconditions	The user is logged in as an administrator and there are no permissions to delete.
Expected Result	The system should display an error message indicating that the selected permission does not exist or is not found.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The user attempts to edit a permission without sufficient privileges.
Case ID	TC_023
Brief Description	Verify that the system prevents users without sufficient privileges from editing permissions.
Detailed Steps	1. Log in to the system as a user with insufficient privileges. 2. Navigate to the "System Manage" section. 3. Click on "Permission Manage". 4. Attempt to select a permission from the list. 5. Click the "Edit" button.
Priority	High
Preconditions	The user is logged in with insufficient privileges to edit permissions.
Expected Result	The system should display an error message indicating that the user does not have sufficient privileges to edit permissions.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The user attempts to delete a permission without sufficient privileges.
Case ID	TC_024
Brief Description	Verify that the system prevents users without sufficient privileges from deleting permissions.
Detailed Steps	1. Log in to the system as a user with insufficient privileges. 2. Navigate to the "System Manage" section. 3. Click on "Permission Manage". 4. Attempt to select a permission from the list. 5. Click the "Delete" button. 6. Confirm the deletion.
Priority	High
Preconditions	The user is logged in with insufficient privileges to delete permissions.
Expected Result	The system should display an error message indicating that the user does not have sufficient privileges to delete permissions.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator confirms resetting a user's password to the default password.
Case ID	TC_025
Brief Description	Verify that the system correctly resets a user's password to the default password when confirmed by the administrator.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "User Manage" section. 3. Select the user "admin" from the user list. 4. Click the "Reset Password" button. 5. In the "Confirm password reset" dialog box, click "YES(Y)" (Yes) to confirm the password reset.
Priority	High
Preconditions	The user is logged in as an administrator and the user "admin" exists in the system.
Expected Result	The password of the selected user "admin" should be reset to the default password (123123) successfully.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator cancels the reset of a user's password.
Case ID	TC_026
Brief Description	Verify that the system correctly cancels the reset of a user's password when the administrator chooses not to confirm.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "User Manage" section. 3. Select the user "admin" from the user list. 4. Click the "Reset Password" button. 5. In the "Confirm password reset" dialog box, click "NO(N)" (No) to cancel the password reset.
Priority	Medium
Preconditions	The user is logged in as an administrator and the user "admin" exists in the system.
Expected Result	The password of the selected user "admin" should not be changed, and the system should display a message or confirmation that the reset was cancelled.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function	Users can view customer list in the customer management page
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Description	
Case ID	TC_027
Brief Description	Verify users can properly view customer list after login
Detailed Steps	1. Open system main interface 2. Enter valid username on login page 3. Enter valid password 4. Click "Login" button 5. After login, click "Customer Management" in main menu 6. Access customer management page 7. Check customer list section
Priority	High
Preconditions	1. User is registered 2. Customer data exists in system
Expected Result	Customer list displays all customer information correctly
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function	Users can add new customers in the customer management
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Description	page
Case ID	TC_028
Brief Description	Verify users can successfully add new customers after login
Detailed Steps	1. Open system main interface 2. Enter valid username on login page 3. Enter valid password 4. Click "Login" button 5. After login, click "Customer Management" in main menu 6. Access customer management page 7. Click "Add Customer" button 8. Enter customer information (name, contact details etc.) in popup form 9. Click "Save" button
Priority	High
Preconditions	1. User is registered 2. User has customer creation permissions
Expected Result	1. Customer is successfully created 2. New customer appears in customer list
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function	Users can edit customer information in the customer
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Description	management page
Case ID	TC_029
Brief Description	Verify users can edit selected customers after login
Detailed Steps	1. Open system main interface 2. Enter valid username on login page 3. Enter valid password 4. Click "Login" button 5. After login, click "Customer Management" in main menu 6. Access customer management page 7. Select a customer from the list 8. Click "Edit" button 9. Modify customer information in edit dialog 10. Click "Save" button
Priority	High
Preconditions	1. User is registered with edit permissions 2. Customer data exists in system
Expected Result	1. Customer information updates successfully 2. Updated data displays in customer list
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function	Users can delete customers in the customer management page
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Description	
Case ID	TC_030
Brief Description	Verify users can delete selected customers after login
Detailed Steps	1. Open system main interface 2. Enter valid username on login page 3. Enter valid password 4. Click "Login" button 5. After login, click "Customer Management" in main menu 6. Access customer management page 7. Select a customer from the list 8. Click "Delete" button 9. Click "Confirm" in the popup dialog
Priority	Medium
Preconditions	1. User is registered with delete permissions 2. Customer data exists in system
Expected Result	1. Customer is successfully deleted 2. Customer no longer appears in customer list
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function	
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Description	Customer management page should display proper prompt when no customers exist
Case ID	TC_031
Brief Description	Verify the page shows user-friendly prompt when empty after login
Detailed Steps	1. Open system main interface 2. Enter valid username on login page 3. Enter valid password 4. Click "Login" button 5. After login, click "Customer Management" in main menu 6. Access customer management page 7. Confirm no customer data exists in system
Priority	Low
Preconditions	1. User is registered 2. No customer data in system
Expected Result	Page displays "No customers available" or similar prompt message
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function	Users can view department list in the department management
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Description	page
Case ID	TC_032
Brief Description	Verify users can properly view department list after login
Detailed Steps	1. Open system main interface 2. Enter valid username on login page 3. Enter valid password 4. Click "Login" button 5. After login, click "Department Management" in main menu 6. Access department management page 7. Check department list section
Priority	High
Preconditions	1. User is registered 2. Department data exists in system
Expected Result	Department list displays all department information correctly
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	Users can add new departments in the department management page
Case ID	TC_033
Brief Description	Verify users can successfully add new departments after login
Detailed Steps	1. Open system main interface 2. Enter valid username on login page 3. Enter valid password 4. Click "Login" button 5. After login, click "Department Management" in main menu 6. Access department management page 7. Click "Add Department" button 8. Enter department information (name, manager etc.) in popup form 9. Click "Save" button
Priority	High
Preconditions	1. User is registered 2. User has department creation permissions
Expected Result	1. Department is successfully created 2. New department appears in department list
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	Users can edit department information in the department management page
Case ID	TC_034
Brief Description	Verify users can edit selected departments after login
Detailed Steps	1. Open system main interface 2. Enter valid username on login page 3. Enter valid password 4. Click "Login" button 5. After login, click "Department Management" in main menu 6. Access department management page 7. Select a department from the list 8. Click "Edit" button 9. Modify department information in edit dialog 10. Click "Save" button
Priority	High
Preconditions	1. User is registered with edit permissions 2. Department data exists in system
Expected Result	1. Department information updates successfully 2. Updated data displays in department list
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	Users can delete departments in the department management page
Case ID	TC_035
Brief Description	Verify users can delete selected departments after login
Detailed Steps	1. Open system main interface 2. Enter valid username on login page 3. Enter valid password 4. Click "Login" button 5. After login, click "Department Management" in main menu 6. Access department management page 7. Select a department from the list 8. Click "Delete" button 9. Click "Confirm" in the popup dialog
Priority	Medium
Preconditions	1. User is registered with delete permissions 2. Department data exists in system
Expected Result	1. Department is successfully deleted 2. Department no longer appears in department list
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	Department management page should display proper prompt when no departments exist
Case ID	TC_036
Brief Description	Verify the page shows user-friendly prompt when empty after login
Detailed Steps	1. Open system main interface 2. Enter valid username on login page 3. Enter valid password 4. Click "Login" button 5. After login, click "Department Management" in main menu 6. Access department management page 7. Confirm no department data exists in system
Priority	Low
Preconditions	1. User is registered 2. No department data in system
Expected Result	Page displays "No departments available" or similar prompt message
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator cancels the modification of a role's details.
Case ID	TC_037
Brief Description	Verify that the system correctly cancels the modification of a role's details.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "Role Manage" section. 3. Select the "Administrator" role from the role list. 4. Click the "Update" button. 5. Modify the "Role Name" and "Description" fields. 6. Change some permissions. 7. Click "Cancel".
Priority	Medium
Preconditions	The changes made should not be saved, and the role details should remain unchanged.
Expected Result	Page displays "No departments available" or similar prompt message
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator attempts to modify a role with a duplicate role name.
Case ID	TC_038
Brief Description	Verify that the system correctly handles an attempt to modify a role with a duplicate role name.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "Role Manage" section. 3. Select the "Administrator" role from the role list. 4. Click the "Update" button. 5. Change the "Role Name" to a name that already exists in the system. 6. Click "Confirm".
Priority	High
Preconditions	The user is logged in as an administrator and there is another role with the same name.
Expected Result	The system should display an error message indicating that the role name is already taken and should not save the changes.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator confirms the deletion of a selected user.
Case ID	TC_039
Brief Description	Verify that the system correctly deletes a selected user when confirmed by the administrator.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "User Manage" section. 3. Select the user "admin" from the user list. 4. Click the "Delete" button. 5. In the "Confirm deletion" dialog box, click "YES(Y)" (Yes) to confirm the deletion.
Priority	High
Preconditions	The user is logged in as an administrator and the user "admin" exists in the system.
Expected Result	The selected user "admin" should be deleted successfully, and the system should display a confirmation message that the user has been deleted.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function	The administrator cancels the deletion of a selected user.
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Description	
Case ID	TC_040
Brief Description	Verify that the system correctly cancels the deletion of a selected user when the administrator chooses not to confirm.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "User Manage" section. 3. Select the user "admin" from the user list. 4. Click the "Delete" button. 5. In the "Confirm deletion" dialog box, click "NO(N)" (No) to cancel the deletion.
Priority	Medium
Preconditions	The user is logged in as an administrator and the user "admin" exists in the system.
Expected Result	The selected user "admin" should not be deleted, and the system should display a message or confirmation that the deletion was cancelled.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator edits the details of a user.
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Case ID	TC 041
Brief Description	Verify that the system correctly updates user details when edited by an administrator.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "User Manage" section. 3. Select the user "admin" from the user list. 4. Click the "Edit" button. 5. Modify the "Real name" field to a new name. 6. Click "Save".
Priority	High
Preconditions	The user is logged in as an administrator and the user "admin" exists in the system.
Expected Result	The details of the selected user "admin" should be updated successfully, and the system should display the updated information.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator cancels the editing of a user's details.
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Case ID	TC 042
Brief Description	Verify that the system correctly cancels the editing of a user's details when the administrator chooses not to save.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "User Manage" section. 3. Select the user "admin" from the user list. 4. Click the "Edit" button. 5. Modify the "Email" field to a new email address. 6. Click "Cancel".
Priority	Medium
Preconditions	The user is logged in as an administrator and the user "admin" exists in the system.
Expected Result	The changes made should not be saved, and the user details should remain unchanged.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator attempts to edit a user with an invalid email format.
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Case ID	TC 043
Brief Description	Verify that the system correctly handles an attempt to save a user with an invalid email format.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "User Manage" section. 3. Select the user "admin" from the user list. 4. Click the "Edit" button. 5. Modify the "Email" field to an invalid email format (e.g., "invalid-email"). 6. Click "Save".
Priority	High
Preconditions	The user is logged in as an administrator and the user "admin" exists in the system.
Expected Result	The system should display an error message indicating that the email format is invalid and should not save the changes.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator attempts to edit a user with an invalid email format.
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Case ID	TC 044
Brief Description	Verify that the system correctly handles an attempt to save a user with an invalid email format.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "User Manage" section. 3. Select the user "admin" from the user list. 4. Click the "Edit" button. 5. Modify the "Email" field to an invalid email format (e.g., "invalid-email"). 6. Click "Save".
Priority	High
Preconditions	The user is logged in as an administrator and the user "admin" exists in the system.
Expected Result	The system should display an error message indicating that the email format is invalid and should not save the changes.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator adds a new user to the system.
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Case ID	TC 045
Brief Description	Verify that the system correctly adds a new user with all required details.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "User Manage" section. 3. Click the "Add" button. 4. Select "Management Department" from the "Department" dropdown. 5. Select "Administrator" from the "Role" dropdown. 6. Enter a valid "Username". 7. Enter a valid "Real Name". 8. Select "Male" from the "Gender" dropdown. 9. Enter a valid "Email" address. 10. Enter a valid "Phone" number. 11. Click "Add".
Priority	High
Preconditions	The user is logged in as an administrator.
Expected Result	The new user should be added successfully, and the system should display the new user in the user list.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator cancels adding a new user.
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Case ID	TC 046
Brief Description	Verify that the system correctly cancels the addition of a new user.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "User Manage" section. 3. Click the "Add" button. 4. Fill in all the required fields with valid data. 5. Click "Cancel".
Priority	Medium
Preconditions	The user is logged in as an administrator.
Expected Result	The new user should not be added, and the system should not display the new user in the user list.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator attempts to add a user with a duplicate username.
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Case ID	TC 047
Brief Description	Verify that the system correctly handles an attempt to add a user with a duplicate username.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "User Manage" section. 3. Click the "Add" button. 4. Fill in all the required fields with valid data, including a username that already exists in the system. 5. Click "Add".
Priority	High
Preconditions	The user is logged in as an administrator and there is an existing user with the same username.
Expected Result	The system should display an error message indicating that the username is already taken and should not add the new user.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator attempts to add a user with an invalid email format.
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Case ID	TC 048
Brief Description	Verify that the system correctly handles an attempt to add a user with an invalid email format.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "User Manage" section. 3. Click the "Add" button. 4. Fill in all the required fields with valid data, except for the "Email" field, which should be in an invalid format (e.g., "invalid-email"). 5. Click "Add".
Priority	High
Preconditions	The user is logged in as an administrator.
Expected Result	The system should display an error message indicating that the email format is invalid and should not add the new user.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator confirms the deletion of a selected role.
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Case ID	TC 049
Brief Description	Verify that the system correctly deletes a selected role when confirmed by the administrator.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "Role Manage" section. 3. Select the "Administrator" role from the role list. 4. Click the "Delete" button. 5. In the "Confirm Delete" dialog box, click "YES(Y)" (Yes) to confirm the deletion.
Priority	High
Preconditions	The user is logged in as an administrator and the "Administrator" role exists in the system.
Expected Result	The selected "Administrator" role should be deleted successfully, and the system should display a confirmation message that the role has been deleted.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator cancels the deletion of a selected role.
Case ID	TC 050

Brief Description	Verify that the system correctly cancels the deletion of a selected role when the administrator chooses not to confirm.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "Role Manage" section. 3. Select the "Administrator" role from the role list. 4. Click the "Delete" button. 5. In the "Confirm Delete" dialog box, click "NO(N)" (No) to cancel the deletion.
Priority	Medium
Preconditions	The user is logged in as an administrator and the "Administrator" role exists in the system.
Expected Result	The selected "Administrator" role should not be deleted, and the system should display a message or confirmation that the deletion was cancelled.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator attempts to delete a role that does not exist.
Case ID	TC_051
Brief Description	Verify that the system handles the attempt to delete a non-

	existent role correctly.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "Role Manage" section. 3. Attempt to select a non-existent role from the role list. 4. Click the "Delete" button.
Priority	Medium
Preconditions	The user is logged in as an administrator and there are no roles to delete.
Expected Result	The system should display an error message indicating that the selected role does not exist or is not found.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator modifies the details and permissions of a role.
Case ID	TC_052
Brief Description	Verify that the system correctly updates the details and

	permissions of a role.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "Role Manage" section. 3. Select the "Administrator" role from the role list. 4. Click the "Update" button. 5. Modify the "Role Name" to a new name. 6. Update the "Description" field. 7. Change the permissions by adding or removing them. 8. Click "Confirm".
Priority	High
Preconditions	The user is logged in as an administrator and the "Administrator" role exists in the system.
Expected Result	The details and permissions of the selected role should be updated successfully, and the system should display the updated information.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator adds a new role to the system.
Case ID	TC_053
Brief Description	Verify that the system correctly adds a new role with all

	required details.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "Role Manage" section. 3. Click the "Add" button. 4. Select "Management Department" from the "Department" dropdown. 5. Enter a valid "Role Name". 6. Enter a valid "Description". 7. Select appropriate "Permissions" (e.g., Dashboard, Order Management). 8. Click "Confirm".
Priority	High
Preconditions	The user is logged in as an administrator.
Expected Result	The new role should be added successfully, and the system should display the new role in the role list.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator cancels adding a new role.
Case ID	TC_054
Brief Description	Verify that the system correctly cancels the addition of a new

	role.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "Role Manage" section. 3. Click the "Add" button. 4. Fill in all the required fields with valid data. 5. Click "Cancel".
Priority	Medium
Preconditions	The user is logged in as an administrator.
Expected Result	The new role should not be added, and the system should not display the new role in the role list.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

Function Description	The administrator attempts to add a role with a duplicate role name.
Case ID	TC_055
Brief Description	Verify that the system correctly handles an attempt to add a

	role with a duplicate role name.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "Role Manage" section. 3. Click the "Add" button. 4. Fill in all the required fields with valid data, including a role name that already exists in the system. 5. Click "Confirm".
Priority	High
Preconditions	The user is logged in as an administrator and there is an existing role with the same role name.
Expected Result	The system should display an error message indicating that the role name is already taken and should not add the new role.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27
Function Description	The administrator attempts to add a role without selecting any permissions.
Case ID	TC 001
Brief Description	Verify that the system correctly handles an attempt to add a role without selecting any permissions.
Detailed Steps	1. Log in to the system as an administrator. 2. Navigate to the "Role Manage" section. 3. Click the "Add" button. 4. Fill in all the required fields with valid data, but do not select any permissions. 5. Click "Confirm".
Priority	Medium
Preconditions	The user is logged in as an administrator.
Expected Result	The system should display an error message indicating that at least one permission must be selected and should not add the new role.
Actual Result	Success
Planned Test Date	2025-06-27
Actual Date	2025-06-27

17 User Guide

17.1 How to Log In to Your Account

The screenshot shows a web browser window titled "Login" for the "Smile Sunshine" application. The interface includes a blue header with the text "Smile Sunshine". Below the header, there are two text input fields: "Username:" and "Password:". A checkbox labeled "Remember me" is positioned below the password field. To the right of the "Remember me" checkbox is a blue hyperlink labeled "Forgot Password?". At the bottom of the form is a large blue button labeled "Login in". Five red arrows with numbers 1 through 5 point to the following elements: 1 points to the Username text field, 2 points to the Password text field, 3 points to the Remember me checkbox, 4 points to the Forgot Password? link, and 5 points to the Login in button.

Step 1:Username(Text Field):Enter your registered account(e.g.,email or employee ID).

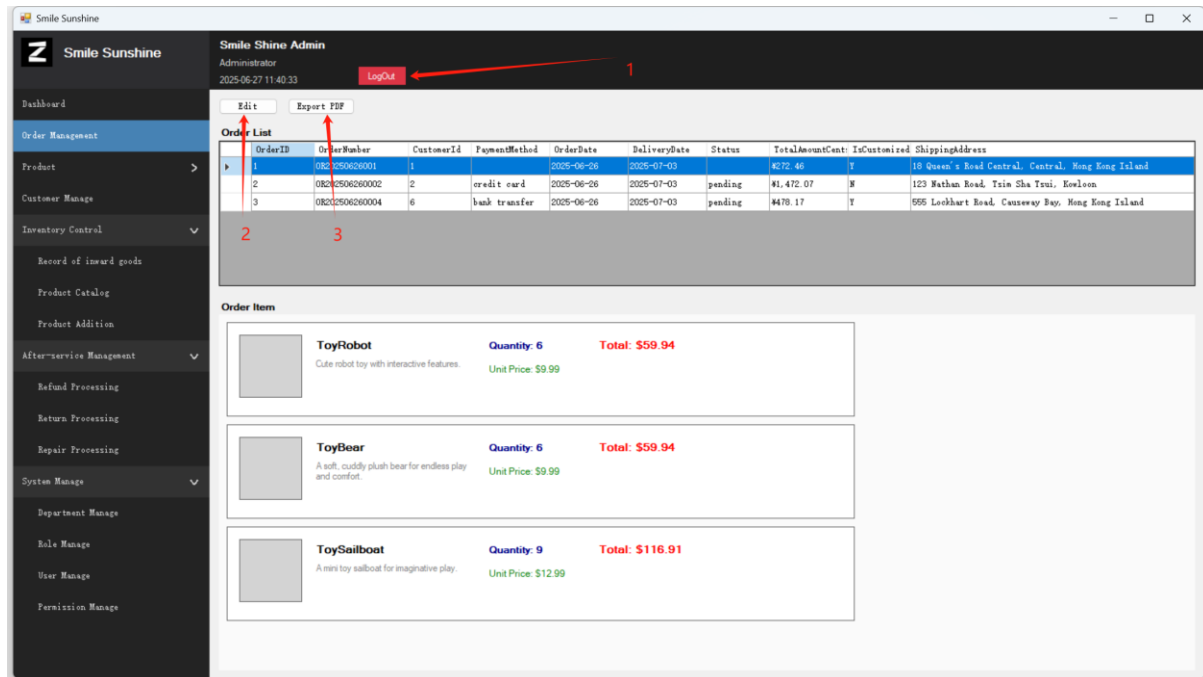
Step 2:Password(Text Field):Type your password(hidden as dots•by default).

Step 3:Remember Me(Checkbox):Check to save your username for future logins(avoid on public devices).

Step 4:Forgot Password?(Link):Click to reset your password.

Step 5:Login Button(Blue Button):Click to submit your credentials.

17.2 Admin System Actions: Logging Out, Editing Orders, and Exporting PDFs

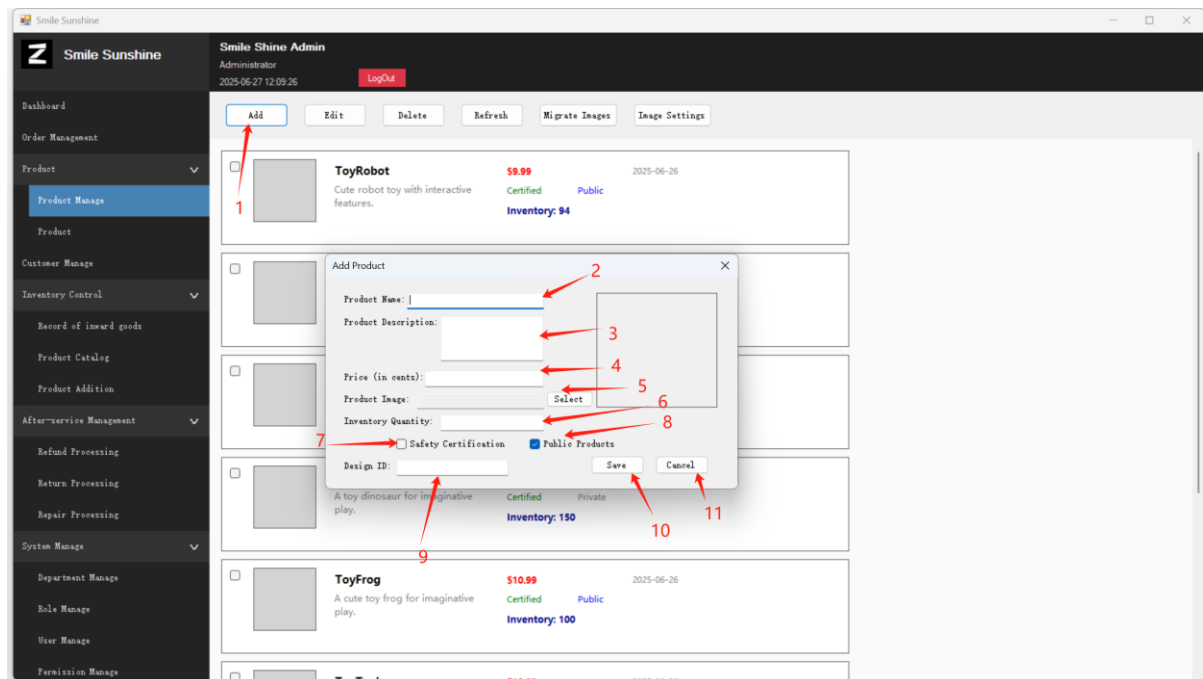


Step 1:LogOut(Red Button,Top-Right):Click to securely sign out of the admin system.

Step 2:Edit(Order List):Click to modify selected order details(e.g.,customer info,items).

Step 3:Export PDF(Order List):Click to generate/download a PDF invoice for the selected order.

17.3 How to Add a New Product to the System



Step 1:Add(Blue Button, Main Interface):Click to open the “Add Product” popup window.

Step 2:Product Name(Required Field):Enter product name(e.g., “ToyRobot”).

Step 3:Product Description: Enter product description(e.g., “Cute robot toy with interactive features”).

Step 4:Price(in cents):Enter product price(in cents,e.g.,“510”equals5.10).

Step 5:Product Image-Select(Button):Click to select and upload product photo.

Step 6:Inventory Quantity: Enter stock quantity(numeric input only).

Step 7:Safety Certification(Checkbox):Check to indicate if the product has safety certification.

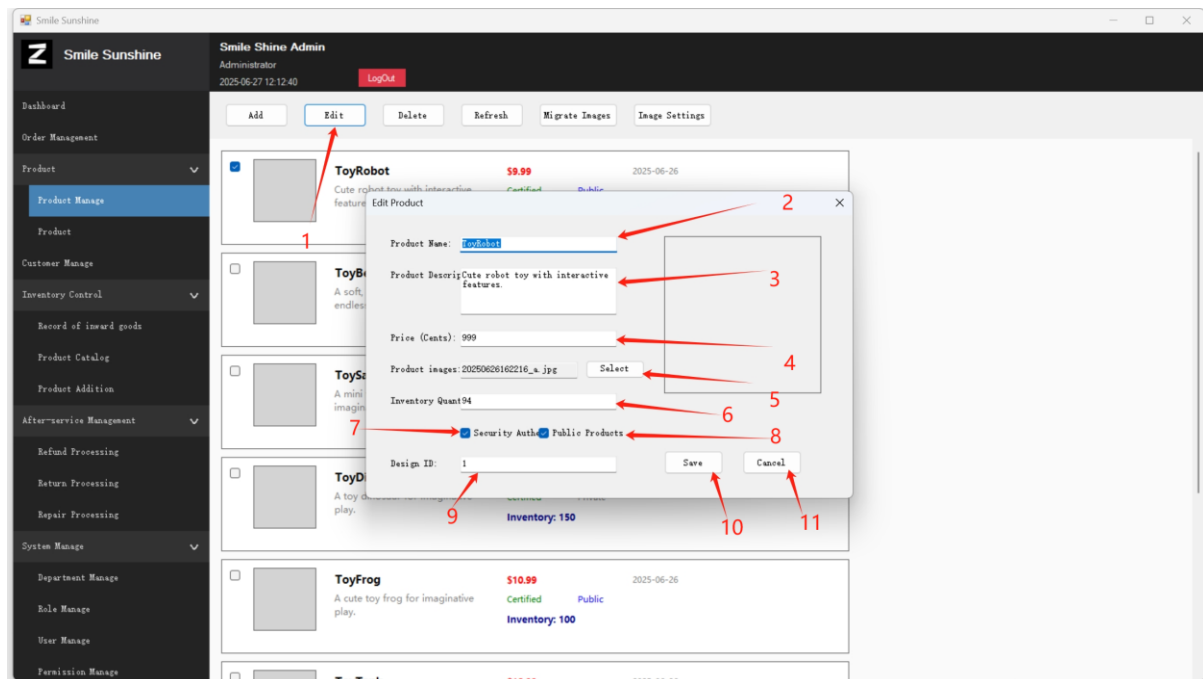
Step 8:Public Products(Radio Buttons):Set product visibility to Public or Private.

Step 9:Design ID(Optional Field):Enter product design ID.

Step 10:Save(Blue Button):Click to confirm and save new product information.

Step 11:Cancel(Gray Button):Click to discard changes and close the window.

17.4 How to Edit Product Information in the System



Step 1:Edit(Blue Button, Main Interface):Click to open the “Edit Product” popup.

Step 2:Product Name(Editable Field):Displays and allows editing of the current product name.

Step 3:Product Description(Editable Field):Edit the product description.

Step 4:Price(in cents)(Editable Field):Edit the product price.

Step 5:Product Image(Display Field):Shows the current image path.

Step 6:Inventory Quantity(Editable Field):Edit the stock count.

Step 7:Security Auth(Checkbox):Check/uncheck to indicate safety certification.

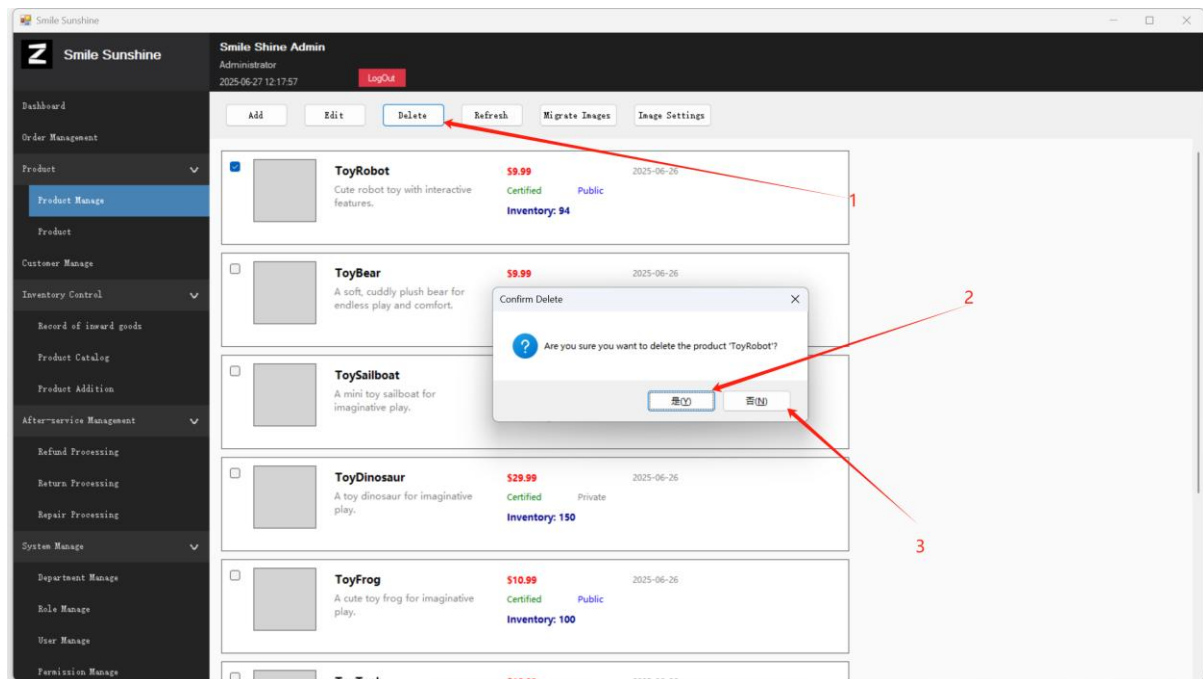
Step 8:Public Products(Radio Buttons):Set product visibility to Public or Private.

Step 9:Design ID(Editable Field):Edit the design reference.

Step 10:Save(Blue Button):Click to save all changes.

Step 11:Cancel(Gray Button):Click to discard changes and close the window.

17.5 Deleting a Product: Steps and Confirmation



Step 1:Delete(Blue Button, Main Interface):Click to open the “Delete Confirmation” popup window(Current product: ToyRobot).

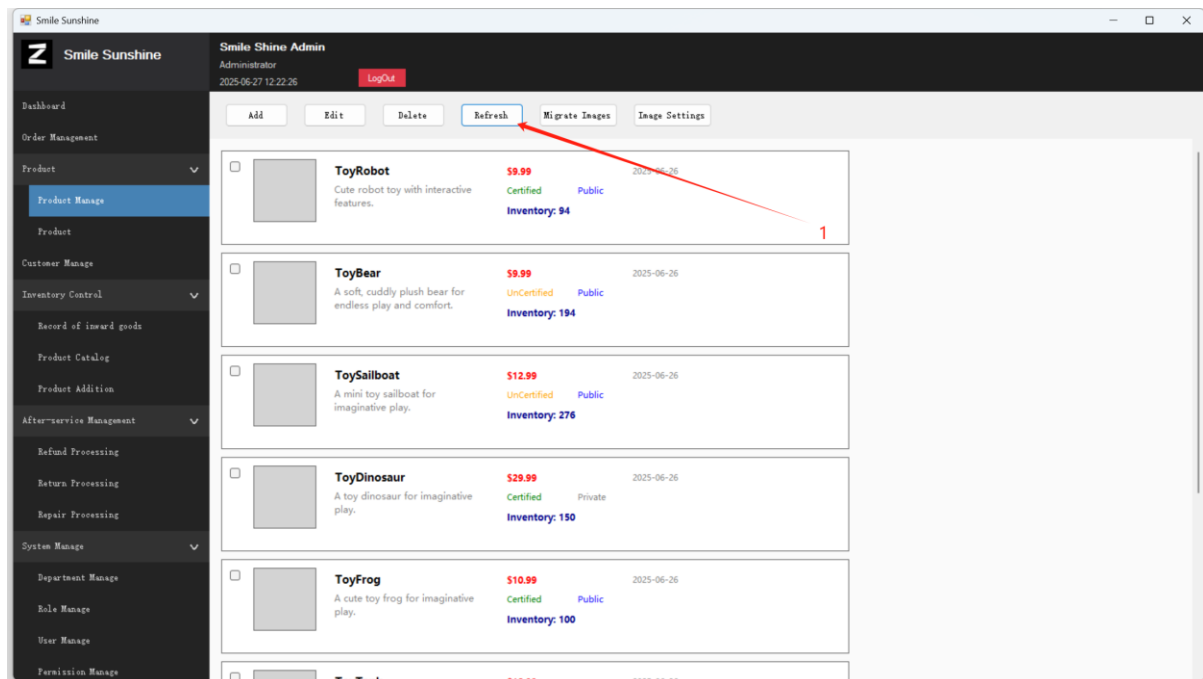
Step 2:Yes(Y)(Red Button): Click to permanently delete the selected product.

- Action: Permanently deletes the selected product.
- Warning: This action cannot be undone.

Step 3:No(N)(Gray Button): Click to close the popup without deleting.

- Action: Closes the popup and returns to the product list view.

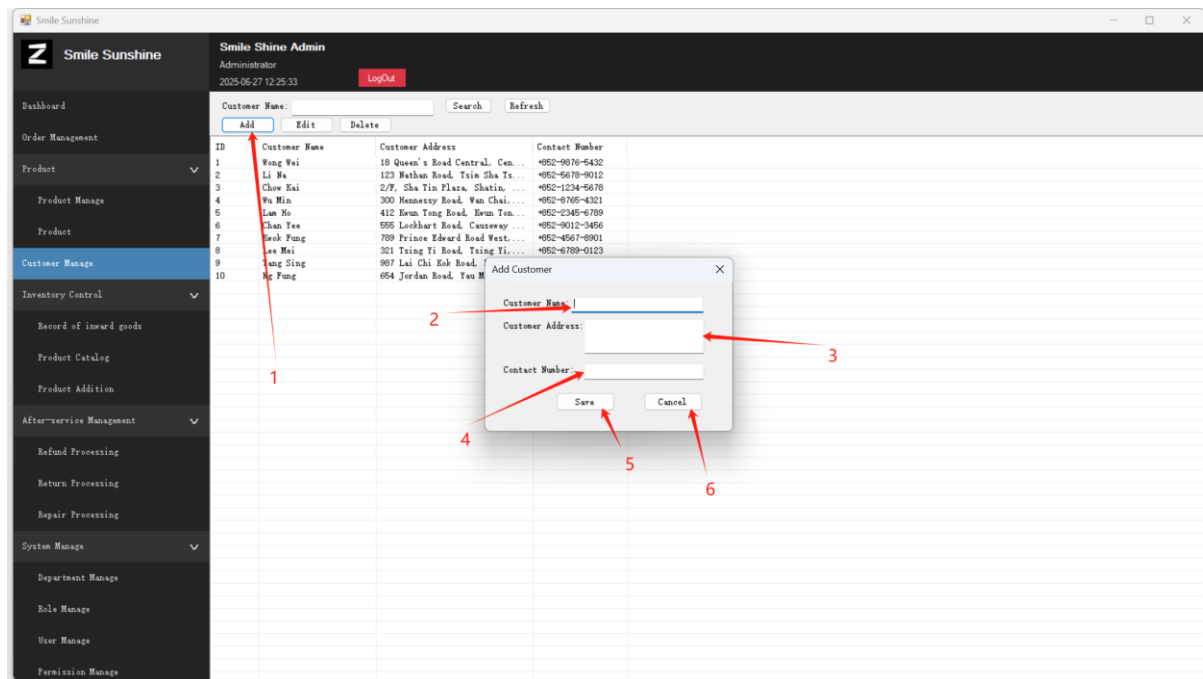
17.6 How to Refresh the Product List and Update Information



Step 1: Refresh (Blue Button, Top Action Bar): Click to reload and update the current product list display.

- Function: Reloads and updates the current product list display.
- Visual Indicator: Red border highlight (as shown in your image).
- Behavior:
 - Fetches latest data from server.
 - Updates all product information (Inventory, Status, etc.).
 - Maintains current sorting/filtering settings.

17.7 How to Add a New Customer: Step-by-Step Guide



Step 1:Add(Blue Button, Main Interface):Click to open the “Add Customer” popup window.

Step 2:Customer Name(Required Field):Enter full customer name(e.g., “Wong Wei”).

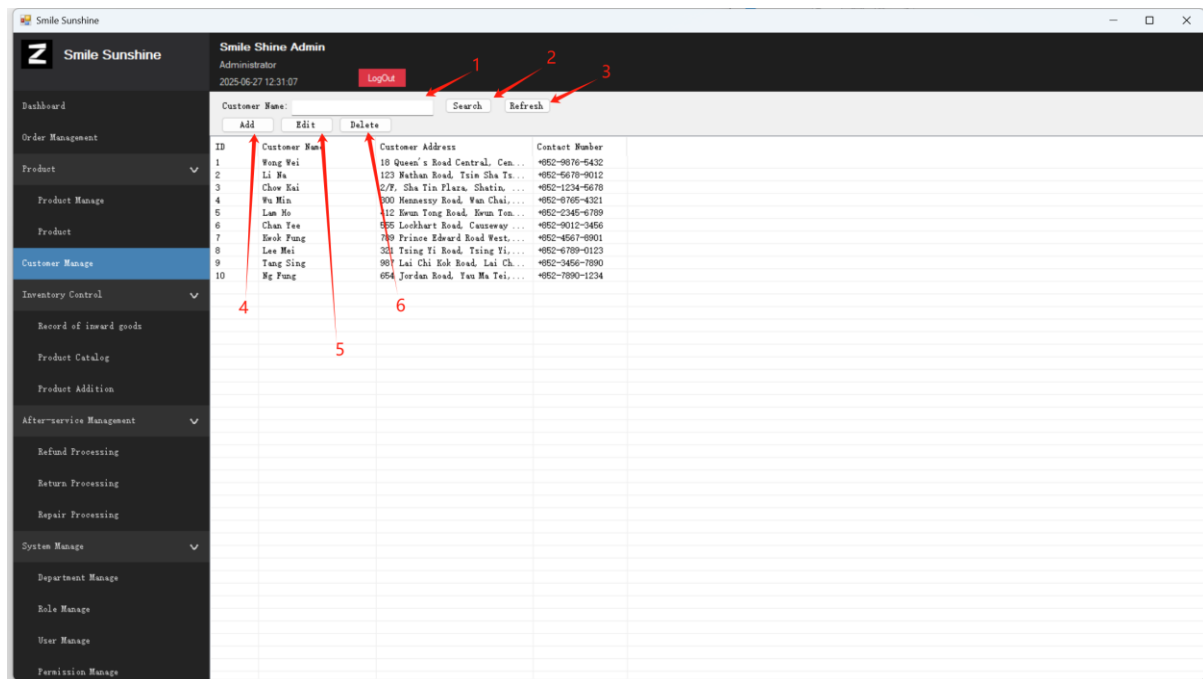
Step 3:Customer Address: Enter complete address(e.g.,“18 Queen’s Road Central”).

Step 4:Contact Number: Enter phone number with area code(e.g.,“+852-9876-5432”).

Step 5:Save(Blue Button):Click to save the new customer.

Step 6:Cancel(Gray Button):Click to discard changes and close the window.

17.8 Customer Management Operations: Search, Add, Edit, and Delete



Step 1:Customer Name(Search Field, Top Right of Customer List):Enter full or partial customer name to filter the list.

- Function: Enter full/partial customer name to filter the list.
- Example: Type “Wong” to find “Wong Wei”.

Step 2:Search(Blue Button, Next to Search Field):Click to execute the name-based search.

- Action: Executes the name-based search.
- Hotkey: Press Enter after typing.

Step 3:Refresh(Gray Button, Next to Search):Click to reload the unfiltered customer list.

- Action: Reloads the unfiltered customer list.

Step 4:Add(Blue Button, Above the Customer Table):Click to open the “Add Customer” popup.

- Function: Opens “Add Customer” popup(as previously described).

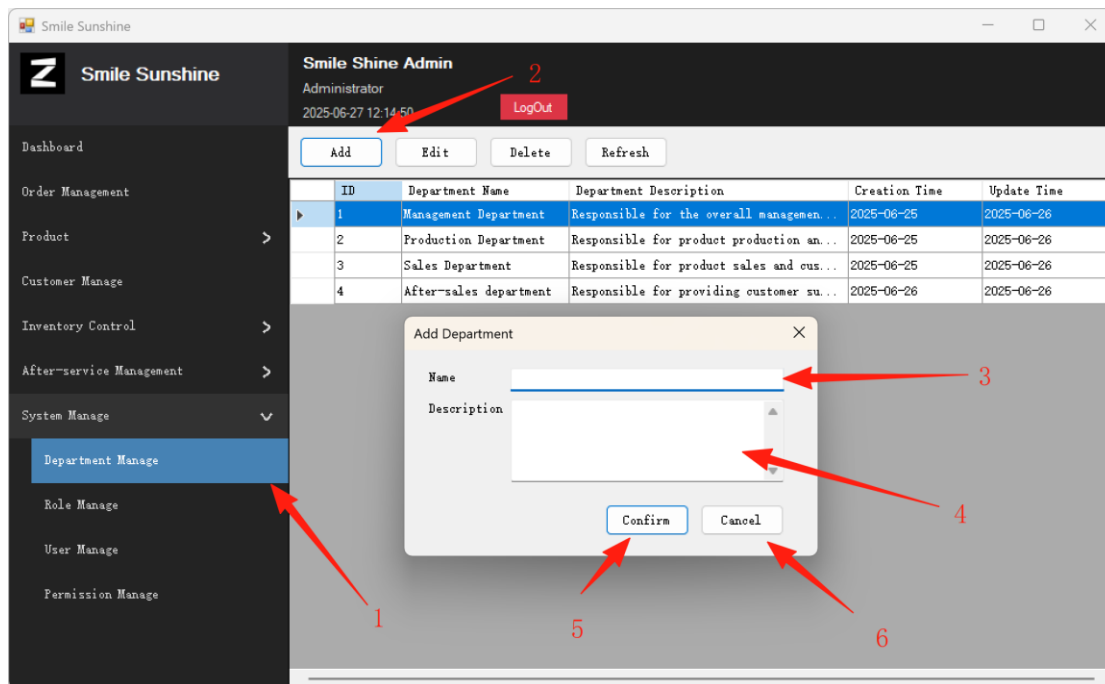
Step 5:Edit(Blue Button, Requires Selecting a Customer First):Click to open the edit window for the selected customer.

- Action: Opens edit window for selected customer.

Step 6:Delete(Red Button, Requires Selection):Click to display confirmation popup before deletion.

- Warning: Displays confirmation popup before deletion.

17.9 How to Add a New Department: Step-by-Step Guide



Step 1: Add (Department Management): Click to open the “Add Department” pop-up window for creating a new department.

Step 2: Department List: Displays a list of existing departments with details such as ID, name, description, creation time, and update time.

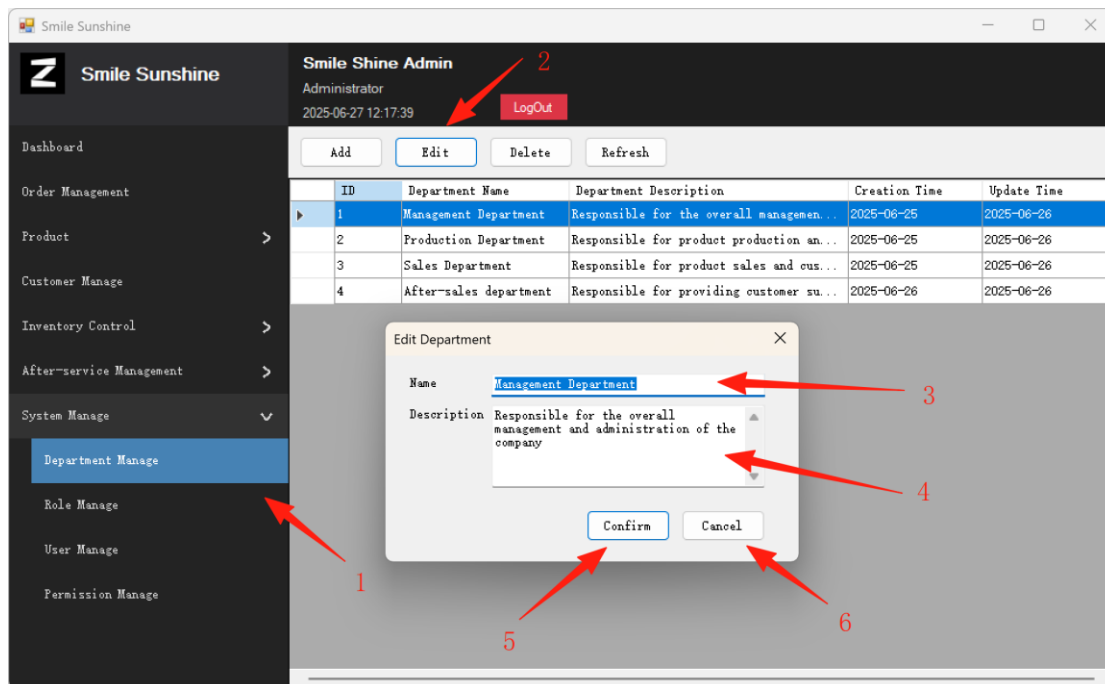
Step 3: Add Department Pop-up: Appears when the “Add” button is clicked, allowing administrators to input new department details.

Step 4: Description (Text Area): Enter a description for the new department in the pop-up window.

Step 5: Confirm (Button): Click to save the new department details.

Step 6: Cancel (Button): Click to discard the input and close the pop-up without saving.

17.10 How to Edit Department Details: Step-by-Step Guide



Step 1: Department Management: Click to access the department management section in the system menu.

Step 2: Edit Button, Top-Right: Click to securely sign out of the admin system.

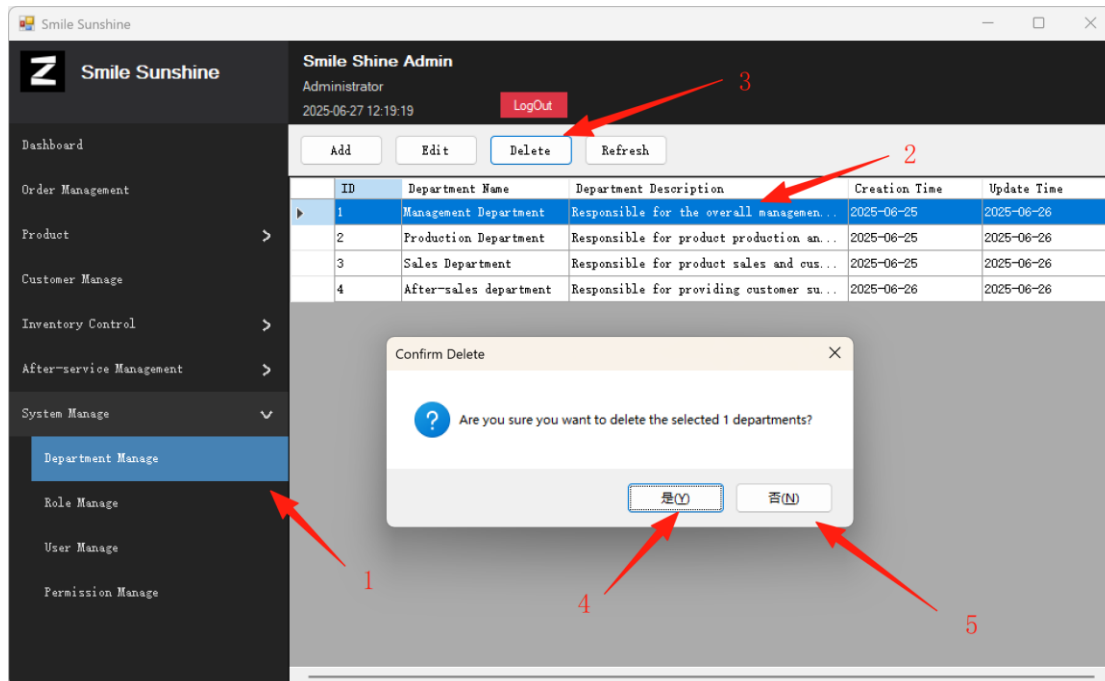
Step 3: Fill the Name: Field in the pop-up window to edit the name of the selected department.

Step 4: Description: Text area in the pop-up window to edit the description of the selected department.

Step 5: Confirm: Button in the pop-up window to save the edited department details.

Step 6: Cancel: Button in the pop-up window to discard the changes and close the pop-up without saving.

17.11 Department Management: Refreshing, Logging Out, and Deleting Departments



Step 1: Department Management: Click to access the department management section in the system menu.

Step 2: Refresh(Department List):Click to refresh the list of departments.

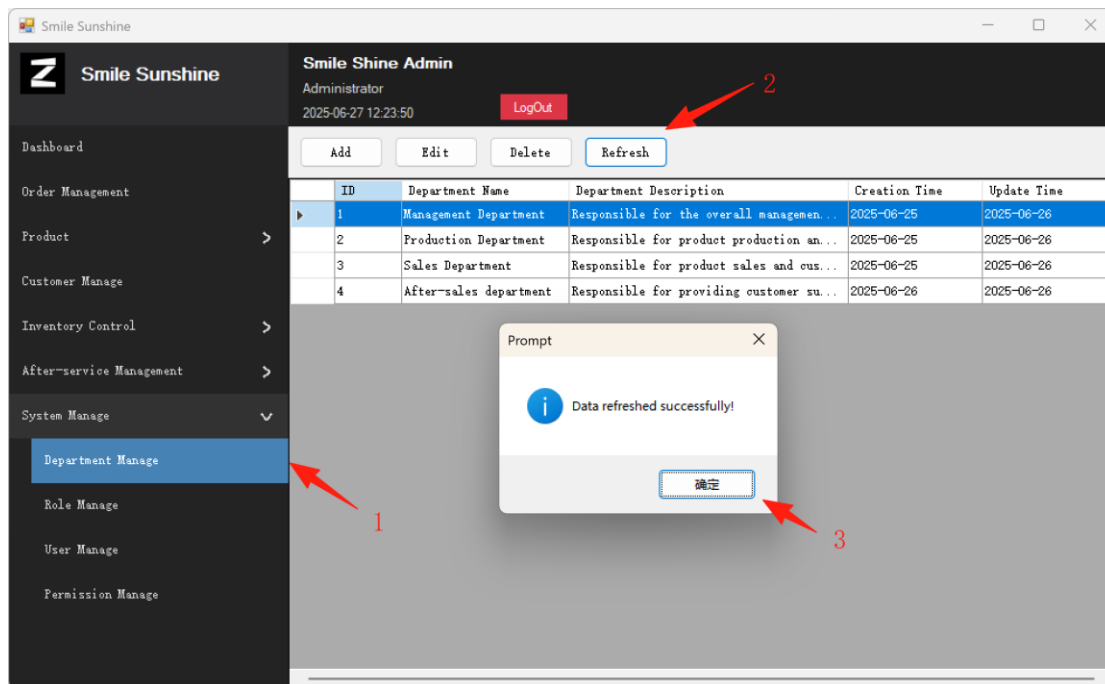
Step 3: LogOut(Red Button,Top-Right):Click to securely sign out of the admin system.

Step 4: Confirm Delete(Dialog Box):Appears when the"Delete"button is clicked,asking for confirmation to delete the selected department.

Step 5:NO(N)(Cancel):Click to cancel the deletion and close the dialog box without deleting the department.

Step 6:YES(Y)(Yes):Click to confirm the deletion of the selected department.

17.12 How to Refresh Department Data and Confirm Updates



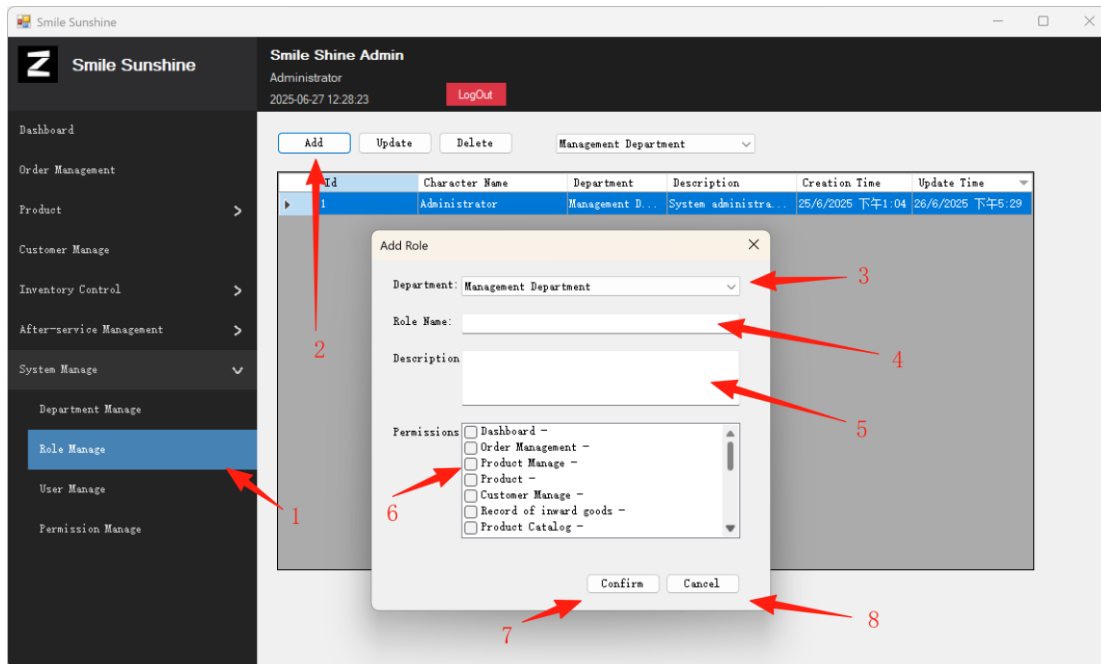
Step 1: Department Management: Click to access the department management section in the system menu.

Step 2: Refresh(Department List): Click to refresh the list of departments, updating any changes made.

Step 3: Prompt(Dialog Box): Appears after refreshing the data, displaying the message "Data refreshed successfully!" to confirm the update.

Step 4: Ensure(Confirm): Click to close the prompt and return to the department management list.

17.13 Creating a New Role: Step-by-Step Guide



Step 1: Role Management: Click to access the role management section in the system menu.

Step 2: Add(Role Management):Click to open the "Add Role" pop-up window for creating a new role.

Step 3: Department: Dropdown menu in the pop-up window to select the department for the new role.

Step 4: Role Name: Field in the pop-up window to enter the name of the new role.

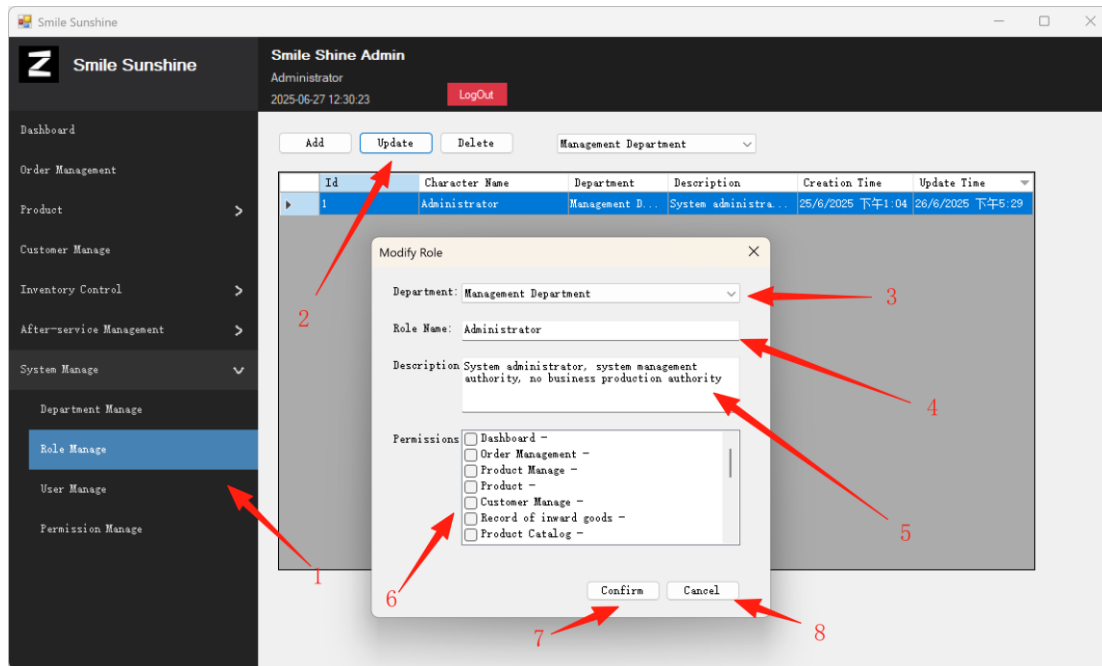
Step 5: Description: Text area in the pop-up window to enter a description for the new role.

Step 6: Permissions: Checkboxes in the pop-up window to select the permissions for the new role.

Step 7: Confirm: Button in the pop-up window to save the new role details.

Step 8: Cancel: Button in the pop-up window to discard the input and close the pop-up without saving.

17.14 How to Update Role Information and Permissions: Step-by-Step Guide



Step 1: Role Management: Click to access the role management section in the system menu.

Step 2: Update(Role Management):Click to open the "Modify Role" pop-up window for editing an existing role.

Step 3: Department: Dropdown menu in the pop-up window to select or confirm the department for the role.

Step 4: Description: Text area in the pop-up window to edit the description of the selected role.

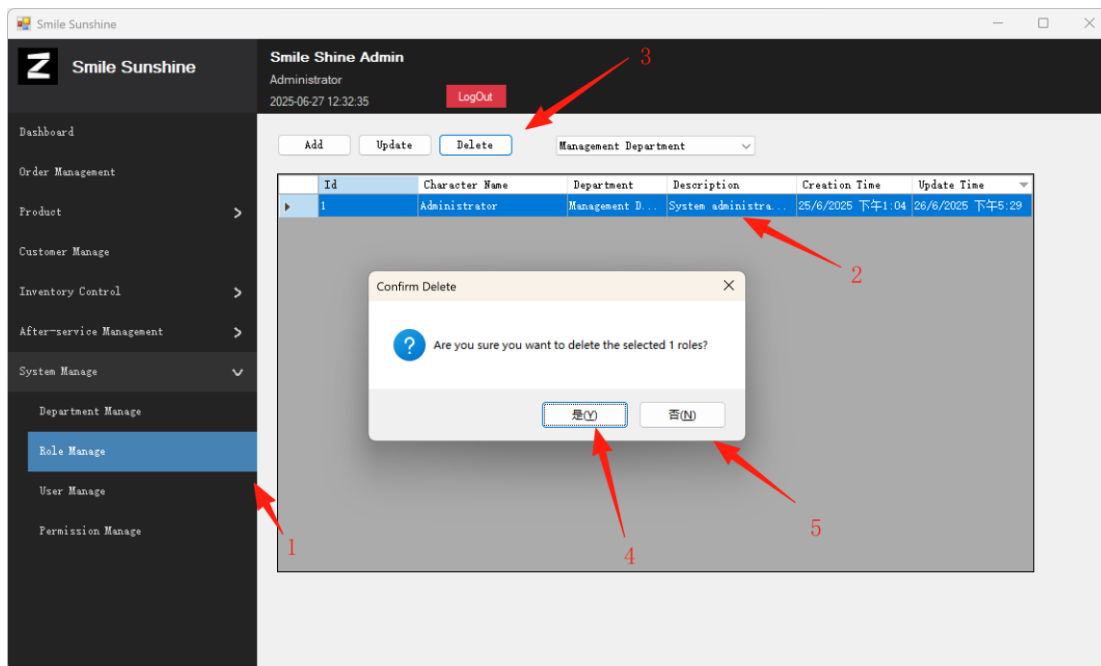
Step 5: Cancel: Button in the pop-up window to discard the changes and close the pop-up without saving.

Step 6: Permissions: Checkboxes in the pop-up window to select or modify the permissions for the role.

Step 7: Confirm: Button in the pop-up window to save the updated role details.

Step 8: Role Name: Field in the pop-up window to edit the name of the selected role.

17.15 Role Management: Deleting a Role and Logging Out



Step 1: Role Management: Click to access the role management section in the system menu.

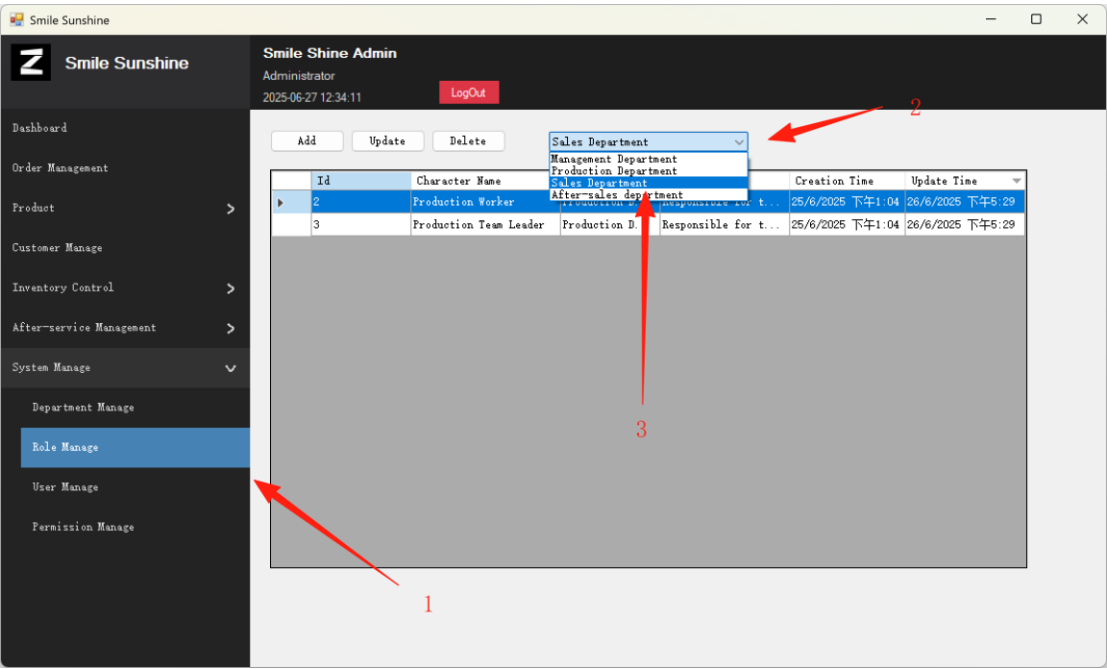
Step 2: Confirm Delete(Dialog Box): Appears when the "Delete" button is clicked, asking for confirmation to delete the selected role.

Step 3: LogOut (Red Button, Top-Right):Click to securely sign out of the admin system.

Step 4: YES(Y)(Yes):Click to confirm the deletion of the selected role.

Step 5: NO(N)(No):Click to cancel the deletion and close the dialog box without deleting the role.

17.16 How to View and Manage Roles by Department

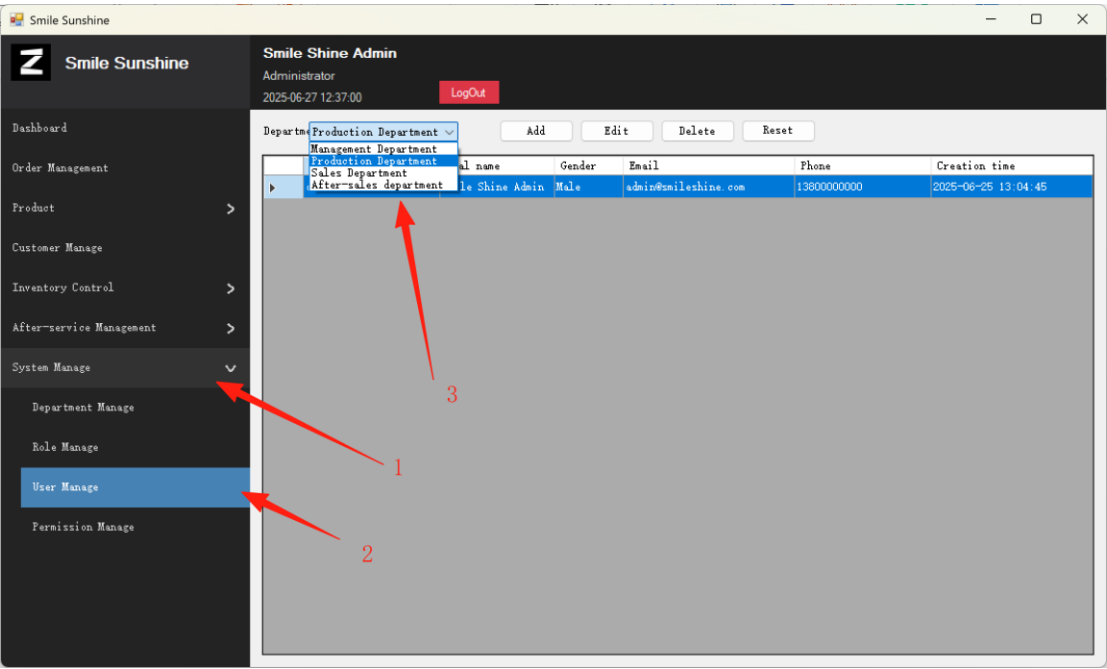


Step 1: Role Management: Click to access the role management section in the system menu.

Step 2: Department Dropdown: Click to select a specific department from the list(e.g., Sales Department, Management Department).

Step 3: Role List: Displays a list of roles associated with the selected department, including details such as ID, character name, department, description, creation time, and update time.

17.17 How to View and Manage Users by Department

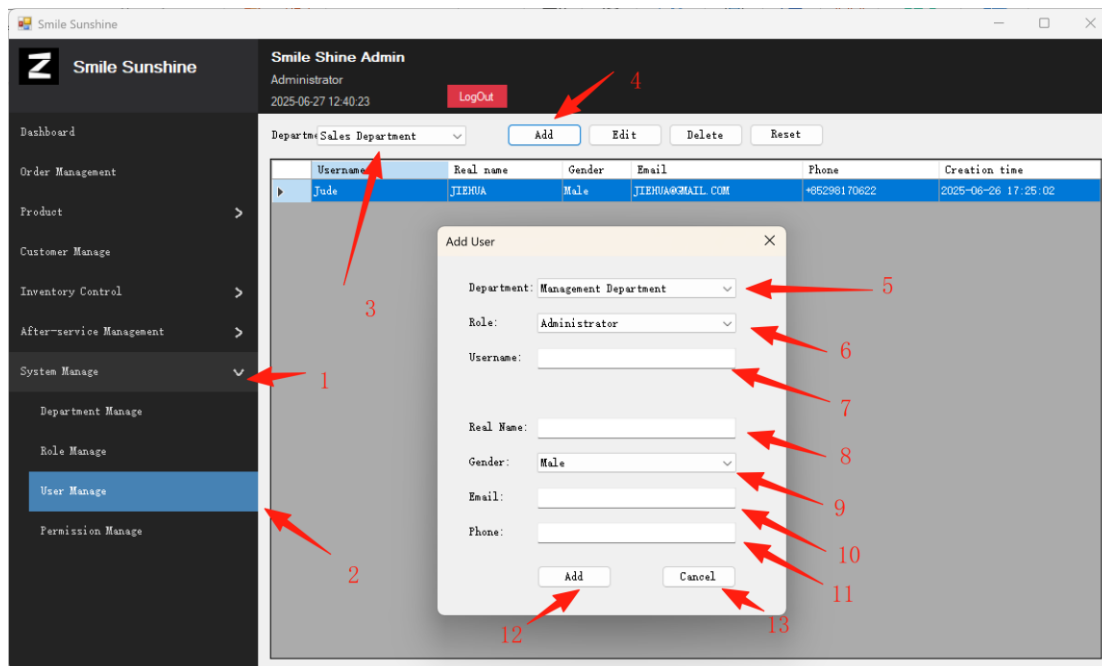


Step 1: User Management: Click to access the user management section in the system menu.

Step 2: Department Dropdown: Click to select a specific department from the list(e.g., Management Department, Production Department).

Step 3: User List: Displays a list of users associated with the selected department, including details such as real name, gender, email, phone, and creation time.

17.18 How to Add a New User Account: Step-by-Step Guide



Step 1: User Management: Click to access the user management section in the system menu.

Step 2: Add(User Management):Click to open the "Add User" pop-up window for creating a new user account.

Step 3: Username: Field in the pop-up window to enter the username for the new user.

Step 4: LogOut (Red Button, Top-Right):Click to securely sign out of the admin system.

Step 5: Department: Dropdown menu in the pop-up window to select the department for the new user.

Step 6: Role: Dropdown menu in the pop-up window to select the role for the new user.

Step 7: Real Name: Field in the pop-up window to enter the real name of the new user.

Step 8: Gender: Dropdown menu in the pop-up window to select the gender of the new user.

Step 9: Email: Field in the pop-up window to enter the email address of the new user.

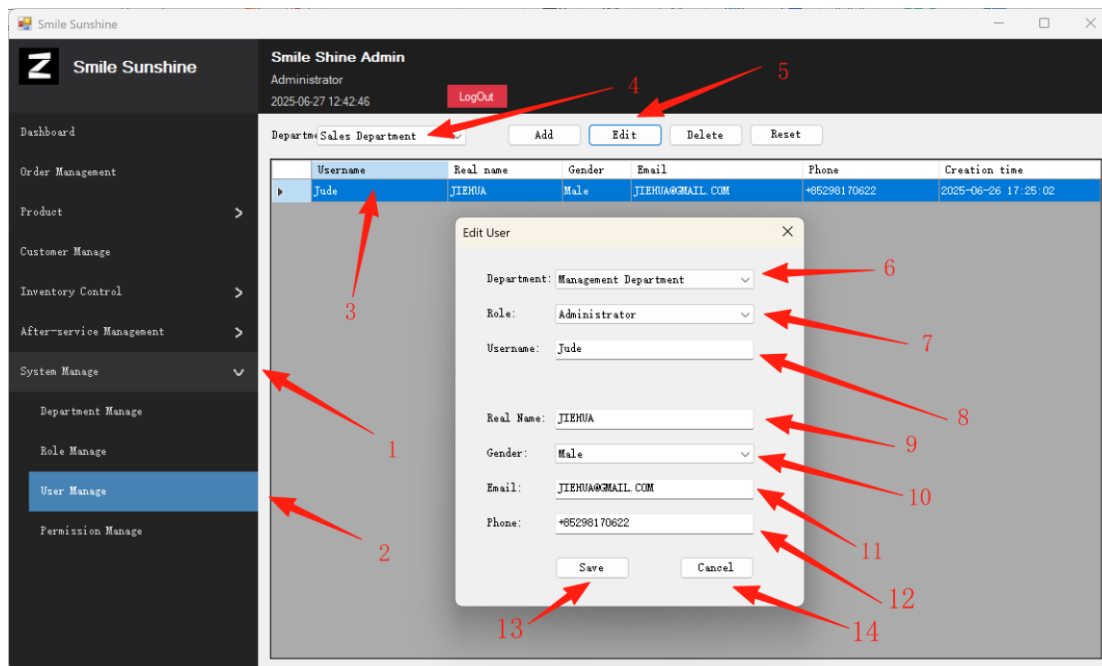
Step 10: Phone: Field in the pop-up window to enter the phone number of the new user.

Step 11: Cancel: Button in the pop-up window to discard the input and close the pop-up without saving.

Step 12: Add: Button in the pop-up window to save the new user details.

Step 13: Confirm: Button in the pop-up window to confirm and save the new user details.

17.19 How to Edit an Existing User Account: Step-by-Step Guide



Step 1: User Management: Click to access the user management section in the system menu.

Step 2: Edit (User Management): Click to open the "Edit User" pop-up window for editing an existing user account.

Step 3: Username: Field in the pop-up window displaying the username of the selected user.

Step 4: LogOut (Red Button, Top-Right): Click to securely sign out of the admin system.

Step 5: Department Dropdown: Dropdown menu in the pop-up window to select or confirm the department for the user.

Step 6: Department: Displays the selected department for the user.

Step 7: Role: Dropdown menu in the pop-up window to select or confirm the role for the user.

Step 8: Real Name: Field in the pop-up window to edit the real name of the selected user.

Step 9: Gender: Dropdown menu in the pop-up window to select or confirm the gender of the selected user.

Step 10: Email: Field in the pop-up window to edit the email address of the selected user.

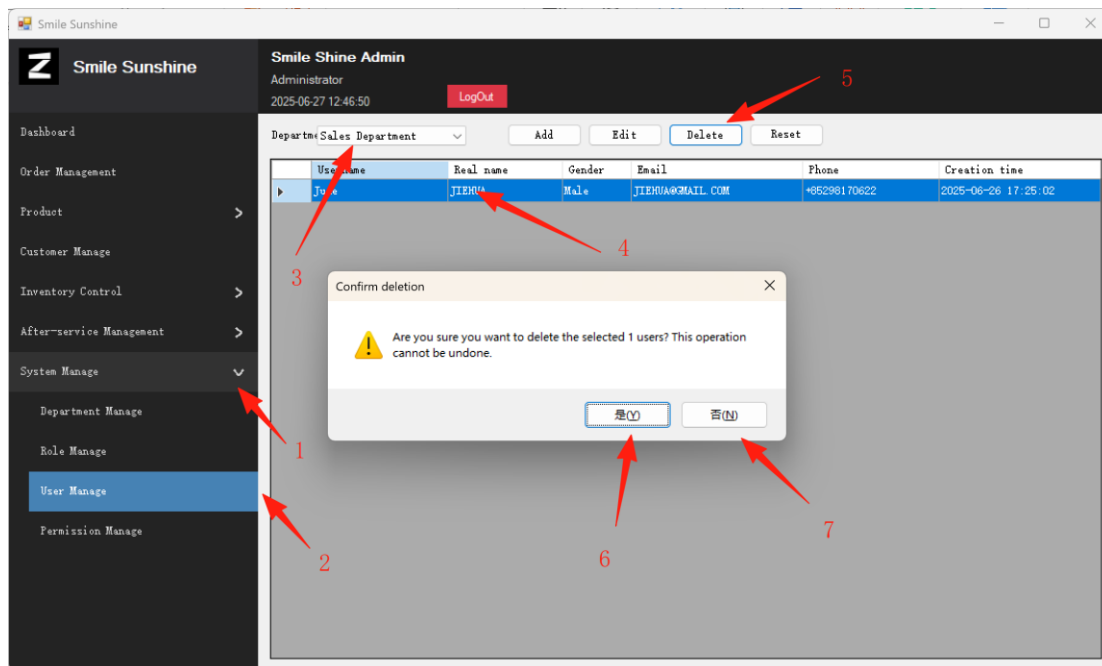
Step 11: Phone: Field in the pop-up window to edit the phone number of the selected user.

Step 12: Cancel: Button in the pop-up window to discard the changes and close the pop-up without saving.

Step 13: Save: Button in the pop-up window to save the updated user details.

Step 14: Confirm: Button in the pop-up window to confirm and save the updated user details.

17.20 How to Delete a User Account: Steps and Confirmation



Step 1: User Management: Click to access the user management section in the system menu.

Step 2: Department Dropdown: Dropdown menu to select a specific department from the list.

Step 3: Username: Displays the username of the selected user in the user list.

Step 4: Real Name: Displays the real name of the selected user in the user list.

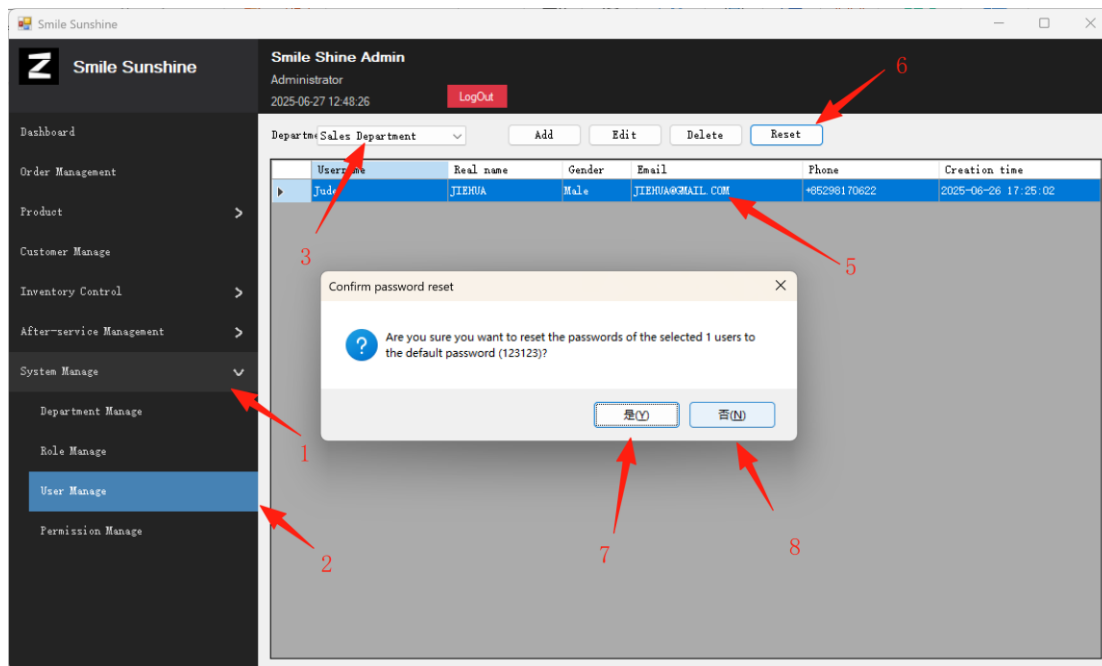
Step 5: Delete(User List):Click to open the "Confirm deletion" dialog box for deleting the selected user.

Step 6: Confirm Deletion(Dialog Box):Appears when the "Delete" button is clicked, asking for confirmation to delete the selected user.

Step 7:NO(N)(Cancel):Click to cancel the deletion and close the dialog box without deleting the user.

Step 8:YES(Y)(Yes):Click to confirm the deletion of the selected user, with a warning that this operation cannot be undone.

17.21 How to Reset a User's Password: Steps and Confirmation



Step 1: User Management: Click to access the user management section in the system menu.

Step 2: Reset(User Management):Click to open the "Confirm password reset" dialog box for resetting the password of the selected user.

Step 3: Username: Displays the username of the selected user in the user list.

Step 4: Real Name: Displays the real name of the selected user in the user list.

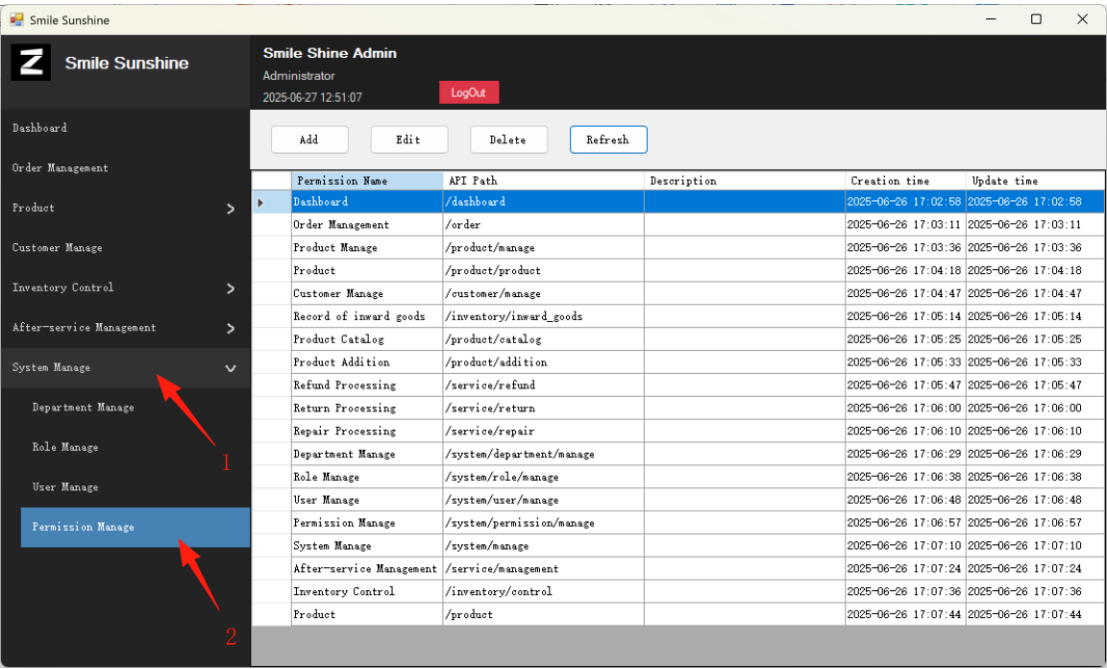
Step 5: Email: Displays the email address of the selected user in the user list.

Step 6: Reset Button: Click to initiate the password reset confirmation dialog.

Step 7: YES(Y)(Yes):Click to confirm resetting the password of the selected user to the default password.

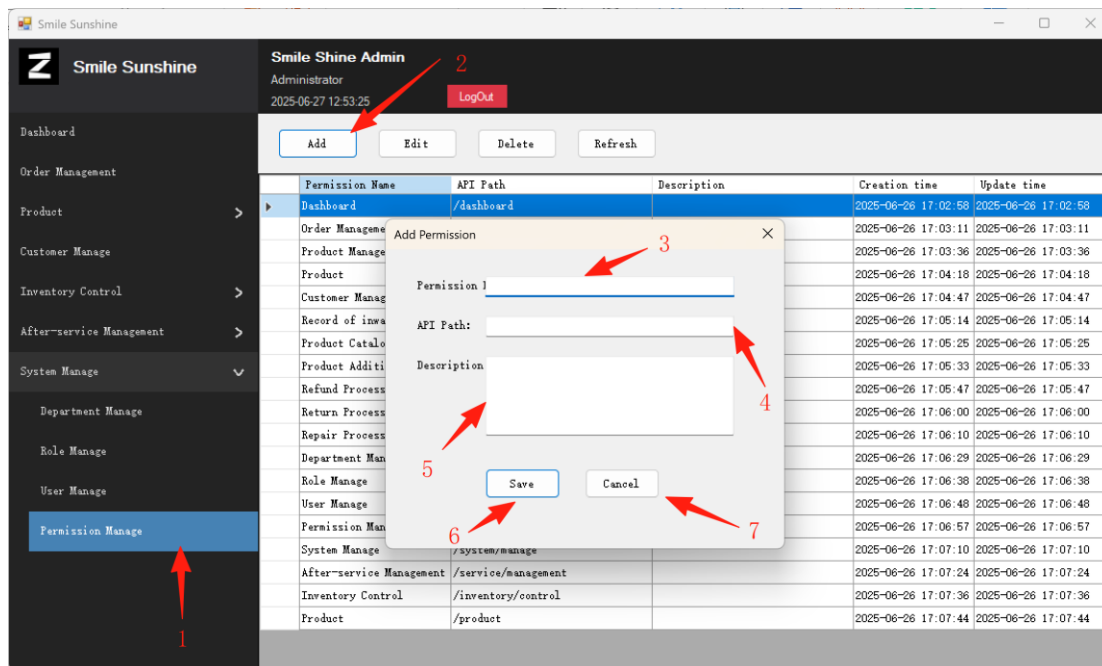
Step 8:NO(N)(No):Click to cancel the password reset and close the dialog box without making changes.

17.22 Accessing Permission Management in the System Menu



Step 1: Permission Management: Click to access the permission management section in the system menu.

17.23 How to Add a New Permission in the System



Step 1: On the left navigation menu, click on System Manage.

From the dropdown, select Permission Manage. This will display the list of existing permissions.

Step 2: At the top of the permission management interface, click the Add button. This will open the "Add Permission" dialog box.

Step 3: Enter a Permission Name in the provided field. This should describe the new permission clearly.

Step 4: Enter the API Path associated with this permission.

Step 5: Provide a Description that explains the purpose of the permission.

Step 6: After filling in all the required fields, click the Save button to add the new permission to the system.

Step 7: If you decide not to add the permission, click the Cancel button to close the dialog box without saving any changes.

18 Project Schedule

N/A

19 Conclusion

When developing a centralized computer management system for Smile & Sunshine Toy Co., Ltd., we encountered many challenges, including those related to data security, user training, ensuring real-time data accuracy, and the difficulty of selecting appropriate programming tools. However, these issues were successfully addressed through careful planning and effective solutions.

First, data security was a key issue, especially given the sensitivity of user data and financial information. Ensuring that the system was protected from unauthorized access and potential security vulnerabilities was our top priority. We implemented strong encryption for data transmission and storage, conducted regular security audits to identify and address vulnerabilities, and integrated role-based access control (RBAC) into the system to ensure that users only access information and functions relevant to their roles.

Second, unnecessary communication between departments should be limited. Excessive communication can easily lead to more detailed errors in the information that really needs to be conveyed.

User training was also a major challenge. Many employees were unfamiliar with the new system, so they had to be equipped with the skills required to use it effectively. To address this, we developed comprehensive training materials, including user guides and video tutorials, and provided hands-on training sessions. This helped shorten the learning curve and enabled employees to adapt to the new system more quickly.

Ensuring real-time accuracy of data, especially in inventory management and logistics tracking modules, is another challenge. Delays or inaccuracies can lead to operational inefficiencies and reduced customer satisfaction. To address this, we use IoT sensors and RFID tags for real-time inventory tracking to obtain the latest cargo information. These technologies ensure data accuracy and timeliness, which improves decision-making and customer service.

We had difficulty choosing the right programming tools. After careful consideration and evaluation of various options, we ultimately decided to use C# and MySQL. C# is a powerful and flexible programming language that is well suited for developing robust and scalable applications. MySQL is a widely used open-source database management system that provides high performance, reliability, and security for storing and managing data. The combination of C# and MySQL enabled us to develop the system efficiently while meeting the complex requirements of Smile & Sunshine Toy Co., Ltd.

Overall, the development of a centralized computer management system was a complex but rewarding project. By addressing challenges related to data security, user training, real-time data accuracy, and choosing the right programming tools, we have significantly improved the operational efficiency and competitiveness of Smile & Sunshine Toy Co., Ltd. The successful implementation of this system not only improved internal processes, but also enhanced customer experience, laying a solid foundation for the company's continued growth and success in the global toy market.

20 Project Log

N/A

21 Appendices

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